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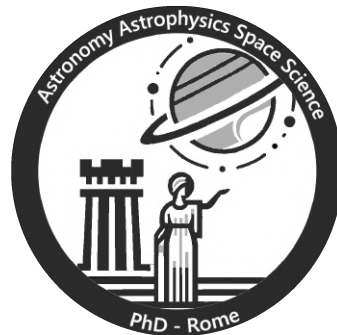
Tracing the properties of  
Lyman continuum emitting galaxies  
over cosmic epochs

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alle radici

che hanno amato  
nutrito  
sostenuto  
i rami

affinché potessero  
raggiungere  
*le stelle*



“Attempting to understand the universe is  
perhaps comparable to doing a great jigsaw puzzle.  
Although possibly it’s as much a mental or intellectual riddle,  
and we may have the answer to it within us.  
We are here, after all.  
We are the universe.”  
Jostein Gaarder



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# Summary

The Epoch of Reionization (EoR) is a critical period in the history of the Universe, occurring roughly during its first billion years. During this time, the hydrogen in the intergalactic medium (IGM) transitioned from a completely neutral to a nearly completely ionized state. This transition was driven by Lyman Continuum (LyC;  $\lambda < 912 \text{ \AA}$ ) radiation emitted by the first luminous sources that formed in the early Universe. However, these sources, the so-called cosmic reionizers, remain elusive. Direct observations of the LyC escape fraction ( $f_{\text{esc}}$ ) during reionization are not possible, as the IGM absorbs these photons before they can reach us. A possible solution is relying on indirect diagnostics – properties of galaxies that correlate with  $f_{\text{esc}}$ , calibrated at lower redshifts where direct LyC detection is possible – to infer  $f_{\text{esc}}$  for galaxies in the early Universe. This thesis aims to investigate the properties of Lyman Continuum emitting galaxies (LCEs) across various cosmic epochs. By doing so, we seek to uncover the mechanisms that facilitated the escape of ionizing photons, and to characterize the nature of the sources that contributed most significantly to the reionization of the IGM.

We first explored the role of CIV emission as a predictor of  $f_{\text{esc}}$  in galaxies at  $z \sim 3$ , using data from the VANDELS ESO public spectroscopic survey. Our results indicated that CIV emitters are characterized by high photon production efficiency ( $\xi_{\text{ion}}$ ) and likely exhibit a large escape fraction of ionizing photons, suggesting that CIV emission can be used to identify analogs of cosmic reionizers at higher redshifts.

We further extended our analysis of LCEs over cosmic time by studying extreme [OIII] emitters at high redshift from the GLASS-JWST program, using JWST/NIRISS observations. These galaxies are dominated by young stellar populations that generate intense ionizing radiation, contributing significantly to the cosmic ionizing budget.

From the same program, we also studied a small sample of galaxies at  $z \geq 4.5$ . We found that many of these galaxies share similar physical properties as the low-redshift LyC leakers, as they are often compact star-forming galaxies with blue UV slopes, reinforcing their potential role as cosmic reionizers.

We then analyzed a larger sample of star-forming galaxies at  $6 \leq z \leq 9$  from the CEERS survey, identifying a population of faint, low-mass galaxies with compact sizes and blue UV slopes. The predicted  $f_{\text{esc}}$  for these galaxies are moderate (less than 0.1). By calculating the total ionizing budget, we found that these galaxies contributed less than one-third to the overall reionization. This supports the idea that faint galaxies, which have not yet been fully characterized with JWST, likely played a key role in reionizing the Universe.

Finally, we examined the impact of mergers and galaxy morphology on LyC escape by combining multiple public JWST datasets, and assembling a large sample of more than 400 galaxies. Our findings reveal that most of the sources are compact and symmetric galaxies, indicating a lack of correlation between morphology and LyC leakage. This challenges previous models that emphasized the role of mergers in enhancing LyC emission.

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# Sommario

L'Epoca della Reionizzazione (EoR) è un periodo cruciale nella storia dell'universo, avvenuto approssimativamente durante il suo primo miliardo di anni. In questa fase, l'idrogeno contenuto nel mezzo intergalattico (IGM) è passato da uno stato quasi completamente neutro a uno quasi completamente ionizzato, a causa della radiazione del Continuum di Lyman (LyC;  $\lambda < 912 \text{ \AA}$ ) emessa dalle prime sorgenti luminose formatesi nell'universo primordiale. Tuttavia, queste sorgenti, i cosiddetti reionizzatori cosmici, rimangono ad oggi un mistero. Le osservazioni dirette della frazione di fuga del LyC ( $f_{\text{esc}}$ ) durante la reionizzazione non sono possibili, poiché l'IGM neutro assorbe questi fotoni prima che possano raggiungerci. Pertanto, per stimare  $f_{\text{esc}}$  nelle galassie dell'universo primordiale, ci si affida a diagnostiche indirette, ovvero proprietà delle galassie che correlano con  $f_{\text{esc}}$  calibrate a redshift più bassi, dove è possibile la rilevazione diretta del LyC. Questa tesi mira a investigare le proprietà delle galassie che emettono Continuum di Lyman (LCE) in diverse epoche cosmiche, cercando di comprendere i meccanismi che hanno facilitato la fuga dei fotoni ionizzanti e caratterizzare le sorgenti che hanno contribuito maggiormente alla reionizzazione dell'IGM.

Abbiamo inizialmente esplorato il ruolo dell'emissione di CIV come predittore di  $f_{\text{esc}}$  nelle galassie a  $z \sim 3$ , utilizzando i dati del programma pubblico VANDELs dell'ESO. I risultati indicano che le galassie con emissione di CIV mostrano un'alta efficienza di produzione di fotoni ( $\xi_{\text{ion}}$ ) e probabilmente un'elevata frazione di fuga di fotoni ionizzanti, suggerendo che l'emissione di CIV possa essere utilizzata per identificare analoghi dei reionizzatori cosmici a redshift più elevati.

Abbiamo poi esteso l'analisi delle LCE studiando emettitori estremi di [OIII] ad alto redshift provenienti dal programma GLASS-JWST, utilizzando osservazioni JWST/NIRISS. Queste galassie, dominate da giovani popolazioni stellari, generano intense radiazioni ionizzanti e contribuiscono significativamente al budget ionizzante cosmico.

Dallo stesso programma, abbiamo inoltre analizzato un piccolo campione di galassie a  $z \geq 4.5$ . Molte di queste galassie condividono proprietà fisiche simili a quelle delle LCE a basso redshift: per lo più galassie compatte, a bassa massa, con formazione stellare attiva e pendenze UV blu, caratteristiche che suggeriscono il loro potenziale ruolo come reionizzatori cosmici.

Successivamente, abbiamo analizzato un campione più ampio di galassie a formazione stellare a  $6 \leq z \leq 9$  provenienti dal programma CEERS, identificando una popolazione di galassie deboli e di bassa massa, con dimensioni compatte e pendenze UV blu. Abbiamo stimato il loro  $f_{\text{esc}}$ , scoprendo che questi valori sono generalmente moderati ( $f_{\text{esc}} < 0.1$ ). Calcolando il loro contributo al budget ioniz-

zante complessivo, abbiamo constatato che queste galassie hanno contribuito per meno di un terzo alla reionizzazione totale, suggerendo che le galassie deboli, non ancora pienamente caratterizzate con il JWST, abbiano probabilmente svolto un ruolo chiave nella reionizzazione dell'universo.

Infine, abbiamo esaminato l'impatto delle fusioni e della morfologia delle galassie sulla fuga del LyC, combinando dati provenienti da diversi programmi pubblici del JWST e assemblando un campione di oltre 400 sorgenti. I nostri risultati mostrano che la maggior parte di queste galassie sono compatte e simmetriche, indicando una mancanza di correlazione tra morfologia e fuga di LyC. Questi risultati sfidano i modelli precedenti, che enfatizzavano il ruolo delle fusioni nell'aumentare l'emissione di LyC.

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# Contents

|          |                                                                                                  |           |
|----------|--------------------------------------------------------------------------------------------------|-----------|
| <b>1</b> | <b>Introduction</b>                                                                              | <b>1</b>  |
| 1.1      | A short history of nearly everything . . . . .                                                   | 1         |
| 1.2      | The evolution of the intergalactic medium . . . . .                                              | 3         |
| 1.2.1    | Constraints on the evolution of the IGM during the EoR . . .                                     | 6         |
| 1.2.2    | Theoretical tools for probing the EoR . . . . .                                                  | 8         |
| 1.3      | The nature of the cosmic reionizers . . . . .                                                    | 8         |
| 1.3.1    | Active Galactic Nuclei . . . . .                                                                 | 8         |
| 1.3.2    | Star-forming galaxies . . . . .                                                                  | 10        |
| 1.4      | The total ionizing budget of star-forming galaxies . . . . .                                     | 11        |
| 1.4.1    | The UV luminosity density ( $\rho_{\text{UV}}$ ) . . . . .                                       | 11        |
| 1.4.2    | The ionizing photon production efficiency ( $\xi_{\text{ion}}$ ) . . . . .                       | 13        |
| 1.4.3    | The escape fraction of ionizing photons ( $f_{\text{esc}}$ ) . . . . .                           | 15        |
| 1.4.4    | The IGM transmission . . . . .                                                                   | 16        |
| <b>2</b> | <b>The escape of Lyman continuum photon from galaxies</b>                                        | <b>19</b> |
| 2.1      | Mechanisms of LyC photons escape . . . . .                                                       | 19        |
| 2.2      | Measuring the LyC escape fraction . . . . .                                                      | 22        |
| 2.3      | Direct measurements of Lyman Continuum escape . . . . .                                          | 23        |
| 2.3.1    | Local LyC emitters . . . . .                                                                     | 23        |
| 2.3.2    | Intermediate-redshift LyC emitters . . . . .                                                     | 25        |
| 2.4      | The “indirect indicators” of the LyC escape fraction . . . . .                                   | 26        |
| 2.4.1    | Ly $\alpha$ line . . . . .                                                                       | 27        |
| 2.4.2    | Other rest-frame UV lines . . . . .                                                              | 29        |
| 2.4.3    | Nebular properties: the O32 emission line ratio . . . . .                                        | 29        |
| 2.4.4    | Morphology . . . . .                                                                             | 31        |
| 2.4.5    | Luminosity and stellar mass . . . . .                                                            | 31        |
| 2.4.6    | Dust attenuation . . . . .                                                                       | 32        |
| 2.5      | Multivariate predictors of the LyC escape fraction . . . . .                                     | 33        |
| 2.6      | Thesis goals and structure . . . . .                                                             | 37        |
| <b>3</b> | <b>CIV emission as a predictor of LyC escape in cosmic-noon galaxies from the VANDELS survey</b> | <b>39</b> |
| 3.1      | Introduction . . . . .                                                                           | 39        |
| 3.2      | The VANDELS CIV emitters . . . . .                                                               | 41        |
| 3.2.1    | Sample selection . . . . .                                                                       | 41        |
| 3.2.2    | Identification of CIV emitters . . . . .                                                         | 42        |
| 3.2.3    | Removal of X-ray AGN/BLAGN and final sample . . . . .                                            | 43        |

|          |                                                                                                                                     |           |
|----------|-------------------------------------------------------------------------------------------------------------------------------------|-----------|
| 3.3      | Spectroscopic properties of the CIV emitters . . . . .                                                                              | 46        |
| 3.4      | Classification of SF and AGN based on UV emission lines . . . . .                                                                   | 48        |
| 3.4.1    | AGN models: Feltre et al. (2016) . . . . .                                                                                          | 48        |
| 3.4.2    | SF models: Gutkin et al. (2016) and Xiao et al. (2018) . . . . .                                                                    | 49        |
| 3.4.3    | Separating SF galaxies from AGN in the UV-BPTs . . . . .                                                                            | 49        |
| 3.4.4    | Spectral stacking . . . . .                                                                                                         | 52        |
| 3.4.5    | Nature of the possible AGNs . . . . .                                                                                               | 53        |
| 3.5      | Physical properties of the SF CIV emitters . . . . .                                                                                | 53        |
| 3.5.1    | The BEAGLE tool . . . . .                                                                                                           | 55        |
| 3.5.2    | Results and comparison to parent population . . . . .                                                                               | 56        |
| 3.5.3    | Average stellar metallicity . . . . .                                                                                               | 57        |
| 3.6      | Discussion: The ionizing properties of CIV emitters . . . . .                                                                       | 59        |
| 3.7      | Summary and conclusions . . . . .                                                                                                   | 62        |
| 3.8      | BEAGLE parameters . . . . .                                                                                                         | 63        |
| <b>4</b> | <b>Characterizing extreme [OIII] emitters in high-redshift galaxies with JWST/NIRISS</b>                                            | <b>65</b> |
| 4.1      | Introduction . . . . .                                                                                                              | 65        |
| 4.2      | JWST/NIRISS imaging and spectroscopy of HFF A2744 . . . . .                                                                         | 67        |
| 4.3      | NIRISS detection of Rest-Optical emission lines in individual sources . . . . .                                                     | 67        |
| 4.4      | Determination of rest-Optical equivalent widths . . . . .                                                                           | 71        |
| 4.5      | Identification of Extreme Emission Line Galaxies with NIRISS WFSS . . . . .                                                         | 73        |
| 4.6      | Conclusions . . . . .                                                                                                               | 75        |
| <b>5</b> | <b>Identifying LyC leakers at <math>z \geq 4.5</math>: first results from the GLASS-JWST program</b>                                | <b>77</b> |
| 5.1      | Introduction . . . . .                                                                                                              | 77        |
| 5.2      | Observations and method . . . . .                                                                                                   | 79        |
| 5.2.1    | JWST/NIRSpec MSA observations and data reduction . . . . .                                                                          | 79        |
| 5.2.2    | Imaging data . . . . .                                                                                                              | 81        |
| 5.2.3    | Emission line and redshift identifications . . . . .                                                                                | 81        |
| 5.2.4    | Dust correction and flux measurements . . . . .                                                                                     | 82        |
| 5.2.5    | Measurements of physical parameters . . . . .                                                                                       | 83        |
| 5.2.6    | Sizes and SFR surface density estimates . . . . .                                                                                   | 84        |
| 5.2.7    | UV- $\beta$ slopes . . . . .                                                                                                        | 85        |
| 5.3      | Results . . . . .                                                                                                                   | 86        |
| 5.3.1    | Possible AGN contamination . . . . .                                                                                                | 86        |
| 5.3.2    | LyC indirect diagnostics . . . . .                                                                                                  | 88        |
| 5.3.3    | O32-R23 diagnostics . . . . .                                                                                                       | 90        |
| 5.4      | Predicting escape fractions of EoR galaxies . . . . .                                                                               | 91        |
| 5.5      | Summary and conclusions . . . . .                                                                                                   | 94        |
| <b>6</b> | <b>The role of faint, low-mass galaxies in cosmic reionization: analyzing LyC escape indicators at <math>6 \leq z \leq 9</math></b> | <b>95</b> |
| 6.1      | Introduction . . . . .                                                                                                              | 95        |
| 6.2      | Data . . . . .                                                                                                                      | 97        |
| 6.2.1    | CEERS-JWST data . . . . .                                                                                                           | 97        |

|          |                                                                                                            |            |
|----------|------------------------------------------------------------------------------------------------------------|------------|
| 6.2.2    | Data from other programs . . . . .                                                                         | 98         |
| 6.3      | Method . . . . .                                                                                           | 99         |
| 6.3.1    | Measurements of physical parameters . . . . .                                                              | 99         |
| 6.3.2    | Dust correction and emission line flux measurements . . . . .                                              | 100        |
| 6.3.3    | UV- $\beta$ slopes . . . . .                                                                               | 100        |
| 6.3.4    | UV half-light radii . . . . .                                                                              | 102        |
| 6.3.5    | AGN contamination . . . . .                                                                                | 103        |
| 6.4      | Results . . . . .                                                                                          | 104        |
| 6.4.1    | Evaluating $f_{\text{esc}}$ . . . . .                                                                      | 104        |
| 6.4.2    | $f_{\text{esc}}$ dependencies . . . . .                                                                    | 105        |
| 6.4.3    | Redshift evolution . . . . .                                                                               | 109        |
| 6.4.4    | Extreme LyC emitters . . . . .                                                                             | 110        |
| 6.4.5    | Ionizing photon production efficiency . . . . .                                                            | 111        |
| 6.5      | The ionizing photon production of bright and faint sources . . . . .                                       | 112        |
| 6.6      | Summary and conclusions . . . . .                                                                          | 114        |
| 6.7      | Appendix A: Properties of the sources . . . . .                                                            | 115        |
| 6.8      | Appendix B: An analysis on the empirical relation calibrated on the<br>LzLCS sample . . . . .              | 119        |
| <b>7</b> | <b>The impact of morphology on the production and escape of<br/>ionizing photons in the early Universe</b> | <b>121</b> |
| 7.1      | Introduction . . . . .                                                                                     | 121        |
| 7.2      | Data . . . . .                                                                                             | 124        |
| 7.2.1    | EIGER data . . . . .                                                                                       | 124        |
| 7.2.2    | CEERS and DD-2750 . . . . .                                                                                | 125        |
| 7.2.3    | JADES . . . . .                                                                                            | 127        |
| 7.3      | Methods . . . . .                                                                                          | 128        |
| 7.3.1    | SED fitting . . . . .                                                                                      | 128        |
| 7.3.2    | Line measurements and $\xi_{\text{ion}}$ estimation . . . . .                                              | 128        |
| 7.3.3    | Mergers identification . . . . .                                                                           | 129        |
| 7.4      | A revised model to infer $f_{\text{esc}}$ using low-redshift analogs . . . . .                             | 132        |
| 7.4.1    | Identifying LzLCS+ analogs of EoR sources . . . . .                                                        | 132        |
| 7.4.2    | A revised Cox model to predict $f_{\text{esc}}$ during the EoR . . . . .                                   | 134        |
| 7.4.3    | Applying the model to known leakers at $z = 3$ . . . . .                                                   | 136        |
| 7.5      | Results . . . . .                                                                                          | 138        |
| 7.5.1    | The predicted $f_{\text{esc}}$ values . . . . .                                                            | 138        |
| 7.5.2    | Merger fraction versus $f_{\text{esc}}$ . . . . .                                                          | 139        |
| 7.5.3    | Merger fraction versus $\xi_{\text{ion}}$ . . . . .                                                        | 141        |
| 7.6      | Summary and Conclusions . . . . .                                                                          | 142        |
| <b>8</b> | <b>Conclusions and future perspectives</b>                                                                 | <b>147</b> |
|          | <b>List of Publications</b>                                                                                | <b>151</b> |
|          | <b>Bibliography</b>                                                                                        | <b>155</b> |





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# Chapter 1

## Introduction

### 1.1 A short history of nearly everything

There are endless ways to describe the origin and evolution of our Universe. We could delve into how humanity discovered the few things we now know for sure, or we could narrate it like a timeless story, one that we might read to a child. This latter approach is the one I prefer, as it reflects my own journey. When I decided to become an astronomer at the age of five, it was this sense of wonder and fascination with the Universe that captured my curiosity. I remember falling in love with the cosmos through a marvelous tale my father told me.

To understand this tale, however, we need some background: we should start with a principle to which all astronomers agree – *the Universe is homogeneous and isotropic*. This implies that, on a large scale, the Universe looks the same no matter where you are or in which direction you look. This principle, called the *Cosmological Principle*, is fundamental to our understanding of the Universe. Thanks to it, we can study the evolution of galaxy properties without tracking a single galaxy throughout its lifetime. This would be impossible since the lifetime of a galaxy far exceeds that of a human or even our solar system. The cosmological principle implies that a galaxy billions of light-years away has properties similar to those of our galaxy or any other galaxy in the Universe.

And so, our story begins with: once upon a time there was nothing. Then a bang occurred – what we now call the *Big Bang*. Approximately 13.8 billion years ago ( $z \sim \infty$ ), from a state of infinite density and temperature, the Universe exploded into existence, marking the beginning of time and space as we know them. This event was followed by a rapid and exponential expansion phase known as *inflation* [186], which increased the Universe’s size by approximately 70 orders of magnitude, smoothing out irregularities and making the early Universe’s curvature more uniform. The model we refer to when discussing the Universe’s evolution is the  $\Lambda$  Cold Dark Matter ( $\Lambda$ CDM) model [187]. According to it, after inflation, the Universe’s first moments were dominated by *nucleosynthesis*, during which the initial atomic nuclei – hydrogen (H), helium (He), lithium (Li), and other isotopes – were formed. During the Big Bang nucleosynthesis, the main characters of our story, matter and light, were tightly coupled through scattering equilibrium. This means that photons were not free to travel in the Universe, as

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they scattered off charged particles, rendering the Universe opaque to radiation. At that time, it was just a dense, hot plasma.

This presents a problem for astronomers, as our knowledge of the Universe primarily comes from the light we receive from it. If matter and light are coupled, we don't receive any photons from that time, so we can only theorize about what was happening.

After the nucleosynthesis, the Universe entered a phase known as the *Recombination Epoch* [362], approximately 380,000 years after the Big Bang ( $z \sim 1090$ ). During this epoch, the Universe cooled sufficiently for protons and electrons to combine, forming neutral hydrogen atoms (HI) that filled the Universe, creating what is known as the *Intergalactic Medium* (IGM). This transformation allowed photons to travel freely through space for the first time, marking the decoupling of matter and radiation and making the Universe, as we said, “transparent” to radiation [e.g., 23]. Today, we observe the remnants of this early radiation as the Cosmic Microwave Background (CMB) radiation, a nearly uniform signal detected in all directions of the sky.

However, following recombination, the Universe was not illuminated as one might expect. Post-recombination, the Universe was filled with HI, so it was hot, dense, and dark. This is why this period is called *the Dark Ages*. During this time, the dark matter contained in the Universe, which we now believe is part of each astronomical structure, started to collapse, leading to the formation of primordial dark matter halos. These halos grew over time through gravitational attraction and eventually merged, forming larger structures. This process led to the gravitational collapse of the baryonic matter as well, so primarily H and He began to collapse. Baryons fell into the gravitational potential wells created by the dark matter overdensities, marking the onset of structure formation at cosmological scales.

The baryonic collapse corresponds to the formation of the very early stars, often referred to as Population III (PopIII) stars. The ignition of hydrogen nuclear fusion reactions in these stars occurred within dense gas clouds where gravitational forces overcame thermal pressure, facilitated by efficient cooling mechanisms such as the presence of  $H_2$  molecules.

PopIII stars were distinct from later generations in several key aspects. They formed from pristine gas composed primarily of H and He. As a result, they were extraordinarily massive, potentially reaching hundreds of times the mass of the Sun. Their short, intense lives ended in supernova explosions that enriched the IGM with the first heavy metals, contributing to the chemical enrichment of subsequent generations of stars and galaxies. This process made later stellar populations more similar to our Sun and the Milky Way. The formation of these stars marked the end of the Dark Ages and the beginning of the *Cosmic Dawn*.

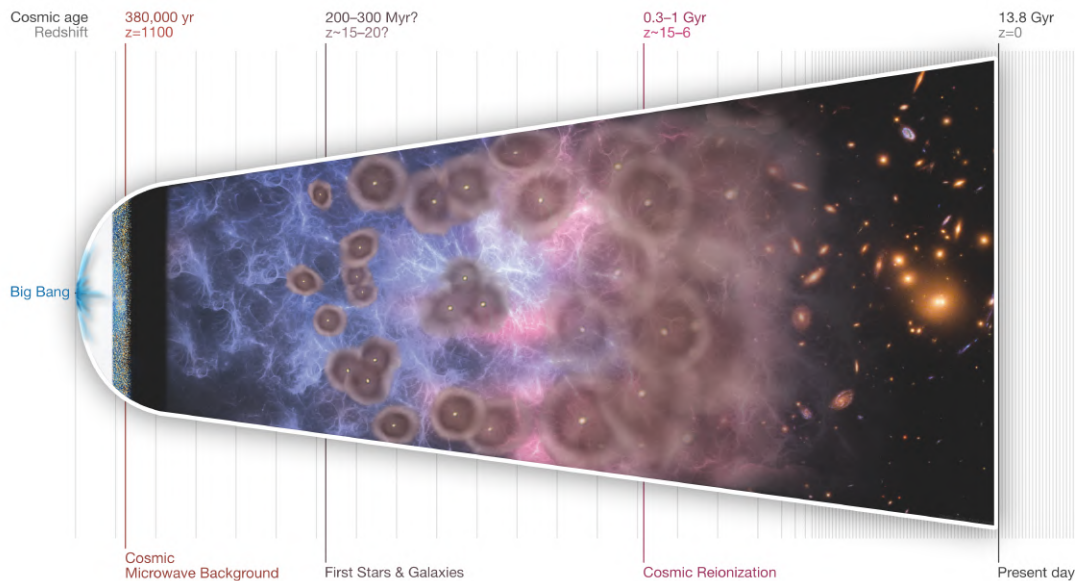
The Cosmic Dawn culminated with the *Epoch of Reionization*, which started around 200 million years after the Big Bang ( $z \sim 15 - 20$ ) and was largely completed around 1 billion years later ( $z \sim 5$ ). During this epoch, the Universe underwent a phase transition as the first galaxies formed and emitted ionizing radiation in the ultraviolet (UV) range, known as Lyman Continuum (LyC) radiation. This radiation was capable of ionizing the HI contained in the IGM, leading

to its transition from a neutral to a fully ionized state.

The period between 2 to 3 billion years after the Big Bang is often referred to as the *Cosmic Noon*, a period of vigorous star formation and galaxy evolution. During this epoch, galaxies formed and matured rapidly, fueled by the abundant reservoirs of gas in the IGM. The cosmic landscape evolved as galaxies grew in size and complexity, shaping the Universe into its present-day form.

Today, 13.8 billion years after the Big Bang, we find ourselves in the *Present Day* epoch. The Universe continues to expand, driven by dark energy, which dominates its energy budget. Galaxies have evolved into the diverse structures we observe today, and while star formation continues, it occurs at a much slower rate compared to earlier epochs.

This simplified history of the Universe is illustrated in Figure 1.1, which also depicts the redshift ( $z$ ) and age of the Universe during each cosmic phase. The figure provides a broad overview of what we discussed in this section. Naturally, numerous unresolved questions persist regarding the origin of the Universe, as well as the formation and evolution of galaxies, topics that remain subjects of active debate within the scientific community. This is why this thesis focuses specifically on the Epoch of Reionization, the Universe's last and least understood major phase transition.



**Figure 1.1:** *Cosmic timeline, from the origin of the Universe in the Big Bang, 13.8 billion years ago, till the present day [Figure from 3].*

## 1.2 The evolution of the intergalactic medium

As mentioned earlier, the Epoch of Reionization (EoR) marks the last phase transition of the Universe, after the so called Dark Ages, during which the IGM underwent significant changes. After the Universe expanded and cooled following the Recombination Epoch, the IGM emerged as gravity acted on primordial mat-

ter fluctuations, leading to the formation of the Universe's large-scale structure. Dark matter, which shapes this structure, exerts a gravitational pull that causes baryonic matter, including gas, to clump together in specific regions. Before reionization, the IGM was neutral and diffuse. The timing and spatial evolution of the Reionization process are closely linked to the formation of the Universe's first radiation sources, such as supermassive black holes and PopIII stars.

The ionization potential of the hydrogen atom is 13.6 eV. Consequently, any UV photons with energies exceeding 13.6 eV (corresponding to wavelengths shorter than 912 Å) can eject the electron from a H atom, thus ionizing it. This energy threshold, above which photons can ionize HI, is known as the *Lyman limit*. Radiation with energies above this limit is referred to as LyC radiation [125].

As the first stars formed, their LyC radiation started to ionize their surrounding medium, creating ionized hydrogen (HII) regions that gradually expanded. Within these regions, free electrons could recombine with protons, forming HI again. When recombination occurs directly to the ground state of hydrogen, a new LyC photon is emitted. This photon may then be absorbed by another hydrogen atom in the surrounding interstellar medium (ISM), causing further ionizations.

Focusing on the two processes of ionization and recombination, the change in the volume filling factor of HII regions ( $Q_{\text{HII}}$ ) is represented by the following equation [189]:

$$\frac{dQ_{\text{HII}}}{dt} = \frac{\dot{n}_{\text{ion}}}{\langle n_H \rangle} - \frac{Q_{\text{HII}}}{t_{\text{rec}}}, \quad (1.1)$$

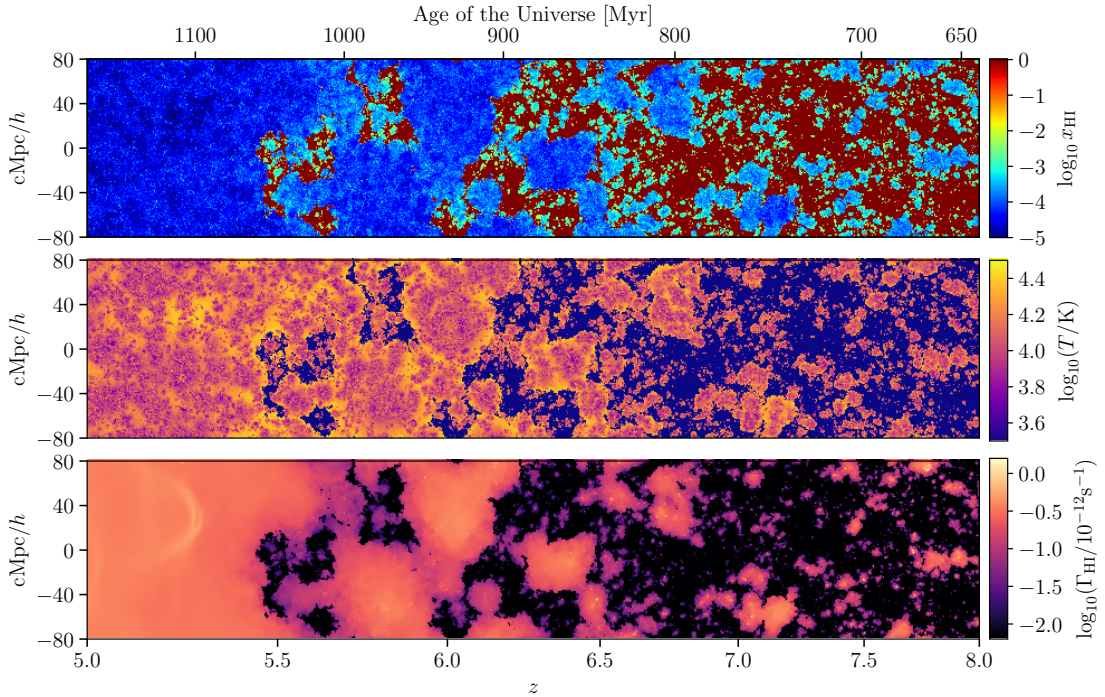
where  $\frac{dQ_{\text{HII}}}{dt}$  represents the rate of change of the volume filling factor of HII, indicating how rapidly the ionized regions are changing in time;  $\dot{n}_{\text{ion}}$  is the rate of ionizing photons emitted by all radiation sources (stars and quasars) per unit volume. The term  $\langle n_H \rangle$  represents the average density of H atoms, which is dependent on the cosmological baryon density  $\Omega_b$ .  $\frac{Q_{\text{HII}}}{t_{\text{rec}}}$  accounts for the recombination of HII back into HI. As  $Q_{\text{HII}}$  increases, the frequency of recombinations rises, which counteracts the growth of ionized regions. The recombination timescale  $t_{\text{rec}}$  indicates how quickly this process occurs and it is given by  $t_{\text{rec}} = (\alpha_B \langle n_e \rangle)^{-1}$ , where  $\alpha_B$  is the case B recombination coefficient, which depends on the temperature of the IGM, and  $\langle n_e \rangle$  is the average electron density.

To account for inhomogeneities in the IGM, the recombination rate also depends on the clumping factor,  $C$ , defined as:

$$C = \frac{\langle n_e^2 \rangle}{\langle n_e \rangle^2}, \quad (1.2)$$

which measures the degree to which gas is concentrated in dense clumps compared to a uniform distribution. A higher clumping factor results in a higher effective recombination rate. This effect is particularly important during the initial stages of the EoR [e.g., 131, 360].

The average volume filling factor of ionized hydrogen,  $Q_{\text{HII}}$  and its evolution only captures the average progress of reionization. This process, however did not occur uniformly: ionized regions (the so-called *bubbles*) initially formed around early radiation sources, gradually expanded and, over time, began to overlap,



**Figure 1.2:** *The evolution of the neutral hydrogen fraction ( $x_{\text{HI}}$ , top panel), gas temperature ( $T$ , middle panel), and hydrogen photoionization rate ( $\Gamma_{\text{HI}}$ , lower panel) from  $z = 5$  to  $z = 8$ , based on the P-GADGET-3 + ATON simulation. These panels illustrate the patchy and delayed nature of reionization, with large neutral hydrogen regions persisting down to  $z < 5.5$ . [Figure from 259].*

creating a patchy reionization pattern. This means that in different regions, reionization proceeded at a different rate. The spatial and temporal evolution of reionization is closely tied to the nature and timing of the ionizing sources. Faint, low-mass objects presumably formed earlier than their brighter counterparts and could have initiated reionization at an earlier stage. In contrast, the formation of fewer, brighter sources may have occurred later. A reionization process dominated by bright sources would produce fewer, larger bubbles and lead to a more inhomogeneous pattern. Conversely, reionization driven primarily by numerous faint sources would likely proceed more uniformly, with smaller and more evenly distributed ionized regions. This distinction not only influences the size and distribution of ionized bubbles but also shapes the timeline of reionization across the Universe [e.g., 205].

Observations and simulations indicate a transition from a highly inhomogeneous IGM at redshifts  $z \sim 20$  to a nearly uniform distribution of ionized gas by the end of reionization. This transition is illustrated in Figure 1.2, which shows the IGM's evolution in terms of neutral hydrogen fraction, gas temperature and photoionization rate as derived from the P-GADGET-3 + ATON simulation [259]. The spatial inhomogeneity of the process is evident.

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### 1.2.1 Constrains on the evolution of the IGM during the EoR

To study the evolution of the IGM during reionization, we can use several independent observational probes. One primary method involves the CMB. Since CMB photons are emitted at recombination, they may scatter off free electrons as they travel through space, making them sensitive to the presence of free electrons in the IGM. This scattering causes a damping of small-scale temperature anisotropies in the CMB. Free electrons generated during the EoR increase the optical depth of CMB photons and by measuring this, we can constrain the timing of the reionization process [see 190, for example]. The most recent measurements suggest that the midpoint of reionization was at  $z_{\text{re}} = 7.7 \pm 0.7$  [368].

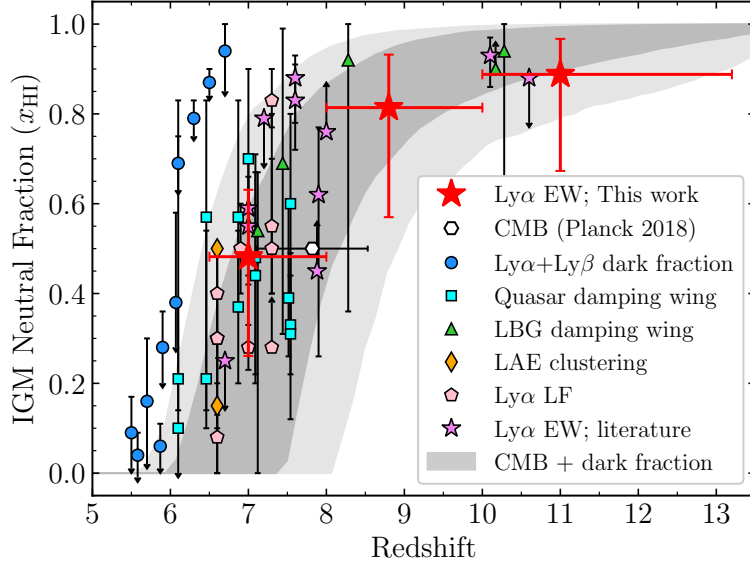
A complementary approach involves studying the Ly $\alpha$  emission in distant galaxies: this emission line is particularly bright in high-redshift, dust-poor, star-forming galaxies [e.g., 448] and is highly sensitive to the neutral content of the IGM. The detectability of Ly $\alpha$  photons is significantly influenced by resonant scattering interactions with HI [see 353, for a review]. In a partially neutral IGM, Ly $\alpha$  photons can be absorbed by HI before they reach the observer. This absorption is expected to lead to a decline in the number of observable Ly $\alpha$  emitters (LAEs) during the EoR. Moreover, the patchy nature of the IGM during the EoR creates substantial spatial variations in the neutral fraction ( $X_{\text{HI}}$ ). In the regions where the IGM is ionized, Ly $\alpha$  photons can escape more easily, enhancing the detectability of LAEs in those areas. Conversely, in regions with a higher neutral fraction, the likelihood of Ly $\alpha$  absorption increases, leading to a decline in detectable LAEs.

Over the years, researchers have used different ways to quantify the change in Ly $\alpha$  visibility and connect it to the change in the IGM neutral hydrogen content  $X_{\text{HI}}$  as a function of redshift. Some studies have explored this by comparing the Ly $\alpha$  luminosity function (LF) with the UV-continuum LF, which will be discussed in Section 1.4.1 [e.g., 204, 210, 256, 352, 354, 530]. The evolution of the Ly $\alpha$  LF is influenced by both galaxy evolution and the opacity of the IGM to Ly $\alpha$  photons, whereas the UV-continuum LF is solely governed by galaxy evolution. These studies found that the Ly $\alpha$  LF rapidly declines from  $z \sim 6.5$  to 7.5, while the UV LF shows a more gradual decrease, indicating an increase in  $X_{\text{HI}}$ .

Alternatively one can measure  $X_{\text{Ly}\alpha}$ , i.e. the fraction of Ly $\alpha$  emitting galaxies among all UV continuum-selected galaxies at a given redshift. Numerous systematic efforts have been conducted over the past decade [e.g., 10, 71, 72, 157, 235, 304, 364, 365, 418, 419, 448, 472, 478, 522]. These studies showed a drop in  $X_{\text{Ly}\alpha}$  above  $z \sim 6$ , providing strong evidence that the Universe was partially neutral at these higher redshifts.

The evolution in the clustering properties of Ly $\alpha$  emitting galaxies can also be used to probe the  $X_{\text{HI}}$ , although such studies require very large areas [353].

Another approach involves analyzing quasar spectra from the EoR and later periods, as initially proposed by [183]. In particular, a strong suppression of flux at wavelengths shorter than Ly $\alpha$ , known as the *Gunn-Peterson trough* is expected in the spectra of bright QSO as the  $X_{\text{HI}}$  increases. This trough was first observed in a quasar at  $z = 6.28$  in the Sloan Digital Sky Survey (SDSS) [29]. The Gunn-Peterson trough is an indicator of the endpoint of reionization, since it saturates



**Figure 1.3:** *Evolution of  $X_{\text{HI}}$  during reionization. Grey shaded regions represent constraints from CMB optical depth and  $\text{Ly}\alpha$  and  $\text{Ly}\beta$  forest dark pixel fractions [304], with dark grey indicating 68% and light grey 95% confidence intervals. IGM neutral fractions are from: CMB optical depth [white hexagon, 368],  $\text{Ly}\alpha$  forest dark pixels [blue circles, 229, 313],  $\text{Ly}\alpha$  damping wing absorption of quasars [cyan squares, 22, 107, 176, 496, 503, 519],  $\text{Ly}\alpha$  damping wing absorption of Lyman break galaxies [green triangles, 104, 202, 482], clustering of  $\text{Ly}\alpha$  emitters [orange diamonds, 355, 439],  $\text{Ly}\alpha$  luminosity function [pink pentagons, 354, 530], and  $\text{Ly}\alpha$  equivalent width [violet stars, 201, 305, 339, 464]. [Figure from 464].*

rapidly when the neutral hydrogen fraction is  $\sim 10^{-4}$ – $10^{-5}$ , offering only upper bounds on the ionization state of the intergalactic medium.

We can measure the density of neutral hydrogen also by analyzing the saturated  $\text{Ly}\alpha$  damping wing [e.g., 374, 381, 442] in the spectra of bright QSO. By examining the strength and profile of this feature in QSOs across different redshifts, we can constrain the neutral hydrogen fraction  $X_{\text{HI}}$ , effectively tracing the timeline of reionization. However, the presence of Damped Lyman-Alpha Absorbers (DLAs) in the host galaxies of quasars creates an effect similar to that of neutral hydrogen in the IGM, making it challenging to distinguish between the two contributions. This degeneracy complicates precise measurements of the hydrogen fraction and thus limits the accuracy of reionization constraints derived solely from QSO spectra.

The *James Webb Space Telescope* (JWST) is now providing new constraints on the patchiness of the IGM during the EoR by detecting faint galaxies and quasars at redshifts greater than 7. We are measuring the  $\text{Ly}\alpha$  damping wing not only to QSOs and Gamma-ray bursts (GRBs) but also in bright galaxies [e.g., 104, 202, 482]. Recent measurements by [342, 464] using samples of LAEs observed with the JWST support a scenario where the IGM becomes increasingly neutral from  $z = 6.5$  to  $z = 13.3$  (Figure 1.3). However such studies also point to significant field-to-field variation due to cosmic variance and to the patchy nature of reionization.



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In Figure 1.3, we show the latest results on the IGM neutral fraction during reionization, including data from multiple studies. The data exhibit substantial error bars, and there is considerable variation among the findings. Nevertheless, they collectively suggest a trend toward a “late reionization” scenario, characterized by a more gradual and patchy reionization process.

### 1.2.2 Theoretical tools for probing the EoR

Cosmological simulations at various scales have provided detailed models of the reionization process. Notable simulations include FiBy [356], FIRE [280], CROC [170], CoDa [271, 344, 345], SPHINX [396, 397], OBELISK [476], Aurora [360], TechnicolorDawn [150], Renaissance [515], Phoenix [505], THESAN [160, 236], SPICE [38], and DELPHI [109] [for a comprehensive review, see 108].

These simulations range from high-resolution, small-volume simulations, such as Renaissance and Phoenix, which focus on the formation of the first stars, to lower-resolution but large-scale simulations like CROC, CoDa, and THESAN, which model a large representative volume of the Universe. Large volumes are essential for obtaining converged reionization histories and robust large-scale IGM predictions [171, 206]. Conversely, smaller volume simulations often serve as galaxy formation models with coupled radiative transfer (RT) rather than comprehensive reionization models.

A representative output from a reionization simulation is shown in Figure 1.2, illustrating the evolution of three parameters: the neutral hydrogen fraction  $x_{\text{HI}}$ , the gas temperature  $T$ , and the hydrogen photoionization rate  $\Gamma_{\text{HI}}$  from  $z = 5$  to  $z = 8$ . These panels underscore the patchy and delayed nature of reionization, with significant regions of neutral hydrogen persisting until  $z \leq 5.5$ . Coherent fluctuations in gas temperature and photoionization rate are particularly evident near the remaining neutral islands.

Thanks to the depth of these simulations, several aspects of the EoR, including its timing and history, are becoming increasingly well-constrained. By comparing the evolving size and distribution of ionized bubbles in these simulations to observed signatures of reionization – such as the Gunn-Peterson trough and the Thomson scattering optical depth from the CMB, as discussed in the previous Section – we can estimate when reionization began and how long it lasted. The large-scale simulations, in particular, allow us to model the gradual overlap of ionized regions, helping to pinpoint the EoR’s timeline.

## 1.3 The nature of the cosmic reionizers

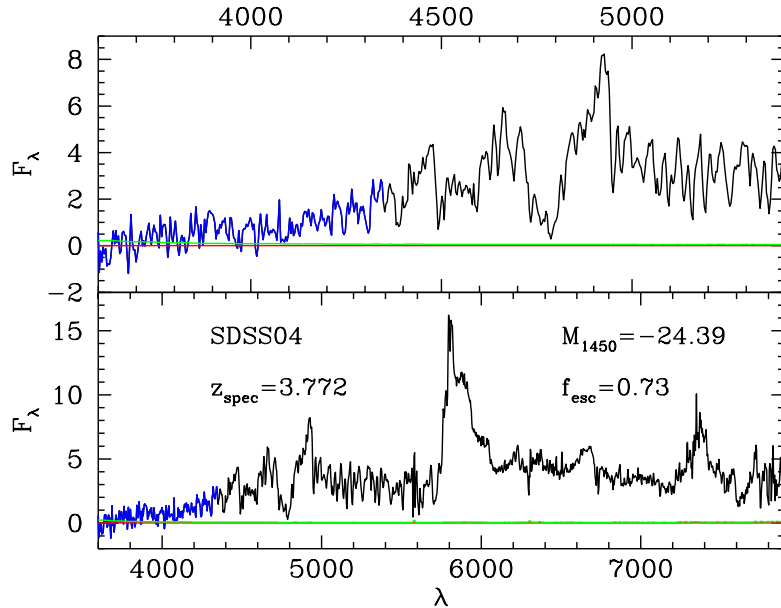
Among the various astrophysical objects capable of producing LyC photons, AGNs and massive stars within star-forming galaxies are considered the leading candidates responsible for reionization [283, 394].

### 1.3.1 Active Galactic Nuclei

AGNs are particularly intriguing as potential contributors to reionization [167, 174, 283]. They are known for their high ionizing emissivity: bright quasars



exhibit an escape fraction  $f_{\text{esc}} = 100\%$  [101], while faint AGNs with  $M_{\text{UV}} \sim -24$  have  $f_{\text{esc}}$  ranging from 44% to 100%, with a median value of approximately 74% [149, 174]. An example of an AGN spectrum is presented in Figure 1.4, showing the UV/optical spectrum of AGN SDSS04 ( $z = 3.772$ ) observed with the *FOcal Reducer/low dispersion Spectrograph 2* (FORS2) at the ESO’s *Very Large Telescope* (VLT).



**Figure 1.4:** *Bottom:* The UV/optical spectrum of AGN SDSS04, obtained with FORS2@VLT, is shown with the  $x$ -axis representing the observed wavelength. *Top:* A zoom of the blue spectral range. The red horizontal lines indicate the zero flux level. The LyC region is shaded in blue, while the green line represents the RMS noise. [From 174].

For many years there has been a significant debate regarding the AGN number density during the reionization epoch. [228] and [98] suggested a decline in AGN distribution during reionization, whereas [167] and [168] found evidence for a substantial AGN population at high redshift. The prevailing view held that AGNs had relatively low number density during this period [158, 306] and therefore likely contributed to a small fraction of the total ionizing budget, as indicated by results from the DELPHI semi-analytic model [111]. The model presented by [149] supported the idea that reionization was “AGN-assisted” rather than “AGN-dominated”, meaning that while AGNs contributed significantly, galaxies remained the primary source of ionizing photons during reionization. AGNs become more influential at lower redshifts ( $z < 4.6$ ), to keep the IGM ionized but their contribution only surpasses that of galaxies at lower redshifts.

However, recent deep surveys conducted with JWST have identified a growing population of moderate-luminosity, broad-line AGNs at  $4 < z < 14$ , powered by accretion onto early massive black holes [e.g., 42, 172, 257, 289]. Findings from [195, 290] suggest that these supermassive black holes (SMBHs) are over-massive compared to local relations between black hole mass and stellar mass ( $M_{\text{BH}} - M_{\star}$ ), although this observation might be affected by selection biases [e.g., 272]. Additionally, the discovery of a previously unidentified population

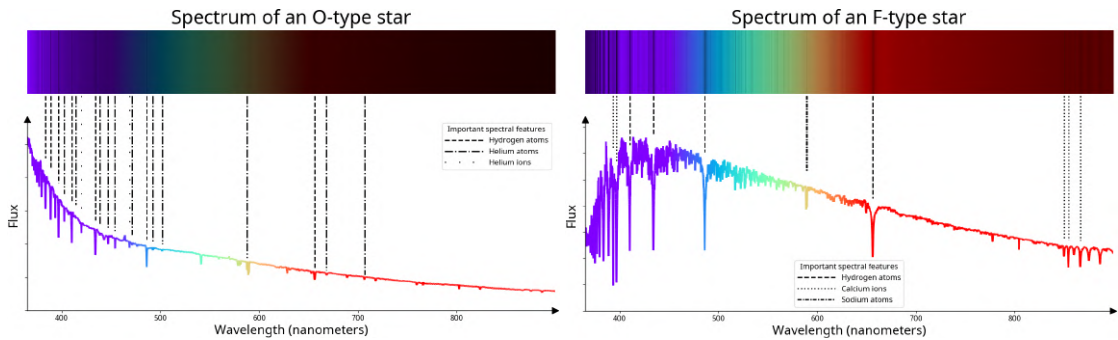
of faint objects at  $z > 4$ , known as “Little Red Dots” (LRDs), further suggests that faint high-redshift AGNs could be more prevalent in the early Universe than previously thought [e.g., 248, 249, 253, 308]. Evidence for the AGN nature (type 1) of LRDs includes their broad Balmer emission lines and compact morphology without visible host galaxies [e.g., 245, 308].

While this thesis assumes that AGNs do not contribute significantly to reionization, we acknowledge that the debate on their role remains open. However, there are several challenges to the idea that AGNs could dominate reionization. Increasing the AGN contribution too much can lead to issues with the timing of helium reionization. If AGNs were the primary drivers of hydrogen reionization, the associated ionizing radiation would likely result in helium (He II) being reionized too early, which is inconsistent with current observational constraints [e.g., 12].

Additionally, the escape fraction of ionizing photons for faint AGNs remains poorly constrained. Even if faint AGNs are more common than currently believed, it is not guaranteed that their ionizing contributions surpass those of stars. For faint AGNs to dominate reionization, their  $f_{\text{esc}}$  would need to be both significant and sustained over cosmic time, which has yet to be demonstrated observationally. This uncertainty underscores the challenges of attributing reionization primarily to AGNs rather than to star-forming galaxies.

### 1.3.2 Star-forming galaxies

Although AGNs have the potential to contribute, the consensus remains that young, hot, massive stars (O and B types) in early star-forming galaxies are the primary drivers of cosmic reionization, and this is the assumption made in this thesis. In Figure 1.5, we present the spectra of an O-type and an F-type star, with the first clearly showing a higher flux in the UV-range compared to the second.



**Figure 1.5:** *Left panel: Spectrum of the O-type star, color-coded following the electromagnetic spectrum. Black lines denote absorption features from various atoms and ions, with strong lines from ionized helium. Right panel: Spectrum of the F-type star, with similar color coding. Absorption lines are shown in black, with prominent features from hydrogen and ionized calcium. Compared to O-type stars, the spectrum has stronger metal lines and reduced flux at the blue end. Credit: IAU OAE/SDSS/Niall Deacon.*

The critical question is how many of the LyC photons produced by these massive stars can escape from their host galaxies’ ISM and circumgalactic medium

(CGM) [394] to reach the IGM. Given the density of star-forming galaxies during the EoR, an average LyC escape fraction ( $f_{\text{esc}}$ ) of around 10% across all galaxies is required [e.g., 149, 393, 525, 526] to match the timing of reionization predicted by the various constraints, discussed in Section 1.2.1.

Another key unknown in this context is whether reionization was predominantly driven by the more numerous faint star-forming galaxies or by the fewer luminous sources [110, 143, 149, 331, 393]. As mention before (Section 1.2), if faint, low-mass galaxies with higher  $f_{\text{esc}}$  dominated the ionizing photon budget, we would expect an early reionization scenario, in which reionization started sooner and also proceeded in a more spatially homogeneous manner [e.g., 110, 143, 149]. Although both types of galaxies likely contributed to the ionizing photon budget, the balance between them is still actively investigated and is one of the goals of this thesis (see Chapter 6).

## 1.4 The total ionizing budget of star-forming galaxies

To clarify the role of these early galaxies, it is imperative to determine their relative contributions to the total ionizing emissivity ( $\dot{n}_{\text{ion}}$ ), which is defined as the number of ionizing photons emitted per unit time and comoving volume as:

$$\dot{n}_{\text{ion}} = \xi_{\text{ion}} \times \rho_{\text{UV}} \times f_{\text{esc}}, \quad (1.3)$$

where  $\xi_{\text{ion}}$  is the ionizing photon production efficiency (number of produced ionizing photons per UV luminosity density),  $\rho_{\text{UV}}$  is the luminosity density (integral of the number of galaxies per UV luminosity per comoving volume), and  $f_{\text{esc}}$  is the LyC escape fraction (the fraction of ionizing photons that reaches the IGM).

### 1.4.1 The UV luminosity density ( $\rho_{\text{UV}}$ )

As discussed in Section 1.3.2, the rest-frame UV emission from galaxies is linked to the presence of young, massive stars [282]. The UV luminosity density,  $\rho_{\text{UV}}$ , is calculated by integrating the galaxy luminosity function,  $\Phi(L) \equiv dn/dL$ , weighted by the UV luminosity  $L$ , typically measured at  $\lambda = 1500 \text{ \AA}$ :

$$\rho_{\text{UV}} = \int_{L1}^{L2} \Phi(L) L dL, \quad (1.4)$$

where the limits are truncated at both ends by the physics of galaxy formation, shaping the functional form of  $\Phi(L)$ .

Traditionally, the luminosity function  $\Phi(L)$  is well-approximated by a Schechter function [417]:

$$\Phi(L) = \phi^* \left( \frac{L}{L^*} \right)^{\alpha} \exp \left( -\frac{L}{L^*} \right) \frac{dL}{L^*}, \quad (1.5)$$

where  $\phi^*$  is the normalization (usually considered  $\leq 10^{-3} [\text{Mpc}^{-3} \text{mag}^{-1}]$ ),  $L^*$  is the characteristic luminosity, and  $\alpha$  is the faint-end slope.

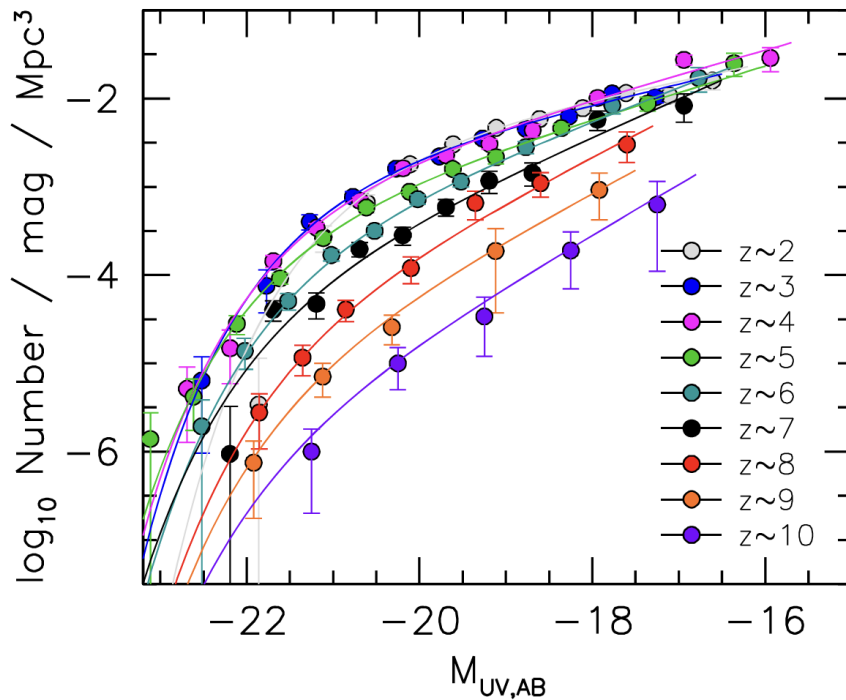
At very high redshifts, excesses in the bright end of the luminosity function have been observed, deviating from the exponential cut-off predicted by the Schechter

function. To model this, particularly for redshifts  $z \sim 6 - 9$ , a double power-law (DPL) function is often used:

$$\Phi_{\text{DPL}}(M_{\text{UV}}) = 0.4 \ln(10) \phi_{\text{DPL}} \left[ \left( 10^{-0.4(M_{\text{UV}} - M_{\text{DPL}})} \right)^{-\alpha-1} + \left( 10^{-0.4(M_{\text{UV}} - M_{\text{DPL}})} \right)^{-\beta-1} \right]^{-1}, \quad (1.6)$$

where  $\beta$  describes the power-law behavior for magnitudes brighter than  $M_{\text{DPL}}$ .

Using a combination of ground-based and Hubble Space Telescope (HST) imaging, [47, 51, 194] have measured the rest-frame UV luminosity function at redshifts  $z \sim 6 - 10$ , employing Lyman break drop-out selection techniques. These studies found that at  $z \sim 6 - 7$ , the bright-end data are well-fit by the DPL model. At higher redshifts,  $z \sim 8 - 9$ , the luminosity function has been measured down to  $M_{\text{UV}} = -17$ , providing critical insights into the evolution of  $\rho_{\text{UV}}$  during the reionization era.



**Figure 1.6:** The best-fit Schechter LFs at  $z \sim 2-9$  obtained from all the noncluster extragalactic legacy HST fields, including the Hubble Ultra Deep Field, the Hubble Frontier Fields parallel fields, and all five CANDELS fields. [Figure from 47].

Figure 1.6 shows the evolution of  $\rho_{\text{UV}}$  with redshift which suggests a smooth decline in the abundance of bright galaxies from  $z \sim 6$  to  $z \sim 10$ . The faint-end slope of the luminosity function appears to steepen, with  $\alpha \approx -2$ , although the extent of this steepening depends on whether a Schechter or DPL model is assumed. These results have been further confirmed by JWST observations up to  $z = 9$ , but above this redshift a stark excess of bright galaxies has been measured by several independent surveys [e.g., 75, 120, 148, 195].

This excess is still not well understood [e.g., 114, 144]. However, we note that, given the relative contribution of bright versus faint galaxies to the total

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ionizing emissivity, and given that the bulk of reionization essentially happened below  $z = 10$ , the unexpectedly numerous bright high-redshift galaxies may have primarily influenced only the early stages of reionization.

#### 1.4.2 The ionizing photon production efficiency ( $\xi_{\text{ion}}$ )

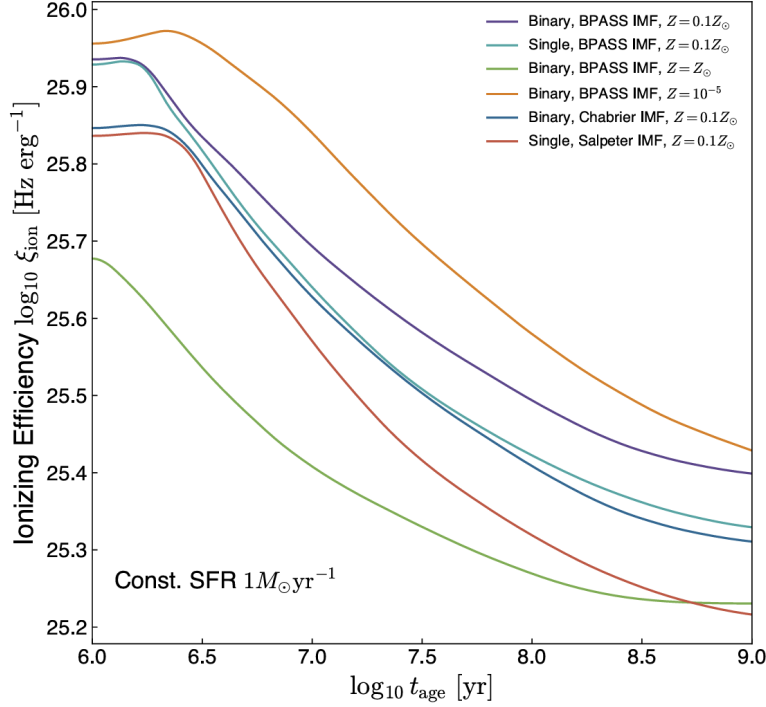
The parameter  $\xi_{\text{ion}}$  [267] gauges the ionization capability of galaxies. As discussed in Section 1.3.2, the primary sources of ionizing photons are hot, massive O and B-type stars. These stars, through various transitions in H atoms, emit photons in both the Lyman and Balmer series, contributing to the ionizing radiation and observable rest-frame UV and optical spectrum.  $\xi_{\text{ion}}$  is defined as follows:

$$\xi_{\text{ion}}[\text{Hz erg}^{-1}] = \frac{N(H^0)[\text{s}^{-1}]}{L_{\nu,1500}[\text{erg s}^{-1}\text{Hz}^{-1}]}, \quad (1.7)$$

where  $N(H^0)$  represents the total number of ionizing photons produced by massive stars per unit time, which can be computed by integrating the intrinsic spectral energy distribution (SED) of the astronomical source below the Lyman limit, and  $L_{\nu,1500}$  is the intrinsic UV luminosity at 1500 Å.

The value of  $\xi_{\text{ion}}$  is determined by several factors. In particular, metallicity plays a crucial role, as stars with higher metallicity lose more mass through stellar winds. This mass loss increases atmospheric opacity and reduces angular momentum, slowing down the internal mixing of fuel and causing high-metallicity stars to burn at lower temperatures over longer periods compared to their low-metallicity counterparts. Metallicity also affects binary star interactions by influencing the efficiency of mass transfer and the nature of post-interaction evolution. In low-metallicity environments, stars experience weaker stellar winds, allowing them to retain more mass. This increases the likelihood of stable mass transfer during binary interactions, where the donor star can transfer substantial amounts of material to its companion. This mass transfer can significantly enhance the rotation rate of the accreting star, boosting internal mixing and fuel consumption. Enhanced mixing prolongs the production of ionizing photons by keeping fresh fuel in the core and sustaining high surface temperatures. Additionally, low-metallicity stars are more likely to form close binaries that survive interactions such as common-envelope phases, further contributing to the population of ionizing photon producers. By using models that account for the fraction of stars in binaries and their orbital distributions, we can better understand how metallicity and binary interactions contribute to ionizing photon production. For example, low-metallicity stellar populations in binary systems can produce 2-3 times more ionizing photons per unit of UV luminosity compared to populations with solar metallicity. As illustrated in Figure 1.7, a stellar population undergoing constant star formation over approximately 100 million years with metallicity of  $Z = 0.1Z_{\odot}$  would yield  $\log \xi_{\text{ion}} \approx 25.46$ , whereas a metallicity of  $Z = 0.005Z_{\odot}$  would result in  $\log \xi_{\text{ion}} \approx 25.60$ . Even when holding metallicity constant, binary interactions can enhance  $\xi_{\text{ion}}$  by approximately 25% at  $t_{\text{age}} = 100$  Myr.

The average production of ionizing photons for a certain stellar population, is also heavily dependent on the Initial Mass Function (IMF), particularly regarding



**Figure 1.7:** The ionizing photon production efficiency,  $\xi_{\text{ion}}$ , as a function of stellar population age for the default BPASS IMF [129], as well as the Chabrier [78] and Salpeter [403] IMFs, all evaluated at a metallicity of  $Z = 0.1Z_{\odot}$ . Additionally, BPASS IMF predictions are included for metallicities of  $Z = 10^{-5}Z_{\odot}$  and  $Z = 1Z_{\odot}$ . Whenever applicable, these predictions account for both single and binary star evolution. [Figure from 392].

the abundance of massive stars. In particular, the BPASS IMF [129] incorporates binary star interactions, especially for massive stars, and assumes a steeper slope for high-mass stars. This model predicts a higher number of massive stars, making it particularly useful for modeling stellar populations in environments where binary interactions are significant. As shown in Figure 1.7, for constant metallicity of  $Z = 0.1Z_{\odot}$ , the BPASS IMF yields a  $\xi_{\text{ion}}$  value that is 25% higher than that produced by the Chabrier IMF [392].

The relationship between ionizing photons and Balmer photons allows for direct constraints on  $\xi_{\text{ion}}$  through spectroscopic measurements of Balmer emission lines, once corrected for dust attenuation [e.g., 416, 431]. This conversion is primarily influenced by nebular conditions such as electron density and gas temperature [350], and relatively insensitive to the choice of stellar population models. According to [267], the production rate of ionizing photons,  $N(H^0)$ , can be expressed as:

$$N(H^0)[\text{s}^{-1}] = \frac{1}{1.36} \times 10^{12} L_{\text{H}\alpha}[\text{erg s}^{-1}], \quad (1.8)$$

where  $L_{\text{H}\alpha}$  is the  $\text{H}\alpha$  luminosity. A key assumption in this derivation is the escape fraction of LyC photons,  $f_{\text{esc}}$ . If  $f_{\text{esc}} > 0$ , the intrinsic  $\xi_{\text{ion}}$  is higher by a factor of  $1/(1 - f_{\text{esc}})$  compared to the value measured from  $\text{H}\alpha$  luminosity.

If we use  $\text{H}\beta$  instead of  $\text{H}\alpha$  as shown by [416], the relationship is:

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$$N(H^0)[\text{s}^{-1}] = 2.1 \times 10^{12} L_{\text{H}\beta}[\text{erg s}^{-1}], \quad (1.9)$$

with the same dependence on  $f_{\text{esc}}$ . The numerical coefficient is derived from [458] for typical conditions found in HII regions.

Since high equivalent widths (EW) of the [OIII] doublet are typically associated with young, metal-poor stellar populations – which are expected to emit large amounts of ionizing photons – another method for estimating  $\xi_{\text{ion}}$  involves using this quantity [82, 467]:

$$\log \xi_{\text{ion}} = (22.12 \pm 0.48) + (1.15 \pm 0.16) \times \log \text{EW}[\text{O III}]\lambda\lambda 4959, 5007[\text{\AA}]. \quad (1.10)$$

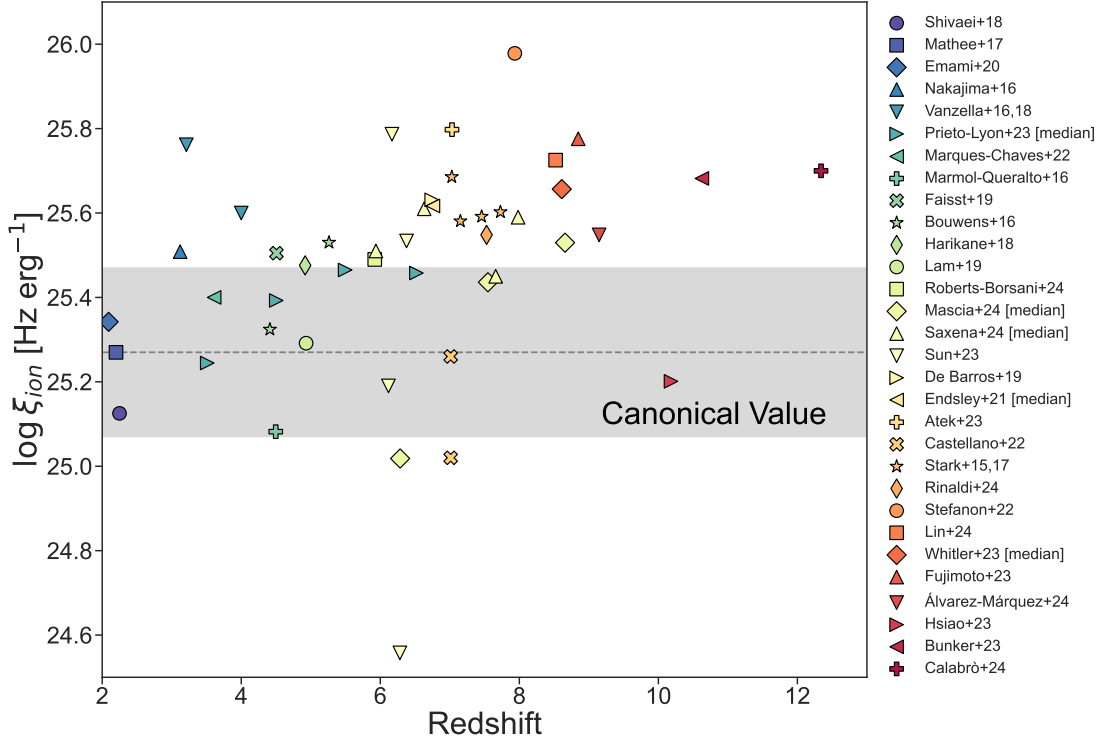
The redshift evolution of  $\xi_{\text{ion}}(z)$ , is still less well constrained compared to the UV luminosity density. At  $z \sim 2$ , estimates from UV-complete samples with available H $\alpha$  spectroscopy suggest  $\log \xi_{\text{ion}} \approx 25.47$  [130]. The MOSDEF survey has also constrained  $\xi_{\text{ion}}$  spectroscopically at  $z \sim 2$ , reporting a lower limit of  $\log \xi_{\text{ion}} \approx 25.06 - 25.34$ , depending on the dust model employed. Higher  $\xi_{\text{ion}}$  values are observed in galaxies characterized by high [OIII]/H $\beta$  ratios or UV continuum slopes of  $\beta < -2.1$  [431]. Additionally, star-forming galaxies at  $z \sim 2 - 3$  frequently show increased Ly $\alpha$  equivalent widths alongside rising [OIII]/H $\beta$  ratios, suggesting a connection between enhanced ionizing photon production and escape mechanisms [466].

In the absence of spectroscopic data, photometric measurements have usually been employed [e.g., 48]. At  $z \sim 4 - 5$  H $\alpha$  nebular emission can be detected via *Spitzer* broadband colors. [48] found a mean  $\xi_{\text{ion}}$  value of 25.27, which was for many years considered as a representative (or canonical) value for high redshift galaxies. Using similar techniques, [113] measured  $\log \xi_{\text{ion}} \approx 25.77$  for  $z \sim 8$  galaxies.

JWST, with its enhanced capabilities, has finally enabled very precise spectroscopic or photometric measurements of the ionizing photon production in galaxies at  $z \sim 6 - 9$  [132, 302, 371, 390, 435], revealing a range of  $\log \xi_{\text{ion}} \approx 25 - 26$ , which likely reflects a diversity of star formation histories. Measurements have been pushed down to  $M_{\text{UV}}$  as faint as  $-15.5$  finding an average  $\log \xi_{\text{ion}} \approx 25.80$  [17]. These new measurements suggest that the mean  $\xi_{\text{ion}}$  values are higher than pre-JWST results. An overview of the  $\xi_{\text{ion}}$  evolution with redshift is shown in Figure 1.8. The role of  $\xi_{\text{ion}}$  in high-redshift galaxies will be further examined in various chapters of this thesis, where we will explore its implications for cosmic reionization.

### 1.4.3 The escape fraction of ionizing photons ( $f_{\text{esc}}$ )

The remaining significant uncertainty in the emissivity equation centers around the escape fraction,  $f_{\text{esc}}$ . It represents the ratio between the flux of photons with  $E_{\gamma} > 13.6 \text{ eV}$  (or  $\lambda < 912 \text{ \AA}$ ) and the flux at  $\lambda = 1500 \text{ \AA}$ , which is not absorbed by Lyman limit systems between the source and the observer [108].



**Figure 1.8:**  $\log \xi_{ion}$  [ $\text{Hz erg}^{-1}$ ] as a function of redshift, compiled from various observational studies [7, 17, 48, 61, 65, 76, 113, 130, 133, 136, 159, 196, 202, 260, 273, 296, 297, 302, 311, 335, 371, 383, 390, 413, 431, 450, 451, 455, 460, 485, 487, 506]. The canonical value of  $\log \xi_{ion} \approx 25.27 \text{ Hz erg}^{-1}$  [from 48], is highlighted in grey for reference.

Following [207],  $f_{\text{esc}}$  can be expressed as:

$$f_{\text{esc}} = \frac{F_{\text{LyC,esc}}}{F_{\text{LyC,int}}}, \quad (1.11)$$

where  $F_{\text{LyC,esc}}$  and  $F_{\text{LyC,int}}$  are the escaped and intrinsic (or emitted) fluxes from a galaxy in the LyC range, respectively. To obtain  $F_{\text{LyC,esc}}$ , we must correct the observed LyC flux density  $F_{\text{LyC,obs}}$  for the photoelectric absorption by the neutral hydrogen in the IGM, so that:

$$F_{\text{LyC,esc}} = F_{\text{LyC,obs}} e^{\tau_{\text{HI,IGM}}}, \quad (1.12)$$

in which  $\tau_{\text{HI,IGM}}$  is the optical depth of neutral hydrogen in the the IGM between the galaxy and the observer. However, a significant challenge that hinders the direct detection of LyC emission from cosmic reionizers is the opacity of the IGM during the EoR, which is discussed in the next Section. The following Chapter is dedicated to understanding the mechanisms of LyC escape and provides a review of the key observational results achieved to date.

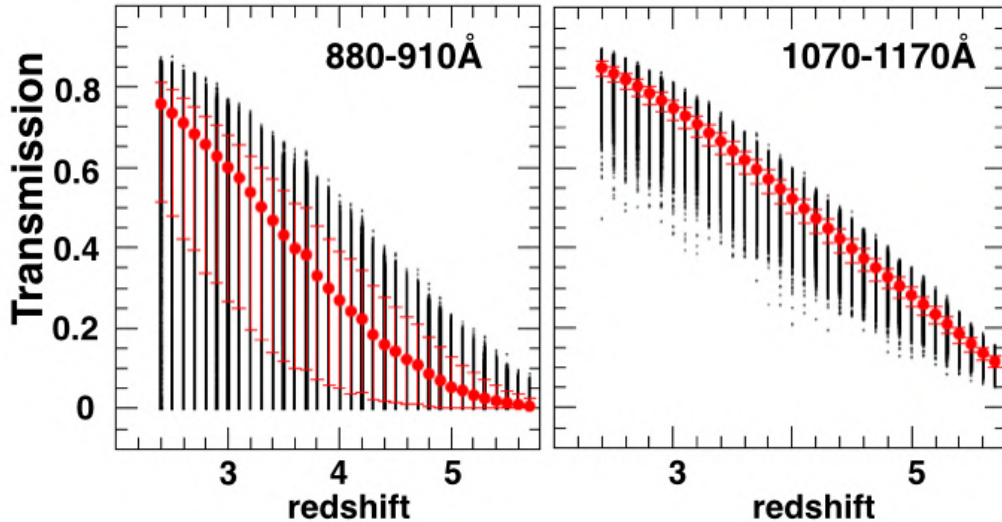
#### 1.4.4 The IGM transmission

Observations of the Ly $\alpha$  forest from QSOs at different redshifts are fundamental for studying the evolution of the IGM. For Lyman Limit Systems (LLSs) that are



optically thick to Ly $\alpha$  photons – particularly the DLAs – the density of neutral hydrogen can be determined by fitting the profile of the saturated Ly $\alpha$  damping wing [e.g., 374, 381, 442]. High-resolution and high signal-to-noise (S/N) observations of the Lyman limit in these QSOs provide valuable insights into the IGM’s temperature and the mean free path for LyC photons [ $\lambda_{\text{mfp}}$ , in Mpc; e.g., 28, 512]. These measurements can also be used to predict the IGM transmission at corresponding redshifts [e.g., 27, 137, 247]. Various analytical models have also been developed to predict the line-of-sight opacity and average IGM transmission at different redshifts [e.g., 208, 457].

In this thesis, we employed the model presented by [208], which uses Monte Carlo (MC) simulations to correct for intergalactic attenuation and estimate the escape fraction of ionizing radiation [e.g., 211, 490, 493]. These simulations are based on an empirical distribution function designed to replicate the observed statistics of the Ly $\alpha$  forest, LLSs, and DLAs. In this approach, a large number of absorbers were simulated along 10,000 random lines of sight across redshifts 0-7, with a redshift interval of  $\Delta z = 0.1$ . Each line of sight spans the wavelength range 600-1300 Å with a resolution of  $\Delta\lambda = 1$  Å. The probability of encountering an absorber follows a Poisson distribution, and the column densities and Doppler parameters are randomly sampled from their empirical distributions. These simulations successfully replicate the observed distributions of absorber number densities at various column densities [e.g., 348, 373–375, 442], and align well with measurements of the mean Ly $\alpha$  transmission [27, 138] and the mean free path of ionizing photons [512] across the redshift range considered. Furthermore, the simulations are consistent with quasar observations [e.g., 138, 395].



**Figure 1.9:** The IGM transmission computed by [208] as a function of redshift for two wavelength ranges: 880-910 Å (left panel) and 1070-1170 Å (right panel). The median transmission values are represented by red filled circles, with error bars indicating the central 68% confidence intervals. [Figure from 491].

Figure 1.9 presents the distributions of median transmission with central 68% intervals in the rest-frame ranges 880-910 Å and 1070-1170 Å. The figure highlights

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the asymmetric distribution with a tail extending toward lower values, primarily due to the presence of DLAs and LLSs. LyC absorption is predominantly caused by systems with  $N_{\text{HI}} > 10^{17} \text{ cm}^{-2}$ , making LyC absorption highly stochastic.

Clearly the direct detection of LyC photons from high-redshift galaxies ( $z \geq 4.5$ ) is infeasible: for this reason researchers have turned to studying the properties of the ISM and conditions at lower redshifts that facilitate the escape of LyC photons and that can be then identified in higher redshift galaxies [e.g., 86, 153, 154, 213, 214, 401, 409, 414, 497]. The next Chapter will explore these findings, focusing on refining our understanding of the reionization process and how ionizing photons escape from galaxies.

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## Chapter 2

# The escape of Lyman continuum photon from galaxies

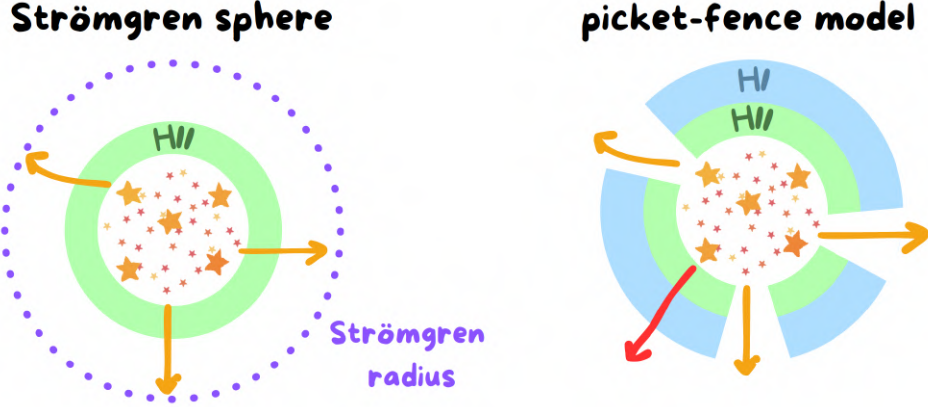
As discussed in Chapter 1, the escape fraction ( $f_{\text{esc}}$ ) is the percentage of ionizing photons produced by a stellar population that successfully reach the IGM. In this Chapter, we examine the physical mechanisms that enable LyC photons to escape from galaxies and present the methods for directly measuring  $f_{\text{esc}}$ , along with a summary of confirmed LyC leakers. Direct observations of LyC escape at  $z \geq 4.5$  (see Figure 1.9) are not feasible due to the IGM opacity (see Section 1.4.4). To overcome this, we determine the LyC escape of cosmic reionizers by identifying lower-redshift galaxies with properties similar to those expected during reionization. By measuring the LyC escape in these lower redshift galaxies, known as LyC emitters (LCEs), and correlating it with key properties of the ISM and the conditions that facilitate LyC photon escape – estimated at high redshifts through spectroscopic and photometric data – we can infer the LyC escape fraction of high-redshift galaxies. This Chapter will focus on these “indirect indicators” and how they can be used to estimate  $f_{\text{esc}}$  during the EoR.

### 2.1 Mechanisms of LyC photons escape

There are several astrophysical conditions within galaxies, such as the structure of the ISM, the presence of young, massive stars, and feedback processes, which ultimately determine whether the LyC photons can successfully reach the IGM. Understanding these conditions is crucial for interpreting the nature of cosmic reionizers and their role in the EoR.

For LyC photons to escape into the IGM, the ISM must have a sufficiently low neutral hydrogen column density so that not all LyC photons get absorbed. In this respect, there are two scenarios that can explain the escape of LyC photons from galaxies [see 527, for a review]: the *density-bounded nebula*, also known as the truncated *Strömgren sphere*, and the *ionization-bounded nebula with holes*, often referred to as the *picket-fence model* [e.g., 379]. These scenarios are schematically illustrated in Figure 2.1, where a model galaxy is represented as a centrally concentrated ensemble of stars surrounded by the HII regions (in green).

The Strömgren radius [Figure 2.1, left panel, 459] defines the boundary around



**Figure 2.1:** Schematic illustrations of LyC escape mechanisms: the Strömgren sphere (left) and the picket-fence model (right). Both scenarios depict a centrally concentrated ensemble of stars surrounded by gas. The ionized gas (HII region) is shown in green, while the neutral gas (HI region) is shown in blue. In the Strömgren sphere scenario, all of the surrounding hydrogen is ionized by the stars, resulting in the truncation of the theoretical Strömgren sphere (the dashed violet circle). In the picket-fence model, LyC photon leakage occurs through channels where the gas has been cleared. In this case, there is sufficient hydrogen to fully form a Strömgren sphere, with HI remaining outside the Strömgren radius.

an ionizing source where the rates of photoionization and recombination are in equilibrium and the medium is totally ionized. In the Strömgren sphere scenario, the LyC flux from a strong star formation episode ionizes all the neutral hydrogen before a complete Strömgren sphere forms, allowing LyC photons to escape into the IGM.

In the picket-fence model (Figure 2.1, right panel), the photoionization rate is insufficient to ionize all the surrounding gas, with HI remaining outside the Strömgren radius. For LyC leakage to occur in this scenario, it is thought that such channels are created by supernovae explosions or stellar winds that clear neutral gas, allowing LyC photons to escape. The fraction of sight-lines that are blocked by these dense HI clouds is described by the covering fraction,  $C_f$ .

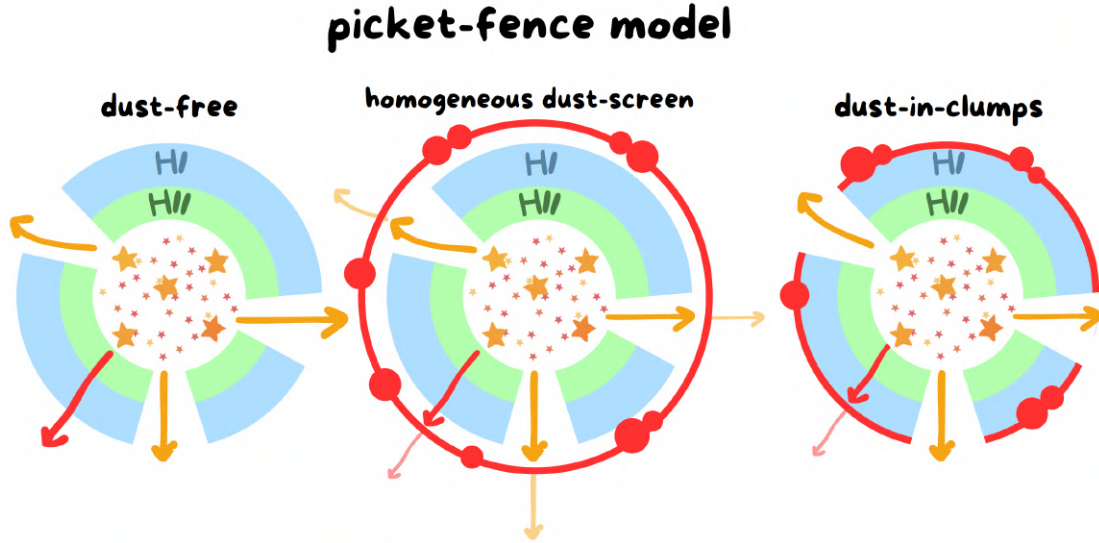
Another important aspect is the distribution of dust relative to the gas clouds. In the picket-fence model [495], two key scenarios are considered, as shown in Figure 2.2. In the *homogeneous dust-screen scenario*, a uniform layer of dust surrounds the entire galaxy (represented by the red circle). In the *dust-in-clumps scenario*, dust is confined within the dense gas clouds of the ISM (shown as the red regions).

Following [165], the solution to the radiative transfer equation in the homogeneous dust-screen geometry allows us to express the emergent flux  $F_\lambda$  as:

$$F_\lambda = F_\lambda^* 10^{-0.4A_\lambda} [C_f \exp(-\tau_\lambda) + (1 - C_f)], \quad (2.1)$$

where  $F_\lambda^*$  is the intrinsic stellar emission,  $\tau_\lambda$  represents the optical depth, given by  $\tau_\lambda = \sigma_\lambda N$ . Here,  $\sigma_\lambda$  is the absorption cross-section and  $N$  is the gas column density. In this model, all the light is attenuated by dust according to  $A_\lambda$ , defined as:

$$A_\lambda = k_\lambda \times E(B - V). \quad (2.2)$$



**Figure 2.2:** *Picket-fence model with varying dust distributions in the ISM. In this model, the ionizing sources are surrounded by HI gas clouds (in red) that are optically thick to LyC radiation. The left panel shows a dust-free model, while the middle and right panels illustrate the two dust configurations explored in this study: a homogeneous dust screen and dust confined within clumps, respectively.*

where  $E(B - V)$  is the relative flux excess in the  $B$  filter relative to the  $V$  filter. This relation is based on the empirical Calzetti law [69] and  $k_\lambda$  is given by the dust-attenuation law. It is important to note that this law is derived from dust reddening observations in local starbursts and might vary with redshift.

For the dust-in-clumps scenario, the emergent flux  $F_\lambda$  is:

$$F_\lambda = F_\lambda^* 10^{-0.4A_\lambda} C_f \exp(-\tau_\lambda) + F_\lambda^* (1 - C_f), \quad (2.3)$$

where  $F_\lambda^* (1 - C_f)$  presents the radiation that escapes directly without any attenuation. In this model, holes are assumed to be free of gas and dust. Without dust or for a covering fraction of unity, both geometries become equivalent.

The covering fraction  $C_f$  can be observationally related to the residual flux of individual lines,  $R_\lambda$ , measured as the ratio of flux at the transition's central wavelength over the continuum. In the case of a homogeneous dust-screen, the covering fraction is given by:

$$R_\lambda = 1 - C_f, \quad (2.4)$$

while for a dust-in-clumps geometry, it depends on both the dust attenuation,  $A_\lambda$ , and the covering fraction:

$$R_\lambda = \frac{1 - C_f}{C_f 10^{-0.4A_\lambda} + (1 - C_f)}. \quad (2.5)$$

Since the escape fraction is proportional to the observed and intrinsic fluxes (Eq. (1.11)), in a dust-screen scenario,  $f_{\text{esc}}$  is expressed as:

$$f_{\text{esc}} = 10^{-0.4A_{912\text{\AA}}} (1 - C_f), \quad (2.6)$$

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and for a clumpy geometry:

$$f_{\text{esc}} = 1 - C_f. \quad (2.7)$$

In the clumpy geometry model, it is assumed that the clumps are optically thick to LyC photons. At a given residual flux, a higher escape fraction is inferred when dust is distributed in clumps compared to a foreground screen.

It is crucial to note that the ISM of real galaxies is far more complex than these simplified models. Several factors such as off-center star formation, significant density variations, and galaxy interactions can create more complex scenarios. Real cases of LyC leakage are likely driven by a combination of these mechanisms.

## 2.2 Measuring the LyC escape fraction

In the previous Chapter, we defined the escape fraction  $f_{\text{esc}}$  as:

$$f_{\text{esc}} = \frac{F_{\text{LyC,esc}}}{F_{\text{LyC,int}}} \quad (1.11),$$

where  $F_{\text{LyC,esc}}$  is the LyC flux that reaches the IGM, and  $F_{\text{LyC,int}}$  is the intrinsic LyC flux emitted by the galaxy. After taking into account the effect of the IGM, we defined the  $F_{\text{LyC,esc}}$  as  $F_{\text{LyC,esc}} = F_{\text{LyC,obs}} e^{\tau_{\text{HI,IGM}}}$  (1.12). We can also estimate the intrinsic flux density  $F_{\text{LyC,int}}$  from the observed flux at 1500 Å,  $F_{1500,\text{obs}}$ , as follows:

$$F_{\text{LyC,int}} = \frac{F_{\text{LyC,int}}}{F_{1500,\text{int}}} F_{1500,\text{obs}} e^{\tau_{\text{HI,ISM}}}. \quad (2.8)$$

where  $\tau_{\text{HI,ISM}}$  is the optical depth of neutral hydrogen in the ISM of the galaxy.

Using these relations,  $f_{\text{esc}}$  can be calculated as:

$$f_{\text{esc}} = \frac{(F_{1500}/F_{\text{LyC}})_{\text{int}}}{(F_{1500}/F_{\text{LyC}})_{\text{obs}}} \times e^{\tau_{\text{HI,IGM}}} \times e^{-\tau_{\text{HI,ISM}}}, \quad (2.9)$$

where  $\tau_{\text{HI,IGM}}$  can be estimated by adopting the IGM model from [208] within the wavelength range where LyC flux is detected at certain redshifts. To compute the absolute value of  $f_{\text{esc}}$ , we should also include a term related to dust attenuation, so that:

$$f_{\text{esc}} = \frac{(F_{1500}/F_{\text{LyC}})_{\text{int}}}{(F_{1500}/F_{\text{LyC}})_{\text{obs}}} \times 10^{-0.4A_{1500\text{Å}}} \times e^{\tau_{\text{HI,IGM}}} \times e^{-\tau_{\text{HI,ISM}}}, \quad (2.10)$$

where  $A_{1500\text{Å}}$  is the dust attenuation at 1500 Å, which can be derived from the relative flux excess in the  $B$  filter relative to the  $V$  filter,  $E(B - V)$  [70]:

$$A_{1500\text{Å}} = 10.33 \times E(B - V). \quad (2.11)$$

The intrinsic ratio  $(F_{1500}/F_{\text{LyC}})_{\text{int}}$  can be determined by assuming the age of the observed stellar population. This ratio can also be expressed as  $(L_{1500}/L_{\text{LyC}})_{\text{int}}$ , the intrinsic luminosity ratio: for typical high redshift stellar populations it can vary between 3 and 7 [e.g., 173, 293, 330, 433].

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Eq. (2.10) can be simplified by excluding the term related to dust attenuation at  $1500 \text{ \AA}$  and assuming that  $A_{1500 \text{ \AA}} = A_{900 \text{ \AA}}$ , allowing us to infer the relative escape fraction,  $f_{\text{esc}}^{\text{rel}}$ . This represents the fraction of ionizing photons that manage to escape the galaxy’s ISM before any interaction with the IGM. This approach is common in the literature because it does not require any assumptions about the dust-attenuation law. Using spectroscopic surveys both in the local and intermediate redshift Universe, where the LyC wavelength is detectable, we can derive the intrinsic escape fraction,  $f_{\text{esc}}^{\text{int}}$ , through SED fitting, using the rest-frame UV [e.g., 213], or by estimating the age of the underlying stellar population using tracers such as the  $\text{H}\beta$  equivalent width (EW), with values provided by the SED fitting [129]. The intrinsic escape fraction represents the fraction of ionizing photons produced by a galaxy that escape its ISM without being absorbed by gas or dust within the galaxy.

By determining  $f_{\text{esc}}^{\text{rel}}$  and  $f_{\text{esc}}^{\text{int}}$ , we can define the absolute escape fraction,  $f_{\text{esc}}$ , as follows:

$$f_{\text{esc}} = \frac{f_{\text{esc}}^{\text{rel}}}{f_{\text{esc}}^{\text{int}}}. \quad (2.12)$$

Unless otherwise specified, this thesis will always refer to the absolute escape fraction when discussing  $f_{\text{esc}}$ .

## 2.3 Direct measurements of Lyman Continuum escape

LyC emission can be observed directly in galaxies at low and intermediate redshifts. There are two primary methods for measuring LyC emission: photometry, which involves analyzing the flux of a source in filters corresponding to  $\lambda < 912 \text{ \AA}$  at the source’s redshift; and spectroscopy, which examines the spectrum of the source within the same wavelength range.

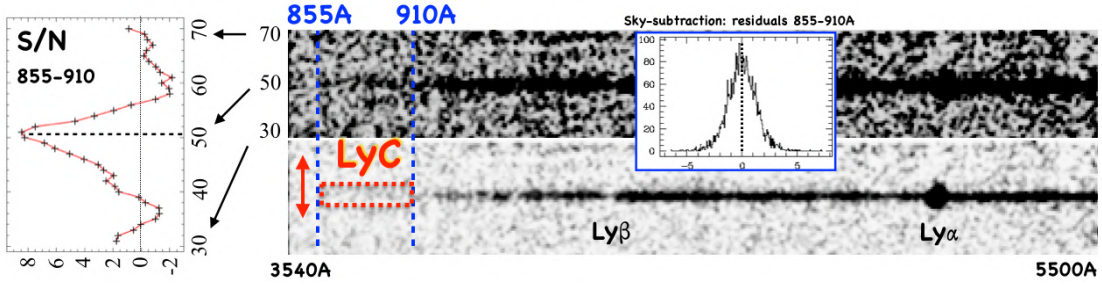
However, detecting LCEs is challenging, as LyC emission is typically much weaker than the galaxy’s other emission. In Figure 2.3, we present the spectroscopic detection of an LCE at  $z = 3.2121$ , Ion2 [112, 487], which has an escape fraction  $f_{\text{esc}} \sim 50 - 100\%$ . As shown, the LyC flux is significantly weaker than the emission at longer wavelengths. Additionally, as discussed in Section 1.4.4, the transmission of the IGM at  $z \leq 4.5$  varies considerably across various lines of sight.

Only recently, studies like the Low-redshift Lyman Continuum Survey (LzLCS, GO 15626, PI A. Jaskot) [153, 154, 402] at low redshift have allowed the first statistically significant LCE samples to be assembled. These samples not only provide direct measurements (or upper limits) of the LyC escape fraction,  $f_{\text{esc}}$ , but also offer a comprehensive set of physical, morphological, and spectroscopic properties of these emitters.

### 2.3.1 Local LyC emitters

Haro 11, at a redshift of  $z = 0.0206$ , was the first local LyC emitter ever discovered, with LyC detection with the *Far Ultraviolet Spectroscopic Explorer* (FUSE) [36].





**Figure 2.3:** Two-dimensional low-resolution *Vi*si**ble** *M*ulti**o**bject *S*pectro**g**raph (VIMOS) UV spectrum of Ion2 at  $z = 3.2121$ . Bottom: 2D spectrum showing the  $\text{Ly}\alpha$  and  $\text{Ly}\beta$  lines. Top:  $\text{LyC}$  flux in the wavelength range of 855–910 Å (marked by vertical dotted blue lines). Left: moving average calculated within a rectangular aperture (855–910 Å  $\times$  1.25'', red-dotted rectangle) along the spatial direction, normalized by its rms. The inset shows the pixel distribution of the background after sky subtraction in the 855–910 Å region (derived from the S/N spectrum). [Figure from 112].

The absolute escape fraction  $f_{\text{esc}}$  for this galaxy, assuming a standard Salpeter initial mass function, is estimated to be around  $\sim 0.04 - 0.1$ . However, subsequent studies have debated the escape fraction for Haro 11. For instance, [264] reported a lower escape fraction of  $f_{\text{esc}} \sim 0.033$ , which aligns with recent HST imaging results from [255]. Haro 11 exhibits a peculiar morphology with three distinct knots, likely due to an ongoing merger event. Since the FUSE aperture covered the entire galaxy, follow-up observations with HST/*Cosmic Origins Spectrograph* (COS) targeted the individual knots (A, B, and C) to study  $\text{Ly}\alpha$  emission and ISM properties [351]. All three knots exhibit modest  $\text{Ly}\alpha$  emission and blue UV continuum  $\beta$  slopes [351, see Section 2.4.6]. HST imaging in the [OII] and [OIII] lines suggested knot A as the likely source of  $\text{LyC}$  leakage due to its high ionization on a large scale [242], though the high  $\text{Ly}\alpha$  escape fraction in knot C indicates it may be the main contributor.

FUSE also played a role in the discovery of Tol 1247-232 [266], another local  $\text{LyC}$  emitter at  $z = 0.048$ , which was further analyzed through HST/COS observations [320]. Like Haro 11, Tol 1247-232 displays a highly disturbed and irregular morphology, indicative of a major merger event, which is typical of starbursts that are considered candidates for  $\text{LyC}$  emission in the local Universe. While earlier studies suggested a relatively high absolute escape fraction,  $f_{\text{esc}} \sim 0.12$  [265], originating from its central region, with a stellar mass of  $1.13 \times 10^9 M_{\odot}$  and a UV magnitude of  $-21$ , subsequent works have reported significantly lower values:  $f_{\text{esc}} \sim 0.024$  [266],  $f_{\text{esc}} \sim 0.045$  [268],  $f_{\text{esc}} \sim 0.015$  [376], and  $f_{\text{esc}} < 0.004$  [83].

Over the years, HST/COS has been extensively used to expand the sample of local LCEs [e.g., 44, 215–218, 268, 291, 376, 501]. Notable campaigns, such as [215–218], detected  $\text{LyC}$  emission in over a dozen galaxies at redshifts  $z < 0.4$ , with  $f_{\text{esc}}$  ranging from a few to 70%. A key approach used in these works involved selecting candidate LCEs based on their [OIII] $\lambda 5007$ /[OII] $\lambda\lambda 3726, 29$  emission line flux ratios (see Section 2.4.3). The use of [OIII] $\lambda 5007$ /[OII] $\lambda\lambda 3726, 29$  as a selection criterion proved successful in multiple HST/COS programs, leading to the identification of several strong LCEs with high  $\text{LyC}$  escape fractions. However, [217] showed that using [OIII] $\lambda 5007$ /[OII] $\lambda\lambda 3726, 29$  alone may not always reliably



predict LyC leakage, indicating the need for a more comprehensive set of selection criteria.

Building on these results, the Low-redshift Lyman Continuum Survey (LzLCS) – an HST Large Program presented by [153, 154, 402] – further expanded the search for LCEs, providing the most comprehensive sample of local LyC emitters to date. The galaxies targeted in this survey, at  $z = 0.2 - 0.4$ , meet one or more of the proposed selection criteria for LCEs: high nebular ionization ( $[\text{OIII}]\lambda 5007/[\text{OII}]\lambda\lambda 3726,29 \geq 3$ ), blue UV slopes ( $\beta < -2$ ), and/or concentrated star formation ( $\Sigma_{\text{SFR}} > 0.1 \text{ M}_{\odot} \text{ yr}^{-1} \text{ kpc}^{-2}$ ). These properties are examined in detail in Section 2.4. The LzLCS sample was augmented with archival COS observations from [215–218, 501]. The combined dataset, referred to as the LzLCS+ sample, includes 88 low-redshift galaxies with LyC observations. These galaxies span a range of stellar masses from  $10^8$  to  $10^{10} \text{ M}_{\odot}$ , and have observed (uncorrected for internal reddening) 1500 Å absolute magnitudes from  $-21.5$  to  $-18.3$ , with either detections or stringent upper limits on LyC emission. With comprehensive spectroscopic and photometric data available for all sources, several promising indirect diagnostics were evaluated to assess their correlation with LyC emission.

### 2.3.2 Intermediate-redshift LyC emitters

Detecting LyC radiation at redshift  $z \sim 3$  is particularly challenging due to the high IGM opacity, which significantly attenuates ionizing radiation (Figure 1.9, Section 1.4.4). While ionizing photons can be detected up to  $z \leq 4.5$ , measurements are affected by the variability in IGM absorption along different lines of sight.

Additionally, the presence of low-redshift interlopers complicates potential LyC detections at high redshifts [493]. Surveys targeting Lyman-break galaxies (LBGs) at  $z \approx 0.7 - 4$  through imaging [6, 30, 173, 175, 182, 223, 321, 328, 330, 399, 438] and spectroscopic studies [293, 319, 370, 428, 432, 456] have been largely unsuccessful in the identification of individual LCEs, although they have allowed us to obtain average  $f_{\text{esc}}$  estimates for star-forming galaxies using stacking techniques.

Despite these challenges, some solid LyC leakers at  $z \approx 2 - 4$  have been identified using imaging (e.g., Wide Field Camera 3 - WFC3/UVIS F275W and F336W filters aboard the HST, Subaru’s Suprime-Cam, and LBC at the LBT telescope) and spectroscopy (e.g., Keck I’s LRIS-B spectrograph).

A notable example is the Sunburst Arc, a lensed compact dwarf galaxy at  $z = 2.2$ , with an average  $f_{\text{esc}}$  of 0.32 considering the LyC-emitting regions, based on 11 gravitationally lensed HST images [246, 387]. It reveals a remarkably small-scale region (tens of parsecs) with significant LyC escape. This compact ( $< 100 \text{ pc}$ ) LyC-leaking area exhibits extreme characteristics, including an exceptionally blue UV slope ( $\beta = -2.9 \pm 0.1$ ), a high ionization state ( $[\text{OIII}]\lambda 5007/[\text{OII}]\lambda 3727 = 11 \pm 3$  and  $[\text{OIII}]\lambda 5007/\text{H}\beta = 6.8 \pm 0.4$ ), strong oxygen emission ( $\text{EW}[\text{OIII}] > 1000 \text{ Å}$ ), and a high Ly $\alpha$  escape fraction. None of these properties are present in the galaxy’s non-leaking regions.

Other examples include Ion2 [ $z = 3.2$ ,  $f_{\text{esc}} = 0.5\text{--}0.9$ , 112, 487, 491], Ion1 [ $z = 3.1$ ,  $f_{\text{esc}} = 0.25\text{--}1$ , 490], and Ion3 [ $z = 3.9$ ,  $f_{\text{esc}} = 0.6$ , 485], which holds the record for the most distant confirmed LCE [227]. These galaxies are characterized

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by strong Balmer and [OIII] emission lines, low mass, and elevated  $\xi_{\text{ion}}$  values (see Figure 1.8). [297, 298] reported robust spectroscopic detection of LyC flux in two of the brightest UV-selected star-forming galaxies: J0121+0025 ( $z = 3.244$ ,  $f_{\text{esc}} \approx 0.4$ ) and J1316+2614 ( $z = 3.6130$ ,  $f_{\text{esc}} \approx 0.9$ ). J1316+2614, with  $M_{\text{UV}} \approx -24$ , holds the record for the largest  $f_{\text{esc}}$  and the brightest LyC emission.

Systematic searches, such as the Lyman Continuum Escape Survey (LACES) [151, 337] and studies by [408] and [384], have provided statistical correlations between individual  $f_{\text{esc}}$  measurements and galaxy properties. Much of our knowledge about LCEs, however, comes from the Keck Lyman Continuum Survey (KLCS), with 13 confirmed LyC detections at  $z \approx 3.1$  [358, 359, 457]. These LCEs are generally compact star-forming galaxies with high star-formation rate (SFR) surface densities, high ionization parameters, strong Ly $\alpha$  emission, and low dust attenuation.

Several recent studies have attempted to further identify LyC leakers at intermediate redshifts. [46] placed strict upper limits on the LyC escape fraction  $f_{\text{esc}} < 0.05$  for Lyman Break Galaxies (LBGs) in the GOODS-S field, based on deep LBC U-band imaging. Similarly, [175] examined a sample of galaxies at  $z \sim 3.3$  in the COSMOS field, obtaining an upper limit on the relative escape fraction  $f_{\text{esc}}^{\text{rel}} < 0.02$ . This was followed by [173], who used galaxies with spectroscopic redshifts in the CANDELS and COSMOS fields to further constrain  $f_{\text{esc}}^{\text{rel}} \lesssim 0.1$  for bright LBGs. More recently, [502] utilized the UVCANDELS HST Treasury Program (HST-GO-15647) to search for leakers in four extragalactic fields, identifying five potential candidates. By stacking galaxies in different redshift bins, they reported a tentative LyC detection at  $z \sim 2.75$ , with a stacked relative escape fraction  $f_{\text{esc}}^{\text{rel}}$  of 0.1. [234] likewise targeted eight strongly lensed galaxies at  $1.3 \leq z \leq 3$  using HST’s SBCF150LP and WFC3/F336W filters, placing upper limits on  $f_{\text{esc}}$  between 0.03 and 0.15. These findings, however, remain tentative, and a consistent picture of LyC leakage at intermediate redshifts remains elusive.

These studies highlight that our understanding of LyC escape has advanced significantly in the past decade, although questions remain. While the fraction of known LyC leakers is relatively high at  $z \sim 0.3 - 0.4$ , confirming LyC emission from galaxies at  $z \sim 3$  and beyond remains challenging. At these redshifts, only a few bright sources with significant LyC leakage have been identified, primarily due to observational limitations and increasing absorption by the IGM.

Furthermore, current studies indicate a diversity in LyC escape fractions across galaxies. Some galaxies exhibit substantial LyC leakage (with escape fractions reaching up to  $\sim 50\text{--}100\%$ ), while others show no measurable escape. This variation suggests that different physical mechanisms, galaxy properties or line-of-sight conditions likely influence the escape fraction, which may vary significantly across redshifts.

## 2.4 The “indirect indicators” of the LyC escape fraction

To enhance our understanding of the conditions enabling LyC emission, it is crucial to investigate how the  $f_{\text{esc}}$  depends on the other physical properties of the galaxies that we can determine. Understanding the properties of LyC leakers is

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crucial for two main reasons. First, a detailed investigation into the mechanisms of LyC escape can clarify the role of factors such as star formation rate, stellar populations, gas distribution, and dust content in influencing the LyC escape fraction. Second, identifying the characteristic properties of LyC leakers allows for a deeper understanding of the nature of cosmic reionizers. In the following, we will provide a brief overview of the Ly $\alpha$  and other rest-frame UV lines, along with the nebular properties, luminosity, morphology, and dust attenuation of the sources, which all present interesting correlation (or anti-correlation) with the LyC  $f_{\text{esc}}$  and thus can serve as “indirect indicators” of LyC leakage.

### 2.4.1 Ly $\alpha$ line

The Ly $\alpha$  recombination emission line is very sensitive to the density of neutral hydrogen along the line of sight: its properties correlate in various ways to the escape of LyC photons [e.g., 218, 497, 498].

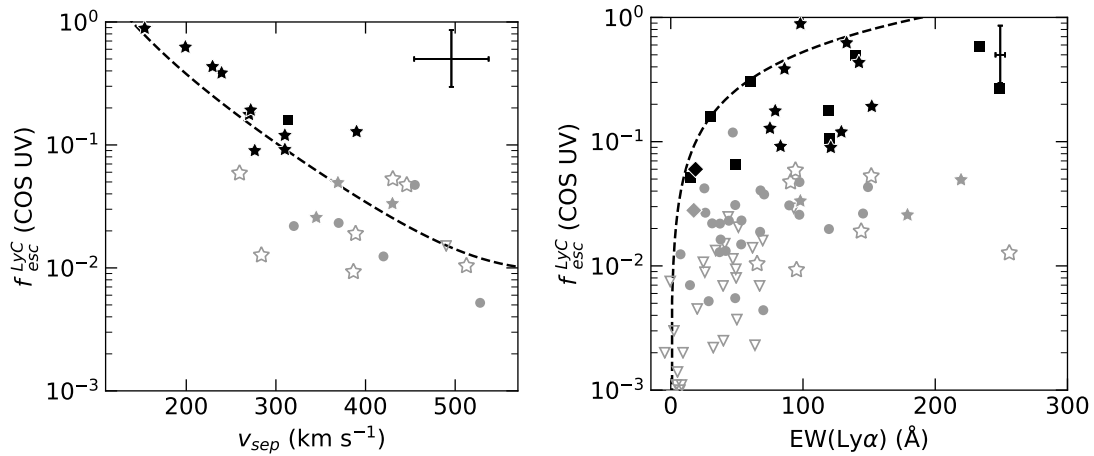
The shape of the Ly $\alpha$  emission line is strongly influenced by the absorption and re-emission processes of neutral hydrogen within the host galaxy. This process, often referred to as “scattering”, is a result of interactions with the surrounding ISM [e.g., 166, 497]. In the Strömgren sphere model, Ly $\alpha$  photons, which have a much higher likelihood of interaction near the line center, undergo multiple scatterings before escaping, resulting in a broadened, double-peaked profile. In most local LyC-leaking galaxies, the Ly $\alpha$  emission show this type of double-peaked Ly $\alpha$  profile, with the narrow separation between the peaks indicating a lower amount of neutral gas, which in turn allows for higher LyC leakage [498].

In the picket fence model, LyC photons can only escape through gaps in the ISM between dense clumps of gas. In this case, Ly $\alpha$  photons can still scatter off the edges of these clumps and escape after a few interactions, without undergoing major radiative transfer effects. This process can result in a narrow, single-peaked Ly $\alpha$  line that retains much of its original shape [e.g., 179]. However, [180] demonstrate that in a clumpy medium, Ly $\alpha$  radiative transfer can also produce double-peaked profiles, even when the medium facilitates LyC escape.

In a model with ionized channels, LyC photons and the central Ly $\alpha$  peak escape directly through open channels in the gas. However, Ly $\alpha$  photons can also scatter in denser, neutral regions, resulting in significant radiative transfer effects. This produces a Ly $\alpha$  profile with a narrow central peak (from the open channels) plus a broader, double-peaked feature emerging from the optically thick regions of gas [e.g., 218, 387, 488].

While the above models offer valuable insights into the behavior of Ly $\alpha$  emission, interpreting the line remains complex due to its sensitivity to additional factors.

Observationally, a very strong correlation is found between  $f_{\text{esc}}$  and  $\text{EW}(\text{Ly}\alpha)$  as shown in Figure 2.4, right panel, for the lzLCS+ sample. It is important to note that part of this trend may be influenced by selection effects, as the LCEs from the sample were chosen based on their strong optical emission lines, among other criteria, to optimize the detection of intrinsic LyC flux. The large scatter is particularly attributed to the large sample size and the higher number of non-detections in the LzLCS+ sample.



**Figure 2.4:** *Left: Correlation between Ly $\alpha$  velocity peak separation  $v_{\text{sep}}$  and  $f_{\text{esc}}$  for strong LCEs (black), weak LCEs (gray) and non-emitters (open gray). LzLCS results are indicated as squares (for strong LCEs), circles (for weak LCEs), and triangles (for non-detections). Right: Correlation between  $\text{EW}(\text{Ly}\alpha)$  and  $f_{\text{esc}}$ . Same symbols as the left panel. The dashed line represents the relationship derived by [358], based on measurements of LyC from stacked spectra of galaxies at  $z \sim 3$ . [Figures from 154].*

Another notable correlation is with the Ly $\alpha$  escape fraction ( $f_{\text{esc}}^{\text{Ly}\alpha}$ ): this quantity is generally higher than the LyC escape fraction, typically by a factor of 2–10, due to the resonant nature of the Ly $\alpha$  line [310, 497]. While LyC photons can only escape when the HI column density along the line of sight is particularly low, Ly $\alpha$  photons have a higher likelihood of escaping due to scattering effects, which can redirect them into paths with lower optical depth. However, the reverse scenario is also possible, albeit less commonly observed: the optical depth for Ly $\alpha$  photons can be so high (often  $\sim 10,000$  times greater than for LyC) that Ly $\alpha$  cannot escape, even in cases where LyC photons can penetrate the ISM [227, 255]. This means that galaxies with significant LyC escape ( $f_{\text{esc}} \gtrsim 10\%$ ) are expected to have at least moderate Ly $\alpha$  escape fractions ( $f_{\text{esc}}^{\text{Ly}\alpha} \gtrsim 20\%$ ) [497], although exceptions may arise depending on the ISM’s structure and distribution.

Finally, in Figure 2.4 we present the anti-correlation between the LyC escape fraction and the separation of the two peaks in the Ly $\alpha$  profile. Smaller peak separations ( $\lesssim 300 \text{ km s}^{-1}$ ) correspond to lower HI column densities, which facilitate both Ly $\alpha$  and LyC escape. This behavior is consistent with radiative transfer models that show narrower peak separations and smaller velocity shifts in media with lower neutral hydrogen column densities.

Although the above correlations are all very significant, we note that Ly $\alpha$  emission cannot be typically employed to indicate LyC escape during the EoR [342]. The neutral IGM becomes more pervasive at higher redshifts, which can significantly reduce the Ly $\alpha$  signal, particularly its blue peak [e.g., 198, 365, 419, 518]. This effect complicates the use of Ly $\alpha$  emission alone to predict LyC  $f_{\text{esc}}$ , as some high-redshift galaxies, including LBGs and LCEs, show little or no Ly $\alpha$  emission [e.g., 227, 492].

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### 2.4.2 Other rest-frame UV lines

The MgII  $\lambda 2800$  doublet has been proposed as a promising proxy for LyC escape [84, 200], as its escape is regulated by resonant scattering in the same low column-density gas that governs Ly $\alpha$  and LyC escape [see also 219, 516]. This is because the ionization potential to doubly ionize magnesium (15 eV) is close to that of neutral hydrogen (13.6 eV). In low-metallicity galaxies, MgII emission can be relatively bright, with fluxes ranging from 10–60% of the observed H $\beta$  flux [185]. The intrinsic flux ratio,  $R$ , between the red and blue peaks of the MgII doublet is set by atomic physics at a ratio of 2, but the MgII  $\lambda 2800$  optical depth along the line of sight can decrease this ratio, reflecting the conditions of the intervening gas [200]. In LCEs, MgII is often observed in emission, with the absence of absorption indicating a low column density of neutral gas that facilitates LyC escape. This suggests that both MgII  $\lambda 2800$  and LyC escape through an optically thin, ionized medium. Based on this assumption, the ratio of the MgII doublet can be used to estimate the column density of MgII, and thus the HI column density, according to the following relation [84]:

$$N_{\text{HI}} = -2.2 \times 10^{17} \text{ cm}^{-2} \ln \left( \frac{R}{2} \right). \quad (2.13)$$

However, this method faces challenges. The assumption of an optically thin ISM may not hold for star-forming galaxies with lower LyC escape fractions, which would require higher signal-to-noise and resolution data to determine the MgII covering fraction. Moreover, the resonant nature of MgII (similar to Ly $\alpha$ ) and other kinematic effects complicate its interpretation, leading some simulations [e.g., 239] to question its reliability as a LyC escape proxy.

More recently, the nebular CIV  $\lambda\lambda 1548, 1551$  emission line, requiring similar high ionization energies to the HeII line ( $E > 47.9$  eV and  $> 54.4$  eV respectively), has gained attention since its presence may be strongly linked to the escape of LyC photons from galaxies [e.g., 301, 409, 415, 423], although this line is generally much fainter than Ly $\alpha$ . The next chapter will focus on CIV as a predictor of high LyC escape.

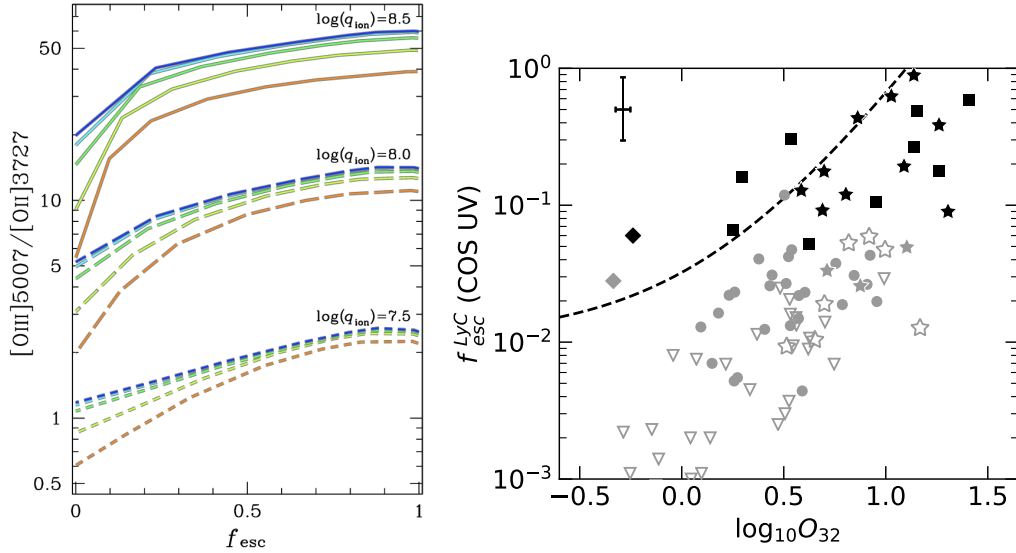
### 2.4.3 Nebular properties: the O32 emission line ratio

The [OIII] $\lambda\lambda 5007$ /[OII] $\lambda 3727$  emission line ratio has been proposed for a long time as an indicator of the LyC escape fraction, due to its association with density-bounded HII regions in galaxies, which can potentially facilitate LyC leakage [218, 224, 292, 333]. This connection arises because the ionization energy of neutral oxygen (13.61 eV) is very close to that of neutral hydrogen, leading to the formation of [OIII] and [OII] lines in regions where hydrogen is also ionized. The results of CLOUDY simulations performed by [333] are shown in Figure 2.5, left panel. For example, one of the best studied leakers at intermediate redshift, Ion2, exhibits an escape fraction  $f_{\text{esc}} > 0.5$  and an [OIII] $\lambda\lambda 5007$ /[OII] $\lambda 3727$  ratio exceeding 10 [112, 487, 488]. [154] proposed an [OIII] $\lambda\lambda 5007$ /[OII] $\lambda 3727$  threshold of 5 as a criterion for identifying potential strong LyC leakers.

However, high  $[\text{OIII}]\lambda\lambda 5007/[\text{OII}]\lambda 3727$  values have been observed in sources with minimal or no LyC leakage [e.g., 25, 218]. For instance, [330] identified a sample of strong  $[\text{OIII}]$  emitters with an average escape fraction  $f_{\text{esc}} < 0.1$ , highlighting the uncertainties in using  $[\text{OIII}]\lambda\lambda 5007/[\text{OII}]\lambda 3727$  as an indirect indicator.

The latest results show indeed a very large scatter in the relation between the two quantities, as evident in Figure 2.5. This large scatter is due to the fact that other mechanisms can also influence the ionization of oxygen [453], indicating that the  $[\text{OIII}]\lambda\lambda 5007/[\text{OII}]\lambda 3727$  ratio can not be solely relied upon as an indirect indicator of LyC leakage [337]. The  $[\text{OIII}]\lambda\lambda 5007/[\text{OII}]\lambda 3727$  ratio is indeed sensitive to factors such as metallicity and ionization parameters [26, 240, 333, 407], as well as the diversity of escape conditions along different lines of sight. Unlike  $f_{\text{esc}}$ , which is inherently a line-of-sight measurement,  $[\text{OIII}]\lambda\lambda 5007/[\text{OII}]\lambda 3727$  represents a more global diagnostic of a galaxy’s ionization structure. Furthermore, variations in  $[\text{OIII}]\lambda\lambda 5007/[\text{OII}]\lambda 3727$  may reflect an evolutionary sequence, with LyC escape becoming more prominent at later stages as massive stars fade and the ISM becomes more porous [154].

In the rest of this thesis, we will adopt the ratio  $[\text{OIII}]\lambda\lambda 5007,4959/[\text{OII}]\lambda 3727$  as O32. This definition does not significantly alter the considerations discussed in this section, as the physical insights derived from  $[\text{OIII}]\lambda\lambda 5007/[\text{OII}]\lambda 3727$  remain consistent.



**Figure 2.5:** Left: O32 ratios and predicted  $f_{\text{esc}}$  from CLOUDY models for different ionization parameters. The solid, long-dashed, and dashed lines represent models with  $\log(q_{\text{ion}})$  values of 8.5, 8.0, and 7.5, respectively. Each line color corresponds to a different metallicity: blue for  $Z = 0.05Z_{\odot}$ , cyan for  $Z = 0.1Z_{\odot}$ , green for  $Z = 0.2Z_{\odot}$ , light green for  $Z = 0.5Z_{\odot}$ , and orange for  $Z = 1.0Z_{\odot}$ . Right: Correlation using the LzLCS+ sample. Same as Figure 2.4, right panel. [Left figure from 333], [right figure from 154].

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#### 2.4.4 Morphology

A common feature observed among the first discovered LCEs is their compact morphology in the rest-frame UV. Many confirmed LyC leakers exhibit a small UV half-light radius,  $r_e$ , both at low [e.g., 154, 213, 215, 216, 218] and intermediate redshift [e.g., 292, 485].

The small  $r_e$  observed in these galaxies suggests that intense star formation is concentrated in a compact region, leading to a high star formation rate surface density ( $\Sigma_{\text{SFR}}$ ), defined as:

$$\Sigma_{\text{SFR}} = \frac{\text{SFR}}{2\pi r_e^2}. \quad (2.14)$$

The observed trends could be explained by the fact that feedback from intense star formation could generate bubbles or chimneys in the ISM of the host galaxy, allowing LyC photons to escape through low-density channels. Observations of several compact LCEs support the link between high  $\Sigma_{\text{SFR}}$  and high  $f_{\text{esc}}$ , suggesting that compactness can indeed be linked to LyC leakage [e.g., 154, 218, 292]. In this context, J1316+2614, at  $z = 3.613$  [297], stands out as the UV-brightest star-forming galaxy known (with  $M_{\text{UV}} = -24.7$ ) and one of the strongest LyC emitters, with an escape fraction of approximately 0.9. This source exhibits a very compact, but resolved, morphology in the LyC and rest-UV with an effective radius  $r_e \sim 0.22$  kpc, and an exceptionally high  $\log_{10} \Sigma_{\text{SFR}}$  ( $\approx 3.47 M_{\odot} \text{ yr}^{-1} \text{ kpc}^{-2}$ ). The high star formation efficiency is likely the primary factor driving the substantial LyC escape, rather than mechanical or radiative feedback alone. Higher efficiencies result in less residual gas to absorb LyC photons, facilitate easier gas removal from the starburst region, and may lead to the formation of more massive stars, thereby enhancing LyC emission [299].

However, compactness may not be a universal characteristic of all LCEs [e.g., 292]. For instance, the well known leakers Haro 11 at  $z = 0.02$  [385] and z19863 at  $z = 3.088$  [184] exhibit extended and irregular structures, likely due to past merger events. Mergers and galaxy-galaxy interactions are known to tidally strip gas and stars, leading to the formation of highly inhomogeneous ISM [100, 444, 473]. Few LyC emitters in the local Universe are close enough to allow spatial mapping of their neutral gas density. Deep 21 cm observations of Haro 11 have revealed pockets of optically thin neutral hydrogen, most likely created by a recent merger event [263].

#### 2.4.5 Luminosity and stellar mass

Although it has long been claimed that LyC leakers are low mass galaxies, known LyC leakers at low and intermediate redshifts exhibit a wide range of stellar masses and UV magnitudes. Low-mass galaxies ( $M_{\star} \sim 10^9 M_{\odot}$ ), with their weaker gravitational potentials, are more prone to the removal of obscuring material by stellar winds and supernovae, thereby enhancing the escape fraction of LyC photons [e.g., 356, 378, 392, 510]. In contrast, more massive galaxies possess the gas reservoirs necessary to sustain high star formation rate (SFR) densities. These

galaxies experience less negative feedback from stellar winds and SNe, allowing for more substantial star formation and greater LyC photon production [513].

The debate over which galaxy mass regime primarily drives reionization is ongoing. Some models, based on empirical data, suggest that relatively more luminous, though still faint, galaxies are the main sources of LyC photons responsible for reionization [331]. Other models propose that less luminous galaxies dominate reionization due to their steeper luminosity functions [e.g., 149]. While observations of local LyC emission [e.g., 153, 154, 218] provide some evidence of increasing  $f_{\text{esc}}$  toward fainter magnitudes, these studies are limited to galaxies with  $M_{\text{UV}} \gtrsim -18$ , leaving the contribution of even fainter galaxies to reionization unconstrained. Radiation hydrodynamical simulations [e.g., 474] and models incorporating massive galaxies, AGNs, and dwarf galaxies [e.g., 110] tend to support the dwarf-galaxy scenario, although with contributions from multiple populations.

#### 2.4.6 Dust attenuation

It was long noticed that some confirmed LyC leakers exhibit very blue UV slopes. For example, galaxies like Ion2 [112, 487, 491], Ion3 [485] and J1316+2614 [297] exhibit  $\beta$  slopes  $< -2$ . The UV  $\beta$  slope is one of the simplest proxies for the dust content in galaxies, since dust absorbs preferentially blue wavelengths [70]. It is defined as the continuum slope of the stellar population in the far-ultraviolet (FUV) spectrum, between rest-frame 1300 Å and 1800 Å, and it is calculated following the relation [67, 318]:

$$F_{\lambda} \propto \lambda^{\beta}, \quad (2.15)$$

where  $F_{\lambda}$  is the flux density at wavelength  $\lambda$ . When a significant amount of dust exists in the ISM, it not only reddens the  $\beta$  slope, leading to less negative  $\beta$  values, but also absorb LyC photons, consequently reducing the number of photons escaping from the galaxy.

Observationally, the  $\beta$  slope is sensitive not only to dust extinction [379] but also to the age of the stellar population, which can also significantly influence the amount of escaping LyC photons. A steeply negative  $\beta$  (i.e., bluer values) typically indicates a stellar population dominated by young stars actively producing LyC photons. In absence of dust, theoretical models suggest that the  $\beta$  values for such young populations range from  $-2.8$  to  $-2.5$  [129, 269, 377].

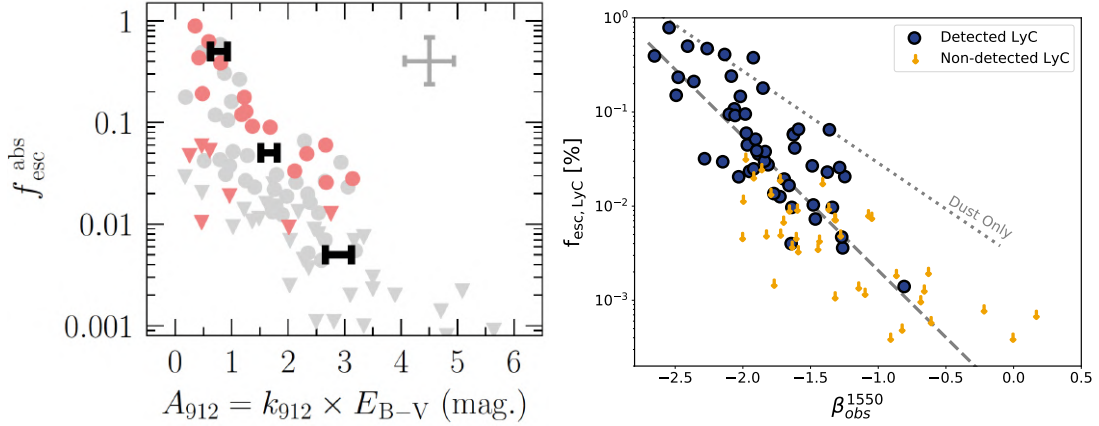
Using the LzLCS+, [86] confirmed the robust correlation between  $\beta$  and the escape fraction of LyC photons (Figure 2.6, right panel). Based on this correlation, they derived a relation for  $f_{\text{esc}}$  based solely on the observed  $\beta$ :

$$f_{\text{esc}} = (1.3 \pm 0.6) \times 10^{-4} \times 10^{-1.22 \pm 0.1 \beta_{\text{obs}}}. \quad (2.16)$$

For instance, the median  $\beta_{\text{obs}}$  for galaxies with detected LyC emission is around  $-2.00$ , while for non-detected galaxies, it is approximately  $-1.63$ .

Furthermore, [402] confirmed the correlation between  $E(B-V)$  and  $f_{\text{esc}}$ , reinforcing the idea that  $\beta$  (or  $E(B-V)$ ) is a valuable indirect indicator of LyC escape. As shown in Figure 2.6, both the correlation identified by [86] and that of [402] illustrate the strong connection between  $\beta$ ,  $E(B-V)$ , and LyC escape.





**Figure 2.6:** Left:  $f_{\text{esc}}$  vs.  $E(B-V)$ , with thick black error bars showing the running median and quantiles. Gray symbols indicate LzLCS data and red symbols represent literature sources [215–218, 501]. Grey bars show median  $1\sigma$  uncertainties. Right:  $f_{\text{esc}}$  vs.  $\beta_{1550,\text{obs}}$  with LyC detections as circles, non-detections as arrows, color-coded by relative escape fraction. Grey dashed line shows the best-fit relation; grey dotted line represents the dust-only attenuation limit. [Left figure from 402], [right figure from 86].

However, the observed relations exhibit considerable scatter, as is the case with many other indirect indicators discussed in previous sections. This scatter underscores the complexity of using  $\beta$  as a standalone predictor of  $f_{\text{esc}}$ , suggesting that while it provides useful insights, it must be interpreted within the broader context of galaxy properties.

## 2.5 Multivariate predictors of the LyC escape fraction

As discussed in Section 2.3.1, several diagnostics correlate significantly with the LyC emission at low and intermediate redshift [154]. Building on these correlations, several groups have refined methods to predict LyC emission in galaxies using a combination of such diagnostics.

Traditional regression techniques, including multivariate linear regression, were first utilized to estimate  $f_{\text{esc}}$  during reionization. [300, 302] employed models calibrated on the LzLCS+ dataset, using predictors such as the O32 ratio,  $H\beta$  EW, the slope  $\beta$ , and the UV half-light radius ( $r_e$ ) (see also Chapters 5 and 6). [273] implemented a logistic regression model to assess the likelihood of a galaxy with  $M_{\text{UV}} < -18$  being a LCE. This model includes  $M_{\text{UV}}$ , the UV slope  $\beta$ , and the O32 ratio as significant variables. However, these methods struggle to effectively handle *censored data* – datasets that include upper limits instead of only exact measurements. Moreover, these methods assume a linear relationship between the variables and  $f_{\text{esc}}$ , which may not always be suitable.

To address these challenges, [225, 226] employed survival analysis techniques, which are more appropriate given the nature of the LzLCS+ dataset, with its broad range of  $f_{\text{esc}}$  values and numerous non-detections (upper limits). Ignoring these non-detections could skew results and lead to biased predictions. Survival analysis, which originated in medical research to study lifetimes, is well-suited to

handling censored data. In medical studies, censored data typically represents individuals who are still alive at the end of a study, meaning their lifespans are unknown beyond the observation period [e.g., 54, 91]. In the astronomical context, the analogy can be made to  $f_{\text{esc}}$  detections: indeed, we model the likelihood of detecting a particular  $f_{\text{esc}}$  given various indirect indicators. A non-detection, represented by an upper limit on  $f_{\text{esc}}$ , parallels a situation where an individual's lifespan exceeds the study period. This method is akin to assessing how deeply we must observe a galaxy to detect LyC radiation, with deeper observations analogous to longer study periods.

To apply this framework, [225, 226] used the *Cox proportional hazards regression model* [99], a semi-parametric method that estimates the probability of an event (the detection of  $f_{\text{esc}}$ ) as a function of multiple variables (the various galaxy properties). The baseline hazard function represents the “default” risk of something occurring at a particular time, serving as a starting point or reference for the likelihood of an event when no other information is considered. The Cox model is particularly advantageous because it does not require the baseline hazard function to follow a specific distribution, offering flexibility in modeling complex datasets.

The hazard function in the Cox model is given by:

$$h(t|x) = h_0(t) \exp \left( \sum_{i=1}^n \beta_i (x_i - \bar{x}_i) \right), \quad (2.17)$$

where  $h(t|x)$  is the hazard function at time  $t$  given the set of independent variables  $x$ ,  $h_0(t)$  is the baseline hazard function, and  $\beta_i$  are the fitted coefficients for each independent variable  $x_i$ .

In this analysis, instead of modeling the probability of an event at time  $t$ , the hazard function represents the probability of a LyC detection in an infinitesimally small window of  $f_{\text{esc}}$  for a set of independent variables, assuming the galaxy was not already detected at a higher value of  $f_{\text{esc}}$ . According to [225], in the following equations we adopt the absorbed fraction of LyC ( $f_{\text{abs}} = 1 - f_{\text{esc}}$ ), and its corresponding lower limits rather than  $f_{\text{esc}}$  and its upper limits.

The survival function  $S(f_{\text{abs}})$ , which gives the probability that  $f_{\text{abs}}$  exceeds a certain threshold, is defined as:

$$S(f_{\text{abs}}) = \exp [-H_{F_0}(f_{\text{abs}}) \cdot \text{ph}(x)], \quad (2.18)$$

where  $H_{F_0}(f_{\text{abs}})$  is the baseline cumulative hazard function:

$$H_{F_0}(f_{\text{abs}}) = \int_0^{f_{\text{abs}}} h_0(f) df, \quad (2.19)$$

and  $\text{ph}(x)$  is the partial hazard function for the set of variables  $x$ :

$$\text{ph}(x) = \exp \left( \sum_{i=1}^n \beta_i (x_i - \bar{x}_i) \right). \quad (2.20)$$

The best-fit coefficients  $\beta_i$  for the Cox proportional hazards model are obtained by maximizing the partial likelihood, which compares  $\text{ph}(x)$  for each detection

with the sum of  $\text{ph}(x)$  for all galaxies with higher  $f_{\text{abs}}$  (lower  $f_{\text{esc}}$ ), including both detections and non-detections. The baseline cumulative hazard function  $H_{F_0}(f_{\text{abs}})$  is then determined using Breslow’s estimator [59], which takes into account the number of detections and the partial hazard values.

Using the Cox model, the median  $f_{\text{abs}}$  is the point where the survival function  $S(f_{\text{abs}})$  reaches 0.5, indicating a 50% probability that the true  $f_{\text{abs}}$  is either above or below this value. By definition, the median predicted  $f_{\text{esc}} = 1 - \text{median predicted } f_{\text{abs}}$ .

The survival function is also used to determine the values at which  $S(f_{\text{abs}})$  reaches 0.159 and 0.841, corresponding to the  $1\sigma$  confidence intervals under the assumption of normally distributed uncertainties. These intervals reflect the scatter in the relationship between  $f_{\text{esc}}$  and the independent variables, which may be caused by both observational uncertainties and intrinsic variability within the galaxy population.

The goodness-of-fit for the models is obtained using the *concordance index* ( $C$ ) [197], which is commonly used in evaluating models that involve censored data. The concordance index considers all possible pairs of data points and compares the observed rank order of  $f_{\text{esc}}$  with the order predicted by the model. For each pair, the model determines if it is:

- concordant ( $n_c$ ): if the data point with a higher observed  $f_{\text{esc}}$  also has a higher predicted  $f_{\text{esc}}$ ;
- discordant ( $n_d$ ): if the data point with a higher observed  $f_{\text{esc}}$  has a lower predicted  $f_{\text{esc}}$ ;
- tied ( $n_t$ ): if their predicted  $f_{\text{esc}}$  values are identical.

The concordance index is thus calculated as:

$$C = \frac{n_c + 0.5n_t}{n_c + n_d + n_t}. \quad (2.21)$$

The concordance index  $C$  ranges from 0 (perfect disagreement) to 0.5 (perfectly random) to 1.0 (perfect concordance) [e.g., 106].

The fiducial model ( $C = 0.89$ ) found by [225] is the one based on stellar mass,  $M_{\text{UV}}$ ,  $\text{H}\beta$  EW, nebular reddening ( $\text{E(B-V)}_{\text{neb}}$ ), oxygen abundance ( $12+\text{O/H}$ ), the O32 ratio, star formation rate surface density ( $\Sigma_{\text{SFR}}$ ), UV reddening ( $\text{E(B-V)}_{\text{UV}}$ ), the  $f_{\text{esc}}^{\text{Ly}\alpha}$ , and the equivalent width of low-ionization state emission ( $\text{EW(LIS)}$ ), to predict  $f_{\text{esc}}$ . However, while several variables correlate with  $f_{\text{esc}}$ , only a few of them are statistically significant [225].

**Table 2.1:** *Best-performing Cox models for LyC predictions. The variables used are categorized into four groups: dust, Ly $\alpha$ , nebular, luminosity and morphology [from 226].*

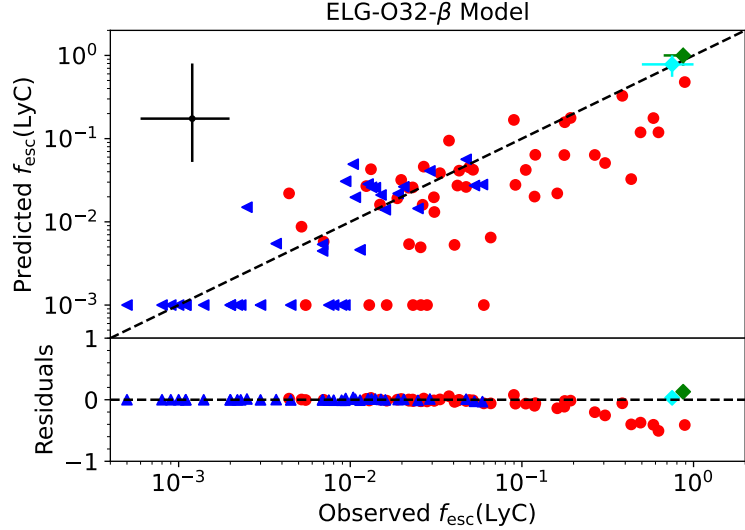
| Model    | Variables                   |                       |                    |            |                            |
|----------|-----------------------------|-----------------------|--------------------|------------|----------------------------|
|          | Dust                        | Ly $\alpha$           | Nebular            | Luminosity | Morphology                 |
| TopThree | $\text{E(B-V)}_{\text{UV}}$ | –                     | $\log(\text{O32})$ | –          | $\log(\Sigma(\text{SFR}))$ |
| LAE      | $\text{E(B-V)}_{\text{UV}}$ | $\text{EW(Ly}\alpha)$ | –                  | $M_{1500}$ | –                          |

**Table 2.1:** *Cox Models for High-Redshift Predictions (continued).*

| Model                         | Variables            |                                    |                            |                       |                             |
|-------------------------------|----------------------|------------------------------------|----------------------------|-----------------------|-----------------------------|
|                               | Dust                 | Ly $\alpha$                        | Nebular                    | Luminosity            | Morphology                  |
| LAE-O32                       | E(B-V) <sub>UV</sub> | EW(Ly $\alpha$ )                   | log(O32)                   | $M_{1500}, \log(M_*)$ | –                           |
| LAE-O32-nodust                | –                    | EW(Ly $\alpha$ )                   | log(O32)                   | $M_{1500}$            | –                           |
| ELG-EW                        | E(B-V) <sub>UV</sub> | –                                  | log(EW([OIII]+H $\beta$ )) | $M_{1500}, \log(M_*)$ | –                           |
| ELG-O32                       | E(B-V) <sub>UV</sub> | –                                  | log(O32)                   | $M_{1500}, \log(M_*)$ | –                           |
| ELG-O32- $\beta$              | $\beta_{1550}$       | –                                  | log(O32)                   | $M_{1500}$            | –                           |
| ELG-O32- $\beta$ -Ly $\alpha$ | $\beta_{1550}$       | $f_{\text{esc}}^{\text{Ly}\alpha}$ | log(O32)                   | $M_{1500}, \log(M_*)$ | –                           |
| R50- $\beta$                  | $\beta_{1550}$       | –                                  | –                          | $M_{1500}, \log(M_*)$ | log( $r_{50, \text{NUV}}$ ) |
| $\beta$ -Metals               | $\beta_{1550}$       | –                                  | 12+log(O/H)                | $M_{1500}, \log(M_*)$ | –                           |

Additionally, due to the presence of the LIS and Ly $\alpha$  terms, this model is not applicable to sources during the EoR. For this reason, a series of other models were also developed that include a smaller number of properties, that can be more readily measured in EoR galaxies: these models are presented in Table 2.1.

The indirect indicators used in these models predominantly include dust, nebular, and luminosity terms. [226] also tested these predictions on a few LyC leakers at intermediate redshifts (such as Ion2 and J1316+2614), where all necessary quantities were available, and the predictions proved to be successful for these sources. The results for the ELG-O32- $\beta$  model are shown in Figure 2.7.



**Figure 2.7:** Predicted  $f_{\text{esc}}$  values using the ELG-O32- $\beta$  model. Red circles indicate LyC detections within the LzLCS+ sample, while blue triangles are upper limits. The dashed line represents a one-to-one correspondence. The two high-redshift LCEs, Ion2 and J1316+2614, are the teal and green diamonds, respectively. The error bar in the upper left corner shows the median uncertainty size for both observed and predicted  $f_{\text{esc}}$ . [Figure from 226].

One key limitation of the above studies is that they tend to under-predict  $f_{\text{esc}}$  for strong LCEs due to potential biases in orientation and the complex interplay of different galaxy properties. In addition, these models are based on a relatively

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small sample of sources (less than 100) located in the low redshift Universe. Applying the conclusions to EoR galaxies, means making the strong assumption that the conditions facilitating LyC escape at  $z \simeq 0.3$  are analogous to those during the EoR, approximately 10 billion years earlier.

## 2.6 Thesis goals and structure

The primary goal of this thesis is to trace the physical properties of LyC emitting galaxies across cosmic time, with the objective of understanding the conditions that allowed the escape of ionizing photons during the EoR. This is achieved by analyzing data from different surveys, focusing on two key epochs, i.e. the cosmic-noon ( $z \sim 3$ ) and the EoR ( $z > 6$ ). The thesis is structured into five main chapters, first exploring the identification of new indirect diagnostics of LyC emission and then analyzing statistical samples of EoR galaxies to determine their role as cosmic reionizers. The chapters are organized as follows:

- **Chapter 3**, [from 301]: This chapter is based on the VANDELS ESO spectroscopic survey, and focuses on  $z \sim 3$  galaxies with strong C IV emission. We investigate whether CIV can be used as a predictor of LyC escape by comparing the galaxies' properties to photoionization models.
- **Chapter 4**, [from 52]: This chapter explores the properties of extreme [OIII] emitters at  $z \sim 3.4$ , observed by JWST's NIRISS instrument. We examine the correlation between [OIII] EWs and the ionization state of galaxies using data from the GLASS-JWST program.
- **Chapter 5**, [from 300]: In this chapter, we analyze 29 gravitationally lensed galaxies at  $z \geq 4.5$  from the GLASS-JWST survey to identify potential LyC leakers using as indirect indicators high [OIII]/[OII] ratios and compact UV sizes. We use NIRSpec and NIRCам data to estimate the physical properties and LyC escape fractions of these galaxies.
- **Chapter 6**, [from 302]: This chapter focuses on the indirect indicators of LyC escape in faint, low-mass galaxies at  $6 \leq z \leq 9$ , using data from several JWST programs. We measure properties such as UV- $\beta$  slopes, O32, EW(H $\beta$ ), star formation rates, and UV compactness, to predict their role in reionizing the Universe.
- **Chapter 7**, (Mascia, in prep.): This chapter investigates how mergers and galaxy morphology might affect LyC photon production and escape. We analyze a sample of 450  $z \sim 5 - 7$  galaxies from various spectroscopic JWST surveys, infer their LyC escape fractions and the photon production efficiency. We then correlate such quantities to their rest-frame UV morphologies.
- The concluding chapter (**Chapter 8**) summarizes the key findings and their implications for understanding cosmic reionization. It also discusses future observational and theoretical work that could further refine our understanding of LyC escape and the role of galaxies in reionizing the Universe.



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## Chapter 3

# CIV emission as a predictor of LyC escape in cosmic-noon galaxies from the VANDELS survey

*The results in this chapter were published and first presented in [301].*

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Recently, intense emission from nebular CIII] and CIV emission lines have been observed in galaxies in the Epoch of Reionization ( $z > 6$ ) and have been proposed as the prime way of measuring their redshift and studying their stellar populations. These galaxies might represent the best examples of cosmic reionizers, as suggested by recent low- $z$  observations of Lyman continuum emitting galaxies, but it is hard to directly study the production and escape of ionizing photons at such high redshifts. The ESO spectroscopic public survey VANDELS offers the unique opportunity to find rare examples of such galaxies at cosmic noon ( $z \sim 3$ ), thanks to the ultra deep observations available. We have selected a sample of 39 galaxies showing CIV emission, whose origin (after a careful comparison to photoionization models) can be ascribed to star formation and not to Active Galactic Nuclei (AGN). By using a multiwavelength approach, we determined their physical properties including metallicity and the ionization parameter and compared them to the properties of the parent population to understand what the ingredients are that could characterize the analogs of the cosmic reionizers. We find that CIV emitters are galaxies with high photon production efficiency and there are strong indications that they might also have a large escape fraction: given the visibility of CIV in the Epoch of Reionization, this could become the best tool to pinpoint the cosmic reionizers.

### 3.1 Introduction

The rest-frame UV and optical emission lines that emerge from star-forming (SF) galaxies are an important indicator of the nature and strength of the ionizing radiation and the conditions in the interstellar medium (ISM) since they can be used to infer gas-phase abundances and ionization parameters. Particularly

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the rest-UV range is currently fundamental to study the early galaxies in the Epoch of Reionization (EoR), as it is accessible with ground-based telescopes and now with the *James Webb Space Telescope* [JWST, 392]. In the last few years, prominent high-ionization nebular emission lines such as OIII]  $\lambda\lambda$  1661,1666, CIV  $\lambda\lambda$  1548,1551 (hereafter simply CIV  $\lambda$  1550), HeII  $\lambda$  1640, and CIII]  $\lambda$  1907 + CIII]  $\lambda$  1909 (hereafter simply CIII]  $\lambda$  1910) have been observed in a few  $z \simeq 6 - 7$  galaxies [287, 451], indicating that these systems are characterized by extreme radiation fields. Whether such properties are really common or not at these early epochs is still a matter of debate [e.g. see 76], but they might become the best probes to detect and characterize the most distant sources, since at  $z > 7$  the Ly $\alpha$  emission line, which is usually the strongest line in young SF galaxies, starts to be suppressed by the increasingly neutral intergalactic medium (IGM) as we approach the reionization epoch [43, 235, 353, 364].

The nebular CIV emission line, requiring comparably high ionization energies as the HeII line ( $E > 47.9$  eV and  $> 54.4$  eV, respectively), has recently attracted attention since it has been shown that its presence might be strongly linked to the escape of Lyman continuum (LyC) photons from galaxies. Indeed [415] recently detected CIV in several confirmed LyC leakers at  $z < 0.7$ , finding CIV emission in all LyC emitters with escape fractions  $f_{\text{esc}} > 0.1$  and suggesting that such strong leakers have CIV/CIII ratios above  $\sim 0.75$ . This makes the CIV emission line a very promising indirect tracer of LyC  $f_{\text{esc}}$ .

Other observations also support this scenario: one of the few solid LyC leakers at high redshift, Ion2 at  $z = 3.2$ , shows the presence of nebular emission in both CIV and CIII], with a rather high ratio CIV/CIII] from deep X-Shooter spectroscopy [488]. [332] indirectly classified  $z \sim 2$  LAEs into strong LyC leaking and non leaking galaxies based on the shape and peak separation of their Ly $\alpha$  profile. The composite spectrum of the potential LyC emitters shows narrow CIV as well as HeII, while there is no sign of these lines in the analogous stack of the non leakers [see also 400]. If confirmed this link would be really important to probe cosmic reionization: at  $z > 5$ , directly detecting the LyC photons escaping from galaxies is impossible because of the IGM opacity [208]. We therefore need indirect indicators, that is to say observational properties that correlate with  $f_{\text{esc}}$  and can be observed in the EoR. Many such indirect diagnostics have been proposed [e.g., 218, 292, 497], but all still present a large scatter [see 154, for a recent comprehensive study on low redshift LyC diagnostics].

To better interpret this intriguing link between nebular emission and the production and escape of ionizing photons, and the important consequences it could have for reionization, detailed investigations need to be conducted at low and intermediate redshift. However, nebular CIV emission is extremely rare in normal SF galaxies at least at low and intermediate redshift. [33] detected CIV and HeII in two nearby SF galaxies and propose that this combination of emission may identify galaxies that produce and transmit a substantial number of high-energy photons contributing to cosmic reionization. [423] also found that both nebular CIV and HeII appear in nearby lower metallicity galaxies ( $< 0.1Z_{\odot}$ ), provided the specific star formation rate (sSFR) is large enough to guarantee a significant population of high-mass stars, although the equivalent widths (EWs) appear smaller



than those measured in individual systems at  $z > 6$ . At intermediate redshift ( $z \sim 2 - 3$ ), although a hint of CIV emission appears in deep stacked spectra of SF galaxies, especially when combining a subsample of galaxies with Ly $\alpha$  in emission [358, 427], individual detections are very rare. [466] investigated a large sample of more than 100 extreme emission line galaxies, selected to have strong OIII, and detected CIV emission in only five sources, but with EWs up to 25 Å. Similarly, at  $z \sim 2$ , [126] analyzed a large sample of possible analogs of EoR sources but only found two objects with strong rest-UV metal line emission including significant CIV which appear comparable to the  $z > 6.5$  galaxies. Only in low-mass galaxies, which are mostly observable thanks to lensing magnification [286, 487, 488, 494], does there seem to be a somewhat higher fraction of sources showing prominent UV emission lines, with the most extreme CIII] emitters also showing nebular CIV emission with rest EWs 3-8 Å [275, 451]. These galaxies show very large sSFRs, indicating that they are in a period of rapid stellar mass growth and have very blue continuum UV slopes, indicating minimal dust content, young ages, and low metallicity [9].

In this context, the VANDELS survey [314, 366] provides the ideal database to search for galaxies showing the rare and faint CIV nebular emission line, since it provides some of the deepest spectra of intermediate redshift SF galaxies available to date. In particular VANDELS spectra can identify this emission line in galaxies in a redshift range ( $\sim 3 - 4$ ) where also the direct detection of LyC flux is possible and thus a confirmation of the link between LyC escape and CIV emission can be directly tested. With this aim we have thus searched for a sub-sample of galaxies at  $z \sim 2.5 - 5$  that show strong nebular CIV emission from the VANDELS survey, in order to determine their physical proprieties and identify their nature as possible analogs of the cosmic reionizers.

The paper is structured as follows. The selection criteria and the sample of CIV emitters in the VANDELS survey are discussed in Sect. 3.2. Sect. 3.3 examines their spectroscopic properties, while Sect. 3.4 analyzes the emission lines' properties to distinguish SF galaxies from Active Galactic Nuclei (AGNs) in our sample. In Sect. 3.5 we investigate the physical properties of the SF CIV emitters and in Sect. 3.6 we investigate how these properties are linked with cosmic reionization physical components. Finally, we summarize our findings in Sect. 3.7.

Throughout this paper, we assume a flat  $\Lambda$ CDM cosmology with  $H_0 = 67.7$  km s $^{-1}$  Mpc $^{-1}$  and  $\Omega_m = 0.307$  [367]. All magnitudes are expressed in the AB system [347].

## 3.2 The VANDELS CIV emitters

### 3.2.1 Sample selection

For our analysis we have used data from “VANDELS: a VIMOS survey of the CDFS and UDS field”, which is a recently completed ESO public spectroscopic survey carried out using the VIMOS spectrograph on the Very Large Telescope (VLT). VANDELS targets are selected in the *UKIDSS Ultra Deep Survey* (UDS)

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(RA = 2:18, Dec = -5:10), and the *Chandra Deep Field South* (CDFS) (RA = 3:32, Dec = -27:48): VANDELS footprints are centered on the HST areas observed by the CANDELS program [177, 252], although given the VIMOS field of view, they are larger. The survey description and initial target selection strategies can be found in [314], while data reduction and redshift determination methods can be found in [366]. Here we give a brief description of the most important aspects relevant to this work.

VANDELS targeted SF galaxies at  $z > 2.4$  as well as massive, passive galaxies in the redshift range  $1 < z < 2.5$ , with integration times ranging from 20 to 80 hours depending on the magnitude to ensure a uniform signal-to-noise ratio (S/N) on the continuum, with most targets observed for 40 hours or more. The EASYLIFE data reduction pipeline was used to reduce the spectra [162], producing the extracted 1D spectra, resampled 2D spectra, and sky-subtracted spectra.

Spectroscopic redshifts were derived using the `pandora.ez` tool [163] by cross-correlation with galaxy templates. The reliability of the redshifts is quantified with a quality flag (QF) as follows: 0 means no redshift could be determined; 1, 2, 3 and 4 mean respectively a probability of 50%, 75%, 95% and 100% of the redshift being correct; finally 9 means that the spectrum shows a single emission line.

The redshift measurement accuracy of spectroscopic observations is typically  $150 \text{ km s}^{-1}$ . For more information, see [366] for details on data reduction and redshift determination.

In this work we have used galaxies from the fourth and final data release (DR4) which is described in [164]. The data release contains spectra of approximately 2100 galaxies in the redshift range of  $1.0 < z < 7.0$ , with over 70 per cent of the targets having at least 40 hours of on-source integration time.

### 3.2.2 Identification of CIV emitters

We base our work on the official VANDELS emission line catalog which will be presented in [462]. Specifically, we use the catalog that was built using Gaussian fit measurements performed with `slinefit`<sup>1</sup>[422], an automated code that models the observed spectrum of a galaxy as a combination of a stellar continuum model and a set of emission and absorption lines. A set of templates is linearly combined to best fit the continuum. The templates were built with the [60] stellar population models and fit to the EAzY [57] default template set using FAST [258]. The code searches for lines around their expected locations given by the VANDELS official redshift. Each line with a S/N higher than 3 is allowed its own offset, with a maximum value set by  $\pm 1000 \text{ km/s}$  while lines with a lower S/N are instead fixed at their expected position. Therefore, for each detected line, the catalog is formed with the following information: flux and error, rest-frame equivalent width ( $EW_0$ ) and error, and the rest-frame full width at half maximum (FWHM) with error. By using the Monte Carlo technique, these errors are determined by randomly perturbing the galaxy spectrum based on its error spectrum, and then computing the uncertainties from the standard deviation of these perturbations compared to

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<sup>1</sup><https://github.com/cschreib/slinefit>

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the original data.

For our analysis we selected galaxies with a reliability flag 3, 4 and 9, which show a CIV emission line with a S/N greater than 2.5. To ensure the reliability of the lowest S/N detections and following [410] we produced a stacked spectrum of the 12 sources with  $2.5 \leq \text{S/N} \leq 3$ , as an additional check to ensure that they are indeed *bona fide* CIV emitters (see also Sect. 3.4.4). In this way we verified that we can safely consider these sources in the sample. We selected a total of 50 galaxies.

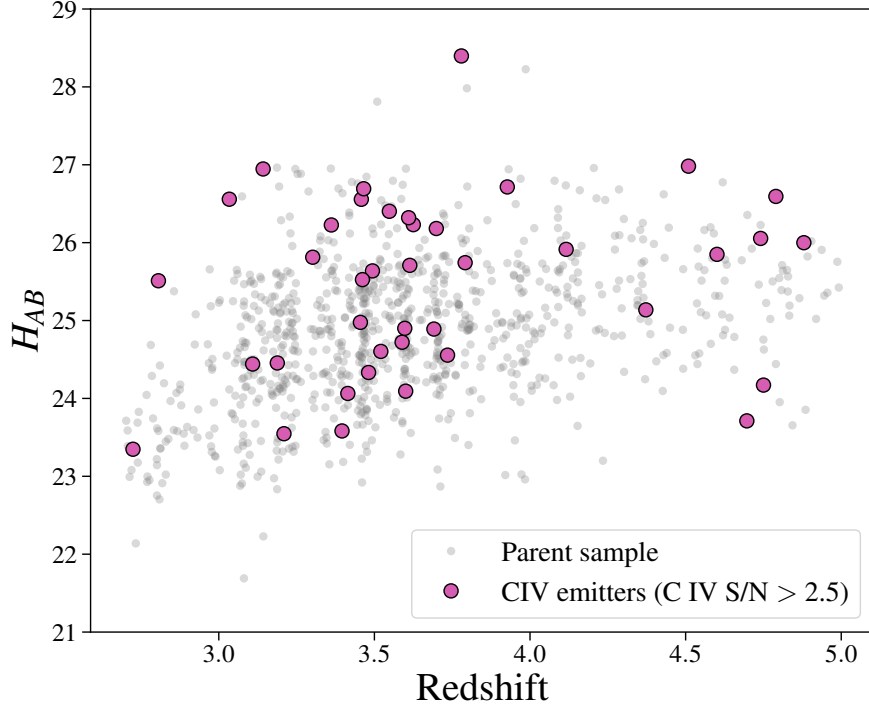
From this list generated by the automatic procedure we removed by eye six sources which present clear problems of sky-residuals in the region of the CIV emission and one source with evidence of merger in the spectrum. Finally from a visual inspection of all VANDELS QF 3, 4 and 9 sources we note that the official catalog failed to detect some galaxies with clear CIV emission lines which were not correctly identified by `slinefit`. This is probably due to the limited velocity offset allowed in the automatic search and to the fact that the reference redshift used for the search is the “VANDELS redshift”, which is not the systemic redshift but is in general driven by the strongest features present in the spectrum. In few cases this redshift is closer to the Ly $\alpha$  redshift or to the ISM redshift depending on their relative strength. A further 8 sources were identified and reanalyzed through `slinefit` by correctly calibrating the redshift (on the CIII] line when possible) which represents better the systemic redshift of the galaxies.

### 3.2.3 Removal of X-ray AGN/BLAGN and final sample

Since we want to study the nature of CIV in SF galaxies, we excluded the sources which are obvious AGN: in particular we excluded 8 X-ray detected sources from the CDFS 7 Ms Source Catalogs [279] and the UDS X-ray observations [250], of which 4 are broad line AGNs (*BLAGNs*: CDFS003360, CDFS019824, UDS017629 and UDS199159) which are easy to recognize from the broad UV emission line, and 4 narrow line AGNs (*NLAGNs*: CDFS005827, CDFS019505, UDS025482 and UDS018960). For an individual analysis of these sources see Bongiorno et al. (in prep.).

In conclusion, from the parent sample made of 933 galaxies at  $2.5 \leq z \leq 5$  without the AGN contribution, we obtain a sub-sample of 43 sources, 20 from the UDS field, and 23 from the CDFS field. The IDs, RA and DEC coordinates, redshifts, related flags and emission line information are presented in Table 3.1. This sample might still contain some NLAGNs with no X-ray counterpart. In the following section, we refine our analysis and employ emission line ratio diagrams to obtain a sample of purely SF CIV galaxies.

In Figure 3.1 we show the distribution of the galaxies’  $H_{AB}$  magnitude as a function of the redshift for the CIV sample (highlighted in purple) and the parent sample (gray symbols), i.e., all other VANDELS sources after removing the AGNs, as classified in Bongiorno et al. (in prep.). For this comparison the parent sample has been cut between redshift 2.5 and 5. As we can see the CIV emitters are a representative subset of the VANDELS population at least in terms of H-band and redshift distribution.



**Figure 3.1:** Distribution of  $H_{AB}$  of the CIV emitters with redshift. The sources cover a redshift range of  $2.8 < z < 5$ , with a median redshift of  $\bar{z} \sim 3.54$ . Of the 43 selected sources, 3 of them do not show any detection in the H (HST/F160W) band, so they are not shown in the figure.

**Table 3.1:** Main features of the CIV emitters. AGN indicates the nature of the sources:  $\checkmark$  for X-ray/BLAGN,  $?$  for a possible AGN and  $\times$  for a purely SF galaxy. The measured HeII, CIII], OIII] and CIV rest-frame line fluxes are in units of  $\times 10^{-18} \text{ erg s}^{-1} \text{ cm}^{-2}$ , and the  $EW_0$  of the CIV line is in units of  $\text{\AA}$ . The sources with the asterisk are the ones reanalyzed with *slinefit* and not contained in [462]. The sources in red are the best candidates to be strong LyC leakers (as discussed in Sec. 3.6).

| ID                | DEC<br>[J2000] | RA<br>[J2000] | z      | AGN          | HeII          | CIII]         | OIII]       | CIV           | $EW_0$<br>[ $\text{\AA}$ ] |
|-------------------|----------------|---------------|--------|--------------|---------------|---------------|-------------|---------------|----------------------------|
| <b>CDFS000303</b> | -27.937461     | 53.069643     | 3.6137 | $\times$     | $< 0.6$       | $1.2 \pm 0.3$ | $< 1.2$     | $1.0 \pm 0.2$ | $3.7 \pm 0.9$              |
| CDFS003839        | -27.875749     | 53.144602     | 3.458  | $\checkmark$ | $1.6 \pm 0.6$ | $< 1.2$       | $< 2.2$     | $3.4 \pm 0.8$ | $14 \pm 3$                 |
| CDFS006327        | -27.850722     | 53.178278     | 3.4937 | $\checkmark$ | $1.4 \pm 0.4$ | $1.0 \pm 0.4$ | $< 0.6$     | $1.7 \pm 0.4$ | $7.8 \pm 1.7$              |
| <b>CDFS012637</b> | -27.795843     | 53.122588     | 3.4655 | $?$          | $< 0.8$       | $1.3 \pm 0.4$ | $< 1.8$     | $1.6 \pm 0.5$ | $6 \pm 2$                  |
| CDFS014277        | -27.785081     | 53.231559     | 3.3616 | $\times$     | $< 1.2$       | $4.4 \pm 1.5$ | $< 1.8$     | $5.9 \pm 0.7$ | $9.8 \pm 1.3$              |
| <b>CDFS021733</b> | -27.724985     | 53.055324     | 3.1087 | $\times$     | $< 1.7$       | $< 3$         | $< 1.9$     | $6.2 \pm 1.4$ | $29 \pm 8$                 |
| CDFS021776        | -27.724619     | 53.098507     | 4.741  | $\times$     | $< 2.0$       | -             | $< 4.0$     | $2.6 \pm 0.8$ | $13 \pm 5$                 |
| <b>CDFS025178</b> | -27.689563     | 53.184370     | 2.806  | $?$          | $< 1.4$       | $< 1.6$       | $< 1.9$     | $4.4 \pm 0.9$ | $5.1 \pm 1.0$              |
| CDFS025760        | -27.696507     | 53.069094     | 4.79   | $\times$     | $92 \pm 5$    | -             | $67 \pm 14$ | $32 \pm 3$    | $20 \pm 2$                 |
| CDFS028933        | -27.848752     | 53.029622     | 3.7795 | $\times$     | $< 1.0$       | $1.9 \pm 0.3$ | $< 1.0$     | $2.2 \pm 0.2$ | $9.4 \pm 1.0$              |

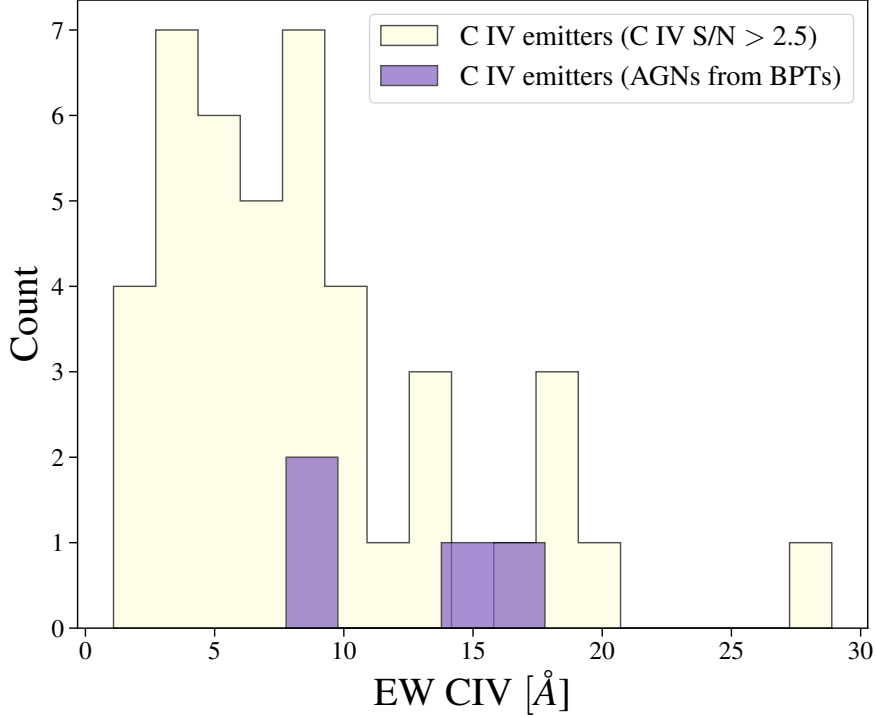
**Table 3.1:** *continued.*

| ID                 | DEC<br>[J2000] | RA<br>[J2000] | z      | AGN | HeII    | CIII]    | OIII]   | CIV       | $EW_0$<br>[Å] |
|--------------------|----------------|---------------|--------|-----|---------|----------|---------|-----------|---------------|
| CDFS029763         | -27.825989     | 53.238443     | 3.105  | ?   | < 0.4   | < 0.5    | 0.8±0.2 | 0.7±0.3   | 8±3           |
| CDFS032490         | -27.754031     | 53.219545     | 3.0747 | ?   | < 0.6   | 1.1±0.4  | < 0.4   | 1.1±0.4   | 9±3           |
| <b>CDFS128455</b>  | -27.745586     | 53.003635     | 3.4813 | ×   | < 1.5   | 3.9±1.2  | < 2.1   | 7±3       | 3.5±1.4       |
| CDFS202937         | -27.963161     | 53.218947     | 3.735  | ✓   | 4.2±0.6 | 3.1±0.8  | < 2.1   | 4.3±0.6   | 8.9±1.3       |
| CDFS203861         | -27.956412     | 53.215551     | 4.7502 | ×   | < 5.0   | -        | < 3.8   | 5.0±1.7   | 19±7          |
| <b>CDFS203989</b>  | -27.955446     | 53.019099     | 3.1366 | ×   | 1.4±0.4 | 4.1±0.5  | 1.8±0.3 | 4.5±0.7   | 8.8±1.4       |
| CDFS208175         | -27.923411     | 53.266337     | 3.462  | ?   | < 1.4   | 1.1±0.4  | < 1.8   | 3.6±1.2   | 3.8±1.3       |
| CDFS209144         | -27.916473     | 53.070682     | 4.4286 | ×   | < 2.0   | -        | < 1.9   | 2.5±0.9   | 8±3.1         |
| <b>CDFS212043</b>  | -27.894033     | 52.994478     | 3.395  | ×   | < 1.6   | 4.2±0.8  | < 1.4   | 2.9±0.9   | 3.4±1.1       |
| <b>CDFS231741</b>  | -27.747690     | 53.287112     | 4.0068 | ×   | < 0.9   | 3.6±1.2  | 2.8±0.6 | 2.9±0.7   | 6.3±1.4       |
| CDFS247279         | -27.676973     | 53.206236     | 3.2959 | ×   | < 1.8   | 4.0±0.6  | < 2.6   | 4.3±0.8   | 7.4±1.5       |
| CDFS009384*        | -27.823526     | 53.078521     | 3.5480 | ×   | 0.8±0.2 | < 1.6    | < 1.5   | 0.33±0.11 | 1.1±0.4       |
| <b>CDFS025002*</b> | -27.688314     | 53.066713     | 3.7915 | ?   | < 1.6   | 1.7±0.8  | < 1.9   | 1.9±0.5   | 4.4±1.3       |
| <b>UDS000184</b>   | -5.275796      | 34.489558     | 4.3722 | ×   | < 2.7   | < 15     | 6.6±1.1 | 6.5±1.3   | 10±2          |
| <b>UDS003724</b>   | -5.255394      | 34.395118     | 3.4616 | ×   | < 1.3   | 3.8±0.9  | < 4.4   | 4.5±0.7   | 7.6±1.2       |
| UDS018872          | -5.171582      | 34.371472     | 4.8799 | ×   | < 3.7   | -        | < 5.4   | 3.3±1.1   | 10±4          |
| UDS020394          | -5.163315      | 34.535879     | 3.3018 | ×   | < 1.1   | 3.9±0.4  | < 5.0   | 2.5±0.8   | 5.1±1.6       |
| UDS021234          | -5.158691      | 34.420972     | 4.6008 | ×   | 2.2±0.6 | -        | < 15    | 5.7±1.0   | 17±4          |
| UDS021291          | -5.158290      | 34.295009     | 3.625  | ×   | < 0.9   | 2.6±1.0  | < 1.3   | 1.9±0.6   | 5.5±1.9       |
| UDS033733          | -5.162411      | 34.406837     | 3.1424 | ×   | < 0.8   | < 1.5    | < 0.6   | 1.4±0.5   | 12±5          |
| UDS034251          | -5.153870      | 34.581079     | 3.61   | ?   | < 1.3   | < 1.6    | < 1.4   | 2.1±0.8   | 6±2           |
| UDS034333          | -5.152631      | 34.404969     | 3.927  | ×   | < 0.8   | < 2.1    | < 2.0   | 2.5±0.6   | 14±4          |
| UDS145830          | -5.336489      | 34.556466     | 3.2094 | ✓   | 8.3±1.0 | 3.5±1.2  | < 3.4   | 5.5±1.3   | 18±6          |
| UDS283510          | -5.368315      | 34.388651     | 4.5092 | ×   | < 2.7   | -        | < 3.7   | 5.3±1.1   | 18±5          |
| UDS313619          | -5.287081      | 34.242560     | 3.188  | ×   | < 2.4   | 10.7±1.8 | 4.6±1.7 | 6.6±1.9   | 7±2           |
| UDS379641          | -5.110200      | 34.334556     | 4.1161 | ×   | < 1.8   | < 4.7    | < 3.5   | 2.9±1.0   | 9±3           |
| UDS395442          | -5.070635      | 34.434154     | 3.589  | ×   | < 1.3   | < 2.5    | < 1.1   | 2.6±0.9   | 2.3±0.8       |
| UDS020932*         | -5.160535      | 34.518914     | 3.6005 | ×   | < 2.4   | < 10     | < 3.1   | 3.2±0.9   | 3.5±1.0       |
| UDS140603*         | -5.361135      | 34.393002     | 3.4146 | ?   | < 3.0   | < 4.2    | < 3.0   | 5.0±1.0   | 5.6±1.2       |
| <b>UDS152744*</b>  | -5.303281      | 34.393005     | 2.7243 | ?   | < 3.2   | < 1.9    | < 2.4   | 5.2±1.1   | 2.4±0.5       |
| UDS312746*         | -5.289354      | 34.243205     | 3.5207 | ×   | < 2.1   | < 6.8    | < 5.4   | 2.2±0.7   | 3.2±1.1       |
| UDS315235*         | -5.282526      | 34.473701     | 3.5976 | ×   | < 2.4   | < 4.1    | < 4.4   | 1.8±0.6   | 2.6±0.9       |

Table 3.1: *continued.*

| ID         | DEC<br>[J2000] | RA<br>[J2000] | z      | AGN | HeII  | CIII] | OIII] | CIV     | $EW_0$<br>[Å] |
|------------|----------------|---------------|--------|-----|-------|-------|-------|---------|---------------|
| UDS388069* | -5.089615      | 34.339181     | 3.4550 | ?   | < 1.7 | < 2.0 | < 1.1 | 2.3±0.4 | 3.7±0.7       |

### 3.3 Spectroscopic properties of the CIV emitters

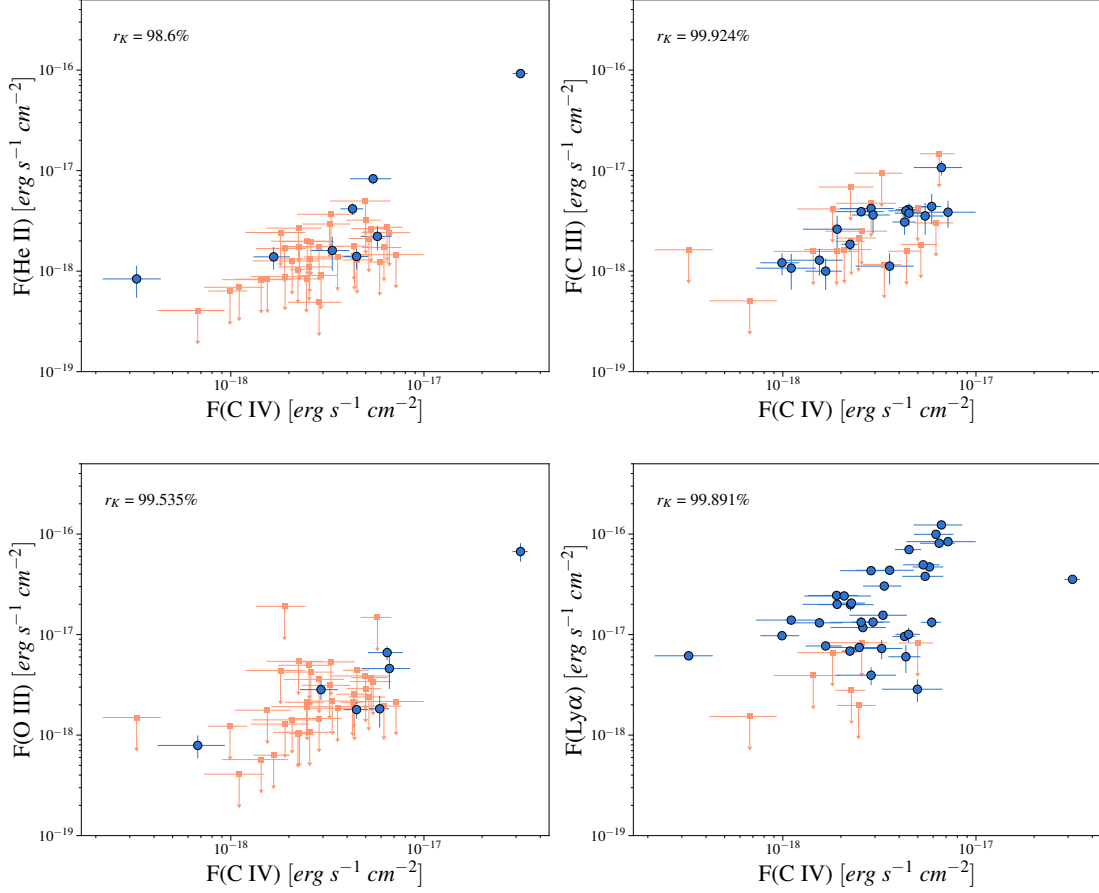


**Figure 3.2:** *Light distribution: measured  $EW_0$  of the CIV emission line for the CIV emitters ( $S/N > 2.5$ ) identified in both CDFS and UDS fields. Purple distribution: measured  $EW_0$  of the CIV emission line for the sample of certain AGNs described in Sec. 3.4.3.*

In Figure 3.2 we present the distribution of the rest-frame equivalent widths ( $EW_0$ s) of the CIV emission line in our sample, which range from 1 to 29 Å. Therefore, although most objects show  $EW_0$  of a few Å, in the VANDELS dataset we also find some galaxies with extreme EWs, analogous to those found in the Epoch of Reionization [287, 452].

In Table 3.1 we report the fluxes of the CIV lines and their  $EW_0$ s. We also present the fluxes for the HeII, OIII] and CIII] emission lines from the catalog of [462] plus the 8 additional sources considered in this work. We then analyze the relation between the CIV emission line and the other UV lines and we present the results in Figure 3.3, where we plot as limits all lines detected with a S/N lower than 2.5. We see a good correlation in all cases. In the insets of Figure 3.3 we also indicate, for each distribution, the probability  $r_K$  that the two variables are dependent, following the generalized Kendall’s  $\tau$  statistic including the upper limits [209]. Other studies at similarly high redshift have investigated analogous

relations between UV lines, although only focusing on the CIII] emission line: for example [420] report that the strengths of HeII, OIII, and SiIII correlate with the flux of the CIII] emission while [102, 294] found that the Ly $\alpha$  EW and the CIII] EW are strongly related [see also 452] although this relation has a really large scatter [e.g., 262] and therefore it appears strong only when using stacks.



**Figure 3.3:** Distribution of the measured HeII, CIII], OIII] and Ly $\alpha$  line fluxes and the C IV line flux for the selected C IV emitters ( $S/N > 2.5$ ) identified in both CDFS and UDS fields (blue points). If  $S/N$  of a line is lower than 2.5, we set  $2|\sigma|$  as upper limit (orange squares). In the figure it is also reported the correlation coefficient  $r_K$  from the generalized Kendall's  $\tau$  statistic.

Even if our emission lines are all quite close in wavelength, to perform a quantitative analysis, dust reddening should be considered. Due to the lack of direct measurements of H $\alpha$  and H $\beta$ , we used the SED fitting results (see Sec. 3.5.1) to estimate the amount of corrections for dust attenuation. From the stellar  $A_v(\text{SED})$  we computed the stellar reddening  $E(B - V)$  and then the nebular  $E(B - V)$  [following 69]: this quantity is on average 0.11. These results are consistent with those reported by [409], in which a  $E(B - V)_{\text{nebular}} \sim 0$  was obtained analyzing the stack of the brightest sub-sample of the VANDELS C IV emitters presented in this work. Since the dust correction inferred would only marginally affect our line ratios, we present only observed values.

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For the CIV emission we must also consider a correction for a component due to stellar emission. Stellar CIV produces a classical p-Cygni profile, that is present only in a small subset of our sources. We consider the results of [409], in which we derived a stellar component to contribute less than 25% on the strong CIV emission<sup>2</sup> by fitting the individual galaxy spectra using SEDs that only contain stellar emission as done in [401]. As we subsequently show in the next section, the p-Cygni profile is missing in the stacked spectrum of all SF galaxies, again indicating that the stellar component is probably very small on average in the total sample and possibly present only in the stronger emitters. In the following analysis, we use the uncorrected CIV, but we indicate in the various cases if and how this correction could alter our results.

### 3.4 Classification of SF and AGN based on UV emission lines

None of the 43 sources in our sample have X-ray individual detections: however some of them might still be faint AGN. In order to identify only the population of SF galaxies, we compare the UV emission properties of our sources to the predictions of photoionization models of AGN and SF galaxies. At variance with the seminal work by Baldwin, Phillips, and Terlevich [21] who first produced diagrams to discriminate AGN from SF galaxies, we use UV emission lines instead of those falling in the optical rest frame.

Specifically we consider the CIV, HeII and CIII] (we do not include the OIII emission since it is only detected in few galaxies). Indeed, these lines are among the most commonly detected UV emission lines in galaxy spectra and they are used to discriminate between photoionization by AGN and shocks in galaxies as well as to investigate the ISM metallicity. The CIV/HeII or CIII]/HeII ratios are sensitive to metallicity, while CIII]/CIV depends mostly on the ionization parameter [e.g., 181, 329, 341]. As we discuss in the next sections, the models of AGN and SF galaxies occupy two distinct regions in the UV diagnostic diagrams (henceforth also called UV-BPTs for simplicity), so these can be used as discriminators to effectively classify our sources.

#### 3.4.1 AGN models: Feltre et al. (2016)

The models developed by Feltre, Charlot, and Gutkin [139] for the emission by the AGN narrow-line regions have been created using CLOUDY [141]<sup>3</sup>. They consider a parameterization of the gas distribution as a cloud of a single type but contaminated by dust and dominated by the radiation pressure. It is useful to highlight that in these models, the line measurements are not corrected for dust attenuation.

For the purposes of our analysis, we use models with  $Z = 0.0001, 0.0002, 0.0005, 0.001, 0.002, 0.004, 0.006, 0.02$  and  $0.04$  with no priors on  $n_H$  in order to

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<sup>2</sup>We note that given our limited resolution and the fact that the stellar component (both emission and absorption) is broad, this might be an overestimate.

<sup>3</sup>These models can be found at <http://www.iap.fr/neogal/agn-models.html>



identify a clear AGN region in all the UV-BPTs. These models were constructed using the relative abundances of heavy elements for solar metallicity [63]. The predicted intensities of 20 optical and ultraviolet emission lines are then produced (in units of  $\text{erg s}^{-1}$ ). We consider the CIV, HeII and CIII] predicted intensities. Since we also want to produce a diagnostic diagram of CIV EW with CIV/HeII which was already used by [334], we need to also derive the EW of the CIV line from the luminosities provided by the models. To determine the line continuum we followed the procedure outlined in [76]. We first renormalize the incident spectra to a chosen fraction of the observed average flux in the R band, i.e., 5950-7250 Å, which corresponds to the observable non ionizing UV flux at  $\sim 1500$  Å, and then compute the EW of the CIV line on the basis of the predicted line flux for each model and of the observed flux at the relevant wavelength.

### 3.4.2 SF models: Gutkin et al. (2016) and Xiao et al. (2018)

It has been argued that the inclusion of interacting binary stars in stellar population synthesis models may hold the key to explain the observations of bright high ionization nebular emission lines in the spectra of high redshift SF galaxies [129, 447]. [514] combine the output of the stellar spectral synthesis model BPASS (Binary Population and Spectral Synthesis) used in conjunction with the photoionization code CLOUDY to predict the nebular emission from H II regions around young stellar populations over a range of compositions and ages.

In total, there are 80262 photoionization models, including separate models for binary and single stars<sup>4</sup>. For our analysis, we consider the photoionization models for binary stars with  $Z = 0.001, 0.002, 0.02$  and  $\log(\text{Age/yr}) = 6, 6.5, 7, 7.5$ . For each model, the predicted luminosities and EWs of 17 optical and ultraviolet emission-lines are given (in units of  $\text{erg s}^{-1}$  and Å, respectively). We retrieved the CIV, HeII and CIII] predicted intensities and EWs.

In addition, we also consider the models developed by Gutkin, Charlot, and Bruzual [188] for the nebular emission from SF galaxies. They have been created using CLOUDY [141]. For our analysis, we use models with  $Z = 0.0001, 0.0002, 0.001, 0.002$  and  $0.02$  with no priors on  $n_H$  and with relative elemental abundances fixed at Solar ratios [63]. The predicted intensities of 20 optical and ultraviolet emission-lines are then produced (in units of  $\text{erg s}^{-1}$ )<sup>5</sup>. As in the previous cases, we consider the CIV, HeII and CIII] predicted intensities.

Similarly to what done with the [139] models, we derive also the EW of the CIV line from the luminosities provided by the models, employing the method described in Sec. 3.4.1 and in [76].

### 3.4.3 Separating SF galaxies from AGN in the UV-BPTs

We now compare the properties of our sample of CIV emitters with the predictions from photoionization models for nebular emission from SF galaxies or AGN narrow-line regions described in the previous sections.

<sup>4</sup>These models can be found at <https://bpass.auckland.ac.nz/4.html>

<sup>5</sup>These models can be found at <http://www.iap.fr/neogal/sf-models.html>

Following the original diagnostics proposed by [139] we shall study the two relations  $\text{CIII}]/\text{HeII}$  vs  $\text{CIV}/\text{HeII}$  and  $\text{CIV}/\text{HeII}$  vs  $\text{CIV}/\text{CIII}]$ . In addition to these, and following the work by [334], who constructed photoionization models to interpret the UV spectra of CIII emitters selected from the VUDS survey at  $z = 2 - 4$ , we also use the relation  $\text{CIV}/\text{CIII}]$  versus  $(\text{CIII}] + \text{CIV})/\text{HeII}$ , and  $\text{CIV EW}$  vs  $\text{CIV}/\text{HeII}$ .

In Figure 3.4 we present the four different diagrams, which for simplicity we refer to UV-BPT1, UV-BPT2, UV-BPT3 and UV-BPT4 from here on. These diagrams give the advantage that objects can be tracked and diagnosed even if only one of the three lines is detected - in fact, if the S/N of a line is  $< 2$  we consider  $2|\sigma|$  as a limit. Whenever both lines considered in a certain line ratio are not detected but are both limits, we do not plot that object in the relevant UV-BPT diagram. This means that only UV-BPT3 and UV-BPT4 contain all the 43 objects in our sample.

In each diagram, the dots with error bars and/or limits indicate the position of the sources in our sample; the green triangles represent the position of the X-ray AGNs which have been excluded from our catalog and the green and magenta dots are the predictions from the models described in the previous subsections, respectively from the AGN models [139], and the SF models [188, 514]. It can be seen that the models occupy regions that are quite well although not completely separated in each of the UV-BPT diagrams. In the UV-BPT1 and UV-BPT4 diagrams we also plot the separation between AGN and SF galaxies proposed by [334] using the distributions of galaxies and AGNs observed in the VUDS survey.

Based on the models considered and on the position of the X-ray detected AGN, we proposed a slightly revised version of the separation between AGN and SF galaxies in the UV-BPT1 diagram which can be written as:

$$\log(\text{CIV}/\text{CIII}]) < 4 \log((\text{CIV} + \text{CIII}])/\text{HeII}) - 2; \quad (3.1)$$

For the other UV-BPT diagrams we consider the following separation lines: for UV-BPT2

$$\begin{aligned} \log(\text{CIII}]/\text{HeII}) &< 0.8 \log(\text{CIV}/\text{HeII}) + 0.5 \quad [\log(\text{CIV}/\text{HeII}) < 0.4] \\ &< 1.5 \quad [\log(\text{CIV}/\text{HeII}) \geq 0.4]. \end{aligned} \quad (3.2)$$

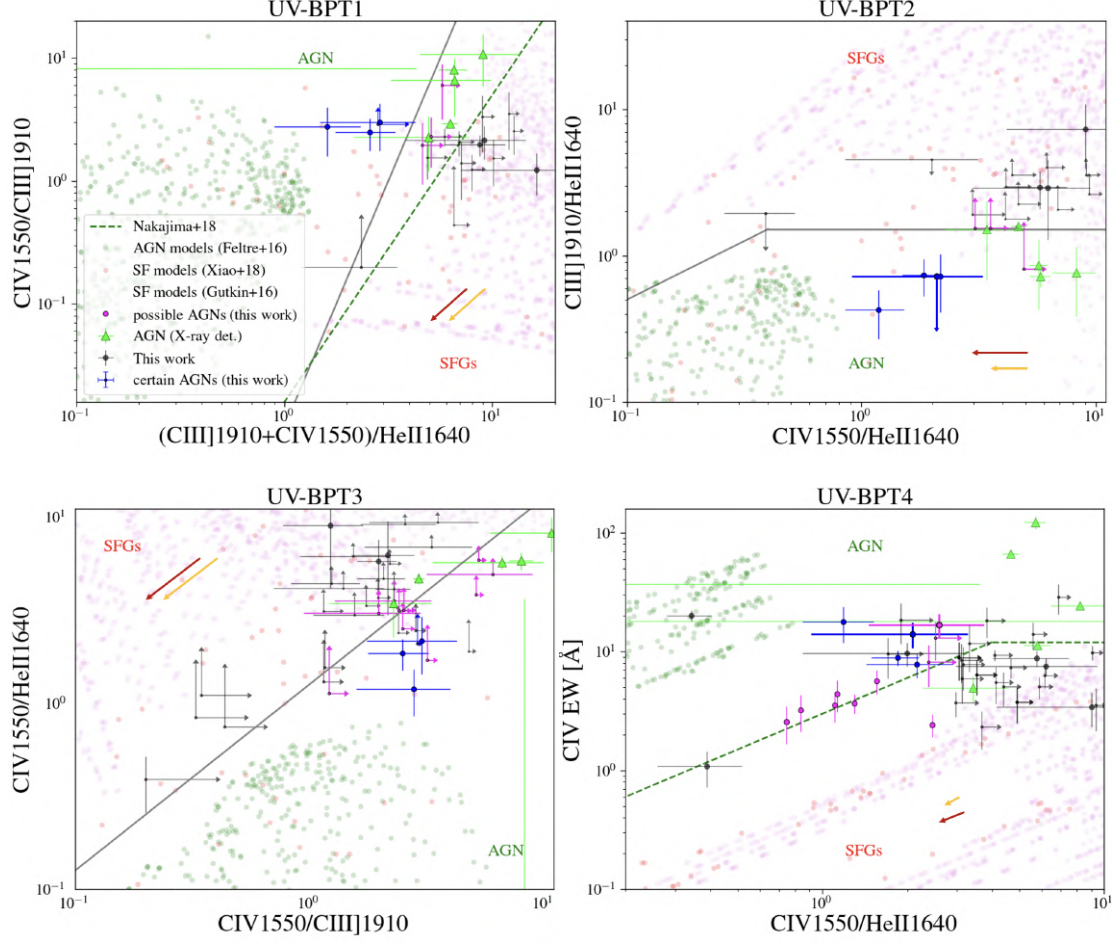
For UV-BPT3 [not considered by 334]:

$$\log(\text{CIV}/\text{HeII}) < \log(\text{CIV}/\text{CIII}]) + 0.1. \quad (3.3)$$

For UV-BPT4 [same as in 334],

$$\begin{aligned} \text{CIV EW} &< 3 \times (\text{CIV}/\text{HeII}) \quad [\text{CIV}/\text{HeII} < 4] \\ &< 12 \quad [\text{CIV}/\text{HeII} \geq 4]. \end{aligned} \quad (3.4)$$

Not all of the 43 sources are in all the diagrams but, when present, their position is always consistently in the AGN or SF part of the diagrams for all cases apart from UV-BPT4. In fact we note that the results from UV-BPT4 are more ambiguous since most sources, including some of the X-ray detected NLAGNs are located in an intermediate region in between the AGN and SF models. In addition, it seems that none of the SF models, can produce CIV EW as large as those observed in our sample. This problem was already highlighted in [410], where we noted that although the low-metallicity binary star models are able to



**Figure 3.4:** UV diagnostic diagrams presenting photoionization models of AGN [dark green dots, 139] and SF galaxies (SFGs) [red dots, 188, 514]. The separating lines consistently divide the two populations, and thus our sample of CIV emitters in: SF galaxies (black dots), confirmed AGNs (X-ray detected, in green), certain AGNs (blue dots) and possible AGNs (magenta), classified according to our criteria. The dashed green line division between ionization due to star formation alone and ionization due to AGN has been presented in [334]. The grey solid line represents our best guess of the possible division between the two regions in all the UV-BPTs. We also plot two arrows indicating the displacement of the data points by imposing a stellar correction of 20% (red one) and 10% (orange one) for the CIV emission [based on 409].

reproduce the UV line ratios of their HeII emitters, they under-predict the HeII EWs. The same applies to the CIV emission which is under-predicted by BPASS, as also discussed in [262, 334]. For the separation between AGN and SF galaxies we therefore base our classification on the bases of the other 3 UV-BPT diagrams. Specifically we find:

- four sources whose position is consistently in the AGN regions in all UV-BPT diagrams: we call these sources “certain AGN” and indicate them with blue dots;
- ten sources whose position is within the AGN regions, but given that the

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CIII and/or HeII emission lines are actually limits, could move into the SF regions. We call these sources “possible AGN” and indicate them in magenta;

- twenty-nine sources whose position is consistently in the SF region for all UV-BPT diagram. We call of course these sources “SF” and indicate them in black.

We note that of the 8 X-ray detected LAGNs, 1 source is consistently located within the SF region, while the other are in the AGN region. This could be due either to a limitation of the models in reproducing all AGN properties or to the fact that in many objects the UV emission line might be of mixed origin.

### 3.4.4 Spectral stacking

To further shed light on the real nature of the ten possible AGNs we apply a spectra stacking technique to infer the average properties of these sources.

Following [410], the stacking is performed by first converting each spectrum to the rest-frame, using the spectroscopic redshift given by the DR4 catalog. We point out that the VANDELS redshifts are not the systemic ones, since they are derived by a match to spectral templates and they are driven by absorption and/or emission lines depending on their strength [164]. For this reason in the stacks the various lines might not appear at the position expected, for example the CIII is not at the exact redshift as we would expect if we had used the systemic redshift for each source. For our analysis of the stacks we neglect this effect. The rest-frame spectra are first normalized using the mean flux density value in the rest-frame wavelength range  $1300 \text{ \AA} < \lambda < 1500 \text{ \AA}$ . The spectra are then resampled to a wavelength grid ranging from 1200 to 2000  $\text{\AA}$ , with a step size of 0.582  $\text{\AA}$  (the wavelength resolution obtained at a redshift of 3.6, which is the median redshift of VANDELS sources in this work). The errors on the stacked spectra are calculated using bootstrapping: we randomly sample and stack the same number of galaxies from the sample 500 times, and the dispersion on fluxes thus obtained gives the errors (Figure 3.5).

We produce three different stacks, one of the possible AGNs (ten objects), one with the certain AGNs (four sources) and one with the purely SF galaxies (twenty-nine sources). The spectra obviously reveal a wide range of emission and absorption features, with the  $\text{Ly}\alpha$  emission line standing out in each of the sub-samples.

The main UV emission lines are fitted with a Gaussian to determine their fluxes and the CIV EW, in order to place the stacks in the various UV-BPT diagrams, and the results are presented in Figure 3.6. We show just one UV-BPT them since the results are consistent for all of them: the blue star represents the sure AGNs, the magenta one the possible AGNs, and the black one the SF galaxies (in accordance with the colors used in Figure 3.4). As in the previous UV-BPTs, the green dots are obtained from the AGN models from [139] and the red dots from the SF models from [188, 514], while the grey lines are the division derived in this work and the green dashed lines refer to [334]. We note again that the AGN stack is not well reproduced by the AGN models but rather sits in an intermediate position.

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In the SF stacked spectrum (Figure 3.5) we note that the HeII emission is basically unresolved ( $\text{FWHM} < 600 \text{ km s}^{-1}$ ). This confirms the nebular nature of the HeII: in fact stellar HeII is expected to be much broader of the order of  $1500\text{--}2000 \text{ km s}^{-1}$  as seen for example in the stacks of the general LBG population [427]. For example [73] used a threshold of  $1200 \text{ km s}^{-1}$  to identify narrow HeII emitters of nebular origin. Thus, we can exclude a stellar origin for our sources.

### 3.4.5 Nature of the possible AGNs

The properties of the stacked spectra are as expected in the sense that the magenta star is at an intermediate position between the other two: however in all diagrams it is more consistent with the SF regions, indicating that the 10 sources are dominated by SF galaxies rather than AGN.

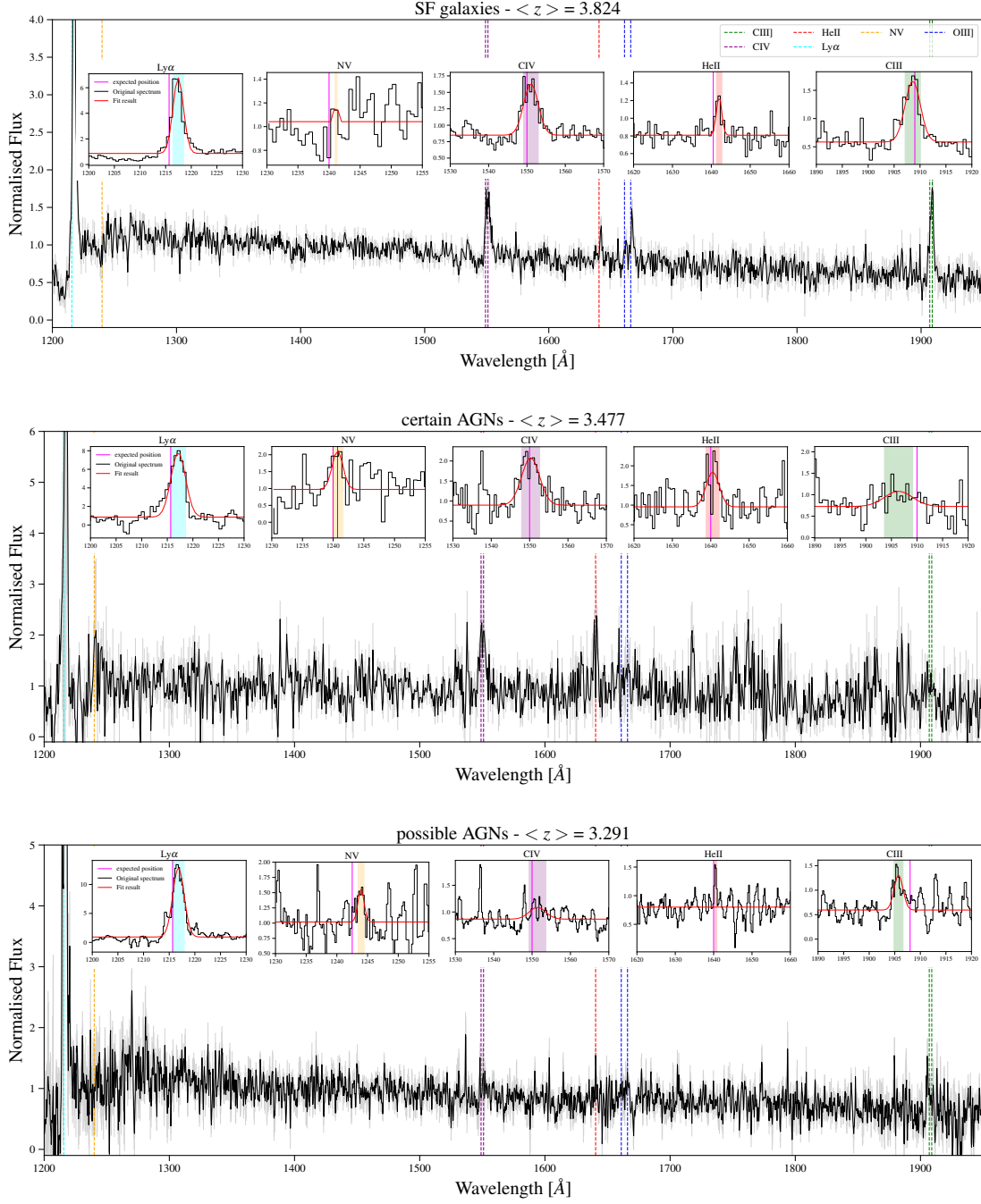
Another check on the nature of the “possible AGN” sample can be obtained by the NV emission. For AGNs, as predicted from theoretical and observational studies [e.g., 191–193], we expect a bright NV line. Therefore, we determine the  $\text{Ly}\alpha\text{NV}$  emission line ratio for our three different stacks. For the SF galaxies stacked spectrum the NV line is not detected at all and we can place a limit  $\text{Ly}\alpha\text{NV} > 30$ . The stack of the possible AGN presents a  $\text{Ly}\alpha\text{NV} = 18.4 \pm 0.7$  that is considerably higher than the certain AGN one ( $\text{Ly}\alpha\text{NV} = 6.0 \pm 0.8$ ), although we have a  $2\sigma$  detection of the NV. This value is higher than the ratios measured in narrow line AGNs [ $\text{Ly}\alpha/\text{NV} \sim 1 - 2$ , 203, 440, 471].

We finally exploit the Chandra X-ray observations: although we have excluded the individually detected X-ray sources since the very beginning, the [279] sources are those detected at  $5\sigma$ , therefore we cannot exclude faint emission from the “possible AGN” sample. We therefore stacked the X-ray data to check if we detect any flux from the combination of the sources. Unfortunately we can do this in a meaningful way only for the 6 CDFS sources which have much deeper X-ray observations [7 Msec, 279] while for the UDS field we only have much shallower observations [250]. To stack the X-ray data we follow the same procedure described in [411]. The stack does not produce any detection, although the limit is not very stringent since there are only 6 sources of which one lies at the border of the Chandra field.

In conclusion, from the position of the stack in the four UV-BPT diagrams, the high  $\text{Ly}\alpha\text{NV}$  ratio and the absence of X-ray emission in the stack, we conclude that the sample of 10 possible AGN must be dominated by SF galaxies, although we cannot exclude a (faint) AGN contribution. In the following we include these 10 sources in the SF sample, bringing it to 39 galaxies in total, to study their physical properties.

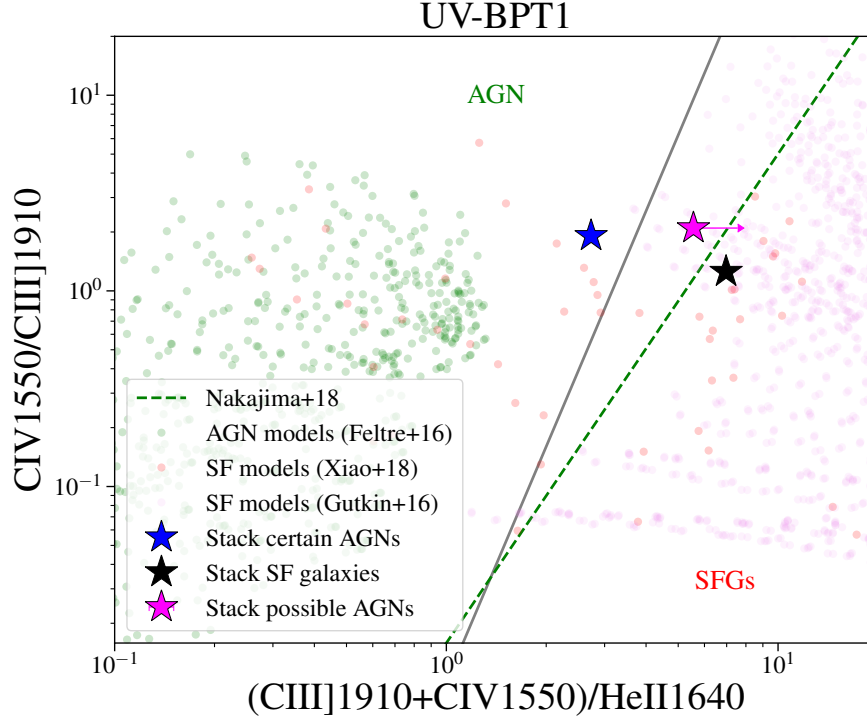
## 3.5 Physical properties of the SF CIV emitters

The combined synergy of both space-based and ground-based telescopes in the VANDELS fields provide an excellent multi-wavelength dataset from ultraviolet (UV) to mid-infrared (MIR), allowing accurate determination of the physical integrated properties of the VANDELS galaxies. We now describe the SED fitting tool



**Figure 3.5:** Stacked spectra of the sources in our sample. Upper panel. Stacked spectrum (black) of all the 29 SF galaxies of the sample. Middle panel. Stacked spectrum of all the certain 4 AGNs of the sample (no X-ray detection). Lower panel. Stacked spectrum of all the possible 10 AGNs of the sample. The noise obtained from bootstrapping is shown in grey. For clarity, the Gaussian fits with  $1\sigma$  regions for the emission lines Ly $\alpha$ , NV, CIV, HeII and CIII in each stacked spectrum considered in this work are also present. Certain and possible AGN composite show clear broad nebular emission line features in their spectra, while the SF stack show much narrower lines.

that we have used to determine the physical properties, the procedure adopted



**Figure 3.6:** UV diagnostic diagram presenting photoionization models of AGNs [139] and SF galaxies [188, 514]. The separating lines consistently divide the two populations and thus the results from the stack spectra for the SF galaxies, the certain AGNs and the possible AGNs (respectively the stars in black, magenta and blue).

and the main results.

### 3.5.1 The BEAGLE tool

The code used for the spectral energy distribution (SED) fitting is the BEAGLE (BayESian Analysis of GaLaxy sEds) software [81] which was chosen for its versatility: through the Bruzual & Charlot libraries it allows us to model in a coherent way the continuous radiation coming from the stars of the galaxies and the nebular component which is modeled with CLOUDY. The code also includes several prescriptions to describe the dust attenuation and the radiation transfer through the intergalactic medium. Stellar formation and chemical enrichment histories are included both with parametric and non parametric prescription. Finally, BEAGLE can simultaneously fit the photometry and the spectra (or individual spectral indices) of the galaxies. BEAGLE was run on all the VANDELS parent sample using v0.24.5 which employs the most recent version of the [60] stellar population synthesis models [see 499, for details]. The full description of the method and the results on the entire sample will be discussed in a companion paper [75]. We provide here only a brief description of the most important parameters adopted: in particular we assume a delayed star formation history (SFH), which is the most flexible parametric SFH used by the code, with a current SF timescale (i.e., duration of the current episode of star formation) of 10 Million years. The escape

fractions  $f_{\text{esc}}$  assumed in the BEAGLE run are set to 0 for all galaxies, regardless of the presence of the CIV emission. We used a [78] Initial Mass Function, metallicity in the range  $-2.2 \leq \log(Z/Z_{\odot}) \leq 0.25$  and we treated the attenuation following the [79] model combined with the [80] prescriptions for geometry and inclination effects.

For each source, BEAGLE is run on the combination of photometry, coming from the official 13 band VANDELS catalog presented in [164], and spectral indices i.e., the fluxes of the HeII, OIII and CIII emission lines as well as the precise spectral windows used (respectively [1634-1654], [1663-1668] and [1897-1919]). After some tests we decided not to provide the measured CIV emission to BEAGLE as a further spectral index to fit, both to have an unbiased estimate of the physical parameters, and also because as just discussed in Sect. 3.4.3, there is a well known inability of stellar population models, even when including binary stars, to reproduce some UV line fluxes [also see 409]. We also tested how the choice of current SF timescale could change our results. In particular this is relevant since the EWs of the emission lines are strongly dependent on the current SF. As a result even when reducing the SF timescale to 5 Myrs, the physical parameters remain very stable with the exception of  $\xi_{\text{ion}}$ , which becomes slightly higher ( $\sim 0.1$  dex) for all galaxies. We note that the inferred  $\xi_{\text{ion}}$  from BEAGLE is almost independent from the initially assumed  $f_{\text{esc}}$ : this value is determined by the intrinsic ionizing SED, which is already adjusted for relatively low metallicities and young ages. The change in  $\xi_{\text{ion}}$  is substantial only for very high  $f_{\text{esc}}$  [see e.g., 297]. The official VANDELS redshift was assumed in all cases for the SED fitting.

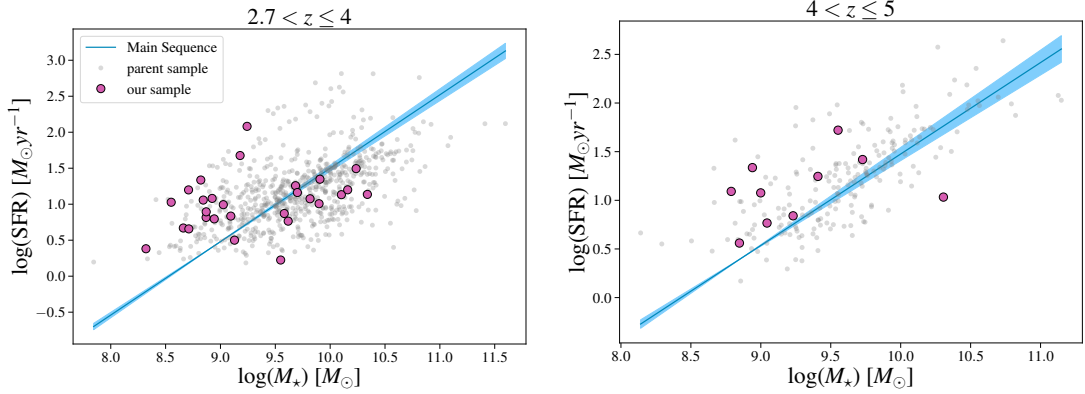
As a result, BEAGLE provides the best SED fit and the best physical parameters, with relative uncertainties: specifically total stellar mass, star formation rate, dust attenuation, age and  $\xi_{\text{ion}}$  were retrieved. For a complete list of the selected physical parameters and the range provided a priori to fit all the sources of the sample, see Table 3.2 in the Appendix.

### 3.5.2 Results and comparison to parent population

Using the derived physical properties we searched for systematic differences between the SF sample of the CIV emitters (39 galaxies) and the parent sample. From the parent sample we have removed all X-ray emitters and all possible AGN as defined in Bongiorno et al. (in prep.). In Figure 3.7 we present the location of our CIV emitters on the main sequence i.e., the stellar mass vs SFR plot: given that our sample spans a rather large redshift range, we split the plot into two redshift bins,  $2.7 < z \leq 4$ , and  $4 < z \leq 5$ . The blue lines are obtained using the main sequence as a function of redshift described in Santini et al. [406]. The VANDELS sources as well as the CIV emitters sub-sample are located more or less on the main sequence of SF galaxies, with perhaps a small bias in the high redshift bin, where the (few) objects are mostly located on or above the MS.

In Figure 3.8 we then show the normalized distributions of the derived stellar masses, the SFRs, the photon production efficiencies  $\xi_{\text{ion}}$  and the stellar ages of the CIV galaxies and the parent population. In each panel we also indicate the mean and median values for the two samples with vertical regions, whose widths represent the uncertainty. From the panels we can see that the CIV emit-





**Figure 3.7:** *Stellar mass vs SFR relation for our CIV emitters (purple dots) and the parent sample (grey dots). The blue line shows the main sequence respectively at  $3 < z \leq 4$ ,  $4 < z \leq 5$  from [406].*

ters tend to have considerable lower masses, slightly lower SFR, higher  $\xi_{\text{ion}}$  and slightly lower stellar ages compared to the parent sample. To assess the significance of these differences, we employ the common Kolmogorov-Smirnov (KS) test. From the p-values (reported in the figures) we conclude that the mass and  $\xi_{\text{ion}}$  of the CIV emitters and the parent sample are statistically different, while the max\_stellar\_age and the SFR are compatible with being randomly drawn. In [76] we find a strong link between the sSFR and  $\xi_{\text{ion}}$ : indeed if we look at our CIV emitters they have slightly higher sSFR compared to the parent population, which could explain the higher  $\xi_{\text{ion}}$ . This link and its implications will be further discussed in that paper. Even using a SF timescale of 5 Myrs, with all values of  $\xi_{\text{ion}}$  being slightly higher, the difference between CIV emitters and the parent sample is still significant.

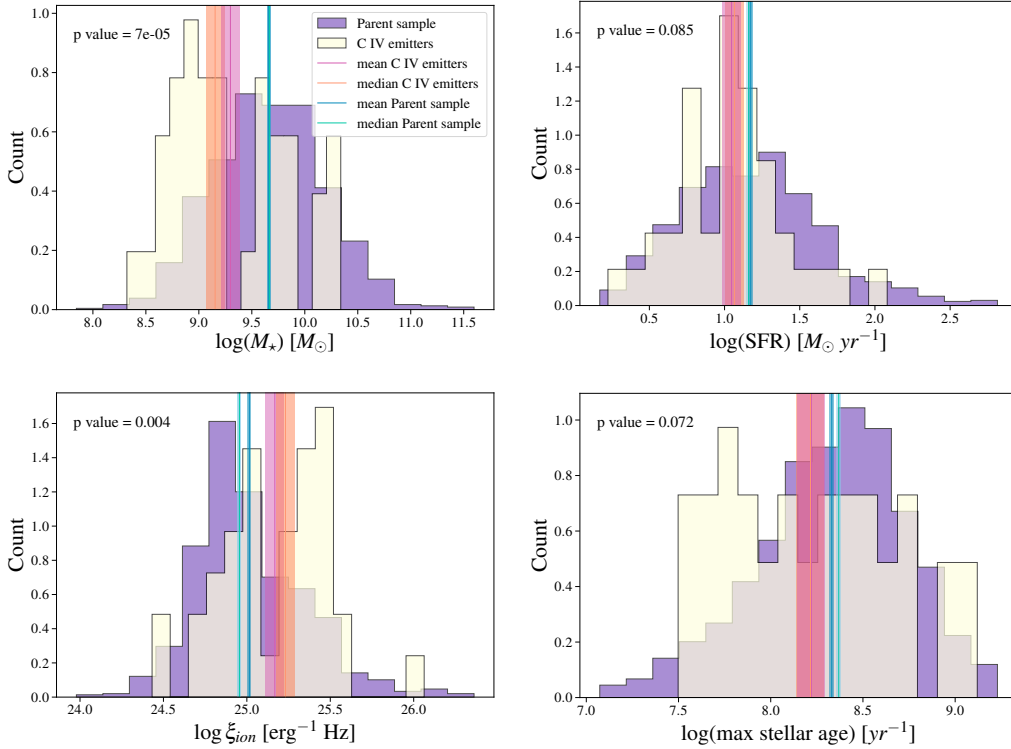
In conclusion our CIV emitters are in general low mass objects which have a higher photon production rate compared to the general SF population at the same redshift.

### 3.5.3 Average stellar metallicity

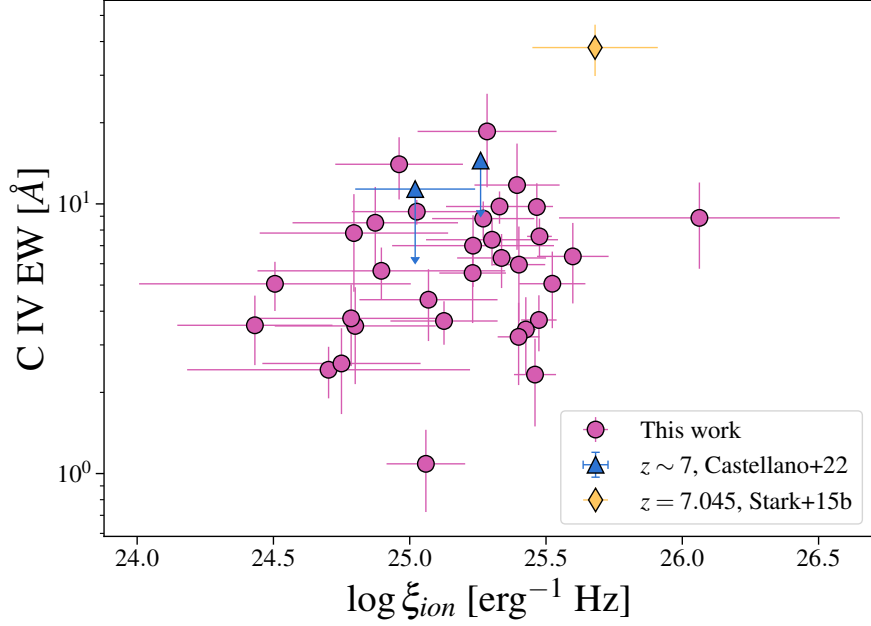
The link between the presence of nebular CIV emission and low metallicity stellar population has been noted both at high and low redshift. For example [424] showed that CIV emission may be ubiquitous in extremely metal-poor galaxies with very high specific star formation rates, although it reaches large EWs only if the ISM is exposed to the hard stellar ionizing spectra produced at extremely low metallicities ( $12 + \log \text{O}/\text{H} \lesssim 7.7$ ).

To derive the stellar metallicity from our spectra, we use the method developed by [64], which is based on stellar photospheric absorption features at 1501 Å and 1719 Å, calibrated with Starburst99 models and are largely unaffected by stellar age, dust, IMF, nebular continuum, or interstellar absorption. The method can be applied to spectra with a S/N of 15-20 in the continuum to avoid large uncertainty, and therefore we cannot derive metallicities for individual sources but only average values from stacked spectra. We produced a stack of the 39 SF

galaxies following the procedure outlined in the previous section. The metallicity we derive is  $Z \sim 0.26Z_{\odot}$  for the sample of 29 purely SF galaxies and a slightly lower (but consistent within the uncertainties) value of  $Z \sim 0.1Z_{\odot}$  for the combined sample of 39 galaxies. Looking at the stellar mass-stellar metallicity relation by [64] we can see that these low metallicities (about  $0.15Z_{\odot}$ ) are perfectly consistent with the general trend for  $z \sim 3$  galaxies with similar masses, i.e., the CIV emitting galaxies do not show lower metallicities compared to their parent sample, as also found by [409]. This is therefore at odds with the low redshift Universe where CIV emitters have almost always an extremely low inferred stellar (and gas phase) metallicities compared to galaxy samples at  $z = 0 - 3$  [e.g., 33, 425]. However in terms of “absolute” metallicity,  $z \sim 3$  CIV emitters have a similar metallicity as the local Universe emitters: this is of course due to the fact that at  $z \sim 3.5$  the average metallicity of SF galaxies is a factor of 0.6 dex lower than in the local Universe [64, 102].



**Figure 3.8:** Derived stellar mass, SFR, ionizing photon production efficiency  $\xi_{\text{ion}}$  and stellar age of the sources identified as SF, determined by BEAGLE (light distribution). For comparison, those of the total sample in the same redshift range are also shown in the Figure (purple distribution). The figure also shows the mean and median with their standard deviations for the distributions and the p-value from the KS test for the main physical properties of the analyzed sample.



**Figure 3.9:** Distribution of the measured  $EW(CIV)$  and the  $\xi_{ion}$  for the selected SF galaxies identified in both CDFS and UDS fields (purple points) compared with  $z \sim 7$  objects from literature: [76] (blue points) and [451] (orange point).

### 3.6 Discussion: The ionizing properties of CIV emitters

In the previous sections we have found that the VANDELS CIV emitters are in general galaxies with low masses and high photon production efficiency compared to their parent population. They have stellar metallicities of  $0.1 - 0.15Z_{\odot}$  consistent with the stellar-mass stellar metallicity relation at redshift 3.5. Most of our CIV emitters show also a prominent  $Ly\alpha$  emission, and we observe a strong correlation between the CIV and the  $Ly\alpha$  flux (shown in Figure 3.3). We now investigate if the CIV emitters have properties that could make them potentially good ionizers.

An investigation of ionizing efficiency is the first step in identifying good candidates for cosmic reionizers. Photon production efficiency expectations depend on several factors, including the initial mass function, the SFH, the evolution of individual stars, the stellar metallicity, and potential binary interactions between stars [e.g., 129, 445, 446, 527–529], but they agree in identifying the expected value of  $\log \xi_{ion}$  around 25.3 to ionize the IGM at  $z > 6$  [393, 394]. This prediction is confirmed both from the local Universe and intermediate and high redshifts’ observations [e.g., 18, 48, 76, 221, 260, 311, 338, 431, 449, 451].  $\log \xi_{ion}$  is also known to have a higher-than-average value when strong  $Ly\alpha$  [e.g., 102, 196, 441] and UV-rest frame emission lines [e.g., 335, 467] are present. Using the results obtained from our sample, we examine the dependence of the  $\log \xi_{ion}$  (calculated, as stated in Sec. 3.5.2, without considering CIV emission) with respect to the  $EW_0$  of the nebular CIV. The results are shown in Figure 3.9. We also compare our values with  $z \sim 7$  objects from literature [76, 451] with CIV detection. According to this analysis, high CIV emission would be accompanied by an increase in  $\xi_{ion}$  values,

which are also within the range expected for cosmic reionization. For a detailed examination of  $\xi_{\text{ion}}$  in relation to the VANDELS sample properties, see [77].

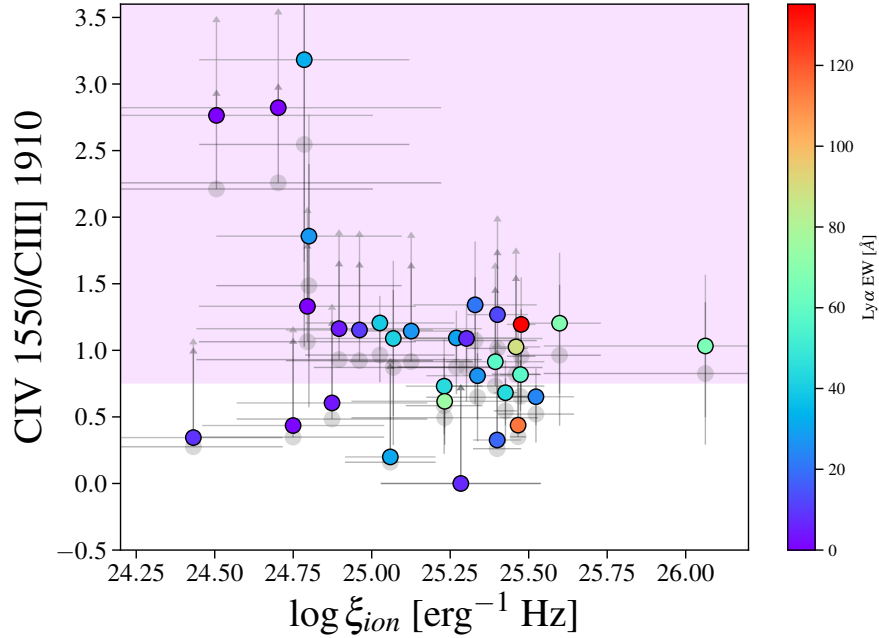
We now turn to the other relevant necessary condition for an ionizer, i.e., the LyC escape fraction. We know that the presence of strong Ly $\alpha$  emission is tightly linked to the escape of LyC photons. Galaxies with moderate values of  $f_{\text{esc}}$  are in general characterized by strong Ly $\alpha$  emission, as observed both in individually detected galaxies [e.g., 30, 153, 166, 358, at high and low redshifts, respectively] and via stacked spectroscopic studies [e.g., 292, 457]. Such a tight relation is also predicted by models which show that both ionizing continuum flux and Ly $\alpha$  photons can escape through the same ionized channels in the interstellar medium [117, 166, 214, 220, 292, 386, 388, 457, 497, 498]. We checked the 37 VANDELS CIV emitters that have their Ly $\alpha$  in the observed spectral range (all those at  $z > 3$ ), and only one has Ly $\alpha$  in absorption, while 24 have Ly $\alpha$  EW larger than 20Å, with several above 100Å. This would imply that a good fraction of the CIV emitters might also be galaxies with non negligible ionizing escape fractions. As shown by [497], the Ly $\alpha$  profiles of strong LyC leakers are in general all double-peaked with a small peak separation, in agreement with our theoretical expectation: unfortunately due to the relatively low resolution of the VANDELS spectra we are not able to determine if this is also the case for our CIV emitters.

We further checked whether the properties of our CIV emitters are similar to the LyC leakers of Schaerer et al. [415]. They define *strong leakers* as those galaxies with a LyC escape fraction above 10% (i.e.,  $f_{\text{esc}} > 0.1$ ) and found that these galaxies show not only the presence of CIV emission but also a high ratio,  $C43 = \text{CIV}/\text{CIII}] > 0.75$ . In Figure 3.10 we show the C43 values for our sample, in the cases when the CIII] emission is measurable (in 10 out of 39 cases, the CIII] falls outside the range included in the VANDELS spectra). For 22 galaxies the C43 ratio is clearly  $> 0.75$  and only in 4 cases it is smaller, with the rest being galaxies in which the relatively high CIII] limit prevents us from drawing any conclusion. In any case at least two thirds of the CIV emitters seem to have properties consistent with the low redshift LyC leakers presented in [415]. Since in [415] the CIV emission was purely nebular in origin, in this Figure we also show how the results would be effected by considering the possible stellar contribution. The grey symbols in Figure 3.10 thus represent the effect of assuming a 20% stellar contribution correction on the sample. We note that only two sources move outside the previously defined leakers' region, but given the high uncertainties on C43, the basic conclusions remain unchanged.

In a companion paper [409] we have investigated a sub-sample of the bright VANDELS CIV emitters by means of spectral stacking to understand if they could indeed be good tracers of hydrogen ionizing LyC photon leakage using a variety of indicators. We found that equivalent width and offset of the Ly $\alpha$  peak from the systemic redshift of galaxies is indicative of significant LyC leakage. These characteristics are consistent with the predictions of e.g. [498] and [116] of high Ly $\alpha$  and LyC escape in galaxies. In addition, the depth and velocity offsets of low-ionization interstellar absorption features in the stack, are indicative of low column density channels through which LyC photons may escape [e.g., 85, 401], with their low covering fractions suggesting LyC  $f_{\text{esc}} \simeq 0.05 - 0.30$ .

In conclusion, a consistent picture emerges, in this work and in the parallel analysis by [409] in that a substantial fraction of galaxies showing nebular CIV emission could be galaxies with significant production and escape of LyC photons. They could therefore be the best analogs of those galaxies that mostly contributed to cosmic reionization at  $z > 7$ . We have shown that CIV emitters do not differ significantly from low-mass galaxies apart from enhanced  $\xi_{\text{ion}}$  (which is probably insufficient to explain the strong UV emission lines associated with CIV). So if the integrated properties (SFR, metallicity, etc.) are equal, the intense UV emission must be instead linked to a peculiar ISM structure. For example, it has been shown that in order to accurately reproduce such high UV emission lines, gas-denser regions and lower-density regions must have different ionizations [32, 415] in order to facilitate the escape of ionizing photons. The Ly $\alpha$  flux at systemic velocity [408] observed in the stack of the bright CIV emitters supports this scenario, suggesting that it is the ISM properties that make the CIV emitting galaxies different from the parent population.

Finally, we identify the best candidate ionizers in our sample by selecting all galaxies having  $C43 = \text{CIV}/\text{CIII}] > 0.75$ ,  $\text{EW}(\text{Ly}\alpha) > 20\text{\AA}$ , and  $\xi_{\text{ion}} > 25$  (the median of the parent sample): in total we have 12 sources with these combined properties and we indicate them in red in Table 3.1. Note that Ly $\alpha$  and CIII] are outside of the observed range respectively for 2 and 10 of the SF sample, so the 12 candidates are selected from a usable sample of only 31 galaxies, thus representing almost 40% of the galaxies.



**Figure 3.10:** Ratio of CIV and CIII] versus the  $\xi_{\text{ion}}$  for individual galaxies in this study. The shaded area limits  $C34 > 0.75$ , that was empirically suggested by [415] to host strong LyC emitting galaxies ( $f_{\text{esc}} > 0.1$ ). The sources are colored using the Ly $\alpha$  EW values. We find that 21 out of 32 galaxies in our sub-sample have CIV/CIII] ratios  $> 0.75$ . In gray we plot the same population assuming an average stellar correction of 20%.

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### 3.7 Summary and conclusions

In this paper we have presented a detailed investigation of SF galaxies at  $2.5 \leq z \leq 5$  which show significant nebular CIV emission, drawn from the VANDELS survey [314, 366]. We summarize our findings as follows:

1. At  $2.5 \leq z \leq 5$ , CIV emitters represent just  $\sim 4\%$  of the VANDELS SF population (933 sources). Their  $EW_0$ s range from 1 to 29 Å (Sect. 3.3). The CIV strength correlates well with the strength of the other UV emission lines.
2. Using the UV-BPTs diagrams, we have identified those sources that are powered by stellar photoionization, indicating massive stars as the only ionization source (Sect. 3.4), opposed to an AGN origin. Out of the 43 initially selected CIV emitters, only 4 are consistently found in the AGN region of the diagrams. For 10 sources which could possibly be AGN, we resort to spectral stacking analysis and confirm they are most probably SF galaxies, bringing the final sample to 39 sources.
3. Using BEAGLE [81], by fitting photometry and the spectral indices we derived the physical properties of the SF CIV emitters (total stellar mass, SFR,  $\xi_{\text{ion}}$ , stellar age). We also derive their average stellar metallicity following [64] method based on weak absorption lines. We find that CIV emitters are low-mass galaxies residing on the main sequence. Their metallicity is consistent with the general SF population at the same redshift, and similar to the metallicity of CIV in the local Universe. CIV emitters have on average similar integrated physical properties when compared to galaxies with no CIV emission at  $2.5 \leq z \leq 5$ , with the exception of a higher  $\xi_{\text{ion}}$  average value.
4. Besides having high production of ionizing photons, there are several indications that CIV emitters have also high escape fractions of LyC photons: this includes the presence of bright Ly $\alpha$  emission and a high CIV/CIII] ratio which was recently proposed as one of the most promising indicators of a possible LyC leakage by [415]. In Sect. 3.6, we identify 12 galaxies as the best analogs to cosmic reionizers, showing the combination of high  $\xi_{\text{ion}} > 25$ , and  $EW(\text{Ly}\alpha) > 20\text{\AA}$  and  $\text{CIV/CIII]} > 0.75$ , indicative of high LyC escape fraction.

Amongst the inferred properties, the most uncertain is obviously the escape fraction: in this study we only have presented indirect evidence that the galaxies could have high  $f_{\text{esc}}$ . Unfortunately, the ultra deep VANDELS spectra only start at 4800 Å, which means that a direct measurement of the LyC flux is impossible with current data. However, at this redshift it is possible to detect the LyC flux at  $\sim 3800\text{--}4000$  Å using blue-sensitive facilities. Currently this is possible with MODS on LBT, but in the near future the upgraded FORS2 on VLT operating from 3400 Å [41], will also be an ideal instrument. We plan therefore to validate our conclusion by observing and possibly directly detecting the LyC emission in our galaxies, and we have already been awarded time on the LBT to follow up the UDS sources. If we indeed confirm that CIV emitters are good ionizers, having both high photon production and high escape fraction, this could have important implications to understand the Epoch of Reionization, where obviously the LyC

flux is not observable directly: indeed  $\text{Ly}\alpha$ , one of the best indirect indicators of LyC escape, loses visibility when the IGM starts to be highly neutral, as it is easily suppressed by neutral hydrogen and is visible only in highly ionized peculiar regions [76]. Nebular C IV emission on the other hand would be easily observable with JWST/NIRSpec in SF galaxies up to very high redshift and could therefore be used as a valid alternative to pinpoint the sources responsible for cosmic reionization.

### 3.8 BEAGLE parameters

**Table 3.2:** *Set of BEAGLE parameters used to fit both the parent and the C IV samples' sources.*

| Parameter             | type      | info           | range            |
|-----------------------|-----------|----------------|------------------|
| sfh_type              | fixed     | value: delayed |                  |
| tau                   | fitted    |                | [7.0, 10.5]      |
| mass                  | fitted    |                | [7.0, 12.0]      |
| max_stellar_age       | fitted    |                | [7.0, 10.0]      |
| formation_redshift    | fitted    |                | [7.011499, 50.0] |
| metallicity           | fitted    |                | [-2.2, 0.25]     |
| sfr                   | fitted    |                | [0.0, 3.0]       |
| current_sfr_timescale | fixed     | value: 7.0     |                  |
| redshift              | fitted    |                |                  |
| attenuation_type      | fixed     | value: CF00    |                  |
| tauV_eff              | fitted    |                | [0.001, 5.01]    |
| mu                    | fixed     | value: 0.4     |                  |
| nebular_logU          | fitted    |                | [-4.0, -1.0]     |
| nebular_xi            | fitted    |                | [0.1, 0.5]       |
| nebular_Z             | dependent |                |                  |
| nebular_sigma         | fixed     | value: 22.93   |                  |





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## Chapter 4

# Characterizing extreme [OIII] emitters in high-redshift galaxies with JWST/NIRISS

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*The results in this chapter were published and first presented in [52].*

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Wide Field Slitless Spectroscopy (WFSS) provides a powerful tool for detecting strong line emission in star forming galaxies (SFGs) without the need for target pre-selection. As part of the GLASS-JWST-ERS program, we leverage the near-infrared wavelength capabilities of NIRISS ( $1 - 2.2\mu\text{m}$ ) to observe rest-optical emission lines out to  $z \sim 3.4$ , to a depth and with a spatial resolution higher than ever before ( $\text{H}\alpha$  to  $z < 2.4$ ;  $[\text{OIII}]+\text{H}\beta$  to  $z < 3.4$ ). In this letter we constrain the rest-frame  $[\text{OIII}]\lambda 5007$  equivalent width (EW) distribution for a sample of  $76\ 1 < z < 3.4$  SFGs in the Abell 2744 Hubble Frontier Field and determine an abundance fraction of extreme emission line galaxies with  $\text{EW} > 750\text{\AA}$  in our sample to be 12%. We determine a strong correlation between the measured  $\text{H}\beta$  and  $[\text{OIII}]\lambda 5007$  EWs, supporting that the high  $[\text{OIII}]\lambda 5007$  EW objects require massive stars in young stellar populations to generate the high energy photons needed to doubly ionize oxygen. We extracted spectra for objects up to 2 mag fainter in the near-infrared than previous WFSS studies with the Hubble Space Telescope. Thus, this work clearly highlights the potential of JWST/NIRISS to provide high quality WFSS datasets in crowded cluster environments.

### 4.1 Introduction

The study of star forming galaxies (SFGs) has benefited from the development of HST/WFC3 Wide Field Slitless Spectroscopy (WFSS), where the low background of space-based observatories provides sensitivity to the dispersed light of sources through a low-spectral resolution grism ( $R = \frac{\lambda}{\Delta\lambda} \sim 100 - 200$ ). The advantage compared to fixed-slit or multi-object spectroscopy is the ability to obtain spectra for every object in the field of view (FOV) without the need for pre-selection and without being encumbered by slit-losses. Substantial work using WFSS on

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HST/WFC3 over the past decade, especially from survey programs such as GLASS [421, 480], WISPS [14, 20] and 3D-HST [323], has seen the detection of line emission for large galaxy samples over a wide redshift range. With the advent of the James Webb Space Telescope (JWST) and NIRISS [124, 509], the observable spectral range available to WFSS now extends further into the near-infrared. The additional wavelength range now allows key emission lines to be traced to higher redshifts. The JWST, GLASS-ERS program (ERS 1324, PI Treu; [479]) is one of the first and deepest extragalactic data sets of the Early Release Science (ERS) strategy, and as part of this program we obtain NIRISS/WFSS observations of the Hubble Frontier Field Abell 2744 [A2744, 276] to study the SFG population.

Through WFSS, it is possible to examine the equivalent widths (EW) of rest-optical emission lines in SFGs, which can be used to understand the conditions of their interstellar medium (ISM) and the nature of their underlying stellar population. WFSS is well leveraged to identify galaxies with strong line emission, including those with weak stellar continua which may be too faint to enter broadband-magnitude selections. The combination of strong nebular line emission and weak rest-optical stellar continuum is characteristic of galaxies going through an upturn/burst of star formation with a young stellar population. The UV-emission from the massive short-lived O/B-type stars powers the strong nebular lines and, without a dominant older stellar population providing a strong stellar continuum in the rest-optical, these galaxies exhibit large EWs (hereafter Extreme Emission Line Galaxies or EELGs). Due to the young stellar population, the SFGs in this EELG phase produce enhanced ionization efficiencies and potentially larger escape fractions of UV ionizing radiation than typical SFGs. This makes EELGs of special interest to astronomers, especially in terms of their potential UV ionizing photon contribution during the epoch of reionization, and they have been studied spectroscopically both locally and at moderate redshifts [ $z < 3$ , e.g., 8, 16, 303, 467, 484]. At higher redshifts ( $z > 5$ ) strong line emitters have been identified through measured flux-excess in photometric surveys [e.g., 134, 436].

In this letter, we use the GLASS-ERS JWST/NIRISS observations of the A2744 to study the rest-optical emission line properties of SFGs at  $1 < z < 3.4$  (where [OIII] is observable). The A2744 region has been well studied and large samples of galaxies with spectroscopic redshifts at our desired epoch have been previously identified. We focus this early paper on the A2744 spectroscopic samples reported in [34, 382] and [480]. Therefore, these objects are the focus of this early work. In the future we will use the same dataset to also search for emission-line galaxies without previous redshift determination. In Section 4.2, we briefly introduce the JWST/NIRISS observations. In Section 4.3, we use NIRISS/WFSS to measure the rest-optical line emission ([OII],  $H\beta$ , [OIII] and  $H\alpha$ ) from which we determine the equivalent widths of these SFGs in Section 4.4. Finally, we discuss the most extreme line emitters, EELGs, in Section 4.5.

We note that since the NIRISS FOV is centered on the Abell 2744 cluster, all objects in our sample will be subject to some degree of magnification due to gravitational lensing; however, in this study we only consider relative quantities for each target (EWs). Therefore our results are independent of magnification, yet gravitational lensing allows us to study fainter galaxies at higher spatial resolution.

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We refer the reader to [391, paper I of this series] and [486, paper VII of this series] for GLASS-ERS studies which provide detailed discussion and utilisation of gravitational lensing. In this study we only measure integrated line properties of galaxies and we refer the reader to [504, paper IV of this series] for a GLASS-ERS study of spatially resolved emission line properties. Where applicable, we use  $H_0 = 70$  km/s/Mpc,  $\Omega_m = 0.3$ , and  $\Omega_\Lambda = 0.7$ . All magnitudes are in the AB system [347].

## 4.2 JWST/NIRISS imaging and spectroscopy of HFF A2744

As part of the GLASS-ERS strategy, we use the WFSS mode of the JWST/NIRISS instrument to conduct 1.0–2.2 $\mu$ m low-resolution spectroscopy (average spectral resolution of  $R \sim 150$ ) over the A2744 region. The NIRISS component of the GLASS-ERS survey utilises a single pointing centered on the core of the Abell 2744 cluster<sup>1</sup>, giving a full area of  $2.2' \times 2.2'$ . The WFSS observations and associated direct-imaging are obtained using the F115W, F150W and F200W blocking filters. For the full details of the GLASS-ERS data acquisition, observing strategy and data reduction we refer the reader to [479] and [391].

At this early stage of the mission, there is ongoing development of JWST, calibration files and pipelines, which means the quality of the extracted data will likely improve over time. The results in this paper are based on a NIRISS data reduction that utilised the following pedigree of data products: `CAL_VER=1.5.2`, `CRDS_VER=11.13.1` and `CRDS_CTX=JWST_0881.pmap`. The EW measurements we present in this letter are wavelength-independent relative-quantities (ratio of line flux to continuum flux density at the same wavelength) and therefore should be relatively unaffected by future changes to the calibration files.

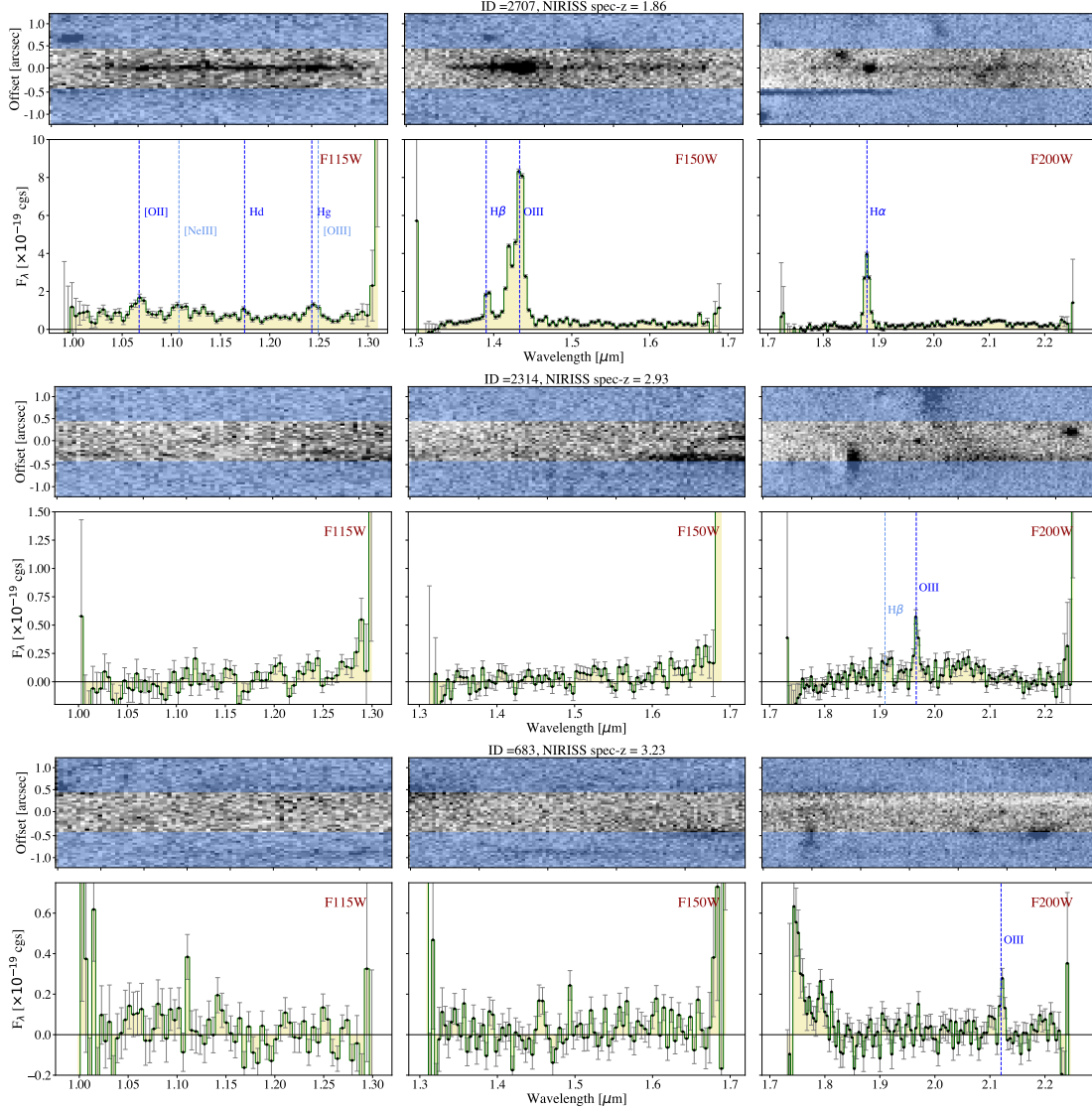
## 4.3 NIRISS detection of Rest-Optical emission lines in individual sources

In this letter, we will focus on NIRISS detections of rest-optical emission lines in a sample of galaxies that have existing spectroscopic redshifts measurements (based on the detection of rest-UV/optical lines) in the range  $1 < z < 3.4$ , where the [OIII] doublet falls within the NIRISS wavelength coverage. This robust sample is collated from several ancillary studies of the A2744 region, utilising both ground-based rest-UV MUSE spectroscopy [34, 382, 86 objects], and space-based rest-optical HST/WFC3 WFSS from [480, 15 objects]. There are 101 spectroscopic candidates within our FOV in our desired redshift range, although we note that 5 may be less robust due to quality flags that mark their redshift as “uncertain” [34] or “tentative/possible” [382, 480].

Our sample represents only a subset of all the emission line galaxies observed with NIRISS and the selection of objects with existing spectroscopic redshifts, where previous redshift determination was reliant on line detection, may bias our sample against weak line emitters or heavily absorbed objects (which would be

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<sup>1</sup>cluster coordinates (J2000) R.A. 00:14:23.4, DEC. -30:23:26



**Figure 4.1:** NIRISS/WFSS *Grizli*-extracted 2D & 1D spectra for three example galaxies in our sample with large  $[\text{OIII}]\lambda 5007$  EW (these measurements are described in Section 4.3), to first demonstrate the typical high quality spectra and then to show how far we can push the NIRISS data. In the 1D panels we mark the location of  $\geq 5\sigma$  (dark blue) and  $\geq 3\sigma$  (light blue) detected emission lines. Top: a galaxy with strong detections in each rest-optical emission line; ID2707 ( $z = 1.82$ ) with a rest-frame  $[\text{OIII}]\lambda 5007$  EW =  $1003 \pm 37 \text{ \AA}$ . The quality of these NIRISS spectra also allows the detection of weaker rest-optical lines including  $\text{NeIII}\lambda 3869$ ,  $\text{H}\delta$  and the blended  $\text{H}\gamma + [\text{OIII}]\lambda 4363$ . Middle: the galaxy in our full sample with the largest EW; ID2314 ( $z = 2.93$ ) with a rest-frame  $[\text{OIII}]\lambda 5007$  EW =  $1901 \pm 787 \text{ \AA}$ . Here the EW signal-to-noise is  $< 5$ , although we note both the line and broad-band continuum are detected above this threshold. Bottom: the faintest galaxy (F150W total mag =  $29.1 \pm 0.2$ ) in our full sample with a  $[\text{OIII}]\lambda 5007$  EW measurement; ID683 ( $z = 3.23$ ) with a rest-frame  $[\text{OIII}]\lambda 5007$  EW =  $446 \pm 109 \text{ \AA}$ .

too reddened and faint to exhibit strong rest-UV emission lines). For example, there may be a population of H $\alpha$  emitters with no UV counterpart that is missed by this pre-selection. Consideration of the selection effects and an analysis with a full statistical sample is beyond the scope of this letter and will be performed in the future. Instead, here we focus on the robust sample as a demonstration of the capabilities of JWST/NIRISS.

We utilise the Grism Redshift and Line Analysis tool [Grizli<sup>2</sup>; 55] to extract and model the Grism spectra associated with each of our targets (see [391] for further details). For 16 objects (all selected from rest-UV spectroscopy), we found no coordinate matches within 1'' in the segmentation map of the composite NIRISS F115W+F150W+F200W direct-image and we remove these from our analysis. From visual inspection of these 16 targets, 7 show no evidence of continuum emission in the direct images at the reported catalogue coordinates and are either examples of extremely blue objects with negligible emission at long wavelength, or more likely, that these sources are not genuine. The latter is supported by each of these 7 having low reliability quality flags in the [382] input catalogue. Visual inspection of the remaining 9 objects reveals each to be a faint-diffuse source near a bright foreground galaxy. In each case the Grizli source extraction has been limited by the significant contamination from the neighbour. We do not attempt to force the extraction of these 9 due to the significant contamination of their dispersed light in both of the orthogonal PRISM images.

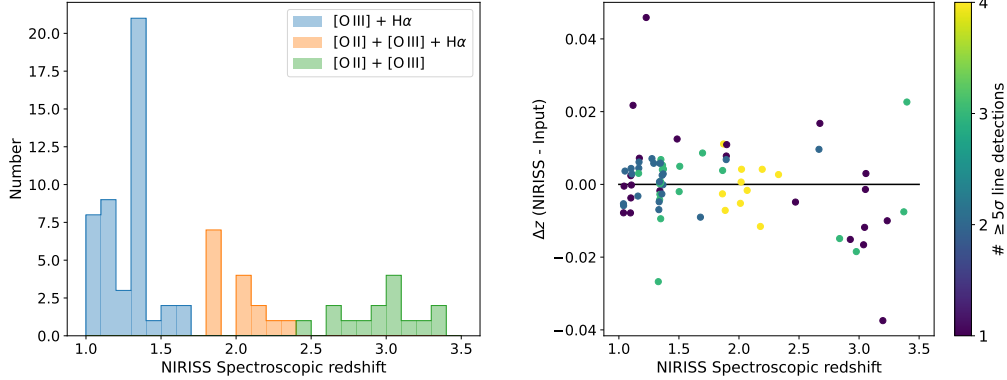
From the 85 matched objects with extracted spectra, we perform the Grizli modelling twice, first with a broad ( $0.5 < z < 5.0$ ) redshift fitting range and again with a narrow ( $\Delta z \pm 0.1$ ) redshift fitting range centred on either the resultant Grizli redshift or the existing spectroscopic redshift if no lines are detected with NIRISS. The first run is used to identify cases when alternative redshift solutions may be present due to new NIRISS line detections, and the second is to aid the identification of weaker lines by reducing the free parameters in the modelling once a confident redshift solution is found.

Within the 2.2arcmin $\times$ 2.2arcmin area, 76 of the 85 galaxies have a NIRISS ( $> 5\sigma$ ) line detection. Of these, 54 have multiple NIRISS line detections. The uncertainty in the line flux reported by Grizli reflects instrumental effects and modelling procedure. We note that uncertainties on faint lines also reflect the presence of additional strong lines in the spectrum which can corroborate the redshift solution. Quoting a line flux limit for the entire survey is not feasible, as the flux limit depends on the level of contamination from nearby sources. Accordingly, we provide the observed median  $5\sigma$  line flux limit, with associated 16<sup>th</sup> and 84<sup>th</sup> percentiles. We measure line flux limits of  $1.5^{+1.9}_{-0.5} \times 10^{-17}$ ,  $1.0^{+1.9}_{-0.4} \times 10^{-17}$  and  $7.6^{+9.1}_{-3.1} \times 10^{-18}$  erg s<sup>-1</sup> cm<sup>-2</sup> for the F115W, F150W and F200W filters respectively using the stated calibration files (these are subject to change in line with calibration file updates).

From the initial run, 23 sources with detected line emission have a primary redshift solution that disagrees with the previous measurement. Upon inspection, we categorise these into three groups: we believe 2 objects<sup>3</sup> have previously mis-

<sup>2</sup><https://github.com/gbrammer/grizli>

<sup>3</sup>NIRISS ID:2808 and 1911. These NIRISS ID's refer to the provided machine readable table which contains the rest-optical properties and coordinates.



**Figure 4.2:** *Left Panel:* The  $1 < z < 3.4$  NIRISS spectroscopic redshift distribution of 76 galaxies with NIRISS-detections ( $> 5\sigma$ ) of  $H\alpha$ ,  $[OIII]$ ,  $H\beta$  or  $[OII]$ . The distribution is split into three colours based on the set of emission lines which are covered by the observations at different redshifts (e.g.,  $[OII]$  is only visible at  $z > 1.7$  and  $H\alpha$  at  $z < 2.4$ ). We note an apparent over-density of 21 independent galaxies in the range  $1.33 < z < 1.37$ . *Right Panel:* The comparison of the NIRISS *Grizli*-derived and input spectroscopic redshifts, colour coded by the number of  $> 5\sigma$  NIRISS line detections. Excellent agreement is found in the sample, with the majority of targets having their previous redshifts confirmed through multiple NIRISS line detections. Only two galaxies (NIRISS ID:2808 and 1911,  $\Delta z > 0.5$ ) are found to have previously mis-identified redshifts solutions.

identified redshifts based on the observed wavelengths of line emission detected with NIRISS and in the previous spectroscopy; 7 objects have a close-probability secondary redshift solution from *Grizli* that agrees with the existing value and is supported by additional line detections in previous spectroscopy, and in the second run we utilise their secondary *Grizli* solution; finally 14 objects have solutions based on weak ( $< 10\sigma$ ) detections of only one (7) or two (7) lines in NIRISS which can allow the redshift solution to remain ambiguous, and for these we rely on the existing spectroscopic redshift for the second run. The 9 galaxies with no NIRISS line-detection either have weak rest-optical line emission, lie at alternative redshifts where no lines are present in our wavelength window, or have considerable contamination from the dispersed light from neighbouring sources such that their lines are no-longer detectable. We note that of the 5 galaxies with less robust redshift quality flags, 3 have their redshift confirmed, 1 is identified to lie at an alternative redshift and 1 had no line detections. This indicates that some of the targets without NIRISS-detections may be due to the quality of the input spectroscopic redshifts. Finally, we inspect one object (NIRISS ID:1710) which had two different spectroscopic redshifts based on claimed  $[OII]$ -line detections from MUSE ( $z = 1.2$ ) and HST/WFC3 WFSS ( $z = 1.9$ ). From our detection of  $H\alpha$  and  $[OIII]$  in NIRISS, we measure  $z_{NIRISS} = 1.16$  and propose that the HST line was actually the mis-identified  $[OIII]$  line.

In Figure 4.1 we present the 2D and 1D NIRISS spectra for 3 objects that demonstrate the capabilities of JWST/NIRISS: ID 2314 which had the largest measured  $[OIII]\lambda 5007$  EW in our sample (determined in the next Section), ID

683 which was the faintest source (F150W  $29.1 \pm 0.2$  mag<sup>4</sup>) with a NIRISS line detection and ID 2707 which shows significant detections in all rest-optical lines considered in this work. The distribution of **Grizli**-derived spectroscopic redshifts for galaxies with NIRISS line detections is shown in the left panel of Figure 4.2; redshifts are taken from the second **Grizli** modelling. In the right panel the previous spectroscopic redshifts and the **Grizli**-derived NIRISS redshifts are compared. Excellent agreement is found between the two spectroscopic redshift measurements, with NIRISS line detections confirming 87% (74/85) of the sample and only identifying 2 galaxies to be at alternative redshifts. We note a small  $\sigma_{\text{NMAD}}^5 = 0.003$  and only 14% (11/76) of those with line detections show notable deviation in spectroscopic redshift ( $\Delta z / (1 + z_{\text{NIRISS}}) > 5\sigma_{\text{NMAD}}$ ).

In summary, we are able to recover accurate redshifts using NIRISS for nearly the entire sample, but note that visual inspection is required in a significant fraction of cases to clearly identify the emission lines. Nonetheless these are early results and the performance of automated redshift and line flux measurements is likely to improve with updated NIRISS calibration and contamination modelling.

#### 4.4 Determination of rest-Optical equivalent widths

To determine individual EW, we start by measuring the integrated line fluxes for emission lines detected at  $\geq 5\sigma$  in NIRISS/WFSS and place upper limits on those below this threshold. In this analysis, we only consider the 76 galaxies that have at least one NIRISS line detection and emission lines within the NIRISS wavelength range. For the objects in the field with NIRISS-detected line emission (66/76 have [OIII] detections), we determine the EW as the ratio of the line flux to the continuum flux density (at the wavelength of the line) where both quantities are “total” values.

We choose to measure the continuum flux density from the broadband imaging rather than directly from the WFSS spectra. This is because the continuum sensitivity in the spectra and the contamination due to overlap with other spectra can lead to large uncertainties on the measured EW for objects with faint continua.

Total line fluxes and their uncertainty are obtained from the **Grizli** extraction pipeline, along with broad-band photometry for each source in the F115W, F150W, and F200W direct images. Photometric fluxes are obtained from a  $0.36''$ -diameter aperture with a total-aperture correction applied and the spectral extraction from **Grizli** is based on the segmentation map obtained from the direct images. Therefore, **Grizli** photometry and spectroscopy are obtained on a self-consistent aperture. Using the self-consistent continuum and emission line fluxes results in galaxy-integrated EW values.

To validate our approach of combining line fluxes from slitless spectroscopy with continuum flux densities from photometry, we select galaxies from the bright-

<sup>4</sup>We note a consistent magnitude offset is observed between the NIRISS F150W total-magnitudes and the associated HST F160W total-magnitudes (F160W - F150W =  $+0.5 \pm 0.1$  mag) for the sources from [34]. We apply this offset correction to the quoted magnitude. This discrepancy is likely to be automatically solved once updated NIRISS in-orbit calibration data become available.

<sup>5</sup> $\sigma_{\text{NMAD}}$  as defined by [57].  $\sigma_{\text{NMAD}} = 1.48 \times \text{median}(|\Delta z - \text{median}(\Delta z)| / (1 + z_{\text{NIRISS}}))$ , where  $\Delta z = z_{\text{NIRISS}} - z_{\text{INPUT}}$

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end of our sample, where the continuum emission can be constrained from the slitless spectroscopy and provide robust EW measurements from the spectra alone for comparison. We find good agreement in the EWs determined using broadband imaging and using slitless spectroscopy alone, i.e. the measurements agree within  $\sim 0.1$  dex ( $1\sigma$ ) with a minor systematic bias of 0.06 dex for 16 objects with  $21 < m_{AB} < 23$ . We further note that the estimated fractional uncertainty associated with the direct measurement of the EW from the slitless spectroscopy alone is consistent with the broadband-derived method.

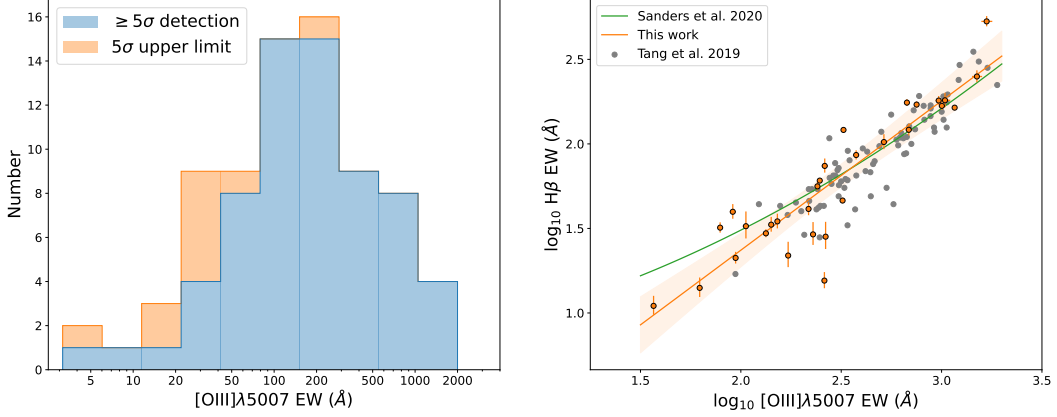
All galaxies for which we identify emission lines are well detected ( $\geq 5\sigma$ ) in the associated imaging, and are thus suitable for equivalent width measurements. The faintest source with a NIRISS detection has a F150W  $29.1 \pm 0.2$  magnitude (compared to a sample median F150W magnitude of 24.57), consistent with the estimated  $5\sigma$  magnitude limit for the pre-imaging [479]. We assume a flat in  $f_\nu$  continuum slope and subtract the flux contribution from any detected emission lines contributing to that filter. EWs are then taken as the ratio of the line flux to the continuum flux density, and are brought to the rest-frame using the NIRISS **Grizli**-derived redshift. The assumed continuum slope and the contribution of non-detected emission lines does not significantly affect the measured EW distribution (e.g., see [53]).

When an emission line is not detected (10/76 do not have [OIII] detections), we report a  $5\sigma$  upper limit on the line flux using the modelled line uncertainty. The  $5\sigma$  EW upper limit is then taken as the ratio of this flux limit and the continuum flux density at the expected location. Here, we rely on the spectroscopic redshift derived from other NIRISS-detected lines ( $H\alpha$ ,  $H\beta$ , [OII]).

We present the distribution of [OIII] $\lambda$ 5007 EWs in the  $1 < z < 3.4$  sample in Figure 4.3, adopting a 1:3 flux ratio between the doublet. We observe a broad dynamic range of [OIII] $\lambda$ 5007 EW spanning  $\sim 10$ – $2000 \text{ \AA}$ , with a clear peak at  $\sim 150 \text{ \AA}$ . We note that our sample is likely incomplete at low EWs due to our line sensitivity and sample selection; we intend to revisit the completeness in future work. We note 14 objects with [OIII] EW measurements are at  $2.4 < z < 3.4$ , an epoch previously unobtainable with WFSS on HST.

We additionally present the comparison of the EWs for  $H\beta$  and [OIII] $\lambda$ 5007 for galaxies where both have  $\geq 5\sigma$  measurements. We observe a clear positive correlation which is in agreement with the trend found by [404, MOSDEF, ground-based] for  $z \sim 2.3$  SFGs and [463, 466] for  $z \sim 2$  SFGs, supporting the accuracy of our measurements. The NIRISS sensitivity provides us higher precision EW measurements and we fit a linear function and report best fit coefficients:  $\log_{10}(\text{EW}_{H\beta}) = 0.88x - 0.39$ , with  $x = \log_{10}(\text{EW}_{[\text{OIII}]\lambda 5007})$ , and measure a scatter of 0.16dex. The [OIII] and  $H\beta$  EW measurements are provided in Table 4.1 for the subset of objects where all rest-optical emission lines are observable and in a machine readable format for the full dataset.





**Figure 4.3:** *Rest-optical EW properties of  $1 < z < 3.4$  SFGs. Left: The  $[\text{OIII}]\lambda 5007$  rest-frame equivalent width distribution of galaxies with NIRISS-detected ( $\geq 5\sigma$ ) line emission. Stacked histograms show  $\geq 5\sigma$  EW measurements (blue) and  $5\sigma$  upper limits (orange), divided between 10 bins of equal logarithmic width. Right: The comparison of  $\text{H}\beta$  and  $[\text{OIII}]\lambda 5007$  EWs, restricted to objects where both have  $\geq 5\sigma$  measurements (in orange). We report a positive trend, in agreement with the overlaid  $z \sim 2$  galaxy sample presented by [463, 466] (in grey). We also overlay the best-fit curve (green) for the  $z \sim 2.3$  [404] sample, who utilise a second order polynomial function. We use a linear function to model our data, which is shown in orange accompanied by the  $2\sigma$  uncertainty for our data.*

## 4.5 Identification of Extreme Emission Line Galaxies with NIRISS WFSS

There is no strict EW threshold for what is considered an EELG, and within literature many studies introduce different EW thresholds and utilise different emission lines ( $\text{H}\beta$ ,  $\text{H}\alpha$ ,  $[\text{OIII}]$ ) in their selection [e.g., 8, 15, 126, 303, 349, 467, 469, 484]. we adopt a rest-frame  $> 750\text{\AA}$   $[\text{OIII}]\lambda 5007$  EW criteria. This threshold is physically motivated by SFGs above this value being observed to exhibit larger ionizing photon production efficiencies than typical SFGs [e.g., 467, at  $z \sim 2$ ] and the potential for high escape fractions of ionizing photons [e.g., 337]. These systems with large EWs are associated with very young stellar populations ( $\lesssim 10\text{Myr}$ , e.g., [468]), created during a recent burst of star formation, where the massive O/B-type stars responsible for the production of the high-energy ionizing UV photons required to produce doubly ionized oxygen still remain in the population. We note that EW thresholds based on Hydrogen recombination lines can be theoretically motivated from population syntheses of a stellar population at given age (e.g., [123], [269]) and are largely insensitive to SFH and metallicity, however, it is still common to choose to use the  $[\text{OIII}]$  line because it allows studies to probe higher redshift samples compared to  $\text{H}\alpha$ , and is typically the strongest rest-optical emission line (e.g., compared to  $\text{H}\beta$ ) which reduces the uncertainty in EW measurements.

We find that out of the 76 initial targets with NIRISS line detections, 12% (9/76) are considered EELGs according to our threshold. This value is larger than the measured rate (3.8%  $> 750\text{\AA}$ ) at  $z \sim 2$  from [53], who utilise HST/WFC3

WFSS. We note that these two values may differ due to the sample selection, given that the quality of the NIRISS/WFSS allows us to extract spectra from sources up to two magnitudes fainter (at  $\lambda \sim 1.5\mu\text{m}$ ) and [53] employ a  $M_{\text{UV}}$  restriction. Additionally, the requirement for our sample to have a spectroscopic redshift based on previous line detections may bias our sample towards stronger line emitters, although we may also miss extreme line emitters with faint continua due to the requirement of a detection in our direct imaging. Future work will consider the completeness in detail.

Our NIRISS/WFSS sample allows us to study the characteristics of these SFGs.  $H\beta$  EW is shown to be a good indicator of the age of a stellar population [e.g., 153, 154] and in this sample we determine a strong relation between the  $[\text{OIII}]\lambda 5007$  and  $H\beta$  EW. This supports that very young stars are required in our large-EW SFGs to produce the high energy photons needed for the production of doubly ionized oxygen.

The identification of the extreme EW population in our analysis selects a sample of efficient ionizing agents. However, this selection will not contain all UV-ionizing sources. Extremely metal-poor galaxies can exhibit the characteristics we have ascribed to EELGs without exhibiting extreme  $[\text{OIII}]\lambda 5007$  EWs, as has been seen in local extremely metal-poor dwarf galaxies [e.g., 212, 222]. Evidence for galaxies with young stellar populations but low  $[\text{OIII}]$  EW has also been recently reported at  $z > 7$  by [132], which has also been attributed to very low metallicity or a large escape fraction of ionizing photons. Therefore, the fraction of SFGs that are EELGs provides a lower threshold to the true abundance of efficient ionizing agents within the SFG population, and further work into the dependence of  $[\text{OIII}]$  emission on metallicity and escape fraction is needed to determine the abundance of the low- $[\text{OIII}]$  EW ionizing sources. We have provided a first study of the rest-optical properties of EELGs based on NIRISS observations, and to higher redshifts than achievable with HST/WFC3 WFSS. We await upcoming JWST/NIRISS programs, including, CANUCS [GTO 1208, PI Willott] and PASSAGE [GO 1571, PI Malkan], to provide greater statistical analysis through large area surveys (involving multiple NIRISS pointings).

**Table 4.1:** *Rest-optical emission line EWs for the  $1.7 < z < 2.4$  (where all lines are observable) NIRISS-detected sources. The columns provided in the machine readable table for the full  $1 < z < 3.4$  sample match those given here. Column (1) lists the IDs of the sources; (2) and (3) show the (J2000) Right Ascension and Declination; (4,5,6) report the **Grizli** maximum probability function redshift and the 16<sup>th</sup> and 84<sup>th</sup> redshift percentiles; the rest-frame equivalent width and uncertainty are given for the  $H\alpha$ ,  $[\text{OIII}]$ ,  $H\beta$  and  $[\text{OII}]$  lines in columns (7-14). When 5 $\sigma$  EW upper limits are required, the EW\_err is set to -99.*

| ID   | RA<br>[Deg] | DEC<br>[Deg] | z<br>[NIRISS]             | $H\alpha$ EW<br>[Å] | $[\text{OIII}]$ EW<br>[Å] | $H\beta$ EW<br>[Å] | $[\text{OII}]$ EW<br>[Å] |
|------|-------------|--------------|---------------------------|---------------------|---------------------------|--------------------|--------------------------|
| (1)  | (2)         | (3)          | (4,5,6)                   | (7,8)               | (9,10)                    | (11,12)            | (13,14)                  |
| 1911 | 3.586765    | -30.390825   | $2.213^{+0.001}_{-0.003}$ | $80.37 \pm 4.77$    | $70.56 \pm 3.54$          | $< 8.13$           | $24.95 \pm 3.78$         |
| 2205 | 3.586972    | -30.387021   | $1.864^{+0.001}_{-0.001}$ | $230.41 \pm 12.12$  | $305.21 \pm 8.46$         | $29.16 \pm 4.50$   | $< 28.36$                |
| 1208 | 3.576590    | -30.399050   | $1.872^{+0.001}_{-0.001}$ | $249.00 \pm 8.08$   | $500.33 \pm 13.71$        | $85.95 \pm 5.35$   | $45.07 \pm 4.18$         |
| 2707 | 3.585922    | -30.380683   | $1.862^{+0.001}_{-0.001}$ | $980.57 \pm 33.70$  | $1337.72 \pm 48.96$       | $167.85 \pm 9.36$  | $106.76 \pm 8.94$        |

**Table 4.1:** *continued.*

| ID   | RA<br>[Deg] | DEC<br>[Deg] | z<br>[NIRISS]             | H $\alpha$ EW<br>[Å] | [OIII] EW<br>[Å]     | H $\beta$ EW<br>[Å] | [OII] EW<br>[Å]   |
|------|-------------|--------------|---------------------------|----------------------|----------------------|---------------------|-------------------|
| 1426 | 3.597289    | -30.396712   | $1.898^{+0.007}_{-0.002}$ | < 123.42             | 149.15 $\pm$ 17.34   | < 62.31             | < 68.84           |
| 951  | 3.583265    | -30.403339   | $1.894^{+0.003}_{-0.002}$ | 212.63 $\pm$ 21.71   | 109.76 $\pm$ 12.03   | < 51.88             | 81.09             |
| 950  | 3.582530    | -30.402313   | $1.895^{+0.002}_{-0.001}$ | < 94.77              | 211.30 $\pm$ 10.08   | < 40.78             | < 111.51          |
| 2616 | 3.591120    | -30.381703   | $1.885^{+0.001}_{-0.001}$ | < 287.23 $\pm$ 3.51  | 428.25 $\pm$ 4.65    | 46.24 $\pm$ 1.71    | 41.77 $\pm$ 2.04  |
| 320  | 3.586433    | -30.409363   | $2.021^{+0.001}_{-0.001}$ | 772.55 $\pm$ 19.82   | 1001.00 $\pm$ 26.51  | 170.83 $\pm$ 7.23   | 56.85 $\pm$ 3.89  |
| 1002 | 3.598541    | -30.401792   | $2.012^{+0.001}_{-0.001}$ | 870.69 $\pm$ 25.02   | 895.90 $\pm$ 26.63   | 175.48 $\pm$ 8.22   | 53.87 $\pm$ 4.53  |
| 451  | 3.594058    | -30.407999   | $2.018^{+0.001}_{-0.001}$ | 2631.01 $\pm$ 145.88 | 2235.78 $\pm$ 140.12 | 530.46 $\pm$ 36.99  | 142.72 $\pm$ 9.48 |
| 1352 | 3.604190    | -30.397187   | $2.068^{+0.001}_{-0.001}$ | 163.38 $\pm$ 4.50    | 105.14 $\pm$ 2.80    | 31.96 $\pm$ 2.24    | 34.00 $\pm$ 2.59  |
| 2289 | 3.579076    | -30.385963   | $2.194^{+0.001}_{-0.001}$ | 607.38 $\pm$ 28.01   | 688.04 $\pm$ 27.85   | 102.58 $\pm$ 10.39  | 70.52 $\pm$ 8.02  |
| 1886 | 3.603156    | -30.391074   | $2.178^{+0.001}_{-0.001}$ | 601.62 $\pm$ 24.57   | 915.38 $\pm$ 35.40   | 121.17 $\pm$ 9.82   | 57.73 $\pm$ 5.88  |
| 2139 | 3.580030    | -30.387832   | $2.329^{+0.001}_{-0.001}$ | 218.86 $\pm$ 4.02    | 347.69 $\pm$ 4.43    | 15.54 $\pm$ 1.70    | 97.50 $\pm$ 2.66  |

## 4.6 Conclusions

The GLASS-ERS program provides some of the earliest and deepest JWST extragalactic observations, and in this Letter we have used deep NIRISS/WFSS to study  $1 < z < 3.4$  SFGs in the A2744 region. From our initial spectroscopic sample of galaxies, we detect line emission in 89% (76/85) of targets with extracted spectra. We find good agreement in the spectroscopic redshift measurements from NIRISS and previous ground- and space-based studies, confirming the redshift of 74 objects and with only 2 objects identified using NIRISS to have an alternative redshift to their previous values. We note there are still subtle differences in the spectroscopic redshifts with 14% of the sample (11/76 objects) having a change in measurement greater than  $\geq 5\sigma_{\text{NMAD}}$  where  $\sigma_{\text{NMAD}} = 0.003$ . We caution that visual inspection was required in a significant fraction of cases (23/76) to clearly identify the emission lines. This will likely improve with updates to contamination modelling and NIRISS calibrations.

We determine the observed rest-frame [OIII] $\lambda$ 5007 EW distribution, ranging from  $\sim 10$ –2000 Å, with a turnover at  $\sim 150$  Å. We identify an EELG sub-sample of 9 (12%) galaxies with  $\text{EW} \geq 750$  Å, an abundance higher than found in previous studies. This discrepancy likely arises from the inherent incompleteness of our sample due to the pre-selection of spectroscopically identified sources and our line sensitivity, although we note the sensitivity of NIRISS allows our sample to include extracted spectra from galaxies up to 2 magnitudes fainter in the near-infrared and to higher redshift than previous HST/WFSS studies. Finally, we find a correlation between the [OIII] $\lambda$ 5007 and H $\beta$  EWs, supporting that very young stellar populations are required to produce the high energy photons needed to produce [OIII].

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This study provides a first demonstration of the capabilities of JWST/NIRISS for studying rest-optical emission lines in high-redshift SFGs. We anticipate that future studies of SFG spectra through NIRISS/WFSS will advance greatly over the first cycles of JWST science, as planned wide area surveys including PASSAGE and CANUCS unlock large galaxy samples, allowing statistical analysis of the rare EELG population.

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## Chapter 5

# Identifying LyC leakers at $z \geq 4.5$ : first results from the GLASS-JWST program

*The results in this chapter were published and first presented in [300].*

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The escape fraction of Lyman-continuum (LyC) photons ( $f_{\text{esc}}$ ) is a key parameter for determining the sources of cosmic reionization at  $z \geq 6$ . At these redshifts, owing to the opacity of the intergalactic medium, the LyC emission cannot be measured directly. However, LyC leakers during the epoch of reionization could be identified using indirect indicators that have been extensively tested at low and intermediate redshifts. These include a high [OIII]/[OII] flux ratio, high star-formation surface density, and compact sizes. In this work, we present observations of 29  $4.5 \leq z \leq 8$  gravitationally lensed galaxies in the Abell 2744 cluster field. From a combined analysis of JWST-NIRSpec and NIRCам data, we accurately derived their physical and spectroscopic properties: our galaxies have low masses ( $\log(M_\star) \sim 8.5 M_\odot$ ), blue UV spectral slopes ( $\beta \sim -2.1$ ), compact sizes ( $r_e \sim 0.3 - 0.5$  kpc), and high [OIII]/[OII] flux ratios. We confirm that these properties are similar to those characterizing low-redshift LyC leakers. Indirectly inferring the fraction of escaping ionizing photons, we find that more than 80% of our galaxies have predicted  $f_{\text{esc}}$  values larger than 0.05, indicating that they would be considered leakers. The average predicted  $f_{\text{esc}}$  value of our sample is 0.12, suggesting that similar galaxies at  $z \geq 6$  have provided a substantial contribution to cosmic reionization.

### 5.1 Introduction

The Lyman-continuum (LyC,  $\lambda < 912$  Å) photons escaping from star-forming galaxies into the neutral intergalactic medium (IGM) can account for the photon budget required to complete reionization only if a substantial fraction of them escape from the galaxies' interstellar and circumgalactic media (ISM and CGM). Given the number density of star-forming galaxies in the Epoch of Reionization

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(EoR), an average LyC escape fraction ( $f_{\text{esc}}$ ) of  $\sim 10\%$  across all galaxies would be required [e.g., 149, 393] to both reionize the Universe by  $z = 6$  and match the Thomson optical depth of electron scattering observed in the cosmic microwave background [CMB, 368].

However, at  $z \geq 4.5$  it is impossible to detect the LyC photons escaping from galaxies, since they are absorbed and scattered by the IGM along the line of sight [208]. Therefore, efforts at low redshift, where LyC can be detected, have been focused on identifying other observable properties that trace physical conditions facilitating the escape of LyC photons. These indirect indicators could then be used in the EoR to identify the cosmic ionizers. Several indirect diagnostics have been proposed [e.g., 218, 292, 497, 517], but they are all characterized by a large scatter. One of the best indicators is the presence of a strong Ly $\alpha$  emission [e.g., 166, 358], often characterized by two emission peaks with a small velocity separation. However, at  $z > 6.5$ , this line is attenuated due to its resonant nature and by the increasingly neutral IGM as we approach the EoR [43, 235, 304, 353, 364]. Emission from the MgII  $\lambda 2800$  doublet has been proposed as a promising LyC proxy [85], as the escape of this line is controlled by resonant scattering in the same low column-density gas as the Ly $\alpha$  [see also 219, 516].

More recently, the nebular C IV emission line, requiring comparably high ionization energies to the He II line ( $E > 47.9 \text{ eV}$  and  $> 54.4 \text{ eV}$  respectively), has attracted attention since its presence might be strongly linked to the escape of Lyman continuum photons from galaxies [e.g., 301, 409, 415, 423], although this line is in general much fainter than Ly $\alpha$ . Another very popular indicator is the emission line ratio  $[\text{O III}]\lambda\lambda 4959, 5007/[\text{O II}]\lambda 3727$  (hereafter, O32), as [333] first found evidence for its correlation with  $f_{\text{esc}}$ : high values of O32 would reflect partially incomplete HII regions, where some LyC photons could escape from [292]. Later on, it was found that this correlation is characterized by a substantial scatter, as highlighted by low-redshift studies [e.g., 154, 218, 337]. As a result, a high O32 flux ratio is still a necessary condition for a significant measurement of  $f_{\text{esc}}$ , although it is not sufficient by itself to define a LyC leaker, as viewing angles might play a role as well as variation in metallicity and ionization parameter [e.g., 26, 240]. Further additional properties, such as low values of Balmer lines' rest-frame equivalent widths ( $EW_0$ ), are thus required. As a matter of fact, measuring low values of EW in these lines could indicate lower optical depth in the HII region [e.g., 37]. However, many local LyC emitters exhibiting high values of Balmer emission line  $EW_0$ s are present in the literature [213, 215, 216, 218]. A possible explanation for this discrepancy lies in the fact that Balmer-line  $EW_0$ s are sensitive not only to HII region size, but also to starburst age and star formation history [6, 39, 529]. It is thus imperative to pair the EW diagnostic with another indicator in order to prevent degeneracy. Finally, another diagnostic for a high  $f_{\text{esc}}$  is the SFR surface density ( $\Sigma_{\text{SFR}}$ ): feedback from star formation can blow bubbles or chimneys into the host galaxy's ISM, linking a high  $\Sigma_{\text{SFR}}$  value to a high  $f_{\text{esc}}$ . This connection is supported by the detection of some compact LyC emitters [LCEs, e.g., 218, 292], even though compactness may not be a defining characteristic of all of them [e.g., 292].

Recently, [154] presented the first statistical test of many of the diagnostics

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just described on a large sample of low-redshift LCEs. Their conclusion is that indicators based on Ly $\alpha$  emission properties (such as peak velocity separation and equivalent width) perform well, and that diagnostics such as the [OIII]/[OII] flux ratio and the SFR surface density,  $\Sigma_{SFR}$ , could also be used to determine if a galaxy is an LCE or not.

With the advent of JWST, we now have the opportunity to constrain LyC diagnostics during the epoch of reionization with the first NIRSpec observations of high-redshift galaxies. In this paper, we make use of NIRSpec spectra obtained in the Abell 2744 cluster region for a sample of galaxies at  $4.5 \leq z \leq 8$  observed as part of the JWST Early Release Science program GLASS [479] and the JWST Director Discretionary Time program [JWST-GO-2756; P.I. Chen; 391]. With the wavelength coverage of  $0.9 - 5.3 \mu\text{m}$  offered by the NIRSpec observations, we can confirm not only the redshifts of tens of photometrically selected candidates using intense emission lines, but we can also measure optical line ratios and rest-frame EWs with high precision. Our unique data set also includes deep JWST/NIRCam images, enabling us to better characterize those cosmic reionizer candidates based on their physical properties.

This paper is organized as follows. We present the data set in Sect. 5.2. We characterize the selected sample and compare the physical and spectroscopic properties with cosmological models and other samples at lower redshifts in Sect. 5.3. In Sect. 5.4, we indirectly infer  $f_{\text{esc}}$  for the high-redshift sample and in Sect. 5.5, we summarize our key conclusions. Throughout this work, we assume a flat  $\Lambda$ CDM cosmology with  $H_0 = 67.7 \text{ km s}^{-1} \text{ Mpc}^{-1}$  and  $\Omega_m = 0.307$  [368] and the [78] initial mass function. All magnitudes are expressed in the AB system [347].

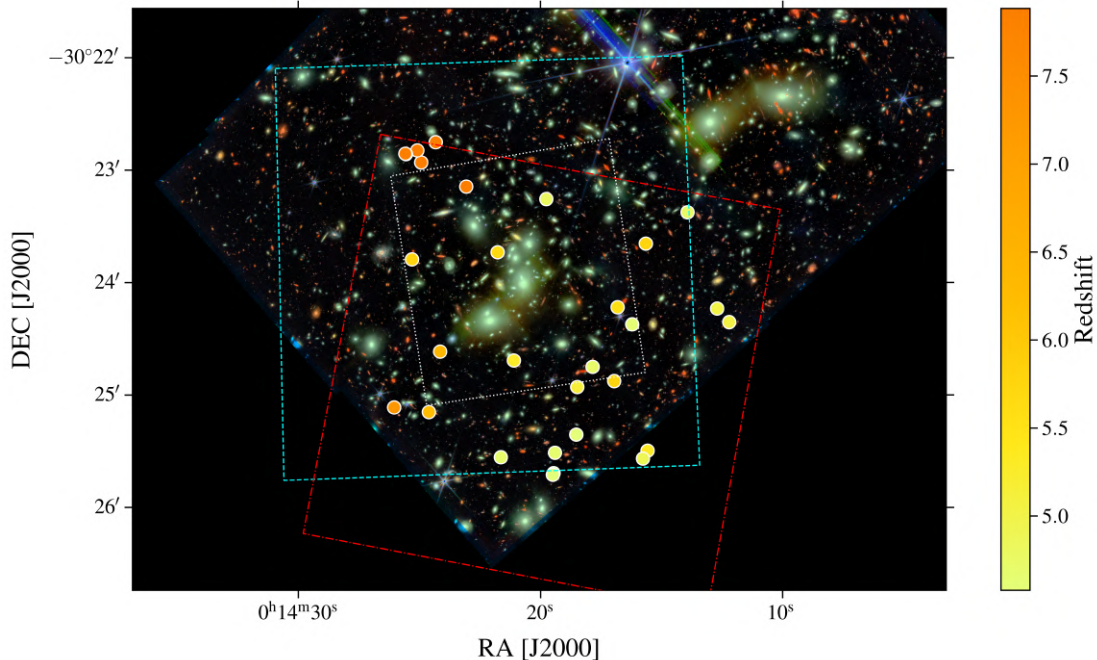
## 5.2 Observations and method

The final target list of the GLASS-JWST program and the way in which targets have been prioritized will be presented in a future paper. However, [325] have already described the observations and data reduction strategy. Here, we present a brief summary, highlighting the points that are most relevant for this work and describing the methods used to study the properties of the galaxies in our sample.

### 5.2.1 JWST/NIRSpec MSA observations and data reduction

Our spectra were acquired through NIRSpec MSA observations in two programs: the GLASS-JWST Early Release Science Program [PID 1324, PI Treu; 479] and a JWST DDT program [PID 2756, PI. W. Chen; 391], which obtained NIRSpec observations for a subset of targets residing over the central regions of the Frontier Field galaxy cluster Abell 2744.

The GLASS-JWST observations were carried out on November 10, 2022, with three spectral configurations (G140H/F100LP, G235H/F170LP, and G395H/F290LP). These configurations cover wavelengths between  $1 - 5.14 \mu\text{m}$ , at  $R \sim 2000 - 3000$ . We exposed each of the three high-resolution gratings for a total of 4.9 hours. Specifically, in this work, we use the G235H/F170LP and G395H/F290LP observations, which contain the bright emission lines we will analyze.



**Figure 5.1:** *Spatial location of the 29 selected sources, color coded by their spectroscopic redshift. They are superimposed on the RGB image of the UNCOVER program, made with NIRCcam filters (blue = F115W + F150W, green = F200W + F277W, and red = F356W + F410M + F444W). The MUSE footprint is shown in white, the NIRSpec GLASS-JWST pointing is shown in cyan, and the NIRSpec DDT pointing is shown in red.*

DDT NIRSpec observations were carried out on October 23 2022, using the CLEAR filter+PRISM configuration, which provides continuous wavelength coverage of  $0.6 - 5.3 \mu\text{m}$  at  $R \sim 30 - 300$  spectral resolution. The on-source exposure time is 1.23 hours.

Data were reduced using the official STScI JWST pipeline (ver.1.8.2)<sup>1</sup> for Level 1 data products and the `msaexp`<sup>2</sup> code for Level 2 and 3 data products, which is based on the STScI pipeline but also includes additional correction routines. In summary, we initially reduced the uncalibrated data using the `Detector1Pipeline` routine and the latest set of reference files (`jwst_1023.pmap`) to correct for detector-level artifacts and convert them to count-rate images. Then, we applied custom preprocessing routines from `msaexp` to remove residual  $1/f$  noise that is not corrected by the IRS2 readout, to identify and remove "snowballs", and to remove bias exposure by exposure before running STScI routines from `Spec2Pipeline` for the final 2D cutout images. To perform WCS registration, flat-fielding, path-loss corrections, and flux calibration, these routines include `AssignWcs`, `Extract2dStep`, `FlatFieldStep`, `PathLossStep`, and `PhotomStep`. Of note, our chosen reference files include an in-flight flux calibration, accounting for NIRSpec's better-than-expected throughput at blue wavelengths. Local background subtraction was performed using a three-shutter nod pattern before the resulting images are drizzled onto a common grid. We optimally extracted the spectra using an inverse-

<sup>1</sup><https://github.com/spacetelescope/jwst>

<sup>2</sup><https://github.com/gbrammer/msaexp>



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variance weighted kernel, which is derived by summing the 2D spectrum along the dispersion axis and fitting the signal along the spatial axis to a Gaussian profile. We visually inspected all kernels to make sure spurious events are not included. As a result, the kernel extracts the 1D spectrum along the dispersion axis. The final step was to verify the default wavelength calibration for the gratings, which is accurate within 1 Å (Williams et al. in prep).

## 5.2.2 Imaging data

Deep NIRCcam images were acquired from the GO program UNCOVER (GO 2561; PI I. Labbe) and included observations in the F115W, F150W, F200W, F277W, F356W, F410M, and F444W filters. The imaging data were reduced using the STScI JWST pipeline and the latest versions of photometric zero points and reference files. A detailed description of all the reduction and calibration steps is presented in [317]. Of the 29 sources analyzed in this work (see next section), 27 were observed within the UNCOVER pointings. Their positions and the UNCOVER footprints are presented in Figure 5.1.

## 5.2.3 Emission line and redshift identifications

The focus of the study is on all sources at  $z \geq 4.5$ . Specifically, we analyzed the spectra of: 13 galaxies with a spectroscopic redshift larger than 4.5 previously confirmed by the MUSE observations [285, 382], all showing Ly $\alpha$  in emission in their optical spectra (the footprint of the MUSE observations is also shown in Figure 5.1); 29 galaxies with a photometric redshift in the range 4.5-8, of which 5 were selected as part of the  $z \simeq 7.9$  candidate protocluster and whose confirmation has been recently presented by [325]. Finally, 23 of the 42 galaxies were observed as part of GLASS-JWST, while 19 were part of the DDT program.

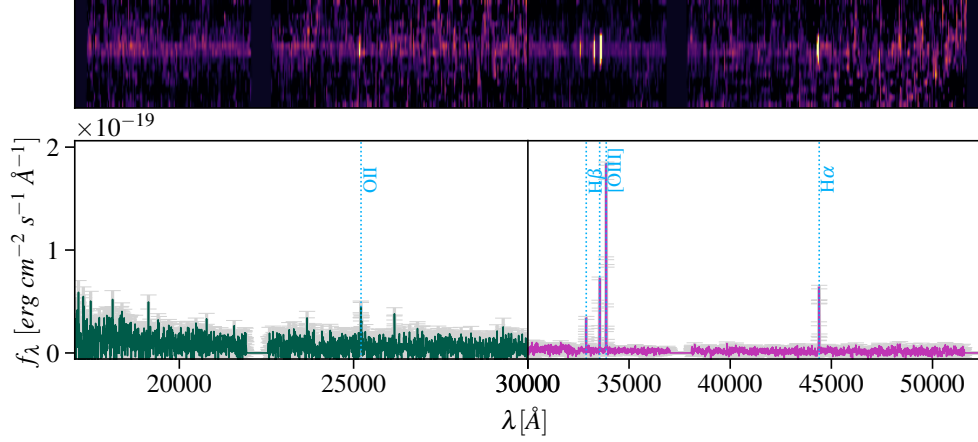
Spectra were visually examined for detectable optical lines using the spectroscopic or photometric redshift information. For photometric sources we determined the spectroscopic redshift when possible using the H $\beta$ , [O III] $\lambda\lambda$ 4959, 5007, and (when present) H $\alpha$  lines. In 29 cases, the [O II] $\lambda$ 3727, [O III] $\lambda\lambda$ 4959, 5007 and H $\beta$  were detected and their line fluxes were measured. We also measured the H $\alpha$  line in 17 out of 29 cases, since it falls within the observed spectrum (examples are shown in Figure 5.2). For this part of our analysis, we used the latest version of the `specutils`<sup>3</sup> packages in PYTHON.

From our initial target list of 42 galaxies, there were eight sources with a spectroscopic redshift confirmed from previous MUSE observations, which were placed in the MSA, but for which we are unable to confirm any emission line from JWST data. Of these, five were observed as part of GLASS-JWST and three during the DDT. Most of these sources are extremely faint (fainter than 28 F150W), thus, their redshift confirmation was solely based on the presence of faint Ly $\alpha$  emission (flux on the order of  $1 - 3 \times 10^{-19}$  erg/s/cm<sup>2</sup>) in the MUSE cubes. We were also unable to detect any emission lines for five galaxies with a photometric redshift between 4.5 and 7: these galaxies are also faint ( $m_{\text{F150W}} \simeq 27.2 - 28$ ).

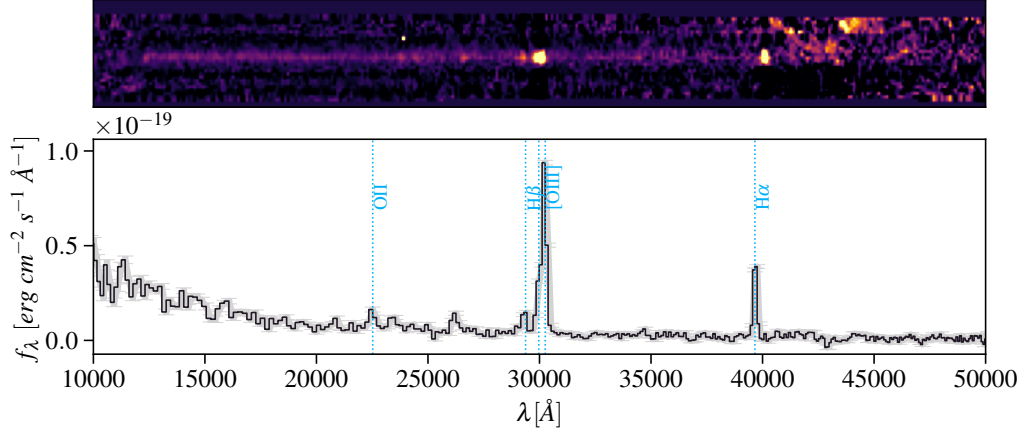
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<sup>3</sup><https://specutils.readthedocs.io/en/stable/index.html>

ID: 110000,  $z = 5.765$



ID: 150015,  $z = 5.041$



**Figure 5.2:** Example 2D and 1D spectra of two representative galaxies in our sample. Top: 2D (top) and 1D (bottom) NIRSpec GLASS-JWST spectrum of 110000 at  $z = 5.765$  (green for the G235H/F170LP and purple for the G395H/F290LP configuration) and  $1\sigma$  uncertainty (gray). Bottom: 2D (top) and 1D (bottom) NIRSpec DDT spectrum of 150015 at  $z = 5.041$  (black) and  $1\sigma$  uncertainty (gray). Vertical dashed lines mark the position of well-detected rest-optical emission lines. The continuum emission is also detected as seen in the 2D spectrum. On the X-axis in the bottom panel, the observed wavelength ( $\text{\AA}$ ) is reported.

However, in these cases it is possible that the photometric redshift was incorrect and this is why we are unable to confirm any emission line in their spectra.

In Table 5.1, we report the GLASS-JWST IDs, the coordinates and the spectroscopic redshifts of all sources together with their  $m_{\text{F150W}}$  magnitudes derived from the imaging data when present.

#### 5.2.4 Dust correction and flux measurements

We measured the total flux of all detected lines (Balmer lines, [OII], and of [OIII]) with a Gaussian fit of each line component. From the flux measurement we subtracted a constant continuum emission, which is estimated as the signal averaged

over regions in which there are no lines near the one being measured. When the signal-to-noise ratio (S/N) of [OII] or  $H\beta \leq 2$ , we set  $2\sigma$  as an upper limit. Prior to carry out a quantitative analysis, it is necessary to consider corrections for dust reddening. For 22 galaxies,  $H\alpha$  and  $H\beta$  are both available: for these, we calculated the correction for dust extinction on the basis of the Balmer decrement, assuming a Calzetti et al. [69] extinction law and an intrinsic ratio  $H\alpha/H\beta = 2.86$  [see e.g., 119, 237, 369], which is valid for an electron temperature of 10000 K. For the other 7 sources where  $H\alpha$  is not in the observed range, we applied the average correction  $E(B - V)_{neb} \sim 0.1$  derived from all other sources.

With the dust corrected values, we calculated the  $R23 = ([OIII] + [OII])/H\beta$  and O32 line fluxes ratios and the  $H\beta$  rest-frame  $EW_0$ s. We list all these values in Table 5.1. It is important to highlight that the O32 values slightly depend on the dust correction. The assumed temperature may be too low for high- $z$  star-forming galaxies [e.g., 103, 336], however the  $H\alpha/H\beta$  ratio varies by less than 5% for a 20000 K temperature [121] and therefore the O32 variations are well below the current uncertainties.

**Table 5.1:** *Sample and spectroscopic measurements.* <sup>a</sup> from MUSE catalog, magnitude  $F140W$  HST. Missing O32 and R23 values are due to [OII] undetection because of its position in un-acquired part of the spectrum.  $H\beta$  and [OIII] are always detected.

| PROG.        | GLASS ID | RA [deg] | DEC [deg] | $z_{spec}$ | $m_{F150W}$ [mag] | $EW_0(H\beta)$ [Å] | R23     | O32     |
|--------------|----------|----------|-----------|------------|-------------------|--------------------|---------|---------|
| <b>DDT</b>   | 10025    | 3.59609  | -30.38581 | 7.875      | 26.7              | 139±30             | 8±3     | 6.6±1.4 |
|              | 40196    | 3.55803  | -30.38962 | 4.760      | 25.5              | 167±42             | > 11    | > 90    |
|              | 50017    | 3.57006  | -30.40369 | 5.550      | 26.3              | 89±17              | 8±3     | 9±2     |
|              | 70002    | 3.60544  | -30.3966  | 5.771      | 27.0              | >210               | > 12    | > 8     |
|              | 80075    | 3.56755  | -30.40623 | 4.632      | 26.9              | 173±38             | 3.4±1.5 | 9±2     |
|              | 100004   | 3.60657  | -30.38093 | 7.884      | 26.9              | > 130              | > 5     | > 5     |
|              | 110003   | 3.59069  | -30.39554 | 5.660      | 25.1              | 99±19              | > 4     | > 8     |
|              | 150015   | 3.55085  | -30.40590 | 5.041      | 25.7              | 268±53             | 4.1±1.8 | 7.2±1.9 |
|              | 150053   | 3.58120  | -30.42853 | 4.580      | 26.7              | 212±51             | > 5     | > 21    |
|              | 160170   | 3.56570  | -30.42613 | 4.895      | -                 | 190±43             | 5±2     | 8±2     |
|              | 160281   | 3.59015  | -30.42593 | 4.715      | 25.6              | 231±52             | 5±2     | 5.3±1.4 |
|              | 160284   | 3.58083  | -30.42526 | 4.700      | 24.8              | 130±27             | > 5     | > 8     |
|              | 160345   | 3.55293  | -30.40389 | 5.020      | 27.2              | 110±44             | > 4     | > 5     |
| <b>GLASS</b> | 10000    | 3.60134  | -30.37923 | 7.884      | 25.5              | 76±13              | 7±2     | 8±3     |
|              | 10021    | 3.60851  | -30.41854 | 7.288      | 25.4              | 104±24             | 12±5    | 13±5    |
|              | 50002    | 3.57700  | -30.41552 | 5.135      | 27.8              | 69±23              | 12±6    | 19±7    |
|              | 50038    | 3.56520  | -30.39426 | 5.773      | 25.5              | 92±19              | 10±4    | 9±3     |
|              | 70018    | 3.58790  | -30.41159 | 5.284      | 27.1              | 155±32             | > 4     | > 12    |
|              | 80070    | 3.58232  | -30.38765 | 4.798      | 26.4              | 135±25             | 12±5    | 53±20   |
|              | 80085    | 3.57435  | -30.41253 | 4.728      | 28.1 <sup>a</sup> | 168±24             | 2.5±1.5 | 19±11   |
|              | 100001   | 3.60385  | -30.38223 | 7.875      | 26.5              | 39±8               | 10±4    | 3.2±1.0 |
|              | 100003   | 3.60451  | -30.38044 | 7.880      | 26.2              | 85±18              | 11±4    | 21±7    |
|              | 100005   | 3.60646  | -30.38099 | 7.883      | 26.7              | 33±15              | 21±12   | 2.9±1.0 |
|              | 110000   | 3.57064  | -30.41464 | 5.765      | 25.8              | 57±13              | 9±4     | 6±2     |
|              | 150008   | 3.60253  | -30.41923 | 6.230      | 25.8              | 141±30             | > 7     | > 20    |
|              | 150029   | 3.57717  | -30.42258 | 4.585      | 26.1              | 205±45             | 8±3     | 17±6    |
|              | 160122   | 3.5649   | -30.42496 | 5.333      | -                 | 149±25             | 5±2     | 18±7    |
|              | 160275   | 3.58107  | -30.42830 | 4.578      | 26.2              | 5±1                | -       | -       |
|              | 400009   | 3.60059  | -30.41027 | 6.376      | 26.8              | 35±7               | -       | -       |

### 5.2.5 Measurements of physical parameters

Following [405], we measured the stellar masses  $M_{\star,obs}$ , the observed absolute UV magnitudes at 1500Å ( $M_{1500,obs}$ ), the star formation rates (SFRs), and dust red-

dening  $E(B - V)$  by fitting synthetic stellar templates to the seven-band NIRCcam photometry and the released HST photometry [74] with ZPHOT [156]. We adopted [60] models and assumed delayed exponentially declining star formation histories –  $\text{SFH}(t) \propto (t^2/\tau) \cdot \exp(-t/\tau)$  – with  $\tau$  ranging from 0.1 to 7 Gyr. The age ranges from 10 Myr to the age of the Universe at each galaxy redshift, while metallicity can assume values of 0.02, 0.2 or 1 times Solar metallicity. For the dust extinction, we used the [69] law with  $E(B - V)$  ranging from 0 to 1.1. We computed  $1\sigma$  uncertainties on the physical parameters by retaining, for each object, the minimum and maximum fitted masses among all the solutions with a probability  $P(\chi^2) > 32\%$  of being correct, fixing the redshift to the best-fit value.

As a final step, we needed to adjust the stellar masses and  $M_{1500}$  and the SFR values for the lensing amplification factor. The magnification factor  $\mu$  was derived by combining the LM-model [34], the model used by [389], with a new spatially more extensive model that fully covers the JWST field of view (Bergamini et al. in prep). The  $\mu$  values range from 1.6 to 12, with a median value of  $\sim 2$ . The results on our sources’ stellar masses,  $M_{\star, \text{true}}$ , their  $M_{1500}$ , and their lensing magnification factors  $\mu$  are reported in Table 5.2.

### 5.2.6 Sizes and SFR surface density estimates

The SFRs were derived from the  $\text{H}\alpha$  emission line using the standard conversion by [243], that is,  $\text{SFR}(\text{H}\alpha)[M_{\odot} \text{ yr}^{-1}] = 7.9 \times 10^{42} L_{\text{H}\alpha}$ , and then corrected for the Chabrier IMF. For those few sources at  $z \geq 6.6$ , where the  $\text{H}\alpha$  line falls outside the observable window we used dust-corrected  $\text{H}\beta$  fluxes instead. To correct for slit losses and possible residual uncertainties on flux calibration, we normalized the spectra to the F444W filter by integrating the spectrum under the F444W bandpass, the closest to the  $\text{H}\alpha$  line in our sample. Thanks to the high resolution of the GLASS-JWST spectra, no correction for [NII] contamination is required for  $\text{H}\alpha$  measurements. For galaxies observed by the DDT programs,  $\text{H}\alpha$  is blended with the [NII] doublet. However our sources are all very-low-mass galaxies and, based on low-redshift results, we expect contamination to be less than 10% [e.g., 135]. Therefore, we did not attempt to make any correction on the fluxes.

The results obtained on the SFR were also compared to the values determined by the SED fitting procedure described in Sect. 5.2.5, finding a reasonable agreement between the two data sets, indicating that there are no systematic issues. Obviously, the SED fitting SFR is heavily dependent on the choice of SFH, and also on the dust correction, while the value derived from the  $\text{H}\alpha$  luminosity is sensitive to short timescales. That is to say that it only probes the presence of short-lived, massive stars, so their ratio is actually an indication of the burstiness of the galaxy.

We measured the size of each galaxy  $r_e$  in the F115W band, corresponding to the UV rest-frame of the galaxies. Adopting forward-modeling technique, we assumed that galaxies are well represented by a Sersic profile [426] as [521]. Then we modeled the appearance of the source in the image plane considering lensing and PSF effects. The source reconstruction was performed via `python` software `Lenstruc` [see details in 519], which is built on `Lenstronomy` [40]. In this way we obtained the intrinsic properties of galaxies in the source plane, hence,

the intrinsic size. Finally, we calculated the SFR surface density using the relation  $\Sigma_{SFR} = SFR/2\pi r_e^2$  [e.g., 154, 331]. The SFR surface densities  $\Sigma_{SFR}$  are listed in Table 5.2.

### 5.2.7 UV- $\beta$ slopes

We measured the UV slope of our galaxies from the NIRCам photometry and/or the previously available HST photometry [74], with the approach detailed in [64]. In brief, we considered all the photometric bands whose entire bandwidths are between a 1216 and 3000 Å rest frame. The former limit is set to exclude the Ly $\alpha$  line and Ly-break, while the latter limit is slightly larger than that adopted in [64] to ensure a larger range.

Then, we fitted the selected photometry with a single power-law of the form  $f(\lambda) \propto \lambda^\beta$ . In practice, we fitted two or three photometric bands amongst HST F814W and JWST-NIRCам F115W, F150W or F200W depending on the exact redshift of the sources. This choice allows us to uniformly probe the spectral range between 1500 and 3000 Å for most of the galaxies. We measured the  $\beta$  and associated uncertainty for each source using a bootstrap method. By using  $n = 500$  Monte Carlo simulations, the fluxes in each band were varied according to their error. The results provided a resultant slope distribution from which we calculated the mean and standard deviation of  $\beta$  for each galaxy. The results on  $\beta$  with associated errors are reported in Table 5.2. We find a median  $\beta$  of  $-2.1$ , with a  $1\sigma$  dispersion of  $0.4$ . The UV magnitudes  $M_{1500}$  that can be derived simultaneously with this approach are consistent with the values obtained from the SED fitting and described in the previous section.

**Table 5.2:** *Physical parameters derived from multi-band photometry. All measurements are corrected for magnification where needed. Median magnification of the lens model by [35], calculated at the position of the source. Measurements are associated with  $1\sigma$  uncertainties. <sup>a</sup> measured from HST photometry.*

| PROG.      | GLASS ID | $\mu$                  | $\log_{10}(M_*)$<br>[ $M_\odot$ ] | $r_e$<br>[kpc] | $\log_{10}(\Sigma_{SFR})$<br>[ $M_\odot \text{yr}^{-1} \text{kpc}^{-2}$ ] | $M_{1500}$<br>[mag]      | $\beta$          |
|------------|----------|------------------------|-----------------------------------|----------------|---------------------------------------------------------------------------|--------------------------|------------------|
| <b>DDT</b> | 10025    | $2.60^{+0.08}_{-0.07}$ | $8.03^{+0.19}_{-0.14}$            | 0.40           | $0.4 \pm 0.2$                                                             | $-19.00^{+0.01}_{-0.03}$ | $-2.08 \pm 0.45$ |
|            | 40196    | $2.95^{+0.21}_{-0.72}$ | $7.87^{+0.13}_{-0.13}$            | 0.33           | $1.6 \pm 1.0$                                                             | $-19.41^{+0.02}_{-0.03}$ | $-2.03 \pm 0.48$ |
|            | 50017    | $2.27^{+0.07}_{-0.16}$ | $8.78^{+0.23}_{-0.69}$            | 0.43           | $0.7 \pm 0.1$                                                             | $-19.13^{+0.10}_{-0.19}$ | $-1.77 \pm 0.25$ |
|            | 70002    | $2.19^{+0.05}_{-0.05}$ | $7.13^{+0.01}_{-0.15}$            | 0.12           | $2.0 \pm 1.3$                                                             | $-18.44^{+0.01}_{-0.01}$ | $-2.54 \pm 0.26$ |
|            | 80075    | $2.02^{+0.06}_{-0.17}$ | $6.76^{+0.23}_{-0.15}$            | 0.07           | $2.3 \pm 1.7$                                                             | $-17.37^{+0.01}_{-0.03}$ | $-2.26 \pm 0.23$ |
|            | 100004   | $1.92^{+0.06}_{-0.08}$ | $9.17^{+0.14}_{-0.28}$            | 0.40           | $0.2 \pm 0.3$                                                             | $-19.76^{+0.11}_{-0.10}$ | $-1.88 \pm 0.44$ |
|            | 110003   | $11.6^{+0.9}_{-1.0}$   | $7.92^{+0.71}_{-0.06}$            | 0.48           | $1.2 \pm 0.5$                                                             | $-20.52^{+0.15}_{-0.03}$ | $-2.51 \pm 0.24$ |
|            | 150015   | $1.79^{+0.05}_{-0.29}$ | $7.87^{+0.34}_{-0.44}$            | 0.38           | $1.2 \pm 0.7$                                                             | $-19.52^{+0.09}_{-0.15}$ | $-2.54 \pm 0.49$ |
|            | 150053   | $1.75^{+0.03}_{-0.14}$ | $7.85^{+0.48}_{-0.45}$            | 0.80           | $-0.1 \pm 0.6$                                                            | $-19.26^{+0.08}_{-0.15}$ | $-1.25 \pm 0.21$ |
|            | 160170   | $1.58^{+0.04}_{-0.13}$ | -                                 | -              | -                                                                         | -                        | -                |

**Table 5.2:** *continued.*

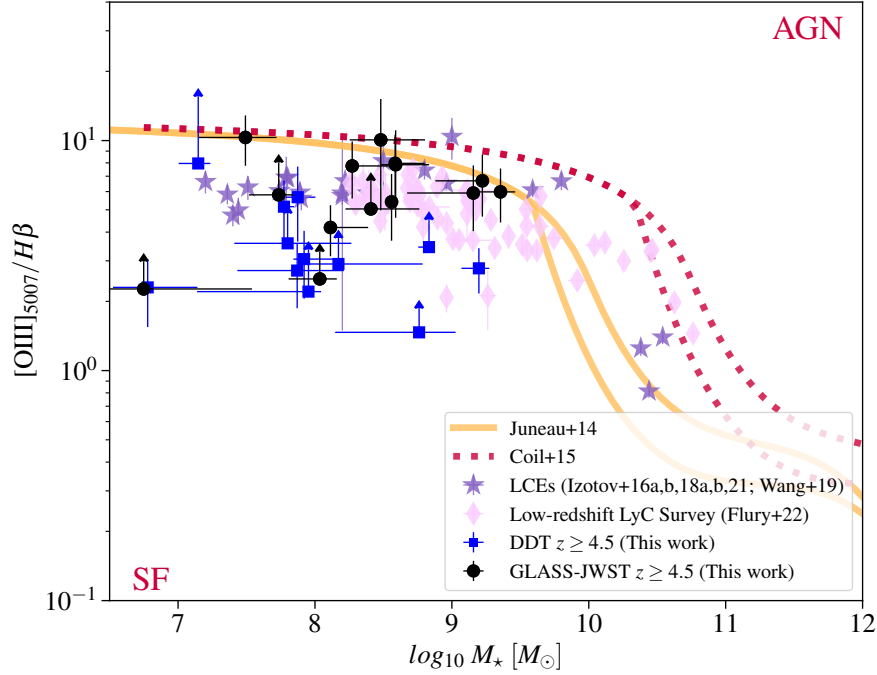
| PROG.        | GLASS<br>ID | $\mu$                  | $\log_{10}(M_*)$<br>[ $M_\odot$ ] | $r_e$<br>[kpc] | $\log_{10}(\Sigma_{SFR})$<br>[ $M_\odot \text{yr}^{-1} \text{kpc}^{-2}$ ] | $M_{1500}$<br>[mag]      | $\beta$          |
|--------------|-------------|------------------------|-----------------------------------|----------------|---------------------------------------------------------------------------|--------------------------|------------------|
|              | 160281      | $1.96^{+0.03}_{-0.15}$ | $7.93^{+0.24}_{-0.08}$            | 1.65           | $-0.3 \pm 0.9$                                                            | $-20.00^{+0.01}_{-0.11}$ | $-2.26 \pm 0.51$ |
|              | 160284      | $1.85^{+0.03}_{-0.15}$ | $8.85^{+0.01}_{-0.10}$            | 0.53           | $1.2 \pm 0.6$                                                             | $-20.16^{+0.02}_{-0.01}$ | $-2.05 \pm 0.48$ |
|              | 160345      | $1.86^{+0.05}_{-0.30}$ | $7.95^{+0.09}_{-0.81}$            | 0.45           | $0.5 \pm 0.1$                                                             | $-18.67^{+0.06}_{-0.12}$ | $-2.49 \pm 0.49$ |
| <b>GLASS</b> | 10000       | $2.13^{+0.07}_{-0.11}$ | $8.11^{+0.27}_{-0.01}$            | 0.20           | $2.0 \pm 1.3$                                                             | $-19.71^{+0.10}_{-0.02}$ | $-2.27 \pm 0.46$ |
|              | 10021       | $2.17^{+0.08}_{-0.03}$ | $8.70^{+0.11}_{-0.35}$            | 0.68           | $0.8 \pm 0.2$                                                             | $-21.18^{+0.08}_{-0.02}$ | $-2.25 \pm 0.48$ |
|              | 50002       | $2.15^{+0.06}_{-0.13}$ | $8.57^{+0.22}_{-0.11}$            | 0.70           | $-1.4 \pm 2.0$                                                            | $-18.78^{+0.06}_{-0.11}$ | $-1.66 \pm 0.66$ |
|              | 50038       | $2.66^{+0.12}_{-0.32}$ | $9.23^{+0.03}_{-0.39}$            | 0.48           | $-1.7 \pm 2.0$                                                            | $-19.76^{+0.03}_{-0.09}$ | $-1.79 \pm 0.25$ |
|              | 70018       | $6.89^{+0.21}_{-0.30}$ | $8.05^{+0.01}_{-0.14}$            | 0.19           | $1.9 \pm 1.3$                                                             | $-18.00^{+0.04}_{-0.04}$ | $-2.02 \pm 0.49$ |
|              | 80070       | $5.45^{+0.26}_{-0.27}$ | $7.42^{+0.21}_{-0.06}$            | 0.59           | $0.4 \pm 0.2$                                                             | $-18.24^{+0.01}_{-0.03}$ | $-1.92 \pm 0.49$ |
|              | 80085       | $2.11^{+0.05}_{-0.11}$ | -                                 | -              | -                                                                         | -                        | $-2.43^a$        |
|              | 100001      | $1.99^{+0.06}_{-0.09}$ | $9.48^{+0.04}_{-0.13}$            | 0.50           | $0.34 \pm 0.3$                                                            | $-20.45^{+0.04}_{-0.02}$ | $-1.63 \pm 0.48$ |
|              | 100003      | $1.96^{+0.06}_{-0.09}$ | $8.29^{+0.31}_{-0.15}$            | 0.15           | $2.3 \pm 1.7$                                                             | $-20.15^{+0.15}_{-0.02}$ | $-2.51 \pm 0.48$ |
|              | 100005      | $1.92^{+0.06}_{-0.08}$ | $8.63^{+0.17}_{-0.27}$            | 0.25           | $1.5 \pm 1.2$                                                             | $-19.85^{+0.11}_{-0.07}$ | $-2.55 \pm 0.48$ |
|              | 110000      | $1.90^{+0.05}_{-0.14}$ | $9.10^{+0.05}_{-0.62}$            | 0.54           | $0.5 \pm 0.2$                                                             | $-20.13^{+0.04}_{-0.17}$ | $-1.70 \pm 0.23$ |
|              | 150008      | $2.60^{+0.03}_{-0.04}$ | $8.41^{+0.35}_{-0.19}$            | 0.40           | $1.5 \pm 0.7$                                                             | $-19.33^{+0.14}_{-0.06}$ | $-2.10 \pm 0.25$ |
|              | 150029      | $1.86^{+0.04}_{-0.14}$ | $8.55^{+0.07}_{-0.01}$            | 0.27           | $1.7 \pm 1.0$                                                             | $-18.75^{+0.01}_{-0.04}$ | $-1.85 \pm 0.20$ |
|              | 160122      | $1.59^{+0.04}_{-0.13}$ | -                                 | -              | -                                                                         | -                        | -                |
|              | 160275      | $1.75^{+0.03}_{-0.14}$ | $7.73^{+0.11}_{-0.14}$            | 0.47           | $-1.3 \pm 1.8$                                                            | $-19.20^{+0.01}_{-0.02}$ | $-2.55 \pm 0.20$ |
|              | 400009      | $9.3^{+1.6}_{-0.6}$    | $6.77^{+0.54}_{-0.26}$            | 0.11           | $1.0 \pm 0.3$                                                             | $-17.05^{+0.14}_{-0.04}$ | $-2.17 \pm 0.25$ |

## 5.3 Results

### 5.3.1 Possible AGN contamination

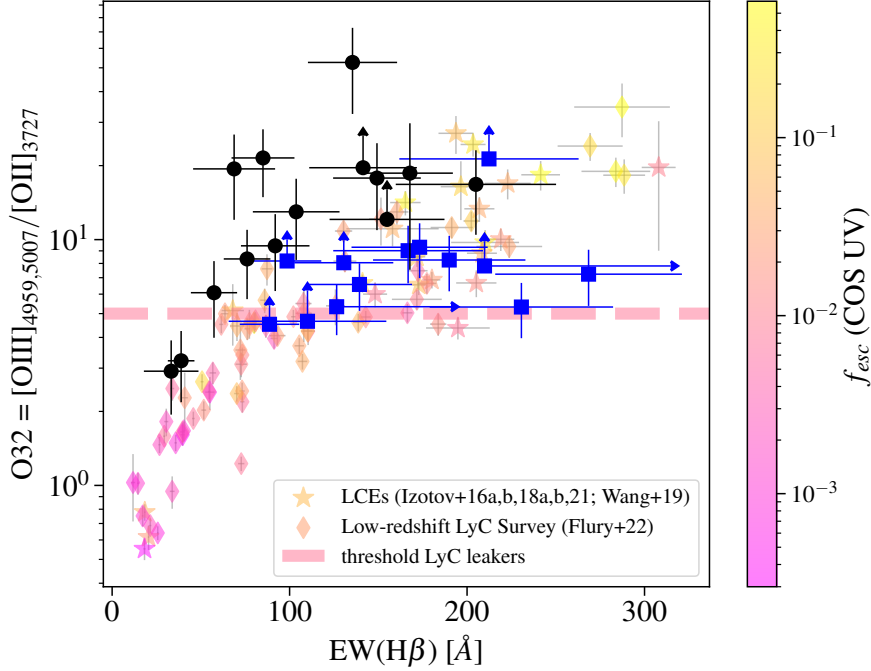
Our sample was selected solely based on known spectroscopic redshift (via faint and narrow Ly $\alpha$  emission) or through photometric redshifts. It could therefore be affected by AGN contamination. Since we are interested in searching for candidate LyC emitters amongst the star-forming population, we first consider whether the primary source of ionization might not be star formation, but AGN activity instead.

We employed the mass-excitation (MEx) diagram, first proposed by [231] to combine the measurements of the [OIII] $\lambda$ 5007/H $\beta$  emission line ratio with the stellar mass to discriminate between AGNs and star-forming galaxies. This diagram was proposed as an alternative to the classical BPT diagram that compares the



**Figure 5.3:** *MEx* diagram for the sample of galaxies analyzed in this paper: black dots and green squares are for the GLASS-JWST and DDT samples, respectively. For reference, we plot also the galaxies at  $z = 0.3 - 0.4$  from [154] (diamonds) and the LCE candidates from previous studies [stars, 213, 215–218, 501]. The two orange demarcation lines from [232] show the boundaries of the AGN-and-star-forming transition region. All objects above the upper line are AGN-dominated; all galaxies below or rightward of the lower line are presumptively dominated by star formation. We also show the separation from [92] (dotted lines), which is the adaptation of the Juneau et al. model for galaxies and AGNs at  $z \sim 2.3$  from the MOSDEF survey.

$[\text{OIII}]\lambda 5007/\text{H}\beta$  to the  $[\text{NII}]/\text{H}\alpha$  emission line ratios [21] when these last two lines fall out of the visibility window and it is not possible to use them to characterize the ionization mechanism in the galaxies – as is the case for our low-resolution DDT spectra where the  $[\text{NII}]$  doublet is blended with  $\text{H}\alpha$ . Due to the high resolution of the GLASS-JWST spectra, for the galaxies with these observations, we used the dust-corrected flux measurement of the 5007 Å component of the  $[\text{OIII}]$  doublet. Instead, for the galaxies observed as part of the DDT program for which the doublet cannot be resolved, we determined the  $[\text{OIII}]\lambda 5007$  flux, assuming the expected line ratio of 1:3 for the two components fixed by atomic physics. As shown in Figure 5.3, the position of our sources in the *MEx* diagram indicates that our sample contains essentially star forming galaxies, lying below or around the division line identified by [92] for  $z = 2.3$  galaxies and AGN from the MOSDEF survey. For reference, we also plot the galaxies at  $z = 0.3 - 0.4$  from the Low-redshift Lyman Continuum Survey from [154] and the compilation of low- $z$  LCEs also used by the same authors [from 213, 215–218, 501].



**Figure 5.4:**  $O32$  vs. rest-frame  $EW_0(H\beta)$ . Symbols are the same as in Figure 5.3. For reference, we also plot the galaxies at  $z = 0.3 - 0.4$  from [154] (diamonds) and the LCEs from [213, 215–218, 501] (stars). Symbols are color coded as a function of their measured  $f_{\text{esc}}$ . The pink line ( $O32 = 5$ ) indicates the threshold for LCEs as predicted by [154].

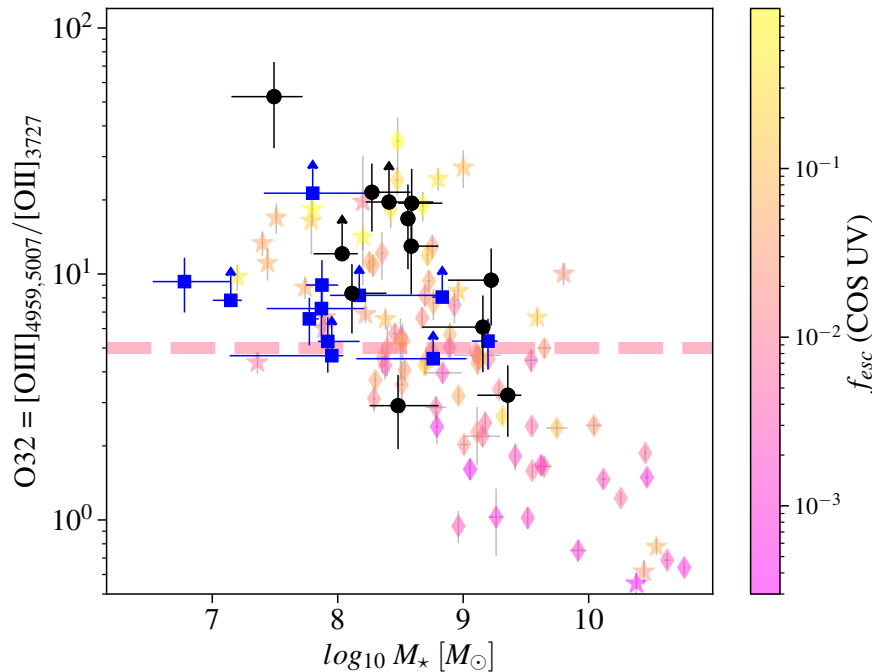
### 5.3.2 LyC indirect diagnostics

As discussed above,  $\text{Ly}\alpha$  is possibly the best indirect diagnostic of LyC escape, since the conditions that favour the escape of  $\text{Ly}\alpha$  photons are often the same that allow for the escape of LyC photons. However, we cannot use this parameter for our sample, since the NIRSpec data do not cover the 1216Å region for most of our galaxies and only a small subset of our sources have been covered by previous MUSE observations (see also Figure 5.1). In addition, for the sources at  $z \geq 7$ , the IGM becomes significantly neutral, thus absorbing the emission even in sources where the line would be intrinsically bright [e.g., 305, 365, 448]. Indeed, [325] recently showed that none of the galaxies in the protocluster candidate at  $z = 7.89$  presents bright  $\text{Ly}\alpha$  emission and estimate an average neutral hydrogen fraction of the IGM in the region to be  $> 0.45$ . Therefore in this work we concentrate on the other most promising diagnostics tested at low redshift by [154] and available for our galaxies: these authors showed that  $O32$ ,  $\beta_{1200}$ ,  $r_{50}$ , and  $\Sigma_{SFR}$  as well as  $EW_0(H\beta)$  and  $M_{1500}$  exhibit some of the strongest and most significant correlations with  $f_{\text{esc}}$ . This would indicate that certain characteristics, such as concentrated star formation, young stellar populations, and high ionization states, play an essential role in the escape of LyC photons.

We began by analyzing the  $O32$  and the rest-frame  $EW_0(H\beta)$  relation, as shown in Figure 5.4. We plot the values for the JWST high-redshift sample and compare them to the local galaxies with measured  $f_{\text{esc}}$  from previous works [154, 213, 215–218, 291, 501]. These values are color-coded as a function of their  $f_{\text{esc}}$  (COS UV)



measured values. We can see that the large majority of the high-redshift sources have values of O32 larger than 5, which has been indicated as a threshold for LyC leakers (LCEs from here onwards) by [154]. We note that [154] define strong leakers as galaxies with an  $f_{\text{esc}} > 0.05$  measured with  $S/N \geq 5$ . Our sample indeed mostly lies in the region of the plot populated by low-redshift leakers. In Figure 5.5, we show O32 as a function of total stellar mass, again comparing our sample to the low- $z$  galaxies: our sample perfectly overlaps with the low- $z$  galaxies at the low-mass end, where we find most of the LCEs.



**Figure 5.5:**  $O32$  vs.  $M_*$ . Symbols are as in Figure 5.4.

[527] suggested that galaxies ought to be identified with high  $f_{\text{esc}}$  by combining the UV slope  $\beta$  with the measurement of the EW of a Balmer line, such as  $H\beta$ , and that the predictions are almost independent of the model assumed (i.e., radiation or density bounded nebula). We know that both  $\beta$  and  $EW_0(H\beta)$  are also good indirect indicators according to [154], although there does not seem to be a direct correlation between the two values: [154] showed that their LCEs do not seem to follow the predictions provided by the models from [527].

We show the properties of our sample in Figure 5.6. Specifically, at a given value of  $EW_0(H\beta)$ , the high-redshift galaxies are, on average, bluer than the majority of the low- $z$  galaxies (consistent with the lower stellar mass probed by our sources); but their slopes are similar to the subset of the low- $z$  sources with moderate-to-high values for  $f_{\text{esc}}$ . The  $\beta$  slopes for our galaxies are perfectly consistent with the average values found for LBGs at  $z \sim 6$  for galaxies with  $M_{\text{UV}}$  in the same range [49]. In the figure, we also show the models from [527] for various values of escape fraction, after correcting the intrinsic  $\beta$  slopes of the models for dust attenuation and assuming the average reddening of our sample derived from the SED fitting,  $E(B - V) = 0.097$  (solid line), and also the maximum and minimum  $E(B - V)$  in

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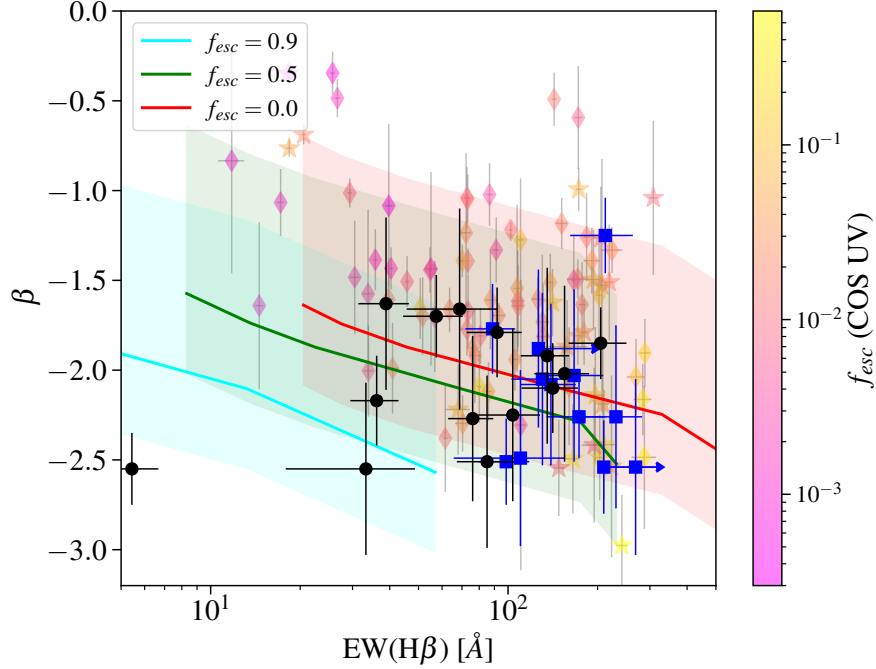
our sample (shaded area). We note that our galaxies show more consistency with model predictions that assume moderate-to-low escape fractions  $0.0 \leq f_{esc} \leq 0.5$ .

Finally, we analyzed the sizes and star formation rate density for our galaxies. Several authors have postulated that concentrated star formation provides the feedback necessary to clear paths in the ISM that allow for the escape of LyC photons. In addition, [292] recently found that stacks of galaxies that are UV-compact ( $r_{UV} \leq 0.30$  kpc) have much higher LyC flux than the average population [see also 218]. We find that, with the exception of one galaxy (ID 160281, which has  $r_e = 1.65$  kpc), our targets are all extremely compact with typical  $r_e$  in the rest-frame UV around  $0.2 - 0.6$  kpc. These values are similar or even lower than those found for the low- $z$  galaxies and LCEs [154]; thus, once again the high-redshift sources have equal properties to the low- $z$  LCEs. Since we know that sizes evolve with redshift and galaxies become progressively more compact, we also compared our sample to the general population of star-forming galaxies (LBGs selected) at  $z \simeq 6$  analyzed by [430]. Specifically, for  $M_{UV} = -18(-20)$ , the average UV rest-frame sizes are  $r_e \simeq 0.38(0.56)$  kpc, respectively. Thus, indeed our galaxies are, from this point of view, as compact as the general UV faint population at the same redshift.

As for the star formation rate densities, the values of  $\log_{10}(\Sigma_{SFR})$  for our galaxies span a very wide range, with an average  $\Sigma_{SFR}$  that is slightly higher than the average expected at their redshift, as measured by [430]. We note that their values are inferred from the UV and then dust corrected. LyC leakers at low and intermediate redshift tend to show high  $\Sigma_{SFR}$  values, as discussed extensively by [331], who actually proposed a physically motivated model in which  $f_{esc}$  would scale as  $\Sigma_{SFR}^{0.4}$ . [154] identify a threshold value of  $\Sigma_{SFR} = 10 M_{\odot} \text{ yr}^{-1} \text{ kpc}^{-2}$  above which their LCE fraction changes from 10% to 60%, and where indeed we find half of our high-redshift galaxies.

### 5.3.3 O32-R23 diagnostics

The O32 vs R23 index diagram (Figure 5.7) is widely used to examine the gas-phase metallicity and ionization state both in the local Universe [e.g., 213, 215, 216, 218, 470] and at high redshift [e.g., 154, 337, 380, 489], as both these indices are sensitive to a combination of these quantities. [337] showed that  $z \sim 3$  LyC leakers tend to populate the upper right part of this diagram. Recently [240] used high-resolution cosmological radiation hydrodynamics simulations to examine the properties of LyC leakers deep into the epoch of reionization and found that simulated high-redshift galaxies populate the same regions of the R23-O32 plane as the  $z \sim 3$  LyC leakers presented in [337] that tend to have low metallicity. Although they conclude that this plane is not the most useful to differentiate between leakers and non-leakers we note that the  $z = 3$  leakers by [337], with measured values, and Ion2 occupy the same region as the  $z = 0.3$  low-redshift leakers. Our galaxies occupy a much broader region, which could reflect either a wider range of metallicity and ionization states or simply the fact that we have large measured uncertainties on the diagnostics.



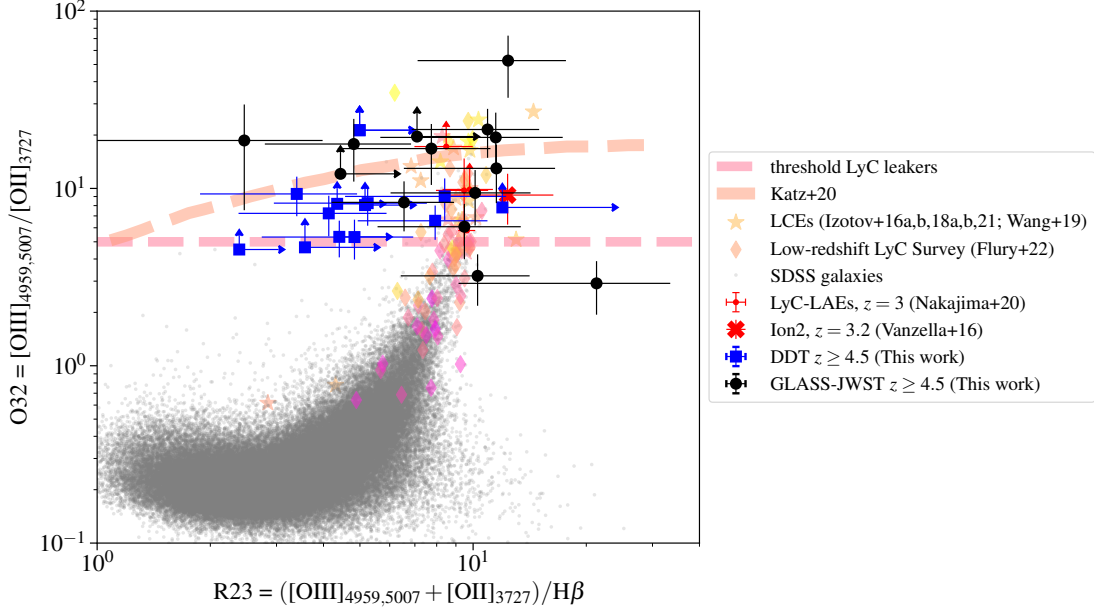
**Figure 5.6:**  $\beta$  vs. rest-frame  $EW_0(\text{H}\beta)$ . Models are from [527] and simulate the expected trend for galaxies with an exponential declining SFR ( $Z = 0.02$ , solid lines) and various values of escape fractions. Symbols are as in Figure 5.4.

## 5.4 Predicting escape fractions of EoR galaxies

In the sections above, we have shown that our high-redshift galaxies have properties that are largely overlapping with those of low- $z$  LCEs. The next step is to try and give an indirect estimate of the  $f_{\text{esc}}$  values for our galaxies, so as to understand if typical low-mass galaxies at  $z \simeq 5 - 7$  could really be the drivers of reionization. We assume that the mechanisms that drive the escape of LyC photons are the same at all redshifts and depend only on the physical properties of the sources.

We proceeded as follows: we used the spectroscopic and physical properties of the 66 galaxies that are part of the [154] sample, with the additional 22 LCEs from previous studies [213, 215–218, 501] to calibrate an empirical relation with the measured  $f_{\text{esc}}$  values. Specifically, we focused on the following properties: O32,  $EW_0(\text{H}\beta)$ ,  $\beta$ ,  $r_e$  (in kpc),  $\Sigma_{\text{SFR}}$ , and  $M_{1500}$ . For all 88 galaxies, these parameters are accurately measured and, of course, accurate measurements of  $f_{\text{esc}}$  are available. To this end, we specifically used the  $f_{\text{esc}}$  derived by the COS UV spectral fits [see definition in 154]. Recently [86] followed a somewhat similar approach, also using the same set of low-redshift observations, but limited to the UV- $\beta$  slope and an indirect proxy. They provided a scaling relation between  $\beta$  and  $f_{\text{esc}}$ , although the relation has appreciable scatter that scales with  $f_{\text{esc}}$ .

We first ran the Spearman rank correlation and found that O32,  $r_e$ ,  $EW_0(\text{H}\beta)$ , and  $\beta$  (in that order) are the properties that are best correlated with  $f_{\text{esc}}$ . We then identified one useful fitting linear relation to obtain an estimate of  $\log_{10}(f_{\text{esc}})$



**Figure 5.7:**  $O32$  vs.  $R23$  diagnostic diagram for our sample. Black dots and green squares show the GLASS-JWST and DDT samples, respectively. For reference, we plot also the SDSS galaxies from [470] as dark grey points. The LCEs from [337] are shown as red dots. The Ion2 at  $z = 3.2$  comes from [112, 487] as a red x sign, the low- $z$  LyC leakers from [213, 215, 216, 218, 501] as stars, and the LCEs at  $z = 0.3 - 0.4$  from the Low-redshift Lyman Continuum Survey [as pink diamonds, 154]. We show also the locus of high-redshift LCEs predicted from cosmological simulations by [240] as an orange dashed line and the threshold of  $O32 > 5$  determined in [154] as a pink dashed line.

on the basis of the other four measured physical properties. A fully data-driven regression analysis was carried out by performing a regularized minimization of the root-mean-square error (MSE), computed between the values provided by the equation and the dataset, for several possible combinations of the above properties. We tested both a scheme in which the error on the escape fraction,  $f_{\text{esc}}$ , is not considered and a scheme in which values are weighted by the error. Since  $O32$  and  $EW_0(H\beta)$  exhibit a very tight correlation (Spearman correlation between them  $> 0.9$ , see also Figure 5.4), the information they provide is redundant and therefore we decided to use only  $O32$ . We checked that the remaining three parameters are reasonably independent (Spearman correlation  $< 0.5$ ) and therefore provide complementary information. We finally found a best-fit relation of the form:

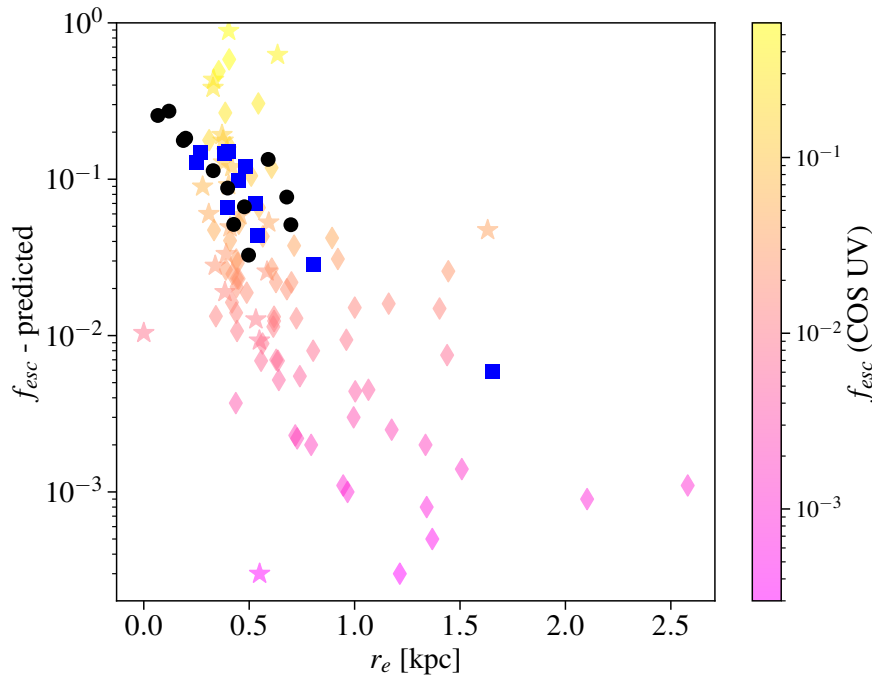
$$\log_{10}(f_{\text{esc}}) = A + B \log_{10}(O32) + C r_e + D \beta. \quad (5.1)$$

After identifying the best type of equation, we repeated the minimization process 100 times, following a bootstrap approach every time a random number of sources (between 1 and 25, to avoid being left with too few sources) is randomly removed from the sample. In this way, we could constrain the confidence interval for the above best fit parameters. We find  $A = -1.92[-2.51, -1.71]$ ,  $B = 0.48[0.38, 0.69]$ ,  $C = -0.96[-1.20, -0.62]$ ,  $D = -0.41[-0.58, -0.31]$ , where the values between the parentheses are in the 95th percentile distribution.

Testing the relation on the residuals, we find that it tends to slightly over-

estimate the  $f_{\text{esc}}$  at very low values (much lower than 0.01), while it tends to underestimate the  $f_{\text{esc}}$  at values that are higher than  $\sim 0.1$ -0.2. This is due to the fact that in the sample used to fit the relation, there are very few galaxies with high  $f_{\text{esc}}$  values and, in general, their measured  $f_{\text{esc}}$  values have higher errors.

We finally applied the above relation to our sample of high-redshift galaxies: out of 29 sources, 3 galaxies do not have the  $r_e$  measurement since they are outside the UNCOVER footprint; for another 2, we do not have an estimate for the O32 parameter. We therefore could apply the best-fitting relation only to 24 sources. In Figure 5.8, we show the predicted  $f_{\text{esc}}$  for our sources as a function of  $r_e$ , as well as the low-redshift comparison sample (for which the  $f_{\text{esc}}$  are measured values). Most of our galaxies have predicted  $f_{\text{esc}}$  values larger than 0.05, meaning that they would be considered strong leakers. The average  $f_{\text{esc}}$  is  $\simeq 0.12$  with the bluest and most compact sources having  $f_{\text{esc}}$  as large as  $0.2 - 0.4$ , which could be lower limits for the reasons discussed above.



**Figure 5.8:** Predicted  $f_{\text{esc}}$  vs. size  $r_e$  for our sample (blue and black symbols) and measured  $f_{\text{esc}}$  vs. size  $r_e$  from the literature. Symbols are as in Figure 5.4.

Clearly, the main limitations of the above analysis are the fact that the low- $z$  sample used to calibrate the relation is small and, most importantly, it is not evenly populated as it contains mostly objects with rather low  $f_{\text{esc}}$ . Also, it would be important to test these predictions at intermediate redshift ( $z \sim 3 - 4$ ), that is, much closer to the EoR, where it is possible to both directly detect LyC emission and determine all other physical and spectroscopic properties. However, at the moment, measurements of  $f_{\text{esc}}$  at intermediate redshift are still sparse and most samples lack near-infrared (NIR) spectroscopic follow-up results. In spite of these limitations, a consistent picture emerges from our results, suggesting low-mass galaxies at  $z \sim 5 - 7$  have mostly properties which indicate moderate values of

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$f_{\text{esc}}$ . Interestingly, the average  $f_{\text{esc}}$  value inferred for our sample is equal to the one predicted by [331] for galaxies in the mass range  $\log_{10} M_{\star} = 8 - 9M_{\odot}$  at  $z \sim 6$ , according to their simple model, where  $f_{\text{esc}}$  scales with  $\Sigma_{\text{SFR}}$ , which was constrained to reproduce the average observed values for the [457] sample at  $z = 3$ .

## 5.5 Summary and conclusions

Thanks to the magnification power of the Abell 2744 cluster, in this paper, we present the first JWST/NIRSpec observations of 29 gravitationally lensed galaxies with photometric or spectroscopic redshift in the range  $4.5 \leq z \leq 8$ . From a combined NIRSpec and NIRCам analysis, we were able to derive accurate physical properties of the galaxies, including  $M_{\star}$ , SFR,  $r_e$ , UV- $\beta$  slopes, and measurements of the most prominent optical emission lines ([OII], [OIII], H $\beta$ , H $\alpha$ ). We summarize our findings as follows:

- Our sample is composed of purely star-forming galaxies, as inferred from the MEx diagram that excludes the presence of any AGN.
- The galaxies in our sample have blue UV slopes (median  $\beta = -2.08$ ) and are mostly very compact with  $r_e \simeq 0.2 - 0.5$  kpc. These properties are consistent with those of the general population of LBGs at similar redshift and with similarly faint  $M_{\text{UV}}$  [49, 430].
- Compared to the low- $z$  sample by [154] as well as to the low- $z$  LCEs from previous studies, our galaxies present properties (in terms of O32,  $EW_0(\text{H}\beta)$ , UV- $\beta$  slopes,  $r_e$ , and  $\Sigma_{\text{SFR}}$ ) that are entirely consistent with those of low- $z$  galaxies with measured  $f_{\text{esc}}$  larger than 0.05 and which are considered to be LyC leakers.
- Using a linear analysis that employs the minimization of MSE, we found a best fitting relation between  $f_{\text{esc}}$  and the three most correlated and independent parameters (O32, UV- $\beta$ , and  $r_e$ ) for the low-redshift sample of 88 galaxies. Applying this relation to the 24 galaxies from our sample where these parameters are all measured, we find that 20/24 have predicted escape fractions larger than 0.05, that is, they would be considered leakers and the average  $f_{\text{esc}}$  of our sample is 0.12.

In conclusion, our results show that indeed the average low-mass galaxies around the epoch of reionization have physical and spectroscopic properties consistent with moderate values of escaping ionizing photons ( $f_{\text{esc}} = 0.1 - 0.2$ ). With upcoming JWST observations from GTO and GO programs, this analysis could be easily extended to larger samples of galaxies with higher masses and/or at higher redshift to determine if the bulk of the reionizing photons have been provided by even more massive and brighter galaxies [as predicted, e.g., by 331, 429] or if we need a more substantial contribution by the fainter and more numerous galaxies with  $M_{\text{UV}} > -18$ , as predicted by most other popular models [e.g., 13, 149, 475].

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## Chapter 6

# The role of faint, low-mass galaxies in cosmic reionization: analyzing LyC escape indicators at $6 \leq z \leq 9$

*The results in this chapter were published and first presented in [302].*

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The Epoch of Reionization (EoR) began when galaxies grew in abundance and luminosity, so their escaping Lyman continuum (LyC) radiation started ionizing the surrounding neutral intergalactic medium (IGM). Despite significant recent progress, the nature and role of cosmic reionizers are still unclear: in order to define them, it would be necessary to directly measure their LyC escape fraction ( $f_{\text{esc}}$ ). However, this is impossible during the EoR due to the opacity of the IGM. Consequently, many efforts at low and intermediate redshift have been made to determine measurable indirect indicators in high-redshift galaxies so that their  $f_{\text{esc}}$  can be predicted. This work presents the analysis of the indirect indicators of 62 spectroscopically confirmed star-forming galaxies at  $6 \leq z \leq 9$  from the Cosmic Evolution Early Release Science (CEERS) survey, combined with 12 sources with public data from other JWST-ERS campaigns. From the NIRCам and NIRSpec observations, we measured their physical and spectroscopic properties. We discovered that on average  $6 < z < 9$  star-forming galaxies are compact in the rest-frame UV ( $r_e \sim 0.4$  kpc), are blue sources (UV- $\beta$  slope  $\sim -2.17$ ), and have a predicted  $f_{\text{esc}}$  of about 0.13. A comparison of our results to models and predictions as well as an estimation of the ionizing budget suggests that low-mass galaxies with UV magnitudes fainter than  $M_{1500} = -18$  that we currently do not characterize with JWST observations probably played a key role in the process of reionization.

### 6.1 Introduction

The Epoch of Reionization (EoR) is a period in the history of the Universe, occurring roughly during its first billion years, when the hydrogen in the intergalactic medium (IGM) transitioned from a nearly completely neutral to a nearly completely ionized state. This transition was driven by the Lyman continuum (LyC;

$\lambda < 912 \text{ \AA}$ ) radiation emitted by the first luminous sources that formed in the early Universe. However, these sources, i.e. the so-called cosmic reionizers, remain elusive: star-forming galaxies can only account for the photon budget to complete reionization if a substantial fraction of the ultraviolet (UV) photons produced by their stellar populations escape from the galaxies' interstellar medium (ISM) and circumgalactic medium (CGM). As a result of the density of star-forming galaxies in the EoR, an average LyC escape fraction ( $f_{\text{esc}}$ ) of 10% across all galaxies is needed [e.g., 149, 393, 525, 526] to maintain the Universe being ionized by  $z = 6$ , and match the Thomson optical depth of electron scattering in the cosmic microwave background (CMB) [368]. At  $z \geq 4.5$ , however, it is impossible to detect the LyC photons escaping from galaxies, since they are absorbed and scattered by the IGM along the line of sight [208], and the LyC can only be detected at low and intermediate redshift [e.g., 151, 154, 213, 215, 216, 218, 297, 298, 457, 485, 488, 501]. To overcome this problem, key properties of the ISM and conditions that facilitate LyC photons escape (the so-called indirect indicators) at lower redshifts have been identified [see 154, for a review] and used to infer the average  $f_{\text{esc}}$  of the cosmic reionizers [e.g., 233, 300, 398, 412].

The relative importance of massive and low-mass galaxies in driving reionization is still a matter of great debate as it is intrinsically related to the timeline and topology of reionization. It is expected that reionization starts earlier, and perhaps proceeds in a spatially more homogeneous manner, when faint and low-mass galaxies with a higher  $f_{\text{esc}}$  dominate ionizing photon budgets over bright galaxies [e.g., 110, 143, 149]. Conversely, a relatively delayed reionization process is predicted when the contributions from faint galaxies ( $M_{1500} \geq -18$ ) are subdominant to that from brighter systems [331, 393]. While both types of galaxies are likely to contribute to the ionizing budget, the balance and interplay between them remain uncertain.

To understand the role of faint and bright sources, we need to determine what their relative contribution is to the total ionizing emissivity ( $\dot{n}_{\text{ion}}$ ), that is, the number of ionizing photons emitted per unit time and comoving volume [see 392, for a detailed review] which is commonly expressed as

$$\dot{n}_{\text{ion}} = f_{\text{esc}} \xi_{\text{ion}} \rho_{\text{UV}}, \quad (6.1)$$

in which  $\xi_{\text{ion}}$  is the ionizing photon production efficiency, that is, the rate of produced ionizing photons per UV luminosity, the  $\rho_{\text{UV}}$  is the integral of the UV luminosity function (the UV luminosity from galaxies per unit volume), and  $f_{\text{esc}}$  is the fraction of ionizing photons that reaches the IGM.

In the above equation, the  $\rho_{\text{UV}}$  of galaxies is relatively well constrained up to  $z \sim 10$  [e.g., 47, 50, 120]. We know that many factors influence the photon production efficiency, including the initial mass function, the stellar metallicity, the evolution of individual stars, and possible stellar binary interactions [e.g., 129, 445, 446, 527–529]. A commonly accepted value is  $\log \xi_{\text{ion}} = 25.3$ , but many recent observations at intermediate and high redshifts [e.g., 18, 48, 75, 76, 221, 260, 311, 338, 431, 449, 451]. [525] have demonstrated that  $\xi_{\text{ion}}$  can vary quite widely as a function of galaxy properties, and a fixed value is just not sufficient to properly capture the scatter in a large population of galaxies. With the James Webb Space Telescope (JWST), we are now able to measure  $\xi_{\text{ion}}$  from the rest-frame optical



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lines [e.g., 80, 416, 431, 467], instead of adapting the same value for the entire galaxy population. The only remaining big uncertainty in the emissivity equation is thus the escape fraction and how it varies with stellar mass or  $M_{1500}$  (which depends on many effects including age, dust attenuation constraints, metallicity, etc.), which is the subject of this work.

In [300] (M23 hereafter), we have shown that at the end of reionization ( $4.5 \leq z \leq 8$ ), star-forming galaxies are often compact (UV half-light radius  $r_e \simeq 0.2\text{--}0.5$  kpc), and with blue UV slopes (median  $\beta = -2.08$ ). Moreover, the analyzed sources present properties (in terms of the  $[\text{O III}]\lambda\lambda 4959, 5007/[\text{O II}]\lambda 3727$  line ratios, O32 hereafter,  $\text{H}\beta$  rest-frame equivalent widths,  $EW_0(\text{H}\beta)$ , UV- $\beta$  slopes,  $r_e$ , and  $\Sigma_{SFR}$ ) consistent with those of low- $z$  galaxies with measured  $f_{esc}$  larger than 0.05. These results suggested that the average low-mass galaxies around the EoR have physical and spectroscopic properties consistent with moderate escape of ionizing photons ( $f_{esc} = 0.1 - 0.2$ ), resulting in a dominance of low-mass, faint galaxies during cosmic reionization. The results of M23 may clarify the role of faint galaxies during reionization, but they were based on a very limited sample of sources. For this work, we used the JWST/Near InfraRed Spectrograph (NIRSpec) and Near InfraRed Camera (NIRCam) observations from the Cosmic Evolution Early Release Science (CEERS) survey of a much larger sample of high redshift galaxies to probe their role as cosmic reionizers during the EoR and put the conclusions of M23 on firmer grounds.

This paper is organized as follows: we present the dataset in Sec. 6.2. We characterize the selected sample in Sec. 6.3, and compare the physical and spectroscopic properties with models in Sec. 6.4. In Sec. 6.5 we estimate the total ionizing budget from our sample and discuss our results, while in Sec. 6.6 we summarize our key conclusions. Throughout this work, we assume a flat  $\Lambda$  cold dark matter cosmology with  $H_0 = 67.7 \text{ km s}^{-1} \text{ Mpc}^{-1}$  and  $\Omega_m = 0.307$  [368] and the [78] initial mass function. All magnitudes are expressed in the AB system [347].

## 6.2 Data

### 6.2.1 CEERS-JWST data

We used JWST/NIRSpec observations from the Cosmic Evolution Early Release Science survey (CEERS; ERS 1345, PI: S. Finkelstein) in the CANDELS Extended Groth Strip (EGS) field [177, 251]. The final list of targets selected for spectroscopic observations during the CEERS program and the way in which targets have been prioritized will be presented in Finkelstein et al. [in prep, see also 146, 147], while the NIRSpec data will be described in Arrabal Haro et al. (in prep.), see also [11]. We also use the CEERS NIRCam imaging in six broadband filters (F115W, F150W, F200W, F277W, F356W and F444W) and one medium-band filter (F410M) over 10 pointings. Details on imaging data reduction and analysis are presented in [19] [see also 146, 147].

In this section, we provide a brief summary, highlighting the most relevant points and explaining the methods we used to study the properties of the galaxies of our sample.

The focus of this study is on all sources at  $6 \leq z \leq 9$ . We selected all the sources with a photometric redshift higher than 5 that have a NIRSpect spectrum obtained either with the three medium-resolution ( $R \approx 1000$ ) grating spectral configurations (G140M/F100LP, G235M/F170LP and G395M/F290LP), which, together, cover wavelengths between 0.7-5.1  $\mu\text{m}$ , or with the PRISM/CLEAR configuration, which provides continuous wavelength coverage of 0.6-5.3  $\mu\text{m}$  with spectral resolution ranging from  $R \sim 30$  to 300.

We visually examined all these spectra for detectable optical lines and measured the systemic redshifts of 70 sources in the chosen range, using the  $\text{H}\beta$ ,  $[\text{O III}]\lambda\lambda 4959, 5007$ , and (when present)  $\text{H}\alpha$  lines. The best redshift solution was determined by fitting single Gaussian functions to the strongest emission lines and combining the centroids of the fits. In 66 cases, the  $[\text{O II}]\lambda 3727, 3729$ ,  $[\text{O III}]\lambda 4959$  and/or  $\text{H}\beta$  were detected and their line fluxes were measured. For the remaining 4 cases, the redshifts were obtained by fitting the  $\text{H}\alpha$  line alone, so they are formally included in our sample but they cannot be used for further analysis since this is the only line present in the spectra. For this part of our analysis, we use MPFIT<sup>1</sup> [295]. We note that with the PRISM's resolution of  $R > 140$  at  $\lambda > 3.4 \mu\text{m}$ , we are able to discern  $\text{H}\beta$  from  $[\text{O III}]\lambda 4959$  and resolve the  $[\text{O III}]\lambda 4959$  doublet, but we do not resolve the  $\text{H}\alpha + [\text{N II}]\lambda 6583$  doublet.

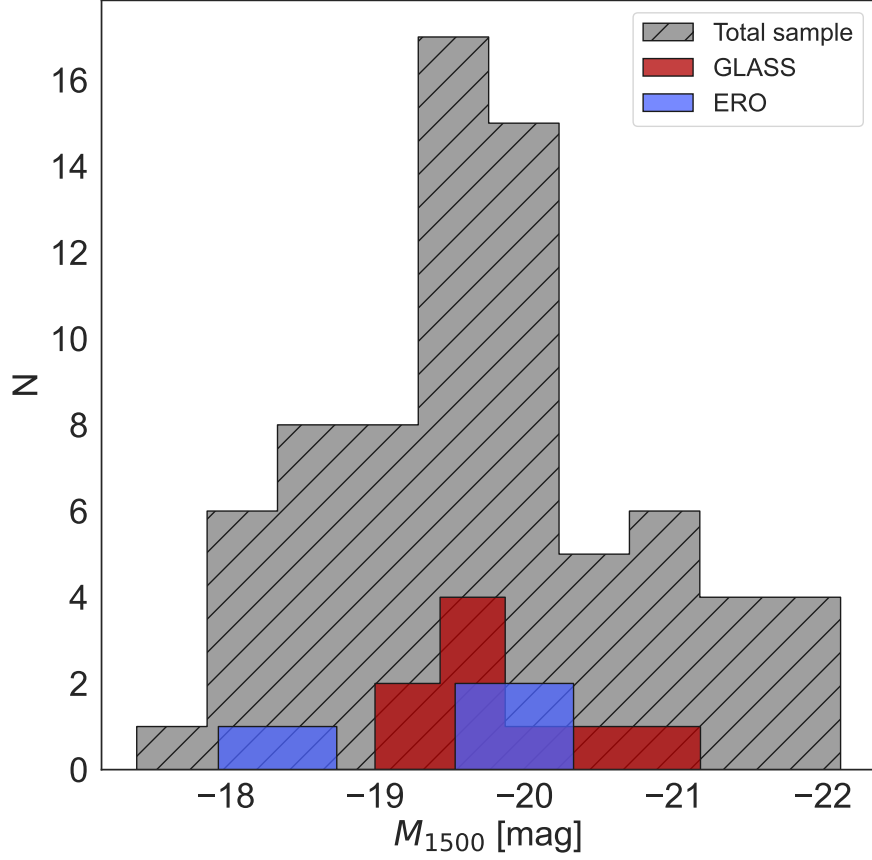
All CEERS MSA IDs, coordinates, and spectroscopic redshifts are reported in Appendix A: Properties of the sources, Table A.1, along with their spectroscopic and physical properties, whose determination is described in the next sections. Some of the sources presented in this work have been already identified and analyzed in previous works, specifically [233] (MSA IDs: 686, 689, 698), [159] (MSA IDs: 2, 3, 4, 7, 20, 23, 24), [11] (MSA IDs: 80025, 80083), [261] (MSA ID: 1019), and [465] (MSA IDs: 3, 23, 24, 44, 407, 498, 499, 686, 689, 698, 717, 1019, 1023, 1025, 1027, 1029, 1038, 1102, 1143, 1149, 1163).

### 6.2.2 Data from other programs

Several additional public sources are used to expand our EoR sample. In M23, we examined a sample of sources observed from the GLASS-ERS program (PID 1324, PI: T. Treu) using three high-resolution ( $R \sim 2000$ -3000) spectral configurations (G140H/F100LP, G235H/F170LP, and G395H/F290LP). For the purpose of this work we specifically selected the 7 sources at  $z_{\text{spec}} > 6$  (GLASS-JWST IDs: 10000, 10021, 100001, 100003, 100005, 150008, 400009), along with 2 additional sources at  $z_{\text{spec}} > 6$  from a DDT program (PID 2756, PI: W. Chen), which were obtained using the PRISM/CLEAR configuration (DDT IDs: 10025, 100004). All these sources are located in the Abell 2744 cluster field.

From the spectroscopic redshift catalog by [343], we selected four more sources from the Early Release Observations (ERO) program on the galaxy cluster SMACS J0723.3-7327 at  $z_{\text{spec}} > 6$  (ERO IDs: 4590, 5144, 6355, 10612). These spectra were acquired with medium resolution spectral configurations (G235M and G395M). The properties we use in this work were derived from [481] and [415]. For all the above sources, IDs, coordinates, spectroscopic redshifts, spectroscopic, and

<sup>1</sup><http://purl.com/net/mpfit>



**Figure 6.1:**  $M_{1500}$  distribution for the analyzed sources at  $6 \leq z \leq 9$  [gray: total sample; red and blue are respectively the GLASS sample and the ERO sample from 300, 343].

physical properties are reported in Appendix A: Properties of the sources, Table A.2.

## 6.3 Method

### 6.3.1 Measurements of physical parameters

We measured the physical parameters of the CEERS sample as described in [405], by fitting synthetic stellar templates with ZPHOT [156] to the seven-band NIRCcam photometry [148, for the sources marked with \* in Table 6.1] and the released HST photometry [454]. Specifically we measured the stellar masses  $M_{\star,obs}$ , the observed absolute UV magnitudes at 1500Å ( $M_{1500,obs}$ ), the dust reddening  $E(B - V)$  and the ages. We adopted [60] models and assumed delayed exponentially declining star formation histories –  $SFH(t) \propto (t^2/\tau) \cdot \exp(-t/\tau)$  – with  $\tau$  ranging from 0.1 to 7 Gyr. The age ranges from 10 Myr to the age of the Universe at each galaxy redshift, while metallicity can assume values of 0.02, 0.2 or 1 times Solar metallicity. For the dust attenuation, we used the [69] law with  $E(B - V)$  which can assume values ranging from 0, 0.03, 0.06, 0.1, 0.15, and from 0.2 to 1.1 in step of 0.1. We computed  $1\sigma$  uncertainties on the physical parameters by retaining,

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for each object, the minimum and maximum fitted masses among all the solutions with a probability  $P(\chi^2) > 32\%$  of being correct, fixing the redshift to the best-fit value. In Figure 6.1 we present the  $M_{1500,obs}$  distribution of the CEERS sources in our sample, which ranges from  $-22$  to  $-18$  AB mag. For reference, we also show the distribution of the  $M_{1500}$  for the GLASS and ERO sources we are considering in this work.

### 6.3.2 Dust correction and emission line flux measurements

We measured the total flux of each detected line (Balmer lines, [OII], and [OIII]) with a single Gaussian fit. From the flux measurement we subtracted a constant continuum emission, which is estimated from a wavelength region adjacent ( $\pm 160$  Å) to the emission line. When the continuum was not well constrained (signal-to-noise ratio  $S/N < 2$ ) from the fit, we estimated it subtracting the line contribution to the F444W photometry, following [159]. When the  $S/N$  of [OII], [OIII], or  $H\beta$  was less than 2, we set  $2\sigma$  as an upper limit.

Prior to carrying out a quantitative analysis, it is necessary to consider corrections for dust reddening. For 28 galaxies,  $H\alpha$  and  $H\beta$  are both available and we calculated the correction for dust extinction on the basis of the Balmer decrement, assuming a [67] extinction law and an intrinsic ratio  $H\alpha/H\beta = 2.86$  [see e.g., 119, 237, 369], which is valid for an electron temperature of 10000 K. The nebular  $E(B - V)$  determined from the Balmer decrement are in agreement with the stellar reddening determined from the SED fitting. Therefore for the 38 sources in the sample without  $H\alpha$ , we converted their stellar  $E(B - V)_{SED}$  to nebular  $E(B - V)$  following [69] and applied the nebular corrections derived from these values.

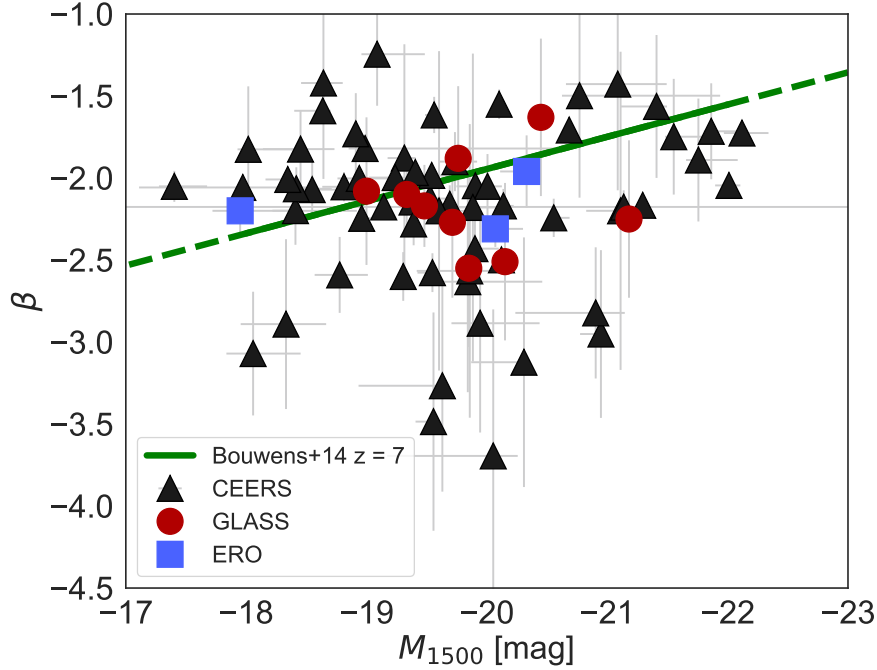
With the dust corrected spectra, we calculated the O32 line ratios and the [OIII] and/or  $H\beta$  rest-frame  $EW$ s. We list all these values in Table Appendix A: Properties of the sources, Table A.1. Within the errors, our measurements are consistent with those from previous works for sources in common [11, 159, 233, 465].

### 6.3.3 UV- $\beta$ slopes

We measured the UV- $\beta$  slope of our galaxies from the NIRCcam photometry and/or the previously available HST photometry [454], with the approach detailed in [64]. We considered all the photometric bands whose entire bandwidths are between 1216 and 3000 Å rest frame. The former limit is set to exclude the  $Ly\alpha$  line and Ly-break, while the latter limit is slightly larger than that adopted in [64] to ensure that we can use more bands.

We then fitted the selected photometry with a single power-law of the form  $f(\lambda) \propto \lambda^\beta$  [67, 318]. In practice, we fit the available photometric bands amongst HST F125W, F140W, F160W and JWST-NIRCcam F115W, F150W or F200W depending on the exact redshift of the sources. This choice allows us to uniformly probe the spectral range between 1500 and 3000 Å for most of the galaxies. We measured the  $\beta$  and associated uncertainty for each source using a Monte Carlo (MC) simulation. For each band, we randomly sampled the flux within  $1\sigma$  for a total of  $n = 1000$  times. The results provided a MC sampling of  $\beta$  from which

we calculated the mean and standard deviation of  $\beta$  for each galaxy. Two of the sources in our sample did not have the necessary data, so we were able to estimate the  $\beta$  slopes only for 64 galaxies. The results on  $\beta$  with associated errors are reported in Table Appendix A: Properties of the sources, Table A.1. We note that for 5 sources different  $\beta$  slopes are published in literature [11, 233], but they were estimated from SED fitting or from the spectra. For 4 out of 5, our values are consistent with the published ones within the uncertainties.



**Figure 6.2:**  $\beta$  vs.  $M_{1500}$ . Black triangles represent the CEERS sources with  $\beta$  slope obtained fitting 3 or 2 photometric bands. Red dots are for the GLASS sample, and blue squares are for the ERO sample. The green line shows the relation at  $z \sim 7$  derived from HST data by [49]. Dashed portions indicate the extrapolation of the relation in our range of  $M_{1500}$ . The Spearman rank correlation coefficient for the correlation between  $\beta$  and  $M_{1500}$  is 0.36, with a  $p$ -value of 0.02.

In Figure 6.2 we show the relation between our measured  $\beta$  slope values and  $M_{1500}$  and the observed trend at  $z = 7$  from [49]. We also plot the  $\beta$  values as function of  $M_{1500}$  for the GLASS and ERO sample. Our results are consistent with the best fit relation from [49] although with a large scatter. We must notice that the galaxies with the bluer slopes (with values around -3) that most deviate from the relation also have the largest uncertainties. Overall we confirm the existence of a broad correlation between  $\beta$  and UV magnitude at  $z \sim 7$  [e.g., 49, 145, 340, 508]. Our average  $\beta$  value at  $z \sim 7$ ,  $\langle\beta\rangle = -2.17 \pm 0.47$ , is in good agreement with [127] ( $\langle\beta\rangle = -2.1 \pm 0.2$  at  $z \sim 7$ ).

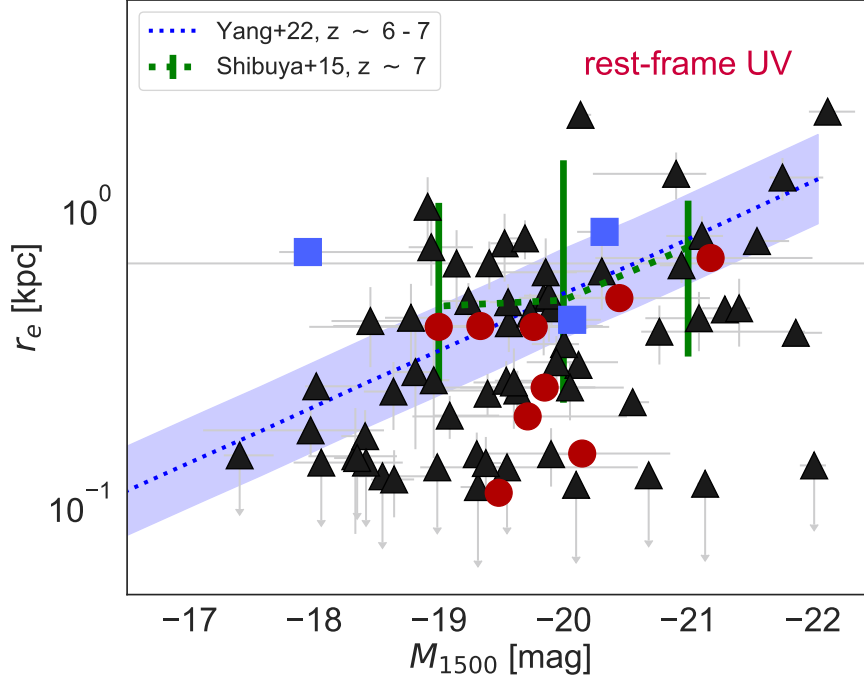
### 6.3.4 UV half-light radii

We measure the  $r_e$  of each galaxy in the rest-frame UV using the python software GALIGHT<sup>2</sup> [118], which adopts a forward-modeling technique to fit a model to the observed luminosity profile of a source. We assume that the galaxies are well represented by a Sersic profile [426]. In the fitting process, we constrain the axial ratio  $q$  to the range 0.1-1, and we fix the Sersic index  $n$  to 1, which is suitable for star-forming galaxies and also adopted by Yang et al. [520] and Morishita et al. [324]. This latter choice is consistent with the median value that we find for a subset of sources with higher S/N for which the fit converges to a finite  $n$  and  $r_e$  when leaving all the parameters free (see also McGrath et al. in prep.). The uncertainties on the sizes were estimated following [520] and re-scaled to the S/N from the photometry. The results obtained with GALIGHT are robust, as shown by previous works [e.g., 241], and in agreement with those estimated using traditional softwares such as GALFIT [363].

For 50 galaxies, we used the NIRCcam photometry to measure  $r_e$  in the F150W band (except for ID:542, for which the size is measured in F200W to improve the fit precision), corresponding to the UV rest-frame of the galaxies. For 19 galaxies where only the HST photometry was available, we measured  $r_e$  using the F160W filter, which has the highest S/N. 14 sources have profile resolutions that are likely unresolved, so we place an upper limit [see 66]. In cases where additional sources are present in the same cutout of a galaxy, we masked them or fitted them with additional Sersic profiles. We list all these measurements in Table 6.1. To determine the minimum size measurable in the F150W band, we followed a similar approach to that recently adopted by [4] in the F444W band. In brief, we performed a set of simulations by creating mock F150W images of galaxies (as observed by CEERS) with a Sersic profile, different magnitudes (from 25 to 28), and different intrinsic sizes from 0.005'' to 0.1'', in steps of 0.005. We then applied PSF fitting with GALFIT, considering unresolved a source if it is undetected ( $S/N < 2$ ) in the residual image. This procedure yields a minimum measurable size of 0.025'' (that is,  $\sim 123$  pc at redshift 8), which we adopt in this work as a lower limit. We will describe these simulations in more detail in [66].

As for the galaxies taken from previous works, for the M23 sample  $r_e$  was measured in the F115W band; for the ERO sample, F200W was considered for the sources at  $z > 7$ , and F150W for the galaxy ID:5144 at  $z = 6.381$  [481]. Typical sizes of our galaxies range from 0.1 to 2 kpc and are consistent with rest-frame UV  $r_e$  measured during reionization by recent works [326, 430, 520]. In Figure 6.3 we show the relation between our measured  $r_e$  and  $M_{1500}$ . Apart for a few outliers, we recover the well known magnitude-size relation: although with a large scatter, our results are consistent with the relation found at  $z \sim 7$  derived from HST data by [430], and the relation at  $z \sim 6 - 7$  from [521] based on photometrically selected galaxies lensed by six foreground Hubble Frontier Fields (HFF) clusters. We note that most potential cosmic reionizers should have very small UV rest-frame dimensions ( $\leq 0.4$  kpc), indicating highly concentrated star formation as for example found by [153] and in a few intermediate redshift leakers such as Ion1 [227] and Ion2 [112].

<sup>2</sup><https://github.com/dartoon/galight>



**Figure 6.3:** Rest-frame UV  $r_e$  vs.  $M_{1500}$ . Symbols are the same as in Figure 6.2. The green line shows the relation derived from HST data by [430] at  $z \sim 7$ , the blue line shows the size–luminosity relation at  $z \sim 6 - 7$  from [521]. The Spearman rank correlation coefficient for the correlation between  $r_e$  and  $M_{1500}$  is  $-0.33$ , with a  $p$ -value of  $0.003$ .

### 6.3.5 AGN contamination

While we recognize that AGN may also play some role in reionization, for example, [283] and [437, 438], a concern with our current dataset is that any AGN identified here may constitute too small a sample, and might be too heterogeneous to properly evaluate their role in reionization. Therefore we exclude them in the current work, in order for us to provide the most robust measurements of the contribution of galaxies (non-AGN) to reionization, while the role of AGN is deferred to future studies with more suitable samples. We first visually examined all spectra to see if there were any broad lines in them. Then, we employed the optical rest-frame spectroscopic diagnostics to distinguish between star-forming galaxies and AGNs. Most of our sources from the CEERS program have redshifts higher than 6.7 (7.07), so their  $H\alpha$  and  $[\text{NII}]$  emission lines cannot be identified due to the long-wavelength limit of NIRSpect G395M at  $z \geq 6.7$ , and  $z \geq 7.07$  for the PRISM. In any case, at lower redshift  $H\alpha + [\text{NII}]$  cannot be resolved with the PRISM. For this reason, we employed the mass-excitation (MEx) diagram [231, 232] with the division line identified by [92] for  $z = 2.3$  galaxies and AGN from the MOSDEF survey, as already done in M23. According to the visual inspection and the position of our sources in the MEx diagram, we conclude that our sample contains one AGN (ID: 1019) and 69 star-forming galaxies. The AGN at  $z = 8.679$  was already identified and discussed by [261].

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## 6.4 Results

### 6.4.1 Evaluating $f_{\text{esc}}$

Assuming that the mechanisms that drive the escape of LyC photons are the same at all redshifts and depend only on the physical properties of the sources, several authors have recently attempted to derive empirical relations between  $f_{\text{esc}}$  values and other observable and/or physical properties that can be measured also at high redshift. In particular, [273] have applied the relation with the  $\beta$  slope derived by [86], which is empirical but based on measurements from  $z \sim 0.3$  galaxies, while [412] applied the relation predicted from cosmological simulations by [88], which relies on the  $\beta$  slope, the  $E(B - V)$ , the  $\text{H}\beta$  line luminosity, the  $M_{1500}$ , the R23, and the O32.

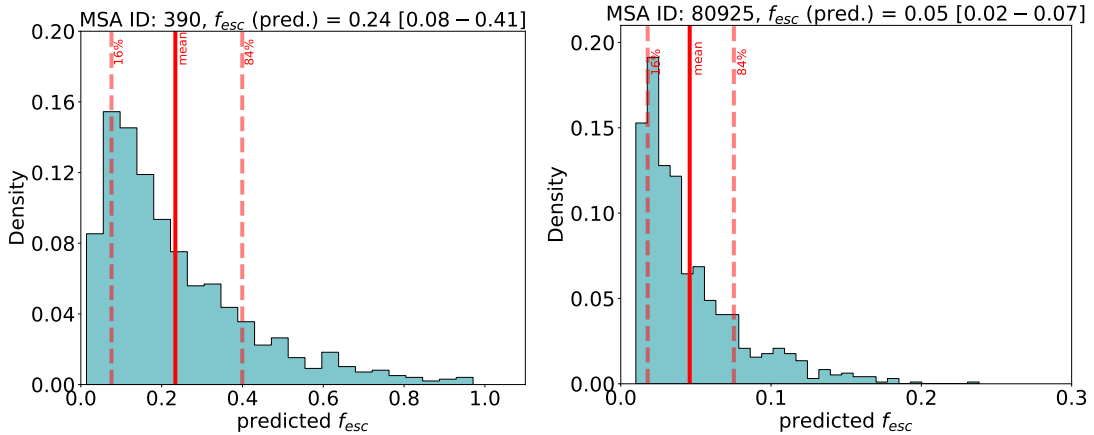
In M23 we presented our own empirical relation calibrated on the [154] Low-redshift Lyman Continuum survey (LzLCS) sample, between  $f_{\text{esc}}$  and  $\beta$  slope,  $r_e$ , and  $\log\text{O32}$  (Eq. 1 in M23). It is based on the weighted least squares linear regression model, as it proved to be the model that minimizes the mean square error (MSE) most effectively. Due to the fact that O32 and  $EW_0(\text{H}\beta)$  exhibit a very tight correlation (Spearman correlation between them  $> 0.9$ ), in M23 we used only one of the two values. However, since in some cases  $\text{H}\beta$  is measurable while O32 is not, here we also present an alternative relation using  $r_e$ ,  $\beta$ , and the  $EW_0(\text{H}\beta)$ . This relation can be used when it is not possible to derive O32 due to a lack of one of the two lines. This new relation has the following form:

$$\log_{10}(f_{\text{esc}}) = A + BEW_0(\text{H}\beta) + Cr_e + D\beta, \quad (6.2)$$

with  $A = -1.92[-2.46, -1.75]$ ,  $B = 0.0026[0.0019, 0.0035]$ ,  $C = -0.94 [-1.14, -0.67]$ ,  $D = -0.42 [-0.59, -0.33]$ , where the values between the parentheses are in the 95th percentile distribution. In Appendix B we present an analysis of the residuals between the measured  $f_{\text{esc}}$  values for the LzLCS sample and those predicted using both relations.

Using either Eq. 1 in M23 or Eq. (6.2), we predicted the  $f_{\text{esc}}$  value for the CEERS 65 star-forming galaxies, in addition to the GLASS+DDT and ERO sources for which we have the  $\beta$  slopes. As already mentioned, the UV half-light radius of 1 source from the CEERS sample could not be determined due to the inability to achieve a good fit of the profile. Moreover, in 2 cases, the  $\beta$  slope could not be measured. Since these quantities appear in both of the proposed equations, we were unable to estimate  $f_{\text{esc}}$  for 3 sources of the CEERS sample. For the remaining sources, we used the M23 equation in 49 cases in which O32 is measured accurately or it is a limit but  $\text{H}\beta$  is not evaluated, and the Eq. (6.2) for the other 13 cases. In total, we predict an  $f_{\text{esc}}$  value for 74 sources from the three samples. Given the uncertainty both on the coefficients of the relations and on the quantities on which  $f_{\text{esc}}$  depends, we estimate the  $f_{\text{esc}}$  errors using  $n = 1000$  MC simulations varying both the coefficients and the individual properties within their uncertainty. The results provide an  $f_{\text{esc}}$  distribution from which we determine the mean  $f_{\text{esc}}$  and the standard deviation for each galaxy, which is taken as the uncertainty. In Figure 6.4 we show two examples of the probability distribution function (PDF) of the  $f_{\text{esc}}$  values resulting from the above Monte Carlo runs,





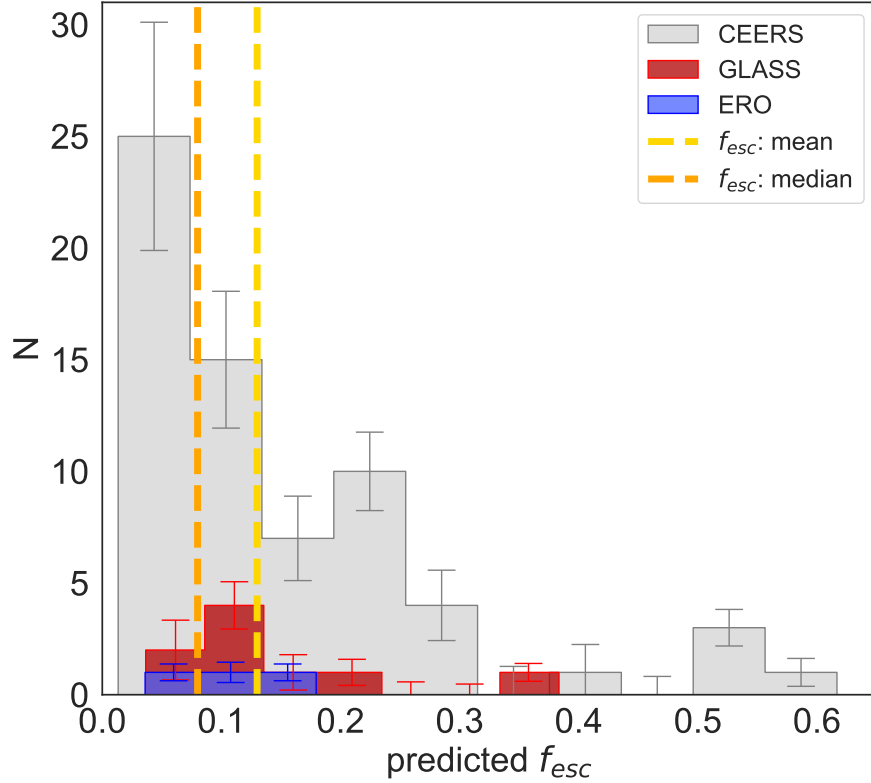
**Figure 6.4:** PDFs of the  $f_{\text{esc}}$  values for two sources from the CEERS sample (MSA IDs: 390 and 80925). The considered  $f_{\text{esc}}$  of the source, which is the mean  $f_{\text{esc}}$  of the distribution, is shown in red. Given the non-normal distribution of the  $f_{\text{esc}}$  values, we have included the 16th and 84th percentiles (red dashed lines) to measure the confidence interval.

for a galaxy with modest inferred mean  $f_{\text{esc}}$  (0.05) and one with a high inferred mean  $f_{\text{esc}}$  (0.24). In Appendix A: Properties of the sources, Table A.1 we report the mean  $f_{\text{esc}}$  and the standard deviation for the CEERS galaxies. In Appendix A: Properties of the sources, Table A.2, we report the same values for the GLASS and ERO sources.

In Figure 6.5 we present the distribution of the inferred mean  $f_{\text{esc}}$  values. Most of our galaxies have modest inferred  $f_{\text{esc}}$ , of the order of 0.10 or below. The average  $f_{\text{esc}}$  for our sample (with the standard error of the mean) is  $0.13 \pm 0.02$ . This value is affected by the high  $f_{\text{esc}}$  (0.3–0.5) inferred for a handful of sources. The median in this case is a more representative value and it is equal to  $0.08 \pm 0.02$ . To evaluate the impact of using the mean  $f_{\text{esc}}$  for each galaxy instead of the full PDF (which is not gaussian but more lognormal), we produced the same distribution shown in Figure 6.5, this time stacking the individual PDF of all galaxies. The resulting distribution is essentially unchanged: computing the mean and median values they are respectively 0.11 and 0.08, confirming that our results are robust. To compute errors for each bin, we generated 100 PDFs, similar to the one illustrated in Figure 6.4, for each galaxy. For each iteration, a value was randomly selected from these PDFs to construct the distribution of the inferred  $f_{\text{esc}}$  values. The error on each bin is defined as the standard deviation of all values within that bin across the 100 obtained histograms.

#### 6.4.2 $f_{\text{esc}}$ dependencies

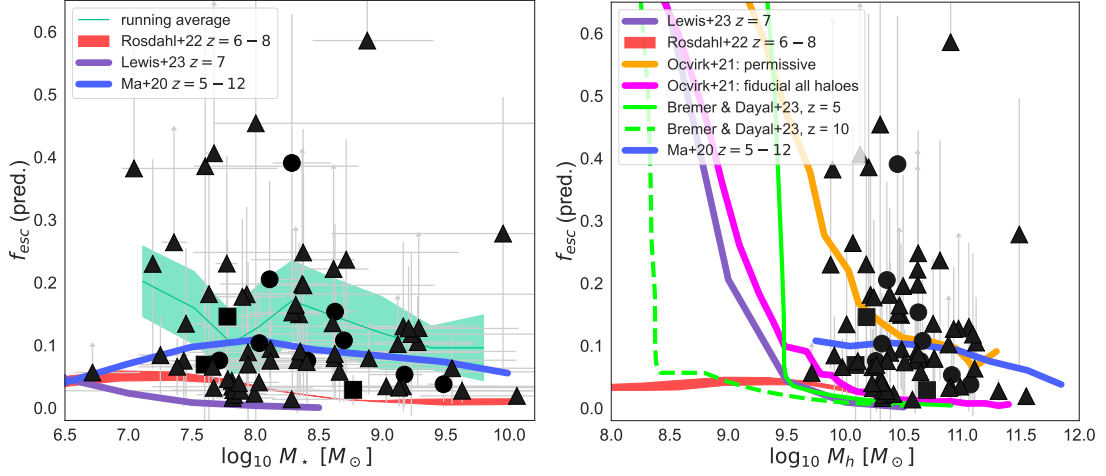
In Figure 6.6, left panel, we plot the predicted  $f_{\text{esc}}$  values versus the stellar mass  $M_*$ . We show average binned values (using a running average) with the shaded area indicating the  $1\sigma$  uncertainty. We find that low-mass galaxies tend to have slightly higher escape fractions, although the relation is rather scattered. For comparison we also plot the prediction by [396] based on SPHINX cosmological radiation-hydrodynamical simulations of reionization. Their simulated values of



**Figure 6.5:** Predicted  $f_{\text{esc}}$  distribution for the analyzed sources at  $6 \leq z \leq 9$  (gray: CEERS sample; red and blue are respectively the GLASS sample and the ERO sample). The mean  $f_{\text{esc}}$  of the sample is shown in yellow, the median  $f_{\text{esc}}$  is presented in orange.

$f_{\text{esc}}$  are generally lower than our predictions, well below 0.1 during most of the EoR, although they also find the same dependence on total stellar mass. We also compare our results with [270], which used RAMSES-CUDATON radiation-hydrodynamical simulations, and [280], whose results are based on FIRE high-resolution hydro-dynamical simulations.

Since most simulations predict  $f_{\text{esc}}$  versus halo mass  $M_h$  relations, we converted our stellar masses  $M_*$  into halo masses  $M_h$  following the relation as a function of redshift derived by [31]. We plot the  $M_h$  versus the predicted  $f_{\text{esc}}$  values in Figure 6.6, right panel. We compare our results to the prediction by [396] (see above) and to those obtained by [346] and [270] using RAMSES-CUDATON simulations. These simulations aim at reproducing the observed Ly $\alpha$  opacity distribution. Their predicted  $f_{\text{esc}}$  are  $\sim 1$  for very low-mass galaxies and drop at  $M_h \geq 10^{9.5} M_\odot$ . We plot both the fiducial and "permissive" model of [346], where this second one allows a more permissive recipe for SF also above the temperature of  $T = 2 \times 10^4 K$ . In [270] the fiducial model of [346] is extended through the inclusion of a physical model for dust production, coupled to the radiative transfer module. Finally we plot the predictions by [58] that are based on DELPHI simulations at  $z = 5$  and  $z = 10$ . In this work, reionization starts at  $z \sim 16$ , is complete at  $z = 5.67$  and it is dominated by faint, low-mass galaxies with  $M_h \leq 10^{7.8} M_\odot$  at  $z \sim 15$  that show  $f_{\text{esc}}$  up to 0.7.



**Figure 6.6:** Left: predicted  $f_{\text{esc}}$  vs. stellar mass ( $\log_{10} M_*$ ). Symbols are the same as in Figure 6.2. To enhance the clarity of the plots, we have colored the entire sample in black. The green line shows the running average for our sample, while the red one is the prediction at  $z = 6 - 8$  from [396]. Right: predicted  $f_{\text{esc}}$  vs. halo mass ( $\log_{10} M_h$ ) estimated with [31] conversion. The magenta and orange lines are the models from [346], the violet one is the prediction from [270], the lime ones are the predictions from [58], and the blue line is the prediction from [280].

Most of the above models predict a very rapid increase of  $f_{\text{esc}}$  with decreasing halo mass, below  $M_h \simeq 10^{10} M_\odot$ , a range which we barely sample with our observations, and a very low almost null  $f_{\text{esc}}$  for the more massive halos, at odds with our inferences. In the range of halo mass observed, simulations are more than  $1 \sigma$  away from our inferred  $f_{\text{esc}}$ .

The strong discrepancy between the  $f_{\text{esc}}$  values we derive from NIRSpect data and the model predictions could be due to a number of aspects:

- It may be that simulations do not adequately capture the bursty nature of star formation. It has been shown that Supernova (SN) feedback plays a critical role in creating regions with higher transparency for LyC escape. As a consequence in the models there is a positive correlation between  $f_{\text{esc}}$  and the SFR measured over the last 10 million years [396]. This suggests that bursty star formation contributes to higher  $f_{\text{esc}}$  values. However accurately quantifying the burstiness of star formation observationally, and comparing it to a simulation’s burstiness is a difficult task. For instance, it has been suggested that  $\text{H}\alpha$  / FUV fluxes could help quantifying SFR burstiness observationally [443], but this requires a fairly sophisticated post-treatment of simulations, and is very sensitive to the details of star formation, feedback (SN and radiative) and ISM modelling. Other probes of burstiness have been and will be proposed [461], and may offer avenues of progress on this topic. Nonetheless, the dominant mechanism for LyC escape remains uncertain. Recent observations of Lyman Continuum Emitters (LCEs) indicate that extreme radiative feedback can clear out gas and contribute to LyC escape [254, 288]. Furthermore, observations of Haro 11 suggest that both SNe and ionizing radiation can play significant roles in LyC escape [315]. A com-

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elling argument is presented by [153], who show that  $f_{\text{esc}}$  correlates more strongly with  $\text{H}\beta$  star formation rates (SFRs) than with UV SFRs. This suggests that radiative feedback might be more influential than burstiness and/or SNe in the context of LyC escape. The intricate interplay between bursty star formation, SN feedback, and radiative processes in LyC escape mechanisms underscores the complexity of this phenomenon.

- Another potential reason for the large discrepancy could be the description of the thermodynamical state of the shock-heated multi phase CGM [483]. For example, in a case in which a clumpy CGM is composed of hot, highly ionized gas surrounding cold dense clumps, if the cold phase is sufficiently dense and the hot phase has high pressure, the clumps may have a small cross-section: with a small total covering fraction, a high  $f_{\text{esc}}$  value could be observed. Insufficient spatial resolution in this case would imply artificially larger clumps, leading to a higher covering fraction and reducing the  $f_{\text{esc}}$ . The complexity of this behavior is being explored in simulations [178]. Observationally, patchiness has been studied in [166, 400, 401], with absorption line measurements from  $z \sim 0.3$  up to  $z \sim 2 - 4$  that suggest the existence of low gas-column-density channels in the interstellar medium, which enables the escape of ionizing photons. We note that the zoomed-in simulations of [280] present some of the highest average escape fraction in our mass range. The authors claim that this has to do with the very high spatial resolution achieved in their work, even higher than SPHINX simulations, allowing LyC to escape through low density channels. We note that it may also be a result of enhanced feedback prescriptions compared to other works. Indeed, their simulations include *continuous* feedback from both OB and AGB stars' winds, on top of the discrete supernova feedback usually implemented. This additional, pre-SN feedback may also play a role in increasing their escape fractions.
- Finally, in the previous Section we have already discussed that the relations that we have used to infer the  $f_{\text{esc}}$  for galaxies in the EoR have been derived and tested using the LzLCS sample that is located at low redshift ( $z = 0.2 - 0.4$ ). Therefore its applicability to the CEERS sample ( $6 \leq z \leq 9$ ) is not straightforward. The large discrepancy with simulations might be due to an overestimate of the  $f_{\text{esc}}$  in the EoR.

The permissive model of [346] has average  $f_{\text{esc}}$  high enough to be comparable to our values. This can be attributed to the permissive run's unique characteristic of permitting star formation in cells with temperatures potentially exceeding  $2 \times 10^4$  K. These higher temperatures inherently lead to greater ionization and increased transparency compared to the fiducial run and hence to larger values of  $f_{\text{esc}}$ . Interestingly, this model is *not* the one favored by [346] as it leads to an overionization of the Lyman- $\alpha$  forest characterized by unrealistically low Lyman- $\alpha$  IGM opacities. Our results align with the predictions of [280] simulations, emphasizing the importance of high resolution in better resolving LyC escape through galaxies. However, as stated before, the absence of radiation-hydrodynamics coupling in their simulations introduces a further layer of complexity that may influence escape fractions.

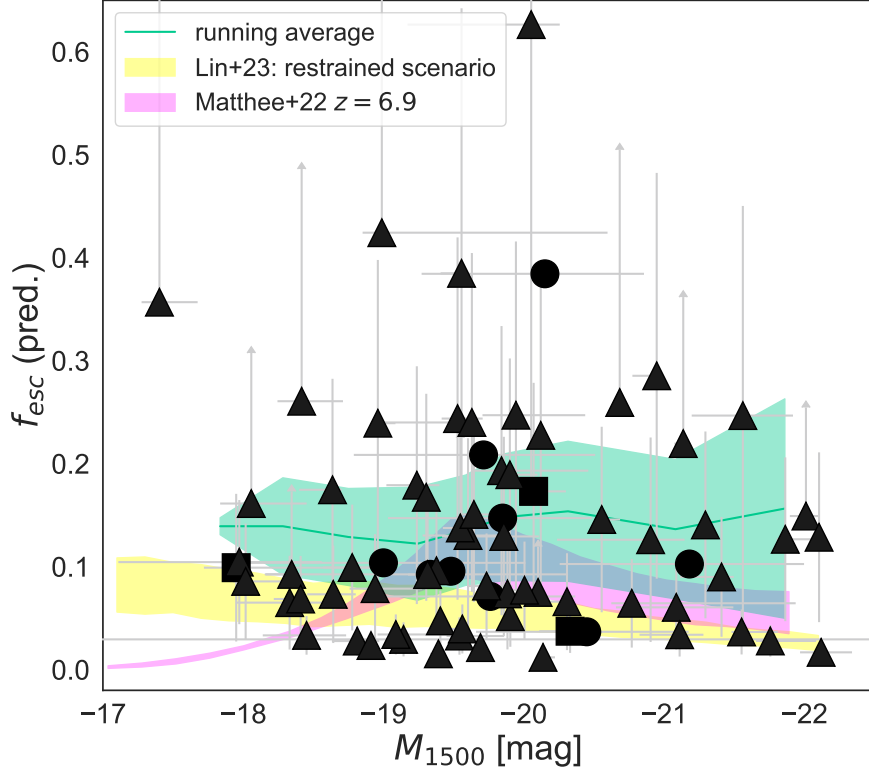
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In Figure 6.7 we plot our predicted average  $f_{\text{esc}}$  values versus the UV magnitude  $M_{1500}$ . We note that Eq. (6.2) and the M23 relation have been derived on the LzLCS which only contains galaxies brighter than  $M_{1500} \simeq -18.5$ . Therefore for our few faintest objects using the above equations might be an incorrect extrapolation. Our average  $f_{\text{esc}}$  is almost constant within the observed magnitude range, although we point out that we might start to be biased at the faintest luminosities (especially for objects with faint emission lines and hence small  $f_{\text{esc}}$ ) due to the spectroscopic flux limit of the CEERS survey. In the same Figure we also show the predicted  $f_{\text{esc}}$  vs.  $M_{1500}$  relationship by [273], who analyzed 3 galaxies at  $z \geq 8$  behind the cluster RX J2129.4+0009. They developed an empirical model based on the LzLCS program, which first defines for a given galaxy a probability of being a LyC-leaker based on  $M_{1500}$ , O32 and  $\beta$  and then infers the  $f_{\text{esc}}$  values from the  $\beta$  slope following [86]. In their *restrained scenario*, they note a consistent rise in the average escape fraction until  $M_{1500} = -16$ , reaching a peak at intermediate UV magnitudes ( $-18.5 < M_{1500} < -16$ ), followed by a decline toward the faint UV magnitude end. While there are not sufficient observations of faint galaxies to study this trend, simulations predict a similar pattern [e.g., 280, 396], for which the decrease for lower mass and fainter objects is linked to inefficient star formation and increased susceptibility to stellar feedback. As a final comparison, we plot the results by [309] who produced a semi-empirical model based on constraints on the escape fractions of bright LAEs at  $z \sim 2$ . These authors find that  $f_{\text{esc}}$  peaks between  $-19 \leq M_{1500} \leq -20$  and then decreases very rapidly at fainter magnitudes (the so-called reionization by the oligarchs). At magnitudes brighter than  $M_{1500} = -19$  our average results are consistent with theirs, within the uncertainties, but we do not observe the strong decrease at fainter magnitudes.

### 6.4.3 Redshift evolution

In Figure 6.8 we plot our predicted  $f_{\text{esc}}$  versus the redshift. We also plot the sources from M23 at redshift lower than 6 which were derived with the same method. The average  $f_{\text{esc}}$  in the three redshift bins,  $5 \leq z < 6$ ,  $6 \leq z < 7$  and  $7 \leq z \leq 9$ , are respectively equal to  $0.11 \pm 0.07$ ,  $0.12 \pm 0.08$  and  $0.14 \pm 0.08$ . The associated error represents the standard deviation of the mean value within each redshift bin. We therefore observe a slight increase of the average  $f_{\text{esc}}$  with redshift, although statistically not significant. A similar trend would be observed using the median values. We also show the predicted  $f_{\text{esc}}$  as function of redshift from [396], derived from Figure 6 of their paper at a median  $M_{1500} = -19$ . As previously discussed, their  $f_{\text{esc}}$  values are generally lower than ours, but they predict a slow increase of the  $f_{\text{esc}}$  with redshift which is very similar to what we observe. Finally, we also present a comparison between our predicted  $f_{\text{esc}}$  values and the prediction from [149] within the UV magnitude range of -16 to -20. This model also predicts lower escape fractions compared to our inferred values, and a similar almost flat trend as a function of redshift.

In the same plot we also show the sample of Ly $\alpha$  emitters at  $z = 6 - 8$  from the JWST Advanced Deep Extragalactic Survey (JADES) presented in [412], which span the same  $M_{1500}$  range as our sources. They predict the  $f_{\text{esc}}$  using an equation proposed by [88], based on the SPHINX simulations which uses six observed

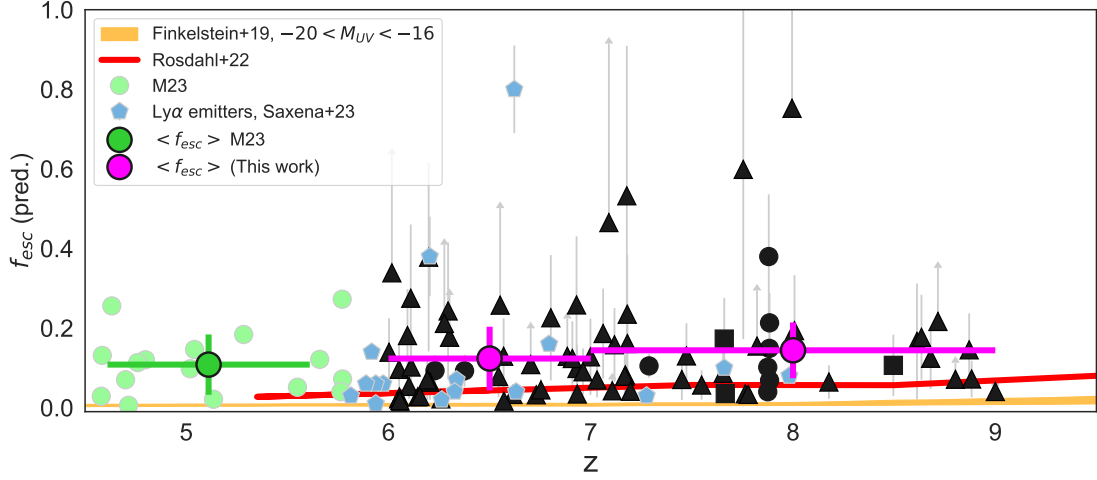


**Figure 6.7:** Predicted  $f_{\text{esc}}$  vs.  $M_{1500}$ . Symbols are the same as in Figure 6.6. The green line shows the running average for our sample, the pink one refers to [309] and the yellow one to [273].

galaxies’ properties to infer the angle-averaged (and not sight-line dependent)  $f_{\text{esc}}$ . We see that non- $\text{Ly}\alpha$  emitters and  $\text{Ly}\alpha$  emitters at the same redshift and in the same UV magnitude range do not show significant differences in the predicted  $f_{\text{esc}}$ , although determined using two different and independent methods. This might be a first indication that in the EoR, when the visibility of  $\text{Ly}\alpha$  emission is increasingly suppressed by the neutral IGM, the  $\text{Ly}\alpha$  line emerging from the galaxies is not a good indicator of the LyC photons’ escape and therefore other indirect indicators are needed. Further investigation of this important issue is in progress and will be presented in a follow-up paper.

#### 6.4.4 Extreme LyC emitters

We analyzed, in more details, the 16 sources from our final sample that show an  $f_{\text{esc}}$  higher than 0.2. The majority of them show an intense O32 or high  $\text{EW}(\text{H}\beta)$  coupled with small  $r_e$  or very blue  $\beta$  slope. We highlight the fact that an extremely blue  $\beta$  is a very good predictor of a high  $f_{\text{esc}}$ : 10 out of 17 sources with  $\beta > -2.5$  have a predicted  $f_{\text{esc}} > 0.2$ . Indeed [86] identified the  $\beta$  slope as one of the best indirect indicators. However this condition does not seem necessary, since there are several sources that have more average  $\beta$  slopes (that is, of the order of  $-2$ ) but for which we predict high  $f_{\text{esc}}$  because they are both extremely compact and have a high O32 or high  $\text{EW}(\text{H}\beta)$ . Similarly of the 8 unresolved sources for which



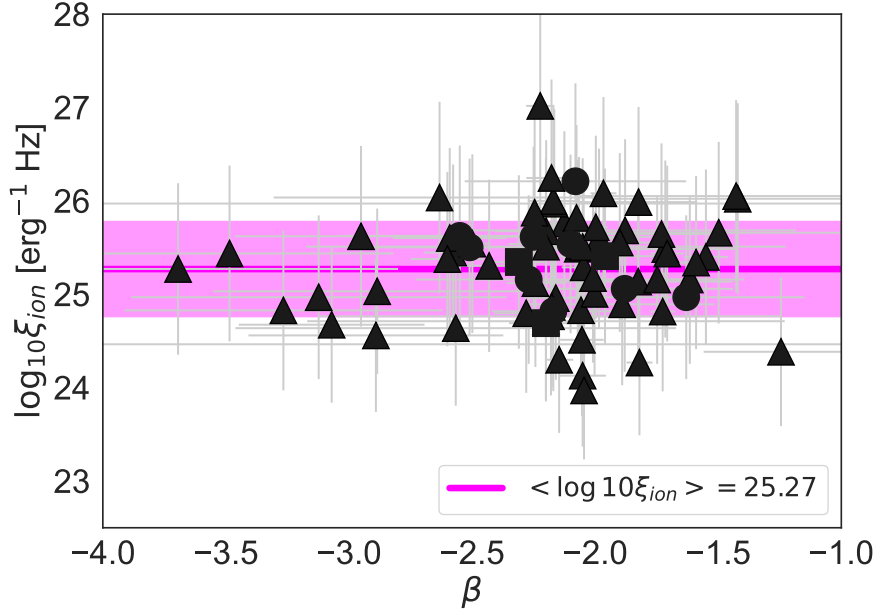
**Figure 6.8:** Predicted  $f_{\text{esc}}$  as a function of redshift. Symbols are the same as in Figure 6.6. The magenta points are the average  $f_{\text{esc}}$  values of our sample at  $z = 6.5$  and  $z = 8$ . The green dots are the M23 sources, while the green point is the average  $f_{\text{esc}}$  value at  $z = 5.1$ . The blue hexagons represent the Ly $\alpha$  emitters from [412]. The red line is the prediction at fixed  $M_{1500} = -19$  from [396], the orange line is the prediction from [149] for sources with  $M_{1500}$  between  $-20$  and  $-16$ .

we are able to infer  $f_{\text{esc}}$ , 5 are extreme leakers ( $f_{\text{esc}} > 0.2$ ). However we have some leakers with  $r_e$  larger than 0.5 kpc. Overall, there is not one single property that stands out as more important [e.g., 154, 409]. This reinforces the idea that more than one indicator is needed to correctly pinpoint the entire population of LyC emitters. The wide range of ISM and physical properties of the extreme LCEs could imply that there could be multiple independent mechanisms and/or conditions that make a galaxy a strong leaker, with no one dominating. For some galaxies it could be the low absorption by neutral gas and dust, for others it could be the presence of cleared channels (e.g., due to a high  $\Sigma_{\text{SFR}}$  or previous AGN activity). Last but not least, line of sight effects are probably also very important [e.g., 86, 117, 529].

#### 6.4.5 Ionizing photon production efficiency

Direct constraints on  $\xi_{\text{ion}}$  can be obtained from the measurement of Balmer emission lines luminosity after correcting for dust attenuation [e.g., 416, 431] or from modelling the contribution of these optical emission lines to the broad band measurements when spectroscopic observations are not available [e.g., 48]. Unfortunately, the H $\alpha$  line is outside the observed range of most of our galaxies (see Sec. 6.3.2) and the H $\beta$  line is also missing from some sources: in addition, there are still some calibration uncertainties on NIRSpec absolute flux (and therefore luminosity). [80] showed that  $\log \xi_{\text{ion}}$  can be measured by using  $\text{EW}([\text{O III}]\lambda\lambda 4959, 5007)$  [see also 467]. The [OIII] lines are clearly detected for all sources, and in addition the EW measurements have less calibration uncertainty compared to the flux. We calculated the  $\log \xi_{\text{ion}}$  values from  $\text{EW}([\text{OIII}])$  following the Eq. 3 from [82]. We obtain an average  $\langle \log \xi_{\text{ion}} \rangle = 25.27 \pm 0.51$ , which is consistent with predictions

from physical models [507, 525] and slightly lower than other measurements at the EoR. For example, [412, 435] estimated  $\log \xi_{\text{ion}}$  from  $\text{H}\alpha$  luminosity, finding respectively average values of 25.56 and 25.44 although their samples included  $\text{Ly}\alpha$  emitters whose photon production efficiency is generally higher, while [76, 132, 372], using SED fitting, obtain an average value of  $\log \xi_{\text{ion}}$  of 25.14, 25.33, 25.7 respectively. In Figure 6.9 we show the distribution of  $\xi_{\text{ion}}$  for our sample. We do not find any correlation with the  $\beta$  slope: our best fit is consistent with the average value also shown in the Figure. At variance with this, [372] find a slight dependence on this property for galaxies at  $z = 3 - 7$ , in the sense that bluer star-forming sources tend to have higher photon production efficiencies [see also 75]. We also do not find any dependence of  $\xi_{\text{ion}}$  on  $M_{1500}$  in accordance to what is found by [132, 372]. We note that the recent results by [17] indicate a higher  $\xi_{\text{ion}}$  for much fainter galaxies ( $M_{\text{UV}} \geq -17$ ) during the EoR.



**Figure 6.9:**  $\log \xi_{\text{ion}}$  vs.  $\beta$ . Symbols are the same as in Figure 6.6. The magenta line shows the mean  $\log \xi_{\text{ion}}$  for our sample. Only detections are shown in this figure. The Spearman rank correlation coefficient for the correlation between  $\log \xi_{\text{ion}}$  and  $\beta$  is 0.05, with a  $p$ -value of 0.63 indicating no correlation between the two quantities.

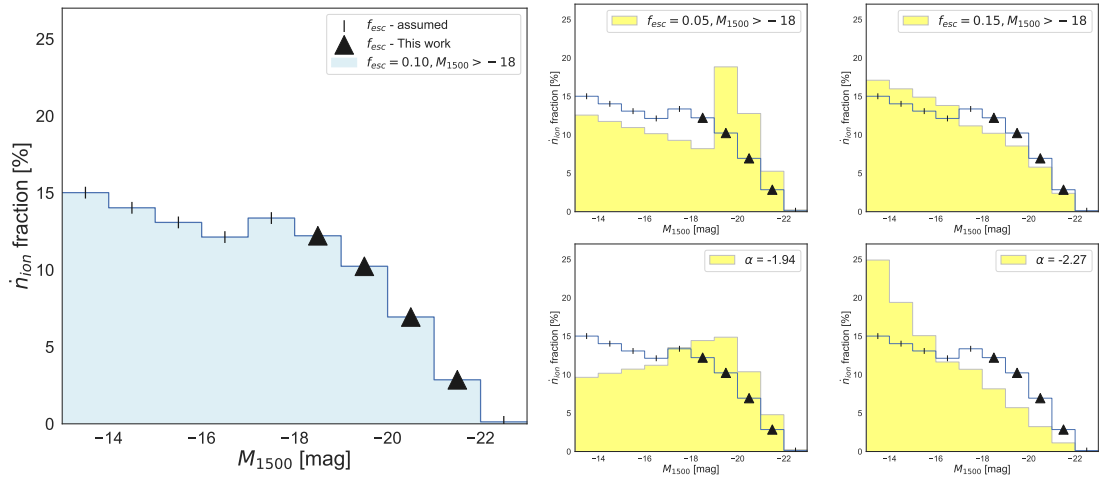
## 6.5 The ionizing photon production of bright and faint sources

Having derived predictions for  $f_{\text{esc}}$  and  $\xi_{\text{ion}}$  for our large sample of galaxies in the EoR, our goal is now to solve Eq. (6.1) and determine the relative contribution of galaxies as a function of  $M_{1500}$ , to establish which sources contributed most to the total ionizing photon production rate at these epochs. We consider: 1)  $\rho_{\text{UV}}$  from the luminosity function (LF) of [47] at our median redshift  $\langle z \rangle \sim 7.2$ . The best-fit  $\alpha$  slope that characterizes the faint-end of the UV-LF is  $-2.06 \pm 0.03$ ; 2)  $f_{\text{esc}}$  as a function of  $M_{1500}$  from the values derived in Figure 6.7 between  $M_{1500}$  -22 and -18 (i.e. the range covered by our observations); we use a fixed value of



0.10 at fainter magnitudes, where we have only few sources, and a value of 0.05 at magnitudes brighter than  $-22$ , where we do not have any observed source in our sample; 3)  $\langle \log \xi_{\text{ion}} \rangle = 25.27$ , which does not vary with  $M_{1500}$ , as found in Sec. 6.4.5. We assume a low luminosity cut at  $M_{1500} = -13$  and a high luminosity cut of  $M_{1500} = -23$  [as in 393].

To estimate the total  $\dot{n}_{\text{ion}}$  we proceeded as follows: we first discretized the  $M_{1500}$  range over  $[-13, -23]$  in bins of width 1 mag. For each of these intervals we calculated  $\rho_{\text{UV}}$  in the considered magnitude bin and multiplied it by the appropriate  $\xi_{\text{ion}}$  and  $f_{\text{esc}}$ . We then summed these values to estimate the total  $\dot{n}_{\text{ion}}$ . The total integrated ionizing emissivity at  $z = 8$  and  $z = 6$  are respectively  $\log \dot{n}_{\text{ion}} = 50.50 \pm 0.38$  and  $50.75 \pm 0.35 \text{ s}^{-1} \text{Mpc}^{-3}$ , consistent with the canonical threshold needed to maintain the Universe ionized at  $z = 7$  [e.g., 171, 281] and in the range of previous determination [50, 149, 393].



**Figure 6.10:** Left: The  $\dot{n}_{\text{ion}}$  fraction of galaxies at  $6 \leq z \leq 9$  as a function of  $M_{1500}$ . Triangles represent the  $\dot{n}_{\text{ion}}$  obtained from the predicted  $f_{\text{esc}}$  as a function of  $M_{1500}$  in each bin. Bars represent the extrapolated value, i.e. the  $\dot{n}_{\text{ion}}$  fraction derived assuming for  $M_{1500} > -18$  a  $f_{\text{esc}}$  value constant and equal to 0.10, and for  $M_{1500} < -22$  a  $f_{\text{esc}}$  value constant and equal to 0.05. Right: in yellow we show the  $\dot{n}_{\text{ion}}$  fraction of galaxies at  $6 \leq z \leq 9$  as function of  $M_{1500}$  changing, respectively, from the top left to the bottom right, the  $f_{\text{esc}}$  values at the faint-end (0.05 and 0.15) and the  $\alpha$  parameter of the  $\rho_{\text{UV}}$  ( $-1.94$  and  $-2.27$ ). The original result is also shown with the same symbols as in the figure from the left.

We then derive the fraction of the total  $\dot{n}_{\text{ion}}$  that is provided by galaxies in each magnitude bin: the results are shown in Figure 6.10 and indicate that the galaxies that we can currently characterize with JWST observations are contributing to only a fraction of the total ionizing budget, i.e. less than 35% of the total. We would therefore need to push our observations at 2-3 magnitudes deeper to characterize the bulk of the cosmic ionizers. We note however that in previous studies [e.g., 149, 393] the fraction of ionizing photons from faint galaxies was even more prominent, with the extreme faint end of the luminosity function dominating the ionizing emissivity [e.g., see Figure 8 of 149]. We also show how the results would vary by changing the faint end  $f_{\text{esc}}$  to values of 0.05 and 0.15 respectively (right top panels of Figure 6.10) and by changing the faint end slope of the UV- $\rho_{\text{UV}}$  within the  $5\sigma$  uncertainties  $[-1.94, -2.27]$ . We see that the contribution of

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JWST sources with  $M_{1500} < -18$  to the integrated ionizing emissivity becomes 40% if we assume a very small  $f_{\text{esc}}$  for the faintest galaxies, or an extremely flat  $LF_{UV}$  at the faint end, but it is never dominant even in these extreme cases.

## 6.6 Summary and conclusions

In this paper, we have presented an analysis of 70 spectroscopically confirmed star-forming galaxies at  $6 \leq z \leq 9$  from the CEERS survey, combined with 12 sources with public data from other JWST early campaigns. Assuming that the mechanisms that facilitate the escape of LyC photons from galaxies remain consistent across all cosmic epochs, we estimated the  $f_{\text{esc}}$  of the observed sources employing two empirical relations based on the most promising indirect indicators of this emission identified at  $z \sim 0.3$ , which are also measurable during the EoR. Using the mean inferred  $f_{\text{esc}}$  as a function of  $M_{UV}$  and the photon production efficiency derived from the [OIII] emission line, we have then evaluated the relative role of faint and bright galaxies and their contribution to the process of reionization. Our main results are the following:

- The majority of our sources show modest  $f_{\text{esc}}$  values, with a mean  $f_{\text{esc}}$  of  $0.13 \pm 0.02$ , and an even lower median of  $0.08 \pm 0.02$ . Just 20% of galaxies have  $f_{\text{esc}} > 0.2$ : the majority of these extreme LyC emitters show an intense O32 or high EW(H $\beta$ ) coupled with small  $r_e$  or very blue  $\beta$  slope. As expected there is no single property that stands out as the best indirect indicator of a high LyC escape.
- The predicted  $f_{\text{esc}}$  has a modest dependence on the total stellar mass  $M_*$  with low-mass galaxies tending to have higher mean  $f_{\text{esc}}$  although the trend is scattered. The relation with  $M_{1500}$  is less well characterized and there is not a significant dependence.
- There is a strong discrepancy between our inferred  $f_{\text{esc}}$  and those predicted by most cosmological hydrodynamical simulations of reionization, which consistently infer much lower  $f_{\text{esc}}$  values for galaxies in the same mass range as the one explored by the JWST observations. We discuss potential causes for the discrepancy such as the failure of simulations to fully account for the bursty nature of star formation, or the limited resolution. Alternatively using relations derived from low-redshift samples to infer  $f_{\text{esc}}$  for galaxies in the EoR might not be correct and could lead to an overestimate of the  $f_{\text{esc}}$  values.
- The average predicted  $f_{\text{esc}}$  have at most a modest increase with redshift from  $z = 5$  to  $z = 8$  raising from 0.11 to 0.14.
- The predicted  $f_{\text{esc}}$  during the EoR does not show a clear dependence on the presence of Ly $\alpha$  emission. This is actually expected since in the EoR the Ly $\alpha$  emission is modulated also by the IGM opacity in the local surroundings and not just by the galaxies' properties as at low redshift;
- With the inferred values of  $f_{\text{esc}}$  and  $\xi_{\text{ion}}$  we derive a total ionizing emissivity  $\log \dot{n}_{\text{ion}} = 50.50 \pm 0.38$  and  $50.75 \pm 0.35 \text{ s}^{-1}\text{Mpc}^{-3}$  at redshift 8 and 6 respectively, i.e. comparable to the threshold needed to maintain the Universe

ionized. Sources brighter than  $M_{1500} = -18$ , which are those we can currently characterize with JWST observations, only contribute less than 35% to the total ionizing emissivity.

The findings of this study provide crucial insights into the reionization epoch, primarily focusing on the characterization of relatively bright sources and indicating that galaxies significantly fainter and less massive than those observed by the initial JWST programs, could potentially play a dominant role in the reionization process. To study significantly large samples of such faint galaxies, ultra-deep observations of galaxy cluster fields will be needed since lensing becomes a necessary tool, as in the recent work by [17] which reaches galaxies as faint as  $M_{UV} = -15$ . In addition further work on the LyC indirect indicators will be needed to validate the fundamental assumption that the physical mechanisms and conditions that facilitate the escape of Lyman continuum photons remain the same over cosmic time. In particular, future work should be aimed at assembling a solid reference sample of Lyman continuum emitting galaxies, analogous in size to the LzLCS survey, but at  $z = 3 - 4$ , i.e. the highest redshift where a direct detection of LyC photons is possible and which is much closer in time to the epoch of reionization. If our derived relations to infer  $f_{\text{esc}}$  were still valid at  $z = 3 - 4$ , then we could be much more confident that they can be also applied in the EoR.

## 6.7 Appendix A: Properties of the sources

This appendix comprises a sequence of tables detailing the properties of the sources included in this study.

**Table 6.1:** *Physical and spectroscopic properties for the CEERS sample. : NIRCam photometry available.  $r_e$  with errors: determined in F150W (or F200W for ID:542) with GALIGHT. Due to its nature as an AGN, ID:1019 (in magenta) is excluded from the final sample [261].*

| MSA ID | RA [deg]   | DEC [deg] | z     | $\beta$         | $EW_0(\text{H}\beta)$ [Å] | O32        | $r_e$ [kpc] | $f_{\text{esc}}$ (pred.) | $\log \xi_{\text{ion}}$ [erg <sup>-1</sup> Hz] | $M_{1500}$ [mag]                         |
|--------|------------|-----------|-------|-----------------|---------------------------|------------|-------------|--------------------------|------------------------------------------------|------------------------------------------|
| 2      | 214.994402 | 52.989379 | 8.803 | -1.55<br>± 0.09 | <178                      | 3.7 ± 0.3  | < 0.12      | > 0.07                   | 25.40 ± 0.94                                   | -20.10 <sup>+0.05</sup> <sub>-0.03</sub> |
| 3      | 215.005189 | 52.996580 | 8.007 | -2.63<br>± 0.67 | 133 ± 81                  | 10.3 ± 0.2 | 0.50 ± 0.18 | 0.20 [0.06, 0.35]        | 26.04 ± 1.02                                   | -19.84 <sup>+0.62</sup> <sub>-0.20</sub> |
| 4      | 215.005365 | 52.996697 | 7.995 | -2.07<br>± 0.17 | 144 ± 121                 | -          | < 0.12      | > 0.18                   | 25.49 ± 0.95                                   | -18.55 <sup>+0.49</sup> <sub>-0.19</sub> |
| 7      | 215.011706 | 52.988303 | 8.871 | -2.24<br>± 0.12 | 296 ± 55                  | 5.3 ± 2.1  | 0.22 ± 0.02 | 0.15 [0.06, 0.25]        | 25.12 ± 0.90                                   | -20.56 <sup>+0.13</sup> <sub>-0.01</sub> |
| 20     | 214.830685 | 52.887771 | 7.764 | -1.25<br>± 0.31 | 57 ± 13                   | 2.2 ± 0.4  | 0.20 ± 0.03 | 0.03 [0.01, 0.06]        | 24.38 ± 0.80                                   | -19.09 <sup>+0.40</sup> <sub>-0.13</sub> |
| 23     | 214.901252 | 52.846997 | 8.883 | -1.42<br>± 0.58 | 187 ± 83                  | 4.2 ± 0.3  | 0.12 ± 0.03 | 0.07 [0.03, 0.11]        | 26.02 ± 1.02                                   | -18.64 <sup>+0.16</sup> <sub>-0.19</sub> |
| 24     | 214.897232 | 52.843854 | 9.000 | -1.88<br>± 0.69 | -                         | > 8        | < 0.12      | > 0.06                   | 25.68 ± 0.98                                   | -19.31 <sup>+0.09</sup> <sub>-0.14</sub> |
| 44     | 215.001115 | 53.011269 | 7.106 | -2.58<br>± 0.12 | 139 ± 11                  | 18.4 ± 4.5 | < 0.13      | > 0.39                   | 25.45 ± 0.94                                   | -19.55 <sup>+0.00</sup> <sub>-0.12</sub> |

**Table 6.1:** *continued.*

| MSA ID | RA [deg]   | DEC [deg] | z     | $\beta$             | $EW_0(\text{H}\beta)$ [Å] | O32              | $r_e$ [kpc]        | $f_{\text{esc}}$ (pred.) | $\log \xi_{\text{ion}}$ [erg $^{-1}$ Hz] | $M_{1500}$ [mag]          |
|--------|------------|-----------|-------|---------------------|---------------------------|------------------|--------------------|--------------------------|------------------------------------------|---------------------------|
| 67     | 215.015597 | 53.011857 | 6.205 | -2.89<br>$\pm 0.52$ | 21 $\pm$ 10               | 0.2 $\pm$ 0.1    | 0.15<br>$\pm$ 0.07 | 0.07<br>[0.02, 0.11]     | 24.56 $\pm$ 0.82                         | -18.33 $^{+0.33}_{-0.38}$ |
| 355    | 214.806482 | 52.878827 | 6.102 | -2.05<br>$\pm 0.09$ | 78 $\pm$ 18               | > 23             | 0.46<br>$\pm$ 0.06 | 0.05<br>[0.02, 0.09]     | 25.29 $\pm$ 0.92                         | -19.90 $^{+0.00}_{-0.02}$ |
| 362    | 214.812689 | 52.881536 | 6.052 | -2.59<br>$\pm 0.23$ | 66 $\pm$ 33               | > 29             | 0.43<br>$\pm$ 0.14 | 0.10<br>[0.04, 0.15]     | 25.60 $\pm$ 0.96                         | -18.78 $^{+0.23}_{-0.21}$ |
| 386    | 214.832184 | 52.885083 | 6.615 | -1.82<br>$\pm 0.24$ | 132 $\pm$ 47              | 1.4 $\pm$ 0.1    | 0.42<br>$\pm$ 0.12 | 0.04<br>[0.01, 0.06]     | 25.99 $\pm$ 1.02                         | -18.45 $^{+0.28}_{-0.31}$ |
| 390    | 214.811038 | 52.868521 | 6.295 | -2.13<br>$\pm 0.07$ | 303 $\pm$ 40              | 123.0 $\pm$ 12.6 | 0.76<br>$\pm$ 0.20 | 0.24<br>[0.08, 0.42]     | 25.76 $\pm$ 0.99                         | -19.53 $^{+0.18}_{-0.11}$ |
| 397    | 214.836197 | 52.882693 | 6.003 | -2.17<br>$\pm 0.03$ | 487 $\pm$ 53              | 12.8 $\pm$ 0.5   | 0.46<br>$\pm$ 0.02 | 0.14<br>[0.05, 0.24]     | 25.97 $\pm$ 1.02                         | -21.29 $^{+0.03}_{-0.00}$ |
| 407    | 214.839316 | 52.882565 | 7.031 | -2.20<br>$\pm 0.21$ | 108 $\pm$ 17              | 1.1 $\pm$ 0.1    | 0.17<br>$\pm$ 0.04 | 0.07<br>[0.03, 0.11]     | 25.68 $\pm$ 0.97                         | -18.41 $^{+0.00}_{-0.19}$ |
| 428    | 214.824551 | 52.868856 | 6.104 | -2.09<br>$\pm 0.03$ | -                         | -                | < 0.15             | -                        | -                                        | -19.13 $^{+0.36}_{-0.25}$ |
| 439    | 214.825364 | 52.863065 | 7.181 | -2.60<br>$\pm 0.15$ | 60 $\pm$ 28               | -                | 0.15<br>$\pm$ 0.03 | 0.16<br>[0.06, 0.25]     | 25.38 $\pm$ 0.93                         | -19.30 $^{+0.00}_{-0.04}$ |
| 476    | 214.805561 | 52.836345 | 6.017 | -2.05<br>$\pm 0.09$ | 17 $\pm$ 3                | > 4              | < 0.15             | > 0.07                   | 24.13 $\pm$ 0.76                         | -17.40 $^{+0.27}_{-0.13}$ |
| 481    | 214.827785 | 52.850615 | 6.932 | -2.15<br>$\pm 0.06$ | < 79                      | 2.1 $\pm$ 0.2    | 0.14<br>$\pm$ 0.03 | 0.09<br>[0.04, 0.15]     | 24.30 $\pm$ 0.78                         | -19.38 $^{+0.12}_{-0.12}$ |
| 496    | 214.864735 | 52.871719 | 6.571 | -2.28<br>$\pm 0.13$ | 25 $\pm$ 9                | -                | 0.24<br>$\pm$ 0.04 | 0.02<br>[0.01, 0.02]     | 24.80 $\pm$ 0.85                         | -19.39 $^{+0.03}_{-0.06}$ |
| 498    | 214.813045 | 52.834249 | 7.180 | -2.50<br>$\pm 0.07$ | 446 $\pm$ 23              | 9.6 $\pm$ 2.2    | 0.30<br>$\pm$ 0.02 | 0.22<br>[0.08, 0.38]     | 24.62 $\pm$ 0.83                         | -20.12 $^{+0.03}_{-0.07}$ |
| 499    | 214.813004 | 52.834170 | 7.171 | -1.83<br>$\pm 0.38$ | 76 $\pm$ 24               | 4.5 $\pm$ 1.5    | 0.25<br>$\pm$ 0.01 | 0.08<br>[0.03, 0.13]     | 25.14 $\pm$ 0.90                         | -18.02 $^{+0.39}_{-0.09}$ |
| 535    | 214.859175 | 52.853587 | 7.117 | -2.08<br>$\pm 0.01$ | 130 $\pm$ 25              | -                | -                  | -                        | 25.82 $\pm$ 0.99                         | -19.64 $^{+0.13}_{-0.08}$ |
| 542    | 214.831624 | 52.831505 | 7.061 | -2.43<br>$\pm 0.18$ | 114 $\pm$ 13              | -                | 0.15<br>$\pm$ 0.03 | 0.19<br>[0.07, 0.31]     | -                                        | -19.90 $^{+0.20}_{-0.03}$ |
| 568    | 214.891863 | 52.869054 | 6.806 | -2.09<br>$\pm 0.11$ | -                         | -                | 0.50<br>$\pm$ 0.28 | -                        | -                                        | -18.45 $^{+0.10}_{-0.13}$ |
| 577    | 214.892861 | 52.865157 | 6.703 | -2.07<br>$\pm 0.13$ | 370 $\pm$ 41              | -                | < 0.14             | > 0.37                   | 25.52 $\pm$ 0.95                         | -18.42 $^{+0.29}_{-0.17}$ |
| 603    | 214.867247 | 52.836737 | 6.059 | -2.17<br>$\pm 0.10$ | 166 $\pm$ 87              | 2.0 $\pm$ 0.1    | 2.05<br>$\pm$ 0.06 | 0.01<br>[0.01, 0.02]     | 26.00 $\pm$ 1.02                         | -20.14 $^{+0.09}_{-0.00}$ |
| 613    | 214.882077 | 52.844346 | 6.731 | -1.98<br>$\pm 0.06$ | 132 $\pm$ 38              | 0.5 $\pm$ 0.1    | 0.26<br>$\pm$ 0.04 | 0.03<br>[0.01, 0.05]     | 25.60 $\pm$ 0.96                         | -19.54 $^{+0.32}_{-0.09}$ |
| 618    | 214.876469 | 52.839412 | 6.050 | -2.16<br>$\pm 0.12$ | 178 $\pm$ 23              | 0.6 $\pm$ 0.1    | 0.79<br>$\pm$ 0.12 | 0.02<br>[0.01, 0.03]     | 24.96 $\pm$ 0.88                         | -19.69 $^{+0.05}_{-0.12}$ |

**Table 6.1:** *continued.*

| MSA ID | RA [deg]   | DEC [deg] | z     | $\beta$             | $EW_0(\text{H}\beta)$ [Å] | O32            | $r_e$ [kpc]        | $f_{\text{esc}}$ (pred.) | $\log \xi_{\text{ion}}$ [erg $^{-1}$ Hz] | $M_{1500}$ [mag]          |
|--------|------------|-----------|-------|---------------------|---------------------------|----------------|--------------------|--------------------------|------------------------------------------|---------------------------|
| 648    | 214.899823 | 52.847647 | 6.054 | -2.05<br>$\pm 0.09$ | 15 $\pm$ 11               | 0.3 $\pm$ 0.1  | 0.28<br>$\pm$ 0.13 | 0.03<br>[0.01, 0.04]     | 24.51 $\pm$ 0.81                         | -18.81 $^{+0.16}_{-0.18}$ |
| 686    | 215.150862 | 52.989562 | 7.754 | -3.69<br>$\pm 0.89$ | 123 $\pm$ 4               | -              | 0.25<br>$\pm$ 0.06 | 0.59<br>[0.17, 1.00]     | 25.27 $\pm$ 0.92                         | -20.05 $^{+0.20}_{-0.88}$ |
| 689    | 214.999052 | 52.941977 | 7.548 | -1.43<br>$\pm 0.65$ | 137 $\pm$ 61              | 8.3 $\pm$ 1.0  | 0.43<br>$\pm$ 0.10 | 0.06<br>[0.02, 0.10]     | 26.05 $\pm$ 1.03                         | -21.09 $^{+0.41}_{-0.43}$ |
| 698    | 215.050317 | 53.007441 | 7.473 | -1.72<br>$\pm 0.29$ | 134 $\pm$ 5               | 20.0 $\pm$ 1.8 | 0.38<br>$\pm$ 0.03 | 0.13<br>[0.05, 0.21]     | 25.48 $\pm$ 0.95                         | -21.86 $^{+0.21}_{-0.06}$ |
| 716    | 215.080349 | 52.993241 | 6.964 | -                   | -                         | -              | -                  | -                        | -                                        | -21.98 $^{+0.07}_{-0.22}$ |
| 717    | 215.081406 | 52.972180 | 6.933 | -1.75<br>$\pm 0.35$ | -                         | 5.4 $\pm$ 0.8  | 0.77<br>$\pm$ 0.08 | 0.03<br>[0.01, 0.05]     | 25.16 $\pm$ 0.90                         | -21.55 $^{+0.26}_{-0.07}$ |
| 749    | 215.002840 | 53.007588 | 7.090 | -1.82<br>$\pm 0.05$ | 142 $\pm$ 12              | > 6            | < 0.13             | > 0.12                   | 24.27 $\pm$ 0.78                         | -18.99 $^{+1.61}_{-0.14}$ |
| 792    | 214.871766 | 52.833167 | 6.259 | -1.73<br>$\pm 0.25$ | 43 $\pm$ 12               | 3.3 $\pm$ 0.3  | 1.01<br>$\pm$ 0.26 | 0.02<br>[0.01, 0.03]     | 25.65 $\pm$ 0.97                         | -18.91 $^{+0.04}_{-0.03}$ |
| 829    | 214.861594 | 52.876159 | 7.168 | -2.05<br>$\pm 0.20$ | 39 $\pm$ 19               | 3.8 $\pm$ 0.2  | 0.35<br>$\pm$ 0.07 | 0.08<br>[0.03, 0.13]     | 25.51 $\pm$ 0.95                         | -20.00 $^{+0.12}_{-0.26}$ |
| 1019   | 215.035391 | 52.890662 | 8.680 | -2.22<br>$\pm 0.06$ | 63 $\pm$ 10               | 12.7 $\pm$ 0.9 | 0.52<br>$\pm$ 0.02 | 0.13<br>[0.05, 0.21]     | 27.02 $\pm$ 1.16                         | -22.10 $^{+0.03}_{-0.02}$ |
| 1023   | 215.188413 | 53.033647 | 7.778 | -2.20<br>$\pm 0.97$ | 49 $\pm$ 18               | 1.0 $\pm$ 0.4  | 0.80<br>$\pm$ 0.13 | 0.03<br>[0.01, 0.05]     | 24.71 $\pm$ 0.84                         | -21.11 $^{+0.17}_{-0.29}$ |
| 1025   | 214.967547 | 52.932953 | 8.716 | -2.18<br>$\pm 0.10$ | 456 $\pm$ 44              | 7.6 $\pm$ 1.2  | < 0.12             | > 0.22                   | 26.25 $\pm$ 1.05                         | -21.14 $^{+0.03}_{-0.06}$ |
| 1027   | 214.882994 | 52.840416 | 7.822 | -1.71<br>$\pm 0.07$ | 127 $\pm$ 28              | 13.2 $\pm$ 2.3 | < 0.13             | > 0.17                   | 25.44 $\pm$ 0.94                         | -20.68 $^{+0.00}_{-0.03}$ |
| 1029   | 215.218762 | 53.069862 | 8.613 | -                   | 32 $\pm$ 4                | 3.4 $\pm$ 0.9  | 0.73<br>$\pm$ 0.08 | -                        | 24.46 $\pm$ 0.81                         | -21.56 $^{+0.36}_{-0.36}$ |
| 1038   | 215.039697 | 52.901597 | 7.196 | -1.62<br>$\pm 0.11$ | 105 $\pm$ 63              | 3.0 $\pm$ 0.1  | 0.41<br>$\pm$ 0.08 | 0.04<br>[0.02, 0.06]     | 25.16 $\pm$ 0.90                         | -19.56 $^{+0.00}_{-0.08}$ |
| 1064   | 215.177167 | 53.048975 | 6.802 | -3.27<br>$\pm 0.65$ | 26 $\pm$ 3                | -              | 0.24<br>$\pm$ 0.04 | 0.24<br>[0.07, 0.41]     | 24.83 $\pm$ 0.86                         | -19.63 $^{+0.03}_{-0.70}$ |
| 1065   | 215.116854 | 53.001081 | 6.192 | -2.18<br>$\pm 0.94$ | 24 $\pm$ 8                | 2.7 $\pm$ 0.6  | 0.51<br>$\pm$ 0.15 | 0.07<br>[0.02, 0.11]     | 24.76 $\pm$ 0.85                         | -19.88 $^{+0.26}_{-0.20}$ |
| 1102   | 215.091047 | 52.954285 | 6.998 | -2.57<br>$\pm 0.90$ | -                         | 6.8 $\pm$ 1.3  | 0.61<br>$\pm$ 0.18 | 0.14<br>[0.03, 0.23]     | 24.64 $\pm$ 0.83                         | -19.86 $^{+0.25}_{-0.44}$ |
| 1115   | 215.162818 | 53.073097 | 6.302 | -                   | 93 $\pm$ 6                | 2.0 $\pm$ 0.1  | < 0.11             | -                        | 25.97 $\pm$ 1.02                         | -                         |
| 1142   | 215.060716 | 52.958708 | 6.962 | -1.56<br>$\pm 0.44$ | -                         | 17.7 $\pm$ 5.2 | 0.46<br>$\pm$ 0.12 | 0.09<br>[0.03, 0.15]     | -                                        | -21.41 $^{+0.09}_{-0.30}$ |
| 1143   | 215.077006 | 52.969504 | 6.93  | -2.89<br>$\pm 0.67$ | 27 $\pm$ 1                | 5.6 $\pm$ 0.2  | 0.30<br>$\pm$ 0.07 | 0.24<br>[0.07, 0.44]     | 25.03 $\pm$ 0.89                         | -19.94 $^{+0.49}_{-0.24}$ |
| 1149   | 215.089714 | 52.966183 | 8.177 | -1.50<br>$\pm 0.62$ | 291 $\pm$ 21              | 7.4 $\pm$ 0.6  | 0.38<br>$\pm$ 0.09 | 0.07<br>[0.02, 0.11]     | 25.66 $\pm$ 0.97                         | -20.77 $^{+1.17}_{-0.15}$ |

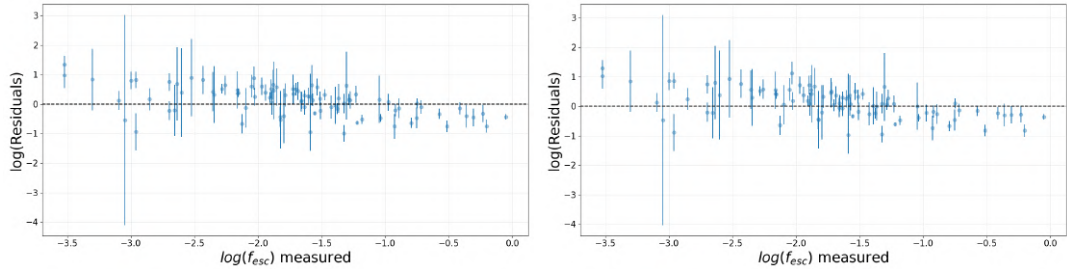
**Table 6.1:** *continued.*

| MSA ID | RA [deg]   | DEC [deg] | z     | $\beta$             | $EW_0(\text{H}\beta)$ [Å] | O32            | $r_e$ [kpc]        | $f_{\text{esc}}$ (pred.) | $\log \xi_{\text{ion}}$ [erg $^{-1}$ Hz] | $M_{1500}$ [mag]         |
|--------|------------|-----------|-------|---------------------|---------------------------|----------------|--------------------|--------------------------|------------------------------------------|--------------------------|
| 1160   | 214.805047 | 52.845877 | 6.569 | -2.20<br>$\pm 0.97$ | $93 \pm 27$               | $> 18$         | 0.26<br>$\pm 0.08$ | 0.13<br>[0.03, 0.22]     | $25.51 \pm 0.95$                         | $-19.60^{+0.22}_{-0.32}$ |
| 1163   | 214.990468 | 52.971990 | 7.450 | -3.12<br>$\pm 0.76$ | $19 \pm 11$               | $0.6 \pm 0.1$  | $0.61 \pm 0.16$    | 0.07<br>[0.02, 0.11]     | $24.97 \pm 0.88$                         | $-20.31^{+0.03}_{-0.44}$ |
| 1414   | 215.128029 | 52.984936 | 6.678 | -2.01<br>$\pm 0.02$ | -                         | -              | -                  | -                        | -                                        | $-20.92^{+0.00}_{-0.01}$ |
| 1518   | 215.006802 | 52.965041 | 6.110 | -2.95<br>$\pm 0.51$ | $124 \pm 87$              | $24 \pm 4.8$   | $0.64 \pm 0.08$    | 0.28<br>[0.08, 0.49]     | $25.62 \pm 0.97$                         | $-20.95^{+0.03}_{-0.17}$ |
| 1558   | 214.830637 | 52.835297 | 6.884 | -2.05<br>$\pm 0.01$ | $17 \pm 2$                | $0.4 \pm 0.1$  | $< 0.14$           | $> 0.57$                 | $23.97 \pm 0.74$                         | $-22.01^{+0.09}_{-0.11}$ |
| 1561   | 215.166097 | 53.070755 | 6.198 | -3.49<br>$\pm 0.67$ | $95 \pm 6$                | $15.1 \pm 3.2$ | $0.48 \pm 0.07$    | 0.38<br>[0.14, 0.66]     | $25.44 \pm 0.94$                         | $-19.56^{+0.03}_{-0.15}$ |
| 2355   | 215.008489 | 52.977973 | 6.112 | -2.06<br>$\pm 0.04$ | $49 \pm 7$                | $3.4 \pm 0.6$  | $0.18 \pm 0.03$    | 0.11<br>[0.04, 0.18]     | $24.83 \pm 0.86$                         | $-17.97^{+0.59}_{-0.86}$ |
| 23642  | 215.230033 | 53.015572 | 6.909 | -2.82<br>$\pm 0.40$ | -                         | $10.1 \pm 1.3$ | $1.30 \pm 0.23$    | 0.13<br>[0.03, 0.23]     | -                                        | $-20.90^{+0.24}_{-0.67}$ |
| 28944  | 214.867500 | 52.836872 | 6.056 | -1.73<br>$\pm 0.05$ | $24 \pm 12$               | $27.2 \pm 5.6$ | $2.11 \pm 0.06$    | 0.02<br>[0.01, 0.03]     | $24.82 \pm 0.85$                         | $-22.12^{+0.22}_{-0.15}$ |
| 31329  | 215.055116 | 53.000850 | 6.144 | -1.89<br>$\pm 0.37$ | $46 \pm 7$                | $9.5 \pm 0.6$  | $1.26 \pm 0.21$    | 0.03<br>[0.01, 0.04]     | $24.89 \pm 0.87$                         | $-21.76^{+0.33}_{-0.10}$ |
| 80025  | 214.806065 | 52.750867 | 7.655 | -1.90<br>$\pm 0.18$ | $83 \pm 9$                | $8.2 \pm 1.6$  | $0.44 \pm 0.07$    | 0.09<br>[0.03, 0.14]     | $25.55 \pm 0.96$                         | $-19.73^{+0.18}_{-0.13}$ |
| 80083  | 214.961276 | 52.842364 | 8.635 | -1.59<br>$\pm 0.07$ | $75 \pm 14$               | $28 \pm 1.6$   | $0.24 \pm 0.06$    | 0.18<br>[0.06, 0.29]     | $25.34 \pm 0.93$                         | $-18.64^{+0.07}_{-0.24}$ |
| 80710  | 214.884985 | 52.836045 | 6.552 | -3.07<br>$\pm 0.38$ | $47 \pm 8$                | -              | $< 0.14$           | $> 0.27$                 | $24.67 \pm 0.84$                         | $-18.06^{+0.39}_{-0.22}$ |
| 80917  | 214.933838 | 52.845785 | 6.155 | -2.18<br>$\pm 0.05$ | $39 \pm 9$                | $0.9 \pm 0.1$  | $0.65 \pm 0.19$    | 0.03<br>[0.01, 0.04]     | $24.87 \pm 0.86$                         | $> -19.14$               |
| 80925  | 214.948680 | 52.853273 | 6.754 | -1.97<br>$\pm 0.05$ | $230 \pm 26$              | $4.9 \pm 0.5$  | $0.65 \pm 0.12$    | 0.05<br>[0.02, 0.08]     | $26.08 \pm 1.03$                         | $-19.41^{+0.00}_{-0.04}$ |
| 80374  | 214.898074 | 52.824895 | 7.178 | -2.25<br>$\pm 0.01$ | $321 \pm 119$             | $> 16$         | $0.26 \pm 0.12$    | 0.23<br>[0.08, 0.38]     | $25.87 \pm 1.00$                         | $-18.96^{+0.13}_{-0.03}$ |
| 80596  | 214.771865 | 52.778189 | 6.544 | -2.00<br>$\pm 0.05$ | $37 \pm 17$               | $15.9 \pm 4.6$ | $0.74 \pm 0.19$    | 0.08<br>[0.03, 0.13]     | $25.00 \pm 0.88$                         | $-18.94^{+0.33}_{-0.18}$ |
| 81063  | 214.799110 | 52.725119 | 6.094 | -2.00<br>$\pm 0.05$ | $59 \pm 23$               | $31.0 \pm 3.7$ | $0.49 \pm 0.07$    | 0.18<br>[0.06, 0.29]     | $25.72 \pm 0.98$                         | $-19.24^{+0.16}_{-0.22}$ |
| 81068  | 214.820507 | 52.737148 | 6.276 | -2.01<br>$\pm 0.05$ | $74 \pm 3$                | $18.1 \pm 6.5$ | $< 0.14$           | $> 0.26$                 | $25.17 \pm 0.90$                         | $-18.35^{+0.00}_{-0.18}$ |

**Table 6.2:** *Physical and spectroscopic properties of non-CEERS galaxies.*

| PROG.        | ID     | RA [deg]    | DEC [deg]   | z     | $\beta$          | $EW_0(H\beta)$<br>[Å] | O32            | $r_e$ [kpc] | $f_{esc}$<br>(pred.) |
|--------------|--------|-------------|-------------|-------|------------------|-----------------------|----------------|-------------|----------------------|
| <b>DDT</b>   | 10025  | 3.59609     | -30.38581   | 7.875 | $-2.08 \pm 0.45$ | $139 \pm 30$          | $6.6 \pm 1.4$  | 0.40        | $0.11 \pm 0.09$      |
|              | 100004 | 3.60657     | -30.38093   | 7.884 | $-1.88 \pm 0.44$ | $> 130$               | $> 5$          | 0.40        | $0.07 \pm 0.05$      |
|              |        |             |             |       |                  |                       |                |             |                      |
| <b>GLASS</b> | 10000  | 3.60134     | -30.37923   | 7.884 | $-2.27 \pm 0.46$ | $76 \pm 13$           | $8 \pm 3$      | 0.20        | $0.21 \pm 0.16$      |
|              | 10021  | 3.60851     | -30.41854   | 7.288 | $-2.25 \pm 0.48$ | $104 \pm 24$          | $13 \pm 5$     | 0.68        | $0.10 \pm 0.09$      |
|              | 100001 | 3.60385     | -30.38223   | 7.875 | $-1.63 \pm 0.48$ | $39 \pm 8$            | $3.2 \pm 1$    | 0.50        | $0.04 \pm 0.03$      |
|              | 100003 | 3.60451     | -30.38044   | 7.880 | $-2.51 \pm 0.48$ | $85 \pm 18$           | $21 \pm 7$     | 0.15        | $0.40 \pm 0.22$      |
|              | 100005 | 3.60646     | -30.38099   | 7.883 | $-2.55 \pm 0.48$ | $33 \pm 15$           | $2.9 \pm 1$    | 0.25        | $0.15 \pm 0.12$      |
|              | 150008 | 3.60253     | -30.41923   | 6.230 | $-2.10 \pm 0.25$ | $141 \pm 30$          | $> 20$         | 0.40        | $0.08 \pm 0.07$      |
|              | 400009 | 3.60059     | -30.41027   | 6.376 | $-2.17 \pm 0.25$ | $35 \pm 7$            | -              | 0.11        | $0.07 \pm 0.06$      |
|              |        |             |             |       |                  |                       |                |             |                      |
| <b>ERO</b>   | 4590   | 110.8593287 | -73.4491656 | 8.496 | $-2.20 \pm 0.15$ | $218 \pm 150$         | $> 14.8$       | 0.71        | $0.08 \pm 0.07$      |
|              | 5144   | 110.8396739 | -73.4453570 | 6.378 | -                | $151 \pm 51$          | $18.6 \pm 3.3$ | 0.92        | -                    |
|              | 6355   | 110.8445942 | -73.4350590 | 7.665 | $-1.96 \pm 0.22$ | $150 \pm 4$           | $8.2 \pm 0.3$  | 0.83        | $0.03 \pm 0.02$      |
|              | 10612  | 110.8339649 | -73.4345232 | 7.660 | $-2.31 \pm 0.11$ | $210 \pm 16$          | $14.8 \pm 1.7$ | 0.42        | $0.14 \pm 0.12$      |
|              |        |             |             |       |                  |                       |                |             |                      |

## 6.8 Appendix B: An analysis on the empirical relation calibrated on the LzLCS sample



**Figure 6.11:** *Left: The difference between the measured values and the values estimated by the M23 empirical relation. Right: Same for the values estimated using for Eq. (6.2) of this paper.*

To accurately assess the reliability of the empirical relation calibrated on the LzLCS presented in two versions (Eq. 1 in M23 and Eq (6.2) of this study) for estimating the  $f_{esc}$  values using indirect indicators, we examined the distribution of the residuals which are defined as the difference between the measured values and the values estimated by our relation. In 6.11, we plot the residuals obtained using both versions of our relation plotted against the real values, in logarithmic scale. To assess the correlation quantitatively, we conducted an ANOVA test on the two residual distributions, yielding a p-value of 0.96. It is evident that

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the two relations exhibit statistical equivalence, as their residual distributions are identical. We can also see that our relation tends to underestimate the  $f_{\text{esc}}$  values for  $f_{\text{esc}} > 0.1$  and overestimate them for  $f_{\text{esc}}$  below 0.01. This outcome is a direct consequence of the initial sample, which is predominantly composed of sources with modest  $f_{\text{esc}}$ , around 0.02-0.05.



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## Chapter 7

# The impact of morphology on the production and escape of ionizing photons in the early Universe

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*The results in this chapter will be presented in Mascia et al. (in prep.).*

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Compact, star-forming galaxies with high star formation rate surface densities ( $\Sigma_{\text{SFR}}$ ) are often efficient Lyman continuum (LyC) emitters at  $z \leq 4.5$ , as intense stellar feedback likely creates low-density channels that allow photons to escape. Irregular or disturbed morphologies, such as those resulting from mergers, can also facilitate LyC escape by creating anisotropic gas distributions. We investigate the influence of galaxy morphology on LyC production and escape at redshifts  $5 \leq z \leq 7$  using observations from various *James Webb Space Telescope* (JWST) surveys. Our sample consists of 436 sources, which are predominantly low-mass ( $\sim 10^{8.15} M_{\odot}$ ), star-forming galaxies with ionizing photon efficiency ( $\xi_{\text{ion}}$ ) values consistent with canonical expectations. Since direct measurements of  $f_{\text{esc}}$  are not possible during the Epoch of Reionization (EoR), we predict  $f_{\text{esc}}$  for high-redshift galaxies by applying survival analysis to a subsample of the Low-Redshift Lyman Continuum Survey (LzLCS), which serves as an analogue for reionization-era galaxies. We find that these galaxies exhibit on average modest predicted escape fractions ( $\sim 0.04$ ). Additionally, we assess the correlation between morphological features and LyC emission. Our findings indicate that neither  $\xi_{\text{ion}}$  nor the predicted  $f_{\text{esc}}$  values show a significant correlation with the presence of merger signatures. This suggests that in low-mass galaxies at  $z \geq 5$ , strong morphological disturbances are not the primary mechanism driving LyC emission and leakage. Instead, compactness and star formation activity likely play a more pivotal role in regulating LyC escape.

### 7.1 Introduction

The Epoch of Reionization (EoR) marks a crucial phase in the Universe's history, beginning with the formation of the first stars at  $z \sim 20 - 30$  [e.g., 23] and

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concluding around  $z \sim 5.5$ , when reionization was complete [e.g., 45]. During this period, the first luminous sources, such as active galactic nuclei (AGNs) and massive stars in the early galaxies, emitted ionizing photons that reionized the neutral hydrogen in the intergalactic medium (IGM). While AGNs have been proposed as potential contributors to reionization [e.g., 111, 149, 283, 284], the consensus is that young, hot stars in star-forming galaxies were the primary drivers [12, 17, 394]. These stars produced ultraviolet (UV) radiation, specifically in the Lyman continuum (LyC,  $\lambda < 912 \text{ \AA}$ ), which ionized the surrounding hydrogen and played a key role in transforming the early Universe.

The ability of these galaxies to provide the necessary number of ionizing photons to reionize the Universe depends on the escape fraction ( $f_{\text{esc}}$ ) of the LyC photons, the percentage of LyC photons that successfully escape the galaxies' interstellar medium (ISM) and circumgalactic medium (CGM) to reach the IGM. An average  $f_{\text{esc}}$  of around 10% across all galaxies is estimated to be necessary to match the reionization history, as predicted by independent constraints including cosmic microwave background (CMB) data [368] and high-redshift galaxy luminosity functions [e.g., 149, 393, 525, 526]. However, directly measuring  $f_{\text{esc}}$  at  $z \geq 4.5$ , where reionization occurred, is impossible due to the increasing IGM opacity at that epoch [208].

One avenue to address this challenge is to study low-redshift galaxies with measurable LyC leakage, to search for properties that are analogous to those of galaxies in the EoR. These lower-redshift galaxies, known as Lyman continuum emitters (LCEs), allow astronomers to measure LyC escape fractions and explore the conditions in the ISM that facilitate LyC photon escape [86, 153, 154, 213, 214, 402, 409, 414, 497]. These measurements are then used to infer the average  $f_{\text{esc}}$  of the cosmic reionizers [e.g., 233, 272, 300, 302, 398, 412], consistently finding an average  $f_{\text{esc}}$  lower than 0.1.

Among the many programs analyzing the properties of low-redshift LyC leakers, the most comprehensive is the LzLCS survey (GO 15626, PI Jaskot) [153, 154, 402]. The LzLCS sample was augmented with archival COS observations from [215–218, 501]. The combined dataset, referred to as the LzLCS+ sample, includes 88 low-redshift galaxies with either detections or stringent limits on LyC emission from *Hubble Space Telescope* (HST)-COS spectra. Using complete spectroscopic and photometric information for all sources, the survey tested various indirect diagnostics to understand their correlation with LyC emission. Their analysis found that indicators based on Ly $\alpha$  emission are particularly effective: LyC emitters tend to have a high Ly $\alpha$  escape fraction and a small velocity peak separation [214]. Additionally, properties such as high [OIII] $\lambda\lambda 4960, 5008$ /[OII] $\lambda 3727$  (O32) flux ratio, high specific star formation rate surface density ( $\Sigma_{\text{sSFR}}$ ), and a blue UV slope  $\beta$  [86] also correlate with a high LyC escape [154].

Although many properties correlate with  $f_{\text{esc}}$ , such relations are extremely scattered: for this reason several authors have refined methods to predict LyC emission using a combination of more properties, the so-called multivariate diagnostics at  $z \sim 0.3$ . [273, 300, 302] employed traditional regression methods, such as multivariate linear regression. However, these models often struggle with censored data – datasets that include upper limits rather than exact measurements.

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Additionally, these methods assume a linear relationship between the variables and  $f_{\text{esc}}$ , a condition that may not always be met. Recently, [225, 226] proposed survival analysis techniques, which are better suited to the nature of the LzLC+ dataset, with its broad range of  $f_{\text{esc}}$  values and numerous non-detections (upper limits). Ignoring these non-detections could lead to biased predictions and inaccurate insights into what differentiates galaxies that emit LyC radiation from those that do not [e.g., 209]. One of the most promising models with high concordance ( $C > 0.7$ ) is the TopThree [226], which incorporates three key predictors: the star formation rate surface density ( $\Sigma_{\text{SFR}} = \text{SFR}/(2\pi r_e^2)$ ), where  $r_e$  is the UV half-light radius, the O32 ratio, and the UV slope ( $\beta$ ). It was already noted that a high  $\Sigma_{\text{SFR}}$  correlates with efficient LyC photon escape [e.g., 154, 199, 331], particularly in compact galaxies where intense star formation feedback creates low-density channels in the ISM. These channels facilitate LyC photon escape, resulting in a higher  $f_{\text{esc}}$ . For instance, J1316+2614, a galaxy at  $z = 3.613$ , is noted as one of the brightest LyC emitters known, with an escape fraction of approximately 0.9. This galaxy’s compact morphology, characterized by a half-light radius of  $r_e \sim 0.22$  kpc and an extremely high  $\log_{10} \Sigma_{\text{SFR}} \approx 3.47 \text{M}_{\odot} \text{yr}^{-1} \text{kpc}^2$ , suggests that intense star formation efficiency plays a critical role in driving LyC escape [297, 299].

Although compactness and high  $\Sigma_{\text{SFR}}$  are common in many LCEs, this is not universally true. Some LCEs exhibit more extended or irregular structures, likely shaped by processes such as galaxy mergers or interactions [e.g., 37]. For example, Haro 11 at  $z = 0.02$  and z19863 at  $z = 3.088$  demonstrate a modest LyC escape despite lacking compact morphologies [e.g., 184, 385]. In addition, [312] reported LyC emission from a merging system at  $z = 1.097$ . Mergers can lead to inhomogeneous gas distributions, where tidal interactions strip gas and stars, resulting in complex morphologies [100, 444, 473]. These irregularities can result in anisotropic gas distributions, with pockets of optically thin neutral hydrogen allowing LyC photons to escape. In Haro 11, deep 21 cm observations have revealed reduced gas density pockets, likely due to recent merger activity, facilitating LyC photon escape [263]. Further supporting the link between mergers and LyC escape, [524] observed offset LyC emission at  $z = 3.797$  and proposed that galaxy mergers could facilitate this escape, as indicated by double-peaked Ly $\alpha$  profiles and interaction signatures in high-resolution imaging. Similarly, [227] analyzed Ion1 ( $z = 3.794$ ), where LyC emission is spatially offset from non-ionizing UV radiation. This suggests that the UV radiation from Ion1 originates in star-forming regions, likely young star clusters or associations. According to the picket-fence model, LyC photons escape through a few optically thin channels in an otherwise optically thick ISM. While non-ionizing UV photons can escape from most star-forming regions, ionizing photons are limited to low HI column density channels, allowing detected LyC photons to travel through these low-absorption sight-lines.

The diversity in LCE morphologies, ranging from compact galaxies like J1316+2614 to more extended and irregular systems like Haro 11, highlights the importance of galaxy structure and gas distribution in determining LyC escape. In particular, the anisotropic coverage of neutral gas, rather than relying solely on simplified one-dimensional indicators like the O32 ratio – which can also be influ-

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enced by variations in metallicity and ionization parameters [e.g., 26, 240] – may play a crucial role in regulating the escape of LyC photons. Studies have shown that LCEs with disturbed morphologies often exhibit weaker correlations between the O32 ratio and  $f_{\text{esc}}$ , suggesting that gas geometry and distribution are critical factors [26].

With the advent of the *James Webb Space Telescope* [JWST, 161], we now have the capability to probe the indirect indicators at unprecedented depths, enabling the measurement of  $f_{\text{esc}}$  for a large and diverse sample of galaxies during the EoR. Thanks to JWST’s Near Infrared Camera (NIRCam), we are also refining our understanding of the morphology of these early sources [e.g., 105, 477]. Current findings indicate that signatures of mergers and interactions are detected in approximately 30% of the galaxies studied at  $z \geq 5.5$ . Recent work by [65] showed that, in the GLASS and CEERS spectroscopic surveys,  $\Sigma_{\text{SFR}}$  at  $z \geq 7$  correlates with the O32 ratio and therefore with indirect estimates of the LyC escape fraction. Recent findings by [434] demonstrate that galaxies during the EoR that are characterized by low stellar masses and bursty SFHs exhibit the highest values of  $\xi_{\text{ion}}$ , driven by the dominance of younger stellar populations. This bursty star formation scenario is consistent with recent results from stacking analysis of the LzLCS sample [155], which suggest that such SFHs provide the optimal feedback and geometric conditions necessary for efficient LyC escape.

This work aims at exploring in a systematic way the impact of galaxy morphology, star formation activity, and gas distribution on the physical processes driving LyC production and escape in galaxies during the EoR.

The paper is structured as follows: in Sec. 7.2, we describe the sample selection. In Sec. 7.3, we present the methodology for line measurements, the SED fitting process, the general properties of the sources and the mergers identification criteria, while in Sec. 7.4 we introduce a revised survival analysis to predict the  $f_{\text{esc}}$  for our sample. In Sec. 7.5, we report the predicted  $f_{\text{esc}}$ , analyze the correlation of the presence of merger signatures with  $f_{\text{esc}}$  and  $\xi_{\text{ion}}$ , and interpret our results. In Sec. 7.6, we summarize our key findings. Throughout this work, we assume a flat  $\Lambda$  cold dark matter cosmology with  $H_0 = 67.7 \text{ km s}^{-1} \text{ Mpc}^{-1}$  and  $\Omega_m = 0.307$  [368] and the [78] initial mass function. All magnitudes are expressed in the AB system [347].

## 7.2 Data

### 7.2.1 EIGER data

We selected 133 sources from the Emission-line Galaxies and Intergalactic Gas in the Epoch of Reionization (EIGER) survey [GTO 1243, PI: Lilly, 238]. This study leverages the first deep  $3.5 \mu\text{m}$  JWST/NIRCam Wide Field Slitless Spectroscopy (WFSS) observations from the EIGER program, focusing on spectroscopically confirmed [OIII] emitters at redshifts  $z = 5.33 - 6.93$ . These sources are located in the field of the bright quasar SDSS J010013.02+280225.8, though the majority are not directly associated with the quasar. A comprehensive list of identified sources and detection methods is available in [238, 307]. The dataset is further

**Table 7.1:** *Summary of JWST datasets and the count of confirmed sources with redshifts  $5 \leq z \leq 7$  included in this study.*

| Prog. name | PI           | Target                   | Prog ID  | $N_{\text{sources}}$ |
|------------|--------------|--------------------------|----------|----------------------|
| EIGER      | Lilly        | SDSS J010013.02+280225.8 | GTO 1243 | 133                  |
| CEERS      | Finkelstein  | CANDELS-EGS              | ERS 1345 | 28                   |
| DD-2750    | Arrabal Haro | CANDELS-EGS              | DDT 2750 | 15                   |
| JADES      | Eisenstein   | GOODS-S                  | GTO 1180 | 90                   |
| JADES      | Eisenstein   | GOODS-N                  | GTO 1181 | 108                  |
| JADES      | Eisenstein   | GOODS-S                  | GO 3215  | 24                   |
| JADES      | Luetzgendorf | GOODS-S                  | GTO 1210 | 30                   |
| JADES      | Luetzgendorf | GOODS-S                  | GTO 1286 | 8                    |

complemented by EIGER NIRCam imaging in three broadband filters (F115W, F200W, and F356W). For details on the imaging data reduction and analysis, we refer to [238] and [307].

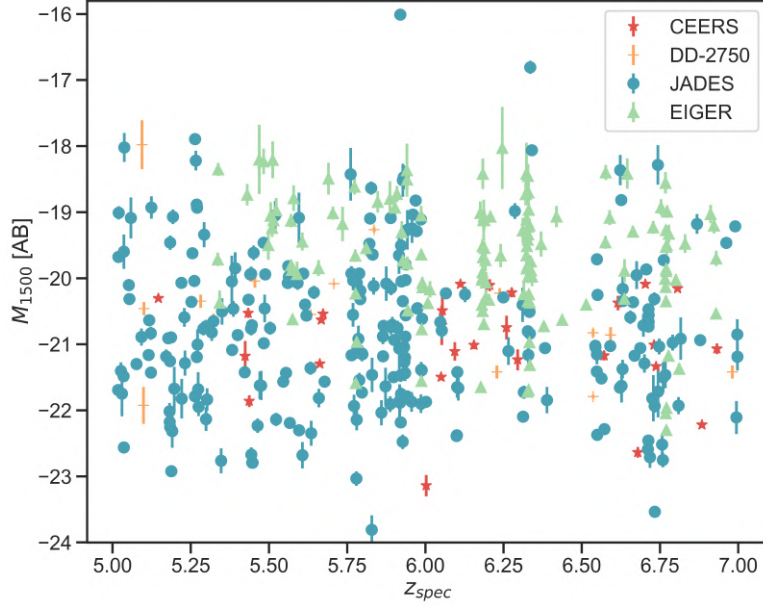
This section provides an overview of our analysis and the methods employed to study the properties of galaxies in our sample. Our focus is on sources in the redshift range  $5 \leq z \leq 7$ , which were identified using NIRCam WFSS in the F356W filter, where the [OIII] and  $H\beta$  emission lines are detectable. For point sources, the spectral resolution is approximately  $R \sim 1500$  at  $3.5 \mu\text{m}$ , with a dispersion of about 1 nm per pixel.

The [OIII] emitters were identified using two complementary approaches: the “backward” and “forward” methods. In the backward approach, emission-line pairs (such as [OIII] $\lambda\lambda 4960, 5008$ ) or triplets (including  $H\beta$ ) were identified by running **SExtractor** on combined continuum-filtered grism images, which were stacked per visit and module. The corresponding galaxy was then located in the direct image based on the estimated wavelength derived from the doublet separation. In the forward approach, **SExtractor** was run on the direct F356W image, and spectra were extracted from the emission-line image (created by subtracting the median-filtered continuum images from the reduced data) at each galaxy’s expected position, using a tracing model to identify line pairs near the trace center.

Emission lines were detected with a minimum signal-to-noise ratio (S/N) of 3. The  $3\sigma$  limiting sensitivity of the spectroscopy varied across the field and with wavelength (by a factor of approximately 2), with the best sensitivity reaching  $0.6 \times 10^{-18} \text{ erg s}^{-1} \text{ cm}^{-2}$  at  $3.8 \mu\text{m}$ . After thorough visual inspection and reconciliation of objects identified by both methods, followed by iterative refinement of search parameters, 133 resolved [OIII] emitting components within the redshift range  $z = 5.33 - 6.93$  have been identified, each with at least two detected emission lines at  $S/N > 3$ . The typical S/N for the [OIII] $\lambda 5008$  line is 14 (ranging from 6 to 70), while  $H\beta$  is detected with  $S/N > 3$  in 68 objects, with 31 of those objects reaching  $S/N > 5$ . Only 3 of the 133 objects were primarily identified via  $H\beta$ , where [OIII] $\lambda 4960$  had  $S/N < 3$ .

### 7.2.2 CEERS and DD-2750

We selected sources from the Cosmic Evolution Early Release Science survey (CEERS; ERS 1345, PI: Finkelstein) in the CANDELS Extended Groth Strip



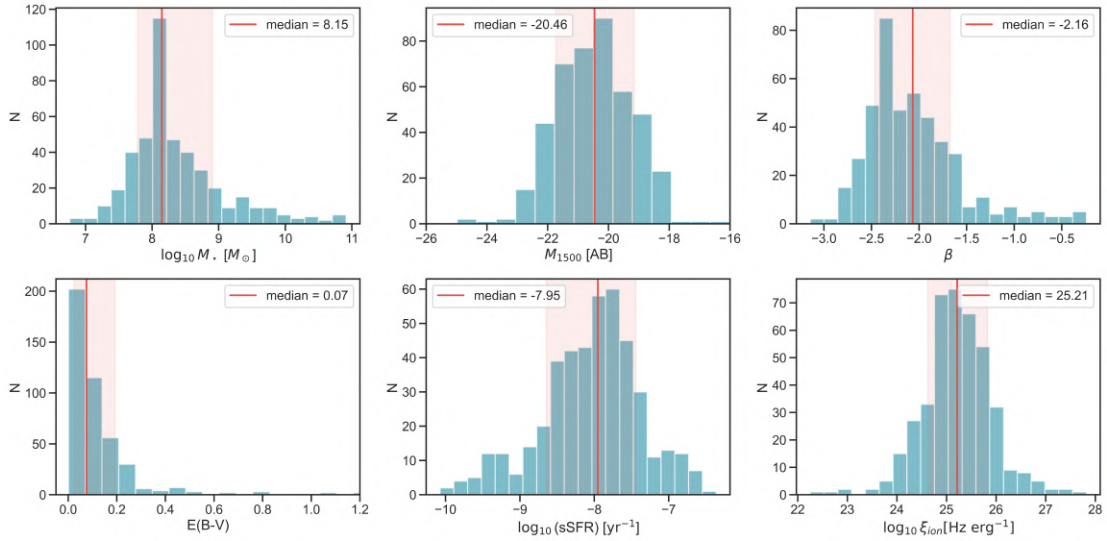
**Figure 7.1:**  $M_{1500} - z$  distribution for the 436 sources at  $5 \leq z \leq 7$  analyzed in this work.

(EGS) field [177, 251]. In particular we selected all the sources with  $5 \leq z_{\text{spec}} \leq 7$  from [302, 342] that have a NIRSpec spectrum obtained either with the three medium-resolution ( $R \approx 1000$ ) grating spectral configurations (G140M/F100LP, G235M/F170LP and G395M/F290LP), which, together, cover wavelengths between  $0.7\text{--}5.1 \mu\text{m}$ , or with the PRISM/CLEAR configuration, which provides continuous wavelength coverage of  $0.6\text{--}5.3 \mu\text{m}$  with spectral resolution ranging from  $R \sim 30$  to 300. The total number of sources is 28.

All sources have available photometry with JWST/NIRCam filters (F115W, F150W, F200W, F277W, F356W, F444W, F410M), as well as HST/ACS filters (F606W, F814W) and HST/WFC3 filters (F105W, F125W, F140W, F160W) [19, 146, 147]. To maintain consistency with the EIGER sample, we applied a cut in the F115W magnitude at 28.5 AB during the selection process.

In addition we also considered sources from a DDT program on the same field (DDT 2750, PI: Arrabal Haro). These sources were observed using the PRISM/CLEAR configuration and possess the same photometric information as the sources from the CEERS sample. We determined the redshifts of the galaxies by analyzing the 1D spectrum with the redshift-fitting module of MSAEXP<sup>1</sup>. This module assesses the goodness of fit across a redshift range of 0 to 15, employing a series of EAZY [56] galaxy and line templates at the nominal NIRSpec/PRISM resolution. Each galaxy was fitted individually, and the resulting solutions were visually inspected. Similar to the CEERS sources, during the selection phase we implemented a cutoff for the F115W magnitude at 28.5 AB. In total we added 15 sources from this sample, in the redshift range of interest.

<sup>1</sup><https://github.com/gbrammer/msaexp>



**Figure 7.2:** *Stellar mass  $\log_{10}(M_*)$ , UV magnitude  $M_{1500}$ ,  $\beta$  slope,  $E(B-V)$ , and  $\log_{10}(sSFR)$  for 436 sources in our sample, derived using **Prospector**. The ionizing photon production efficiency,  $\log_{10} \xi_{ion}$ , has been calculated for all sources with detectable  $H\alpha$  or  $H\beta$  using the relations proposed by [267, 416]. The figure includes the median values along with their standard deviations (16th and 84th percentiles) for each distribution.*

### 7.2.3 JADES

We selected sources from the JWST Advanced Deep Extragalactic Survey [JADES, GTO 1180, GTO 1210, GTO 1286 and GO 3215, 128] in the Great Observatories Origins Deep Survey [GOODS, 169] South field and in the GOODS North field. All sources have available photometry with JWST/NIRCam filters (F115W, F150W, F200W, F277W, F356W, F444W, F410M), as well as HST/ACS filters (F606W, F814W) and HST/WFC3 filters (F105W, F125W, F140W, F160W) [316].

We restricted the sample to sources with a spectroscopic redshift  $5 \leq z \leq 7$  from the Data Release 3 [115]. The sources have a NIRSpec spectrum obtained either with the three medium-resolution grating spectral configurations (G140M/F100LP, G235M/F170LP and G395M/F290LP), or with the PRISM/-CLEAR configuration. We included all sources within the specified redshift range flagged as “A”, “B”, or “C”, indicating emission lines detected in the grism or prism configuration, respectively, or secure redshifts visually identified from spectral breaks and/or low signal-to-noise ratio (S/N) emission lines. The final number of sources selected is 152 in the GOODS-S field and 108 in the GOODS-N field.

The list of the data sets along with their targeted fields is presented in Table 7.1. When combined with the other samples, our final dataset comprises a total of 436 sources observed during the epoch of reionization. Fig. 7.1 illustrates the distribution of our total sample across the  $z - M_{1500}$  plane. All  $M_{1500}$  values were computed using **Prospector** (see Sec. 7.3).

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## 7.3 Methods

### 7.3.1 SED fitting

The physical properties of the EIGER sources, presented in [307], have been estimated using the **Prospector** code [230]. To ensure consistency in the derivation of physical properties across our entire sample, we utilized the same code, incorporating a nebular treatment based on Cloudy version 13.03 [62, 140], to fit the SEDs of sources from public programs. This approach allowed us to fit the available multiband JWST and HST photometry uniformly across the dataset.

Consistent with the methodology outlined by [307], the free parameters in our modeling included the total formed stellar mass, stellar metallicity, star formation history, dust attenuation, gas-phase metallicity, and ionization parameter. The redshift was fixed based on the spectroscopic measurement, and we assumed a [78] initial mass function along with MIST isochrones [87, 122]. The star formation history followed a delayed- $\tau$  model,  $\psi(t) = \psi_0 t e^{-t/\tau_T}$ . Both stellar and nebular emissions were attenuated by dust using a simple screen model with a [69] law.

We employed uniform and wide priors in our model: the stellar mass was allowed to vary between  $10^6$  and  $10^{11} M_\odot$ , stellar metallicity between  $[Z/H] = -2.0$  and  $+0.2$ , dust optical depth  $\tau$  between 0 and 2, stellar age between 1 Myr and the age of the Universe at the galaxy’s redshift, and the star formation timescale  $\tau_T$  between 100 Myr and 20 Gyr. Gas-phase metallicity was modeled in the range  $12 + \log(\text{O}/\text{H}) = 6.7$  to  $9.2$ , while the ionization parameter  $U$  spanned from  $-3$  to  $+1$  on a logarithmic scale, allowing for higher values to provide greater flexibility in the fits.

The distribution of the main properties of the sources in this sample are presented in Fig. 7.2. The resulting SED fits suggest that in our galaxies dust attenuation is minimal, with a median  $E(B - V) = 0.07$ . The galaxies’ observed UV magnitudes range from  $M_{\text{UV}} = -16.1$  to  $-24.6$  (with a typical  $M_{\text{UV}} = -20.46$ ), and their stellar masses span three orders of magnitude, from  $\log_{10}(M/M_\odot) = 6.76$  to  $10.9$ , with a median mass of  $10^{8.15} M_\odot$ .

### 7.3.2 Line measurements and $\xi_{\text{ion}}$ estimation

The catalog of emission lines detected in the EIGER sample is available [238], providing fluxes and equivalent widths (EWs) for the  $\text{H}\beta$  and  $[\text{OIII}]$  doublet lines.

For the remaining samples, we measured emission lines using LIME<sup>2</sup> [142], a versatile tool for analyzing medium-resolution and prism/clear 1D spectra. The tool requires the source’s 1D spectrum, spectroscopic redshift, and a predefined list of lines to fit. Our analysis focused on key emission lines, including  $[\text{OII}]$ ,  $\text{H}\beta$ ,  $[\text{OIII}]$ , and  $\text{H}\alpha$ . LIME estimates the continuum and computes direct integration and Gaussian fluxes for lines with a signal-to-noise ratio (S/N) above 3, as well as their EWs.

Before any quantitative analysis we corrected the line fluxes for dust reddening. Since  $\text{H}\alpha$  is unobserved for most sources, we used stellar  $E(B-V)$  values derived from SED fitting to calculate the dust attenuation for each emission line, applying

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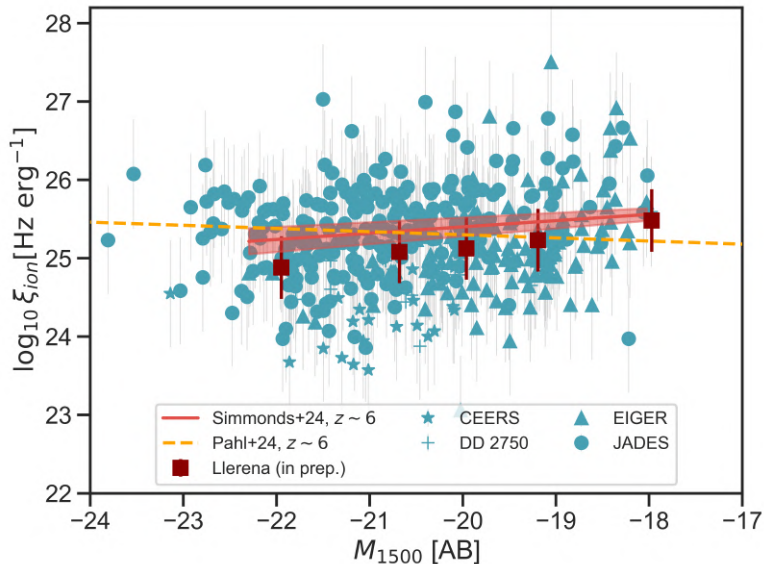
<sup>2</sup><https://lime-stable.readthedocs.io/en/latest/>



the [69] extinction law. The total attenuation in magnitudes was given by  $A_V = 0.44 \times E(B - V) \times \kappa$ , where  $\kappa$  is the wavelength-dependent attenuation coefficient for each line. The corrected fluxes were obtained by multiplying the observed fluxes by  $10^{A_V}$ , with  $\kappa_{H\beta} = 4.60$ ,  $\kappa_{[\text{OIII}]}$  = 4.46, and  $\kappa_{[\text{OII}]} = 5.86$  [68]. We then computed O32 line ratios and rest-frame EWs for the [OIII] doublet and/or  $H\beta$ .

Following [267] and [416], we inferred  $\log \xi_{\text{ion}}$  from the dust-corrected  $H\alpha$  or  $H\beta$  luminosity, as these lines are clearly detected in 413 out of 436 sources (95% of the sample). The distribution of  $\log \xi_{\text{ion}}$  is shown in Fig. 7.2, with a median  $\log \xi_{\text{ion}}$  of 25.21.

In Fig. 7.3 we present the  $\xi_{\text{ion}}$  versus  $M_{\text{UV}}$ . We observe that  $\xi_{\text{ion}}$  increases with  $M_{\text{UV}}$ , consistent with the predictions at  $z \sim 6$  from [371, 434], and Llerena (in prep.). In our sample, fainter galaxies tend to have higher  $\xi_{\text{ion}}$  values compared to brighter galaxies, although there is considerable scatter. This finding is in contrast to recent results by [357], which suggest that fainter galaxies are less efficient at producing ionizing photons than their brighter counterparts.



**Figure 7.3:**  $\log \xi_{\text{ion}}$  versus  $M_{\text{UV}}$ . Each data point corresponds to an individual galaxy from our sample, with different symbols representing the various surveys. We include the curves from [357, 434], as well as the median data points from Llerena (in prep.) at  $z \sim 6$ .

### 7.3.3 Mergers identification

In this study, we aim to determine whether galaxy mergers create channels that facilitate the production and escape of LyC photons during reionization. To investigate this, we analyze NIRCам F115W images, available for all the surveys utilized in this work. These images trace the rest-frame UV at  $\lambda_{\text{mean}} \sim 1181.4 \text{ \AA}$  for galaxies at  $z = 5$  and  $\lambda_{\text{mean}} \sim 886 \text{ \AA}$  for galaxies at  $z = 7$ . Following [477], we create a segmentation map for each object to reduce noise and enhance low surface-brightness regions, particularly useful for galaxies with low signal-to-noise ratios [361]. We apply a  $6 \times 6$  uniform filter to the images and use the `photutils`

package to generate the map, requiring at least 5 connected pixels with flux  $2\sigma$  above the background. The binary detection mask (MD) is defined as the segmentation area corresponding to the target, excluding neighbors. From MD, we derive the galaxy radius  $R_{\text{max}}$ , the maximum pixel distance from the centroid [more effective than the Petrosian radius for disturbed or low S/N morphologies, 361].

For all the images, we also measure the morphological parameters required for merger identification [97, 278]. Following [105], to identify mergers during the EoR, we rely on three well-established morphological parameters: the Gini coefficient ( $G$ ), the second-order moment of brightness ( $M_{20}$ ), and asymmetry ( $A$ ) [see 93, for a review].

The Gini coefficient ( $G$ ) quantifies the distribution of light across a galaxy's pixels [2]. A higher  $G$  value indicates a more uneven light distribution, with mergers typically showing greater central concentration of light due to the dynamical processes involved [e.g., 278]. Mathematically, it is expressed as:

$$G = \frac{1}{\bar{X}n(n-1)} \sum_i^n (2i - n - 1)X_i, \quad (7.1)$$

where  $X_i$  is the intensity of the  $i$ th pixel,  $n$  is the total number of pixels assigned to the galaxy, and  $\bar{X}$  is the mean intensity.

The second-order moment of brightness ( $M_{20}$ ), introduced by [277], measures the spatial concentration of the brightest 20% of a galaxy's light. Mergers are expected to have higher  $M_{20}$  values due to the presence of off-center bright regions resulting from tidal interactions. The parameter is given by:

$$M_{20} = \log_{10} \left( \frac{\sum_i M_i}{M_{\text{tot}}} \right), \text{ with } \sum_i f_i < 0.2f_{\text{tot}}, \quad (7.2)$$

where  $M_i$  is the second-order moment of the  $i$ th pixel and  $M_{\text{tot}}$  is the total second-order moment of the galaxy's light.

The asymmetry parameter ( $A$ ), as defined by [1] and [95], measures how symmetric a galaxy appears after rotating its image by 180 degrees. Mergers typically have higher asymmetry values due to the disrupted and irregular structures caused by the merging process:

$$A = \frac{\sum |I - I_\pi|}{\sum I} - A_{\text{bkg}}, \quad (7.3)$$

where  $I$  is the original image,  $I_\pi$  is the image rotated by 180 degrees, and  $A_{\text{bkg}}$  is the asymmetry of the background.

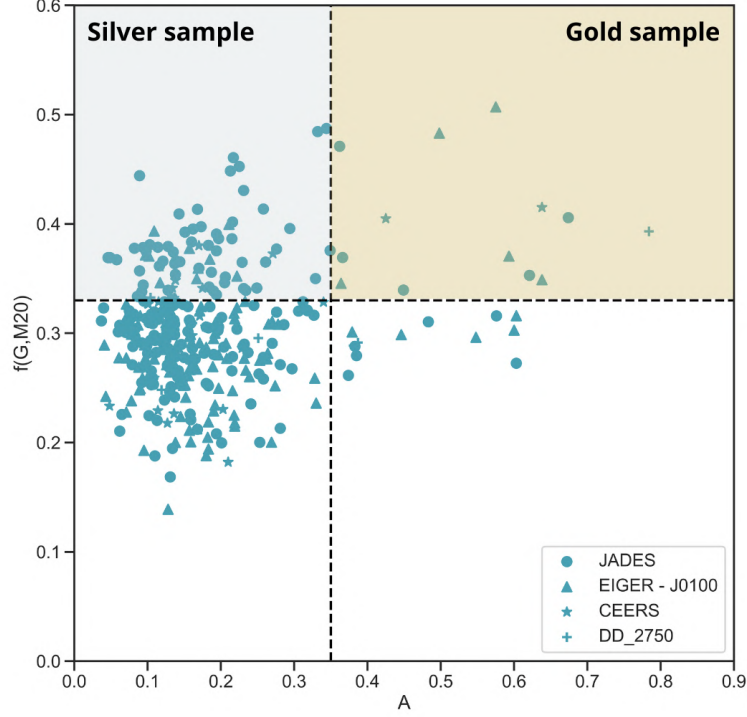
We successfully performed these measurements for 420 out of 436 sources (96% of the sample).

To identify mergers during the EoR, we apply two commonly used criteria from [97] and [278]:

$$f(G, M_{20}) = G + 0.14M_{20} > 0.33, \quad (7.4)$$

$$A \geq 0.35. \quad (7.5)$$

These criteria, initially developed for galaxies at  $z \lesssim 1.2$  [96], have been shown to be applicable at higher redshifts [e.g., 94, 105, 477, 500].



**Figure 7.4:** Visual representation of our classification results. Symbols are the same as in Fig. 7.3. The two dashed lines indicate the threshold values determined from Eqs. (7.4) and (7.5).

In accordance with [105], we define two merger samples:

- Silver sample: galaxies that satisfy the Gini- $M_{20}$  criterion (Eq.7.4).
- Gold sample: a subset of the silver sample, consisting of galaxies that also meet the asymmetry criterion (Eq.7.5).



**Figure 7.5:** Non-merger and merger candidates from the various surveys considered in this study, classified into Silver and Gold samples. Each row shows two representative sources for each category: non-merging systems, Silver sample mergers (selected based on the  $f(G, M_{20})$  criterion, Eq. (7.4)), and Gold sample mergers (satisfying both  $f(G, M_{20})$  and  $A$  criteria, Eqs. (7.4) and (7.5)). The cutouts, in the F115W filter, show the rest-frame UV at the redshift of each galaxy. The source ID, redshift, asymmetry ( $A$ ), and Gini- $M_{20}$  ( $f(G, M_{20})$ ) values are provided for each stamp.

The Gold sample identifies mergers with more pronounced asymmetry, indicat-

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ing more disruptive merger events, while the Silver sample captures mergers based primarily on light concentration. The results of this classification are illustrated in Fig. 7.4, with the two dashed lines indicating the threshold values calculated from Eqs. (7.4) and (7.5). Each point on the plot represents a galaxy from our sample. A visual comparison between merger and non-merger candidates in our sample is provided in Fig. 7.5, where the associated  $f(G, M_{20})$  and  $A$  are also listed for reference. To ensure the accuracy of their classifications, we conducted a thorough visual inspection of all sources in both the Gold and Silver samples, finding no ambiguities.

The distribution of merger and non-merger signatures in our sample (Fig. 7.4) offers important insights into the nature of galaxies during the EoR. Quantitatively, only 13 out of 436 galaxies are part of the Gold sample (merger fraction  $f_m = 0.03 \pm 0.01$ ), and 78 belong to the Silver sample (merger fraction  $f_m = 0.17 \pm 0.01$ ). Interestingly, 18 galaxies show highly asymmetric shapes, though they do not display clear signs of mergers. The vast majority – 311 galaxies – are compact systems. This distribution indicates that during the EoR, galaxies tend to be more compact rather than have major morphological disturbances.

Our findings are consistent with previous studies. For instance, [105] reported a merger fraction for sources brighter than  $M_{UV} = -20.1$  for the Gold sample at  $z \sim 6$  of  $f_m = 0.06 \pm 0.03$  and for the Silver sample of  $f_m = 0.22 \pm 0.04$ , based on high-resolution NIRC2 JWST data in the low-to-moderate magnification ( $\mu < 2$ ) regions of the Abell 2744 cluster field. Our merger fractions are consistent with these findings, further suggesting that galaxy mergers may not be the dominant mechanism driving galaxy assembly during the EoR. Additionally, their analysis revealed no significant differences in specific star formation rates between merger and non-merger signatures, reinforcing the notion that interactions may not play a major role in regulating star formation during this period.

## 7.4 A revised model to infer $f_{\text{esc}}$ using low-redshift analogs

### 7.4.1 Identifying LzLCS+ analogs of EoR sources

The LyC emitters from the LzLCS+ survey exhibit a wide range of physical properties, making them a diverse sample for studying LyC escape [153]. Their stellar masses vary from approximately  $10^{7.2} M_\odot$  to  $10^{9.3} M_\odot$ , and their  $M_{UV}$  range from  $-21.33$  to  $-18.29$ . The sample also shows considerable variability in dust content, reflected by  $\beta$  slopes spanning from  $-2.73$  to  $-1.6$  and reddening values,  $E(B-V)$ , between 0.013 and 0.206. Additionally, their O32 ratios range from  $\log_{10}(\text{O32}) = 0.31$  to 1.54. Because LzLCS+ galaxies span a much wider range of properties than those observed during the EoR, only a fraction of them can be considered true analogs of the sources currently detected by JWST in that era.

To account for these differences, we aim to refine our predictions for  $f_{\text{esc}}$  during reionization by comparing our sample of galaxies at  $5 \leq z \leq 7$  with the LzLCS+ emitters, as well as other intermediate-redshift LyC leakers for which we have crucial information such as the  $\beta$  slope, stellar mass,  $M_{UV}$ , and dust reddening  $E(B-V)$ . In Fig. 7.6, we show the comparison between various properties of the

galaxies, including  $\beta$  slope vs. stellar mass,  $\beta$  vs.  $M_{\text{UV}}$ , and  $\beta$  vs. dust reddening  $E(B-V)$ . The grey-shaded area represents the density distribution of the sources in our sample, with the contours in each panel indicating the 99% interval of the distributions for reionization-era galaxies. Based on the four properties shown in the figure, we find that, out of a total sample of 88 LyC emitters at  $z \sim 0.3$ , only 51 galaxies match the properties of sources during the epoch of reionization, as they consistently fall within the 99% distribution in each panel of the figure. The mean properties of this subsample indicate that the analogs are dust-poor ( $E(B-V) = 0.1$ ), have a blue spectral slope ( $\beta = -2.1$ ), a high O32 ( $\sim 9$ ), a low stellar mass ( $\log M_{\star} = 8.6 M_{\odot}$ ), a  $M_{\text{UV}}$  of approximately  $-19.65$ , and are compact in the UV ( $r_e \sim 0.559$  kpc). Additionally, we also show known LyC leakers at  $z \sim 3$  from the literature [e.g., 112, 152, 234, 244, 297, 485, 487, 490, 491, 523], highlighting that even at this redshift, these galaxies exhibit a wide range of properties, not all of which make them analogs of the sources observed during the reionization.

In Fig. 7.7 we present the results of a Kendall  $\tau$  rank correlation analysis conducted to evaluate the relationship between the predictors identified by [226] and  $f_{\text{esc}}$  values from the LzLCS+ sample. We followed the approach outlined by [154]<sup>3</sup>, computing the Kendall  $\tau$  rank correlation coefficient using the [5] method, which accounts for censored data. We perform this analysis both for the entire sample of 88 sources and for the refined subsample of 51 galaxies. Notably, the correlation between  $f_{\text{esc}}$  and  $M_{\text{UV}}$  is stronger for the subsample. Conversely, the correlations with other indicators, such as  $\log M_{\star}$ ,  $\beta$ ,  $\log_{10}(\text{O32})$ , and  $\log_{10}([\text{OIII}]+\text{H}\beta)$  EWs, show only minor variation, suggesting that their predictive power remains relatively stable across both samples. However, the correlation involving the Ly $\alpha$  EW shows the greatest variation.

We examined whether the reduction in sample size could explain the changes in Kendall  $\tau$  rank correlations. To do this, we generated 200 random subsamples, each containing 51 sources (the same size as our subsample), and calculated the correlation coefficients for each. We found that, for most of the indirect indicators, the decrease in correlation strength observed in the 51-source subsample can be attributed to the reduced sample size, suggesting that the weakening of these correlations is largely a statistical effect. However, the observed change in correlation strength for Ly $\alpha$  EW was significantly larger than expected from sample size reduction alone. Furthermore, the increase in correlation with  $M_{\text{UV}}$  could not be explained by the reduced sample size. Therefore, the variations in  $M_{\text{UV}}$  and Ly $\alpha$  EW correlations likely reflect real physical differences, rather than just the impact of a smaller sample size.

These findings align with those of [90], who explore the connection between LyC leakage and Ly $\alpha$  emission in seven gravitationally lensed Ly $\alpha$  emitters at  $z \sim 2$ . Their galaxies have  $-21 \leq M_{\text{UV}} \leq -19$  and exhibit low dust content, with  $\beta$  slopes ranging from  $-1.9$  to  $-2.4$ . While their galaxies have similar properties to the LzLCS+ sample, their  $f_{\text{esc}}$  values are significantly lower ( $f_{\text{esc}} \leq 0.065$ ). Furthermore, the Ly $\alpha$   $f_{\text{esc}}$  for these targets is higher than that of local galaxies at the same Ly $\alpha$  EW. Based on the relation calibrated on Ly $\alpha$  properties using the LzLCS+ sample, at least half of their sources should be classified as LCEs. These

<sup>3</sup><https://github.com/sflury/kendall>

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findings suggest that the mechanisms governing LyC leakage may differ at high redshift, emphasizing the need for caution when applying low-redshift estimators to high-redshift sources. Indirect  $f_{\text{esc}}$  estimators like Ly $\alpha$  EW may have complex dependencies beyond just LyC escape, as they can also be influenced by Ly $\alpha$  production, the intrinsic Ly $\alpha$  EW, and factors like star formation history and continuum strength. Given that galaxy properties evolve with redshift, it remains uncertain how these changes impact diagnostic reliability for  $f_{\text{esc}}$ .

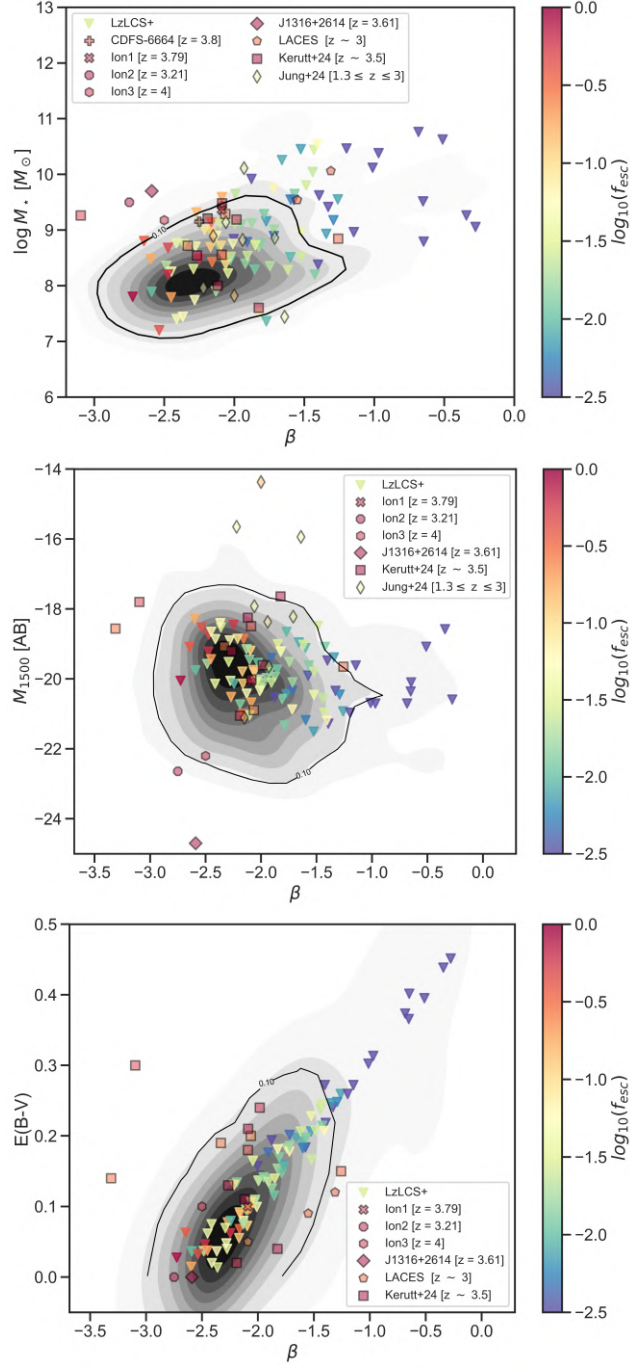
#### 7.4.2 A revised Cox model to predict $f_{\text{esc}}$ during the EoR

By identifying the subset of LzLCS+ galaxies that exhibit properties similar to high-redshift sources, we recalibrate the Cox proportional hazards model – a semi-parametric method that estimates the probability of an event (the detection of  $f_{\text{esc}}$ ) as a function of multiple variables (the various galaxy properties) [225, 226] – to more accurately predict  $f_{\text{esc}}$  during the EoR. This selection allows for a more reliable comparison and predictive framework, thereby improving our ability to model the LyC escape properties of galaxies in the early Universe. In the Appendix A: revised Cox models, we outline the methodology presented in [225].

The concordance index, defined as a measure to assess goodness-of-fit, ranges from 0 (perfect disagreement) to 0.5 (perfectly random) to 1.0 (perfect concordance) [e.g., 106]. From the models with a concordance coefficient  $C > 0.7$ , as identified by [226], we select those applicable to reionization-era sources. Specifically, we focus on two models that: (a) can be applied to all sources in our sample, and (b) do not include a morphology-dependent term. The exclusion of a morphology term aligns with the goal of this study, which is to investigate whether LyC escape is influenced by the morphology of galaxies and the presence of merger signatures with high  $f_{\text{esc}}$ . Since all the models in [226], including those with a morphology term, are statistically equivalent, this selection does not introduce any significant bias. Thus, we choose the ELG-EW and ELG-O32 models, which rely on key galaxy properties such as E(B-V),  $M_{\text{UV}}$ , stellar mass, and, respectively,  $\log_{10}([\text{OIII}]+\text{H}\beta)$  EWs and  $\log_{10}(\text{O32})$ .

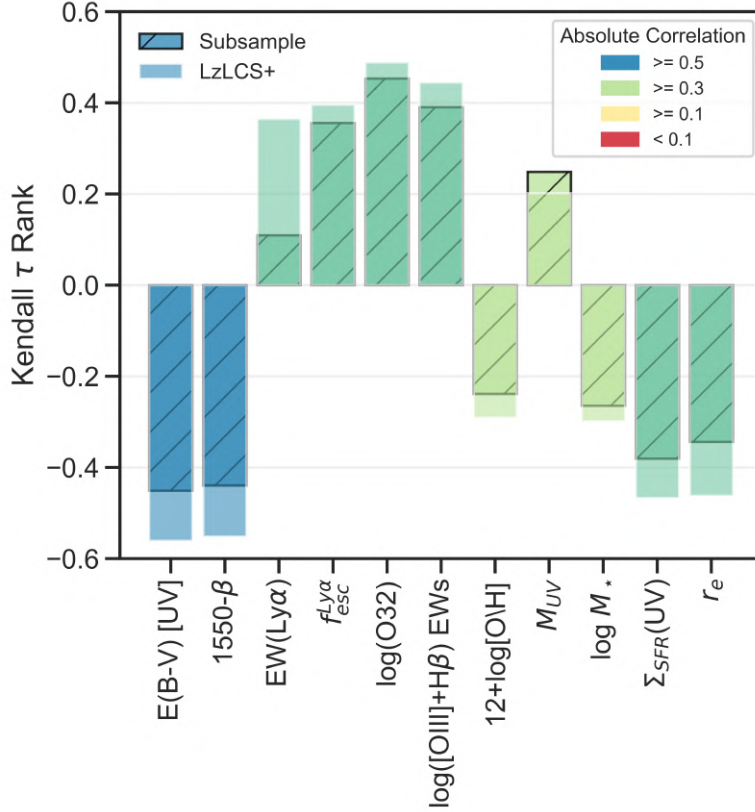
The ELG-EW model is particularly useful for sources where O32 cannot be computed due to the absence of one of the oxygen lines, while the ELG-O32 model provides an alternative for cases where both [OII] and [OIII] lines are available. Following the methodology outlined in Chapter 2, we computed the baseline hazard function  $\text{HF}_0(f_{\text{abs}})$  and the survival function  $S(f_{\text{abs}})$ , which allowed us to estimate the median  $f_{\text{abs}}$  (and consequently the median  $f_{\text{esc}}$ ), along with the 16th and 84th percentiles to establish confidence intervals for our predictions.

In Appendix A: revised Cox models, we provide two tables (Tables A.1 and A.2), one for each model, which, following [226], provide the goodness-of-fit statistics for the subsample of LzLCS+ galaxies we are considering, the fitted coefficients ( $\beta_i$ ) for each included variable, and the reference values  $\bar{x}_i$ , which are the mean of the  $x_i$  values of the “analogs”, where  $x_i$  represents the input variable ( $M_{\text{UV}}$ , E(B-V),  $\log_{10}(M_*/M_\odot)$ , and  $\log_{10}(\text{EW}([\text{OIII}]+\text{H}\beta)/\text{\AA})$  or  $\log_{10}(\text{O32})$ ). Finally, the tables list the baseline cumulative hazard function,  $\text{HF}_0$ , calculated by the `lifelines` Cox Model fitting routine [106] for each of the observed  $f_{\text{esc}}$  values for LCEs in the LzLCS+ subsample.



**Figure 7.6:** Comparison of galaxy properties between  $z \sim 0.3$  LyC emitters from the LzLCS+ [153] and high-redshift sources, shown as a density plot. Panels show  $\log M_*$  vs.  $\beta$  slope (upper),  $M_{UV}$  vs.  $\beta$  (middle), and  $E(B-V)$  vs.  $\beta$  (lower). Of the 88 LyC emitters, 51 galaxies are identified as analogs to high-redshift sources, falling within the 99% confidence interval in all panels. We also include known  $z \sim 3$  leakers from the literature: Ion1 [490], Ion2 [112, 487, 491], Ion3 [485], J1316+2614 [297], CDFS-6664 [523], LACES [152], and samples from [244] and [234]. All the known leakers are color-coded by their  $\log_{10}(f_{\text{esc}})$ .





**Figure 7.7:** Results of the Kendall  $\tau$  rank correlation analysis between various predictors identified by [226] and  $f_{\text{esc}}$  values for both the entire LzLCS+ sample (88 sources) and the subsample of 51 galaxies. The hatched bar represents the refined subsample, while the other corresponds to the full sample.

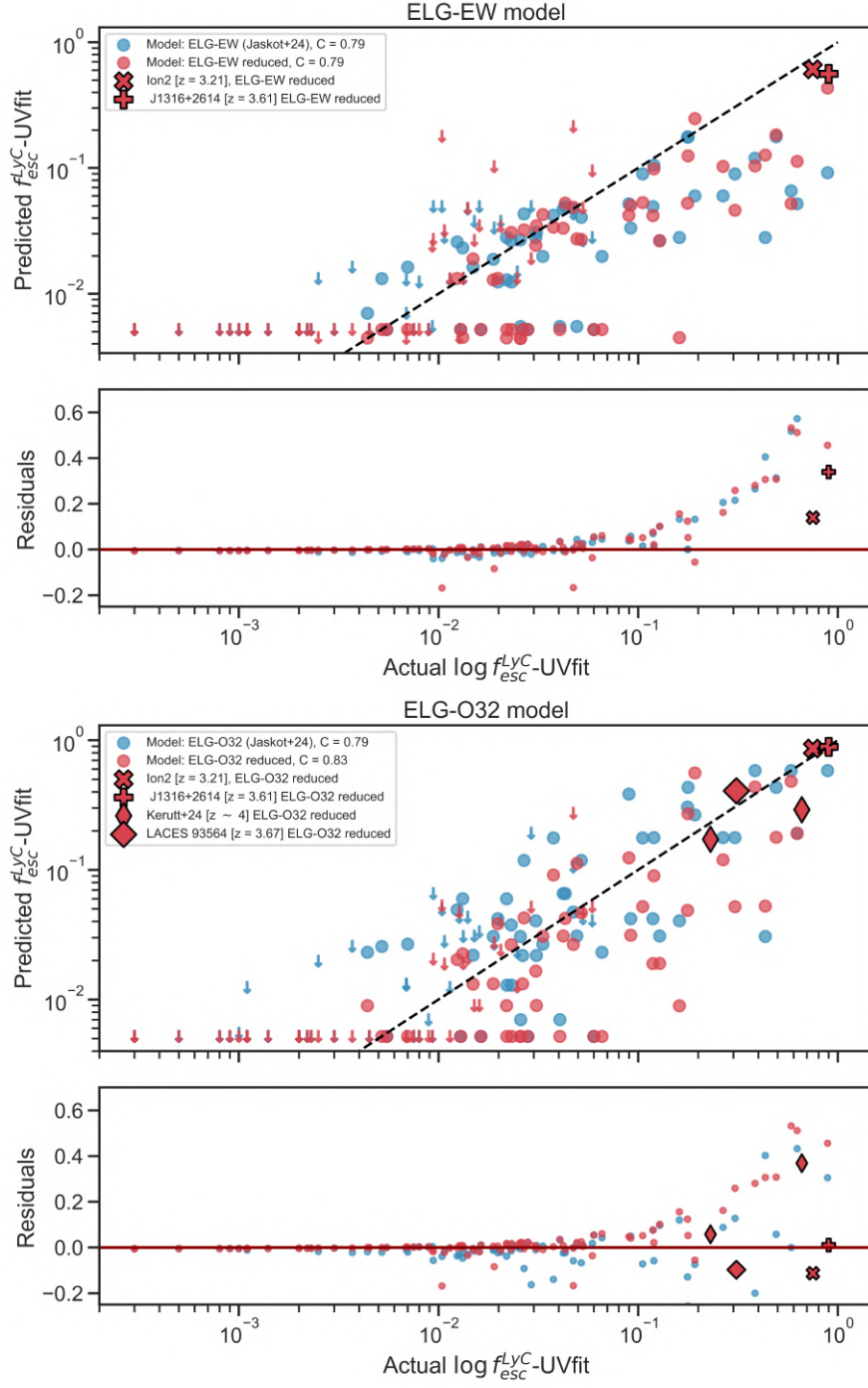
Fig. 7.8 illustrates the comparison between the predicted  $f_{\text{esc}}$  values using the full LzLCS+ sample and the subsample of galaxies that match the properties of reionization-era sources. Both predictions are consistent, with residuals showing similar trends across the two samples.

#### 7.4.3 Applying the model to known leakers at $z = 3$

We further test our model to assess its capacity to predict  $f_{\text{esc}}$  for galaxies at high redshift by applying it to known LyC leakers at  $z \sim 3$ . We compile samples of  $z \sim 3$  leakers with reported detections of global absolute LyC  $f_{\text{esc}}$  values that also include all the relevant input variables:  $E(B-V)$ , stellar mass,  $M_{\text{UV}}$ , and either O32 or [OIII]+H $\beta$  EWs. This results in two distinct samples at  $z \sim 3$ :

- **ELG-EW Sample:** This sample includes Ion2 [ $z = 3.2$ ,  $f_{\text{esc}} = 0.5\text{--}0.9$ , 112, 487, 491] and J1316+2614 [ $z = 3.6130$ ,  $f_{\text{esc}} \approx 0.9$ , 297, 299].
- **ELG-O32 Sample:** This sample consists of Ion2, J1316+2614, two sources from [244] that have been observed in the JADES program ( $z = 3.46$ ,  $f_{\text{esc}} = 0.23 \pm 0.05$  and  $z = 4.43$ ,  $f_{\text{esc}} = 0.69 \pm 0.10$ , respectively), and one source ( $z = 3.67$ ,  $f_{\text{esc}} = 0.31 \pm 0.03$ ) from the Lyman Continuum Escape Survey [LACES, 152].



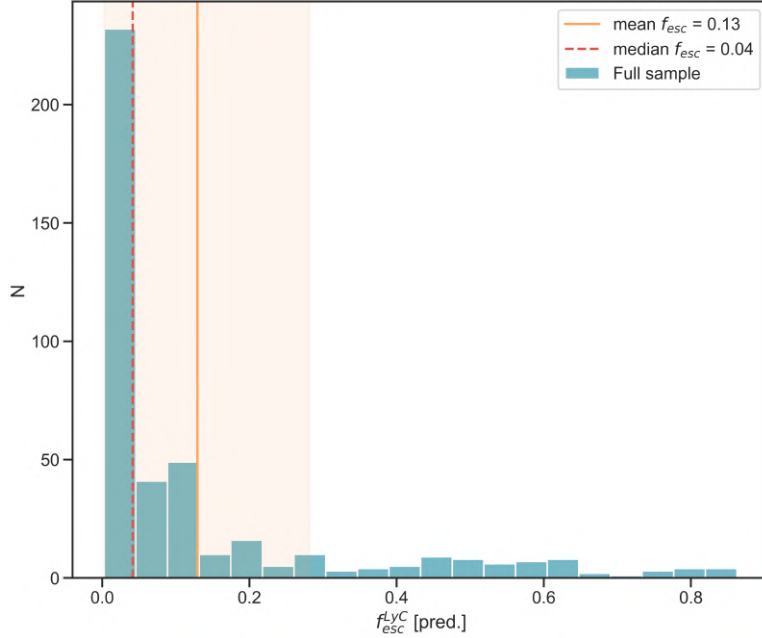


**Figure 7.8:** Detected (or upper limit)  $f_{\text{esc}}$  values versus the predicted  $f_{\text{esc}}$  values using Cox models, calibrated using the full LzLCS+ sample and the subsample of galaxies resembling reionization-era sources. Left panel: Predictions using the ELG-EW model. Right panel: Predictions using the ELG-O32 model. The Cox models from [226] are shown in blue, while the predictions calibrated on the subsample are in red. Predictions applied to known LyC leakers at  $z \sim 3$  are also shown in red [112, 152, 244, 297, 487, 491]. Upper limits are marked with downward-pointing arrows.

Our predictions, shown in Fig. 7.8, revealed a strong positive correlation with measured values at this intermediate redshift, with Pearson’s  $r = 0.812$ , indicating that our model effectively reflects the observed data. Although the  $p$ -value of 0.095 suggests marginal significance, this reinforces the validity of our predictions. A primary limitation of our assessment is that we are testing our model exclusively on strong leakers at  $z \sim 3$  (with  $f_{\text{esc}} \geq 0.2$ ), as these are the only detections available with the complete dataset of indirect indicators needed. Consequently, expanding the number of confirmed LyC leakers (or upper limits) at intermediate redshifts is essential for further refinement of our model.

## 7.5 Results

### 7.5.1 The predicted $f_{\text{esc}}$ values



**Figure 7.9:** Predicted  $f_{\text{esc}}$  distribution for the analyzed sources at  $5 \leq z \leq 7$ . The mean  $f_{\text{esc}}$  of the sample is shown in red and it is equal to  $0.13^{0.25}_{0.002}$ , where the errors show the 16-84th percentiles. The median  $f_{\text{esc}} = 0.04$  is presented in orange.

Using either the revised ELG-O32 or ELG-EW models, we predicted the  $f_{\text{esc}}$  values for most sources in our sample, successfully predicting all but 9 out of a total of 436 sources. The 9 sources for which predictions were not made lack the O32 ratio and the [OIII] + H $\beta$  EW. To obtain the most accurate estimates, we applied the revised ELG-O32 model (which has a concordance value of  $C = 0.83$ ) whenever possible. For sources without detectable [OII] or [OIII] lines, such as those in the EIGER sample, we applied the ELG-EW model ( $C = 0.79$ ).

From the total sample of 427 sources with predicted  $f_{\text{esc}}$ , 206 were predicted using the ELG-O32 model, while the remaining 221 sources were predicted using the ELG-EW model. The distribution of predicted  $f_{\text{esc}}$  values is shown in Fig. 7.9.

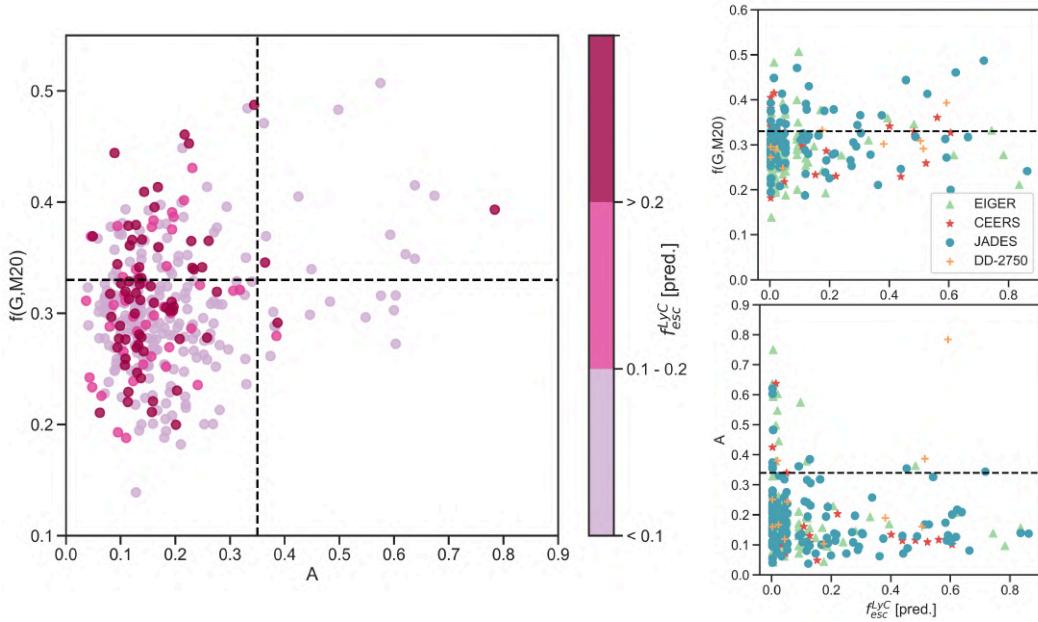
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Most of our galaxies exhibit modest inferred  $f_{\text{esc}}$  values, generally around 0.10 or below. The average  $f_{\text{esc}}$  for the sample, including the standard deviation as the 16th and 84th percentile values of the distribution, is  $0.13^{0.25}_{0.002}$ . This mean is skewed by the relatively high  $f_{\text{esc}}$  values ( $> 0.4$ ) inferred for a modest fraction (53/427) of our sources. As a result, the median provides a more representative statistic, with a value of 0.04.

This distribution aligns well with previous findings [300, 302], both in terms of the shape of the distribution, mean, and median values. The models used in those studies were derived through linear regression, incorporating key predictors such as the O32 ratio or the  $H\beta$  EW, the  $\beta$  slope, and the UV half-light radius  $r_e$ . Our findings are also broadly consistent with Lin et al. [273], who applied a logistic regression model to estimate the probability of a galaxy with  $M_{\text{UV}} < -18$  being a LCE, using  $M_{\text{UV}}$  and  $\beta$ , and O32. They found that at  $z = 8$ , the average  $f_{\text{esc}}$  varies with  $M_{\text{UV}}$ , peaking at intermediate luminosities ( $-19 < M_{\text{UV}} < -16$ ) and reaching about 0.04 for brighter galaxies ( $M_{\text{UV}} < -19$ ). Similarly, [413] observed a comparable trend in a smaller sample of  $\text{Ly}\alpha$  emitters, where a few sources exhibit high  $f_{\text{esc}}$  while the majority have  $f_{\text{esc}} \leq 0.10$ . Their model, based on the SPHINX<sup>20</sup> simulation [89], used six parameters ( $\beta$ , E(B-V),  $H\beta$  luminosity,  $M_{\text{UV}}$ , R23, and O32) to predict LyC escape. It is important to note that all these methods are independent subject to their own caveats – whether in the treatment of upper limits, assumptions in simulations, or the handling of non-detections as non-LCEs – with the commonality being that they were tested on the LzLCS+ sample (which only probes down to  $M_{\text{UV}} \sim -18$ ) and rely on a combination of multiple predictors. However, the models do not always align perfectly. The [89] model, derived from simulated galaxies, substantially disagrees with the LzLCS+ sample, as shown by [226], and also diverges from Cox’s predictions for  $z \geq 6$  galaxies. Therefore, while these methods may yield similar population averages, they differ in their predictions for individual galaxies. Despite these differences, the methods demonstrate the strength of multivariate predictions and the value of combining different diagnostics to achieve a more robust understanding of LyC escape.

### 7.5.2 Merger fraction versus $f_{\text{esc}}$

In our sample of galaxies at redshifts  $z = 5 - 7$ , we examine the presence of merger signatures and their connection to the predicted LyC escape fraction,  $f_{\text{esc}}$  (Fig. 7.10, left panel). By color-coding the predicted  $f_{\text{esc}}$  with respect to merger and non-merger signatures, we found that high  $f_{\text{esc}}$  sources are equally distributed between both groups. In particular, we focused on  $f_{\text{esc}}$  above 0.2 as a threshold of confidence. This choice is supported by our previous works [300, 302], which demonstrated that although there are uncertainties associated with these  $f_{\text{esc}}$  values, those within this range are more reliable. Specifically, 2 of the 13 mergers in the Gold sample ( $15 \pm 11\%$ ), 23 of 78 mergers in the Silver sample ( $29 \pm 7\%$ ) and 59 of the 345 non-mergers ( $17 \pm 8\%$ ) exhibit high LyC leakage. However, the larger uncertainties in some panels, particularly those with lower populations, prevent us from drawing firm conclusions. Furthermore, in Fig. 7.10, right panel, we observe no significant connection between morphological parameters  $A$  and  $f(G, M_{20})$  and



**Figure 7.10:** Left panel: Visualization of the galaxy sample, color-coded according to their predicted  $f_{\text{esc}}$  values in three ranges:  $f_{\text{esc}} < 0.10$  (pale violet),  $0.10 \leq f_{\text{esc}} \leq 0.20$  (magenta), and  $f_{\text{esc}} > 0.20$  (dark red). The spatial distribution follows the same layout as Fig. 7.4. Right panel: Trends for the different observational programs in our sample, showing the predicted  $f_{\text{esc}}$  as a function of the  $f(G, M_{20})$  and  $A$  values. This panel shows the relationship between these quantities and  $f_{\text{esc}}$ , independent of the morphological classification.

$f_{\text{esc}}$  (Pearson’s  $r$  of  $-0.08$  and  $-0.04$ , respectively, with corresponding  $p$ -values of  $0.17$  and  $0.46$ ). It is important to remember that the predictions in the two models we are using are independent of any morphological terms, in contrast to other models [225, 300, 302] previously used to predict  $f_{\text{esc}}$  during the EoR.

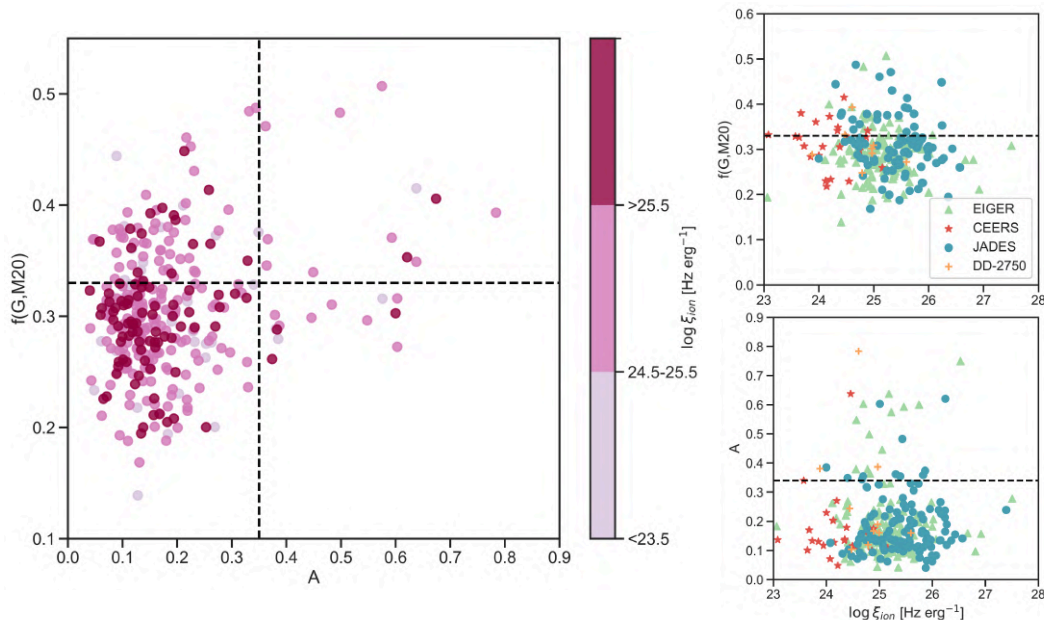
Despite the strengths of our classification, there are certain limitations that must be acknowledged. One key limitation is that our classification relies on rest-frame UV imaging, which may not adequately capture the presence of merger signatures at  $z \sim 7$ . Applying our classification to the optical rest-frame morphology might have resulted in a different outcome, although [477] found that galaxy morphologies remain consistent across wavelengths from the rest-frame optical to UV. Another limitation arises from the uncertainty in classifying merger versus non-merger candidates based solely on parametric classification. While we conducted careful visual inspections for the sources in the Gold and Silver samples, this method may introduce bias and overlook small morphological features indicative of ongoing interactions or mergers. Finally, it is important to consider the timing of mergers, as the formation of channels that facilitate LyC escape may occur specifically during the coalescence phase. This implies that what we identify in our sample could represent pre-merger conditions, rather than fully merging systems.

The results from [38] SPICE simulations support the idea that internal processes, rather than interactions, play a more critical role in driving LyC escape. Their simulations show that bursty stellar feedback can create low-density chan-

nels within galaxies that allow LyC photons to escape, regardless of whether the galaxy is undergoing a merger. This feedback-driven morphology, especially in dispersion-dominated systems, enhances LyC escape by producing irregular structures through internal processes rather than external interactions. In line with these findings, [155] present direct evidence for bursty star formation as a mechanism that enhances  $f_{\text{esc}}$  in LCEs compared to non-LCEs.

These findings align with our results, which show a low occurrence of merger signatures among possible cosmic reionizers. It appears that LyC escape at these redshifts is predominantly governed by intrinsic factors related to the compactness and isolated nature of these galaxies, as the majority of the sources in our sample exhibit compact and symmetric morphologies in their rest-frame UV [e.g., 274, 300, 302, 327]. This observation is consistent with results from the LzLCS+ survey, where [154] and [225] identified compactness as a strong predictor of  $f_{\text{esc}}$ , emphasizing its role as a key driver of the strong dependence of  $f_{\text{esc}}$  on  $\Sigma_{\text{SFR}}$ . Moreover, in a study of a leaker at  $z = 3.088$ , [184] discussed how optically thin channels, potentially created by merger-driven outflows, enable LyC escape. However, they caution that the complex outflow dynamics typical of high-redshift systems limit the likelihood of star-formation-driven feedback alone creating such channels.

### 7.5.3 Merger fraction versus $\xi_{\text{ion}}$



**Figure 7.11:** *Left panel: Visualization of the galaxy sample, color-coded according to their measured  $\log \xi_{\text{ion}}$  values in three ranges:  $23.5 \leq \log \xi_{\text{ion}} \leq 24.5$  (pale violet),  $24.5 \leq \log \xi_{\text{ion}} \leq 25.5$  (magenta), and  $\log \xi_{\text{ion}} > 25.5$  (dark red). The spatial distribution follows the same layout as Fig. 7.4. Right panel: analog of Fig. 7.10, but for  $\log \xi_{\text{ion}}$ .*

We also investigate the relationship between mergers and the ionizing emissivity,  $\xi_{\text{ion}}$ .

Mergers are typically thought to temporarily boost star formation rates (SFR) by compressing gas and triggering starbursts, which in turn could increase the production of ionizing photons [e.g., 24, 322]. Higher SFR typically corresponds to a younger, massive stellar population, which emits more ionizing photons due to its high luminosity in both the UV and ionizing continua. Specifically, a strong correlation between specific star formation rate (sSFR) and  $\xi_{\text{ion}}$  has been observed [75], making sSFR one of the most reliable indicators of a galaxy’s ionizing efficiency, especially in the context of cosmic reionization [e.g., 75, Llerena in prep.]. [511], studying a sample of Ly $\alpha$  emitters at  $z > 7$ , further support this connection, demonstrating that mergers can trigger episodic bursts of star formation, which in turn lead to high intrinsic Ly $\alpha$  emission. These bursts create low-density channels in the ISM, facilitating the escape of both Ly $\alpha$  and ionizing photons, and potentially enhancing  $\xi_{\text{ion}}$  during merging events. However, compact, low-mass galaxies with relatively high SFRs have also been shown to exhibit elevated  $\xi_{\text{ion}}$ , independent of whether they are undergoing mergers [75].

Our results are presented in Fig. 7.11, where we color code the galaxies, in the mergers space, by their value of  $\xi_{\text{ion}}$ . We can see that high  $\xi_{\text{ion}}$  values are distributed across both merging and non-merging morphologies. Specifically, we apply a threshold of  $\log \xi_{\text{ion}} = 25.5$ , and find that 2 of the 13 mergers in the Gold sample ( $15 \pm 11\%$ ), 16 of 78 mergers in the Silver sample ( $21 \pm 6\%$ ), and 100 of the 345 non-mergers ( $29 \pm 14\%$ ) have elevated ionizing emissivity. Moreover, in Fig. 7.11, right panel, our investigation do not reveal any clear correlation between morphological parameters and  $\xi_{\text{ion}}$  (Pearson’s  $r$  of -0.04 and 0.04, respectively, and  $p$ -values of 0.45 and 0.47). The absence of a strong correlation between merger signatures and higher  $\xi_{\text{ion}}$ , coupled with the similar distribution of ionizing emissivity in both classified mergers and non-mergers, suggests that merger morphologies do not influence the variations in  $\xi_{\text{ion}}$ .

## 7.6 Summary and Conclusions

- We assembled a sample of 436 spectroscopically confirmed sources at redshifts  $5 \leq z \leq 7$  from various JWST surveys. The majority of these sources are low-mass galaxies, with stellar masses around  $\sim 10^8 M_{\odot}$ , low dust content ( $E(B-V) \sim 0.07$ ), blue UV slopes ( $\beta \sim -2.16$ ), and ionizing photon production efficiencies of  $\log \xi_{\text{ion}} \sim 25.2$ .
- We found that low and intermediate redshift LyC emitters do not always serve as analogs for cosmic reionizers. In our analysis, we identified a subsample of 51 LzLCS+ emitters that act as analogs for the sources we are characterizing during the EoR.
- Notably, within this subsample, there is no significant correlation between the  $f_{\text{esc}}$  and the Ly $\alpha$  EW. This suggests that the mechanisms driving LyC leakage may differ at high redshift, highlighting the need for caution when applying low-redshift estimators to high-redshift sources.
- Utilizing this subsample of analogs, we propose two refined Cox models to predict  $f_{\text{esc}}$  during the EoR, based on  $M_{\text{UV}}$ ,  $E(B-V)$ ,  $\log_{10}(M_{*})$ , and  $\log_{10}(\text{EW}([\text{OIII}] + \text{H}\beta))$  or  $\log_{10}(\text{O32})$ .

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- We applied these models to a few confirmed LyC leakers at  $z = 3$  and found that they accurately predict the escape fractions of these sources.
  - When we applied the new models to the galaxies observed during the EoR, we discovered that the majority exhibit modest escape fractions (approximately 0.04). This reinforces previous conclusions [302], suggesting that these galaxies contributed less than one-third to the overall reionization, regardless of the presence or absence of mergers. Further exploration of the faint, low-mass galaxy population at high redshift is essential to fully understand their role in cosmic reionization.
  - Our analysis reveals no strong correlation between predicted  $f_{\text{esc}}$  and the presence of merger signatures, with most galaxies in the sample exhibiting compact, symmetric morphologies rather than disturbed or irregular structures typically associated with mergers.
  - The  $\log \xi_{\text{ion}}$  values remain within expected ranges for galaxies in our sample, showing no significant influence from mergers.

One of the major challenges in this work is the limited statistical sample of LyC leakers at intermediate redshift. At  $z \sim 3$ , the transmission of the IGM becomes highly stochastic, allowing for the detection of only the strongest LyC leakers. This highlights the importance of building a statistically significant sample of leakers at  $z \sim 3$ . The LyC22 program (GO 1869, PI D. Schaerer) will provide the first robust sample of LCEs at the highest redshift where LyC can still be directly detected, offering a unique opportunity to recalibrate the Cox model at  $z \sim 3$ . This will significantly enhance the predictive power of  $f_{\text{esc}}$  during the epoch of reionization.

## Appendix A: revised Cox models

In Tables A.1 and A.2, we provide the information necessary to predict  $f_{\text{esc}}$  using the models discussed in this paper.

Table A.1: Revised ELG-EW Model

| $C = 0.79$                                |                        |                          |                |
|-------------------------------------------|------------------------|--------------------------|----------------|
| Model Parameters                          |                        |                          |                |
| Variable                                  | $b_i$                  | $\bar{x}_i$              |                |
| $M_{1500}$                                | -0.69                  | -19.649                  |                |
| $\log_{10}(M_*/M_\odot)$                  | -0.31                  | 8.559                    |                |
| $E(B-V)_{UV}$                             | -13.25                 | 0.103                    |                |
| $\log_{10}(EW([OIII]+H\beta)/\text{\AA})$ | 3.28                   | 2.941                    |                |
| Baseline Cumulative Hazard                |                        |                          |                |
| $f_{\text{abs}}$                          | $HF_0(f_{\text{abs}})$ | $f_{\text{abs}}$ (cont.) | $HF_0$ (cont.) |
| 0.111                                     | 0.011825               | 0.9732                   | 0.72714        |
| 0.3753                                    | 0.025212               | 0.9736                   | 0.790193       |
| 0.4162                                    | 0.039181               | 0.9752                   | 0.790193       |
| 0.5089                                    | 0.053583               | 0.9769                   | 0.855688       |
| 0.5667                                    | 0.069132               | 0.978                    | 0.926178       |
| 0.6948                                    | 0.085736               | 0.9795                   | 0.926178       |
| 0.7341                                    | 0.102827               | 0.9802                   | 1.004294       |
| 0.8079                                    | 0.120801               | 0.981                    | 1.004294       |
| 0.8223                                    | 0.141086               | 0.9812                   | 1.110345       |
| 0.8719                                    | 0.16315                | 0.9837                   | 1.221632       |
| 0.8803                                    | 0.185655               | 0.984                    | 1.221632       |
| 0.8815                                    | 0.209659               | 0.9849                   | 1.221632       |
| 0.8947                                    | 0.234525               | 0.986                    | 1.221632       |
| 0.9083                                    | 0.260956               | 0.9867                   | 1.221632       |
| 0.9102                                    | 0.288869               | 0.9868                   | 1.444606       |
| 0.9411                                    | 0.288869               | 0.9876                   | 1.688703       |
| 0.9471                                    | 0.288869               | 0.9886                   | 1.688703       |
| 0.9481                                    | 0.322321               | 0.9893                   | 1.688703       |
| 0.9507                                    | 0.356882               | 0.9906                   | 1.688703       |
| 0.9527                                    | 0.392632               | 0.9911                   | 1.688703       |
| 0.9569                                    | 0.430956               | 0.9931                   | 1.688703       |
| 0.9579                                    | 0.472974               | 0.9955                   | 1.688703       |
| 0.9624                                    | 0.517262               | 0.9956                   | 3.460356       |
| 0.9667                                    | 0.564117               | 0.9975                   | 3.460356       |
| 0.9691                                    | 0.614575               |                          |                |
| 0.9693                                    | 0.668437               |                          |                |
| 0.9709                                    | 0.668437               |                          |                |



Table A.2: Revised ELG-O32 Model

| $C = 0.83$                 |                               |                          |                       |
|----------------------------|-------------------------------|--------------------------|-----------------------|
| Model Parameters           |                               |                          |                       |
| Variable                   | $b_i$                         | $\bar{x}_i$              |                       |
| $M_{1500}$                 | -1.63                         | -19.649                  |                       |
| $\log_{10}(M_*/M_\odot)$   | -0.76                         | 8.559                    |                       |
| $E(B-V)_{UV}$              | -13.87                        | 0.104                    |                       |
| $\log_{10}(O32)$           | 6.55                          | 0.843                    |                       |
| Baseline Cumulative Hazard |                               |                          |                       |
| $f_{\text{abs}}$           | $\text{HF}_0(f_{\text{abs}})$ | $f_{\text{abs}}$ (cont.) | $\text{HF}_0$ (cont.) |
| 0.111                      | 0.003188                      | 0.9732                   | 0.732829              |
| 0.3753                     | 0.009485                      | 0.9736                   | 0.832722              |
| 0.4162                     | 0.016245                      | 0.9752                   | 0.832722              |
| 0.5089                     | 0.024456                      | 0.9769                   | 0.939902              |
| 0.5667                     | 0.033316                      | 0.978                    | 1.057705              |
| 0.6948                     | 0.042431                      | 0.9795                   | 1.057705              |
| 0.7341                     | 0.051797                      | 0.9802                   | 1.188196              |
| 0.8079                     | 0.061708                      | 0.981                    | 1.188196              |
| 0.8223                     | 0.0772                        | 0.9812                   | 1.381873              |
| 0.8719                     | 0.096801                      | 0.9837                   | 1.590329              |
| 0.8803                     | 0.116622                      | 0.984                    | 1.590329              |
| 0.8815                     | 0.137886                      | 0.9849                   | 1.590329              |
| 0.8947                     | 0.159401                      | 0.986                    | 1.590329              |
| 0.9083                     | 0.182399                      | 0.9867                   | 1.590329              |
| 0.9102                     | 0.206052                      | 0.9868                   | 1.913137              |
| 0.9411                     | 0.206052                      | 0.9876                   | 2.32817               |
| 0.9471                     | 0.206052                      | 0.9886                   | 2.32817               |
| 0.9481                     | 0.238704                      | 0.9893                   | 2.32817               |
| 0.9507                     | 0.273793                      | 0.9906                   | 2.32817               |
| 0.9527                     | 0.315661                      | 0.9911                   | 2.32817               |
| 0.9569                     | 0.359126                      | 0.9931                   | 2.32817               |
| 0.9579                     | 0.406118                      | 0.9955                   | 2.32817               |
| 0.9624                     | 0.455871                      | 0.9956                   | 4.960163              |
| 0.9667                     | 0.51674                       | 0.9975                   | 4.960163              |
| 0.9691                     | 0.58189                       |                          |                       |
| 0.9693                     | 0.648165                      |                          |                       |
| 0.9709                     | 0.648165                      |                          |                       |



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## Chapter 8

# Conclusions and future perspectives

The Epoch of Reionization (EoR) represents a key phase in the Universe’s history, marking the transition from a neutral Intergalactic Medium (IGM) to a nearly fully ionized state, driven by ionizing photons from early luminous sources. In this thesis, we have explored the properties and mechanisms of Lyman Continuum-emitting galaxies (LCEs) across different cosmic epochs, with the final aim of understanding of the role of star-forming galaxies in the reionization. Specifically, the primary focus of this work has been to understand how we can estimate the LyC escape fraction ( $f_{\text{esc}}$ ) in the early Universe, when direct LyC observations are hindered by the IGM opacity.

To address these challenges, we first studied galaxies at intermediate redshift ( $z \sim 2 - 4$ ), which can serve as analogs for cosmic ionizers. At these redshifts, we can directly investigate both the Lyman Continuum (LyC) emission and the physical and spectroscopic properties of these galaxies.

In particular, we first conducted an in-depth study of star-forming galaxies at redshifts  $2.5 \leq z \leq 5$  that exhibit significant nebular CIV emission, selected from the VANDELs public spectroscopic survey. These CIV emitters represent a small fraction – around 4% – of the VANDELs star-forming galaxy population within this redshift range. The rest-frame EWs of these CIV emitters span from 1 to 29 Å. The strength of CIV emission correlates well with other UV emission lines (HeII, OIII, and CIII]) suggesting a consistent origin of ionization across these features. Using UV BPT diagrams, we identified the ionizing source in most of the CIV emitters as stellar photoionization rather than AGN activity. Out of the 43 initial CIV emitters, only four are found in the AGN region of these diagnostic diagrams. For ten additional sources, suspected to potentially host AGN, we used spectral stacking analysis and confirmed their classification as star-forming galaxies. This brings the final sample of CIV-emitting star-forming galaxies to 39. We employed BEAGLE to simultaneously fit photometric data and spectral indices, deriving physical properties such as stellar mass, star formation rate (SFR), and ionizing photon production efficiency ( $\xi_{\text{ion}}$ ). The stellar metallicity of these sources was also estimated following a method based on weak absorption lines. Our results indicate that the CIV emitters are low-mass galaxies residing on the main sequence, with metallicity values comparable to the general star-forming galaxy population at similar redshifts. Importantly, we found that CIV emitters exhibit a higher average  $\xi_{\text{ion}}$  compared to galaxies without CIV emission, high-

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lighting their elevated ionizing photon production. There are multiple indications that these CIV emitters may also have high LyC  $f_{\text{esc}}$ , for example the presence of strong Ly $\alpha$  emission and a high CIV/CIII] ratio, which recent studies have suggested as a promising indicator of LyC leakage. In particular, we identified 12 galaxies that stand out as likely analogs to cosmic reionizers, characterized by  $\log \xi_{\text{ion}} > 25 \text{ erg}^{-1}\text{Hz}$ ,  $EW(\text{Ly}\alpha) > 20 \text{ \AA}$ , and  $\text{CIV/CIII]} > 0.75$ , which are all indicative of high LyC escape fractions.

In a similar fashion, we then studied extreme [OIII] emitters also at cosmic noon, to understand their link to cosmic ionizers. We used deep NIRISS/WFSS data from the GLASS-ERS program to explore the rest-frame optical emission lines of star-forming galaxies at  $1 < z < 3.4$  in the Abell 2744 cluster field. We detected line emission in 89% of the targets with extracted spectra, identifying a population of extreme [OIII] emitters with  $EW([\text{OIII}]\lambda 5007) \geq 750 \text{ \AA}$ , accounting for 12% of the sample. Their strong [OIII] and H $\beta$  emission lines indicate the presence of very young, high-energy stellar populations capable of producing the ionizing photons necessary to reionize the Universe. The abundance of these emitters in our sample is higher than in previous studies, likely due to the increased sensitivity of JWST, which allows for the detection of fainter sources and higher redshift galaxies than was possible with HST/WFSS. These findings demonstrate the capability of JWST to uncover rare populations of extreme emitters, advancing our understanding of their role in the ionizing budget during cosmic reionization.

In the second part on this thesis, we employed indirect diagnostics to infer LyC  $f_{\text{esc}}$  for galaxies during the EoR, to estimate their contribution to the total ionizing photon budget.

We initially studied a small sample of 29 gravitationally lensed galaxies observed with JWST/NIRSpec in the Abell 2744 cluster field at  $4.5 \leq z \leq 8$ . Through a combined analysis of NIRSpec and NIRCам data, we were able to accurately derive key physical properties of these galaxies, including stellar mass ( $M_{\star}$ ), SFR, effective radius ( $r_e$ ), UV- $\beta$  slopes, and prominent optical emission lines such as [OII], [OIII], H $\beta$ , and H $\alpha$ . Our sample consists solely of star-forming galaxies, as confirmed by the MEx diagram which excludes any AGN. The galaxies are characterized by blue UV slopes (median  $\beta = -2.08$ ) and are compact, with  $r_e \simeq 0.2 - 0.5 \text{ kpc}$ , consistent with the general population of Lyman-break galaxies at similar redshift and faint UV magnitudes. Our galaxies show properties (O32,  $EW_0(\text{H}\beta)$ , UV- $\beta$  slopes,  $r_e$ , and  $\Sigma_{\text{SFR}}$ ) consistent with low-redshift galaxies known to have moderate measured LyC escape fractions ( $f_{\text{esc}} \geq 0.05$ ). Assuming that the mechanisms facilitating the escape of LyC photons remain consistent across cosmic epochs, we derived a relation between  $f_{\text{esc}}$  and the properties which are most correlated to LyC emission, namely O32, UV- $\beta$ , and  $r_e$  in the Low-redshift LyC Survey. We then applied this relation at high redshift and we found that 20 out of 24 galaxies in our sample have  $f_{\text{esc}}$  values greater than 0.05, with an average  $f_{\text{esc}}$  of 0.12. These results suggest that the average low-mass galaxies near the EoR exhibit physical and spectroscopic properties consistent with moderate ionizing photon escape fractions.

To put our results on more solid statistical grounds, we then assembled a larger sample of 82 spectroscopically confirmed star-forming galaxies at  $6 \leq z \leq 9$

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mostly from the CEERS survey. Similarly to our previous work, we estimated  $f_{\text{esc}}$  values using two empirical relations derived from indirect indicators identified in the Low-Redshift Lyman Continuum Survey, which are measurable during the EoR. The majority of the sources display modest  $f_{\text{esc}}$  values, with an average  $f_{\text{esc}} = 0.13 \pm 0.02$  and a median of  $f_{\text{esc}} = 0.08 \pm 0.02$ . Approximately 20% of galaxies show extreme LyC escape values ( $f_{\text{esc}} > 0.2$ ). While  $f_{\text{esc}}$  shows a modest correlation with stellar mass – where lower-mass galaxies tend to have slightly higher escape fractions – the trend is scattered, and there is no significant dependence on  $M_{\text{UV}}$ . A significant discrepancy arises between our inferred  $f_{\text{esc}}$  values and those predicted by cosmological simulations, which generally suggest much lower escape fractions for galaxies in the same mass range as observed with JWST. This inconsistency could be due to the limitations of simulations in capturing the bursty nature of star formation or their limited resolution. Alternatively, it could indicate that applying low-redshift-derived  $f_{\text{esc}}$  relations to high-redshift galaxies overestimates the escape fraction. Additionally, our results show only a modest increase in  $f_{\text{esc}}$  with redshift, from 0.11 at  $z = 5$  to 0.14 at  $z = 8$ . Interestingly, we found no clear correlation between the inferred  $f_{\text{esc}}$  and the presence of Ly $\alpha$  emission during the EoR, likely because Ly $\alpha$  is affected by the opacity of the IGM in high-redshift galaxies, unlike at lower redshifts where galaxy properties dominate. Finally, using our inferred  $f_{\text{esc}}$  and  $\xi_{\text{ion}}$  values, we estimate the total ionizing emissivity to be  $\log \dot{n}_{\text{ion}} = 50.50 \pm 0.38 \text{ s}^{-1} \text{ Mpc}^{-3}$  at  $z = 8$  and  $\log \dot{n}_{\text{ion}} = 50.75 \pm 0.35 \text{ s}^{-1} \text{ Mpc}^{-3}$  at  $z = 6$ , values sufficient to sustain ionization in the Universe. However, galaxies brighter than  $M_{\text{UV}} = -18$ , which JWST currently observes, contribute less than 35% to the total ionizing emissivity.

In the final chapter, leveraging data from several JWST surveys, we assembled a sample of 436 predominantly low-mass ( $\sim 10^8 M_{\odot}$ ), star-forming galaxies, with typical  $M_{\text{UV}} \sim -20.5$ , at redshifts  $5 \leq z \leq 7$  to try to assess the relation between the ionizing properties and the morphology. We re-derived the LyC predictions, this time using Cox proportional hazards models based only on a subsample of the Low-Redshift Lyman Continuum Survey that are really analogs of EoR galaxies. We confirmed that early galaxies likely have modest  $f_{\text{esc}}$  values around 0.04. Our morphological analysis reveals that mergers may not be a significant driver of LyC leakage at high-redshift. Instead, compactness and high star formation rate surface densities appear more pivotal especially in driving high values of  $\xi_{\text{ion}}$ .

Building on our results, we can foresee several avenues for future research that could further improve our understanding of LCEs and their role in cosmic reionization.

Expanding the sample of confirmed LyC leakers at  $2 \leq z \leq 4.5$  is clearly the most urgent task. Currently, we are basing our predictions on the assumption that the physical mechanisms and conditions facilitating LyC photon escape are consistent over cosmic time. However, with 10 billion years between  $z \sim 0$  and the reionization epoch, this assumption may not hold true. Confirming that the correlation between  $f_{\text{esc}}$  and the multivariate predictors work at least up to  $z \sim 4$  (less than 1 billion years after the end of reionization) will strengthen our conclusions. Moreover, as shown in this thesis, not all low-redshift leakers are analogs of the cosmic reionizers. Increasing this sample would also enable a more precise

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analysis of how indirect indicators of the  $f_{\text{esc}}$  evolve with redshift. While Ly $\alpha$  has been ruled out as a reliable diagnostic, exploring additional indirect indicators remains essential. In this respect, the LyC22 JWST program (GO 1869, PI D. Schaerer) will provide JWST/NIRSpec spectroscopic measurements, along with additional properties derived from HST, SUBARU, and JWST observations for a sample of nearly 100 sources at  $z \sim 3$  for which we have both direct detections and upper limits on  $f_{\text{esc}}$  from existing HST or Keck observations.

In parallel, future studies should prioritize developing models that reveal how channels in the interstellar medium form to allow LyC photons to escape. Known LyC leakers display a range of morphologies, underscoring the importance of identifying which escape mechanisms prevail, under what conditions, and which models best represent cosmic reionizers. By pinpointing the dominant escape processes across galaxy types, we can better determine which galaxies most effectively contributed to reionization. For this purpose, the The Lyman-alpha and Continuum Origins Survey (LaCOS) HST program (GO 17069, PI: M. Hayes), will conduct spatially resolved, panchromatic imaging of a subsample of sources from the Low-Redshift Lyman Continuum Survey. This program will map the stellar populations, dust obscuration, star-formation history, and Ly $\alpha$  emission, providing detailed insights into the ISM's physical state and its regulation of LyC transmission.

In this thesis, we found that bright galaxies likely contributed less than 35% to the ionizing emissivity, though this estimate assumes an average  $f_{\text{esc}}$  of 0.1 for faint galaxies – a hypothesis that requires testing by probing fainter magnitudes. JWST is the best instrument to detect extremely faint galaxies (down to  $M_{\text{UV}} > -18$ ) during the EoR, especially when coupled with gravitational lensing. The ALT JWST program (GO 3516, PIs: J. Matthee and R. Naidu) is delivering a substantial sample of lensed, faint ( $M_{\text{UV}} \sim -16$ ) galaxies in the Abell 2744 cluster field from the reionization era, with NIRCAM/WFSS providing [OIII] and H $\beta$  detections. Follow-up observations to obtain their rest-frame UV and optical data would supply the complete set of indirect indicators necessary to predict  $f_{\text{esc}}$  and refine their contribution to the ionizing photon budget.

While this thesis assumes that AGNs had a secondary role in reionization, primarily due to their lower numbers compared to star-forming galaxies, recent JWST observations have hinted at a higher than expected number density of AGNs at high redshift. This raises critical questions regarding their contribution to reionization. Future studies should examine whether AGNs played a more significant role, particularly given the changing AGN population beyond  $z > 4$ .

Finally, current simulations of cosmic reionization typically stop around  $z \sim 5$ , when reionization is thought to be nearly complete. However, these models do not fully capture how galaxies evolve beyond this point, especially down to  $z \leq 4.5$ , where LyC can be observed directly. Extending these simulations to lower redshifts will help bridge gaps in our understanding of post-reionization evolution and provide a more continuous picture of how galaxies transition from early reionizers to their lower-redshift analogs.

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# List of publications

This section provides a comprehensive list of all my scientific articles to date, including both first-author and contributing-author papers that have been published, submitted, or are in preparation.

## First Author/Major Contribution

1. Y. Liu, **S. Mascia**, et al., Two Lyman Continuum leaker candidates at  $z \sim 3$  in the Abell 2744 Cluster Field, (in prep.)
2. **S. Mascia**, et al., Little impact of mergers and galaxy morphology on the production and escape of ionizing photons in the early Universe, (in prep.)
3. **S. Mascia**, et al., The GLASS-JWST Early Release Science Program. IV. Data release of 263 spectra from 245 unique sources, *Astronomy and Astrophysics*, 690, A2 (2024)
4. **S. Mascia**, et al., New insight on the nature of cosmic reionizers from the CEERS survey, *Astronomy and Astrophysics*, 685, A3 (2024)
5. G. Prieto-Lyon, C. Mason, **S. Mascia**, et al., Early Results from GLASS-JWST. XXIII. The Transmission of  $\text{Ly}\alpha$  from UV-faint  $z$  3-6 Galaxies, *The Astrophysical Journal*, 956, 136 (2023)
6. **S. Mascia**, et al., Closing in on the sources of cosmic reionization: first results from the GLASS-JWST program, *Astronomy and Astrophysics*, 672, A155 (2023)
7. **S. Mascia**, et al., Insights into the reionization epoch from cosmic-noon C IV emitters in the VANDELS survey, *Astronomy and Astrophysics*, 674, A221 (2023)
8. K. Boyett, **S. Mascia**, et al., Early results from GLASS-JWST. VI. Extreme rest-optical equivalent widths detected in NIRISS Wide Field Slitless Spectroscopy, *The Astrophysical Journal*, 940, L52 (2022)

## Other Publications

1. B. Vulcani,..., **S. Mascia**, et al., Not just  $\text{PAH}_{3.3}$ : why galaxies turn red in the Near-Infrared, submitted to *A&A* (October 2024)
2. L. Napolitano,..., **S. Mascia**, et al., The dual nature of GHZ9: a remote X-ray source balancing AGN and star formation at  $z=10.145$ , submitted to *ApJ* (October 2024)

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3. L. Napolitano,..., **S. Mascia**, et al., Seven wonders of Cosmic Dawn: JWST confirms a high abundance of galaxies and AGNs at  $z \simeq 9 - 11$  in the GLASS field, submitted to A&A (October 2024)
  4. E. Merlin,..., **S. Mascia**, et al., ASTRODEEP-JWST: NIRCam-HST multi-band photometry and redshifts for half a million sources in six extragalactic deep fields, accepted, A&A (October 2024)
  5. H. Weida,..., **S. Mascia**, et al., Characterizing the Average Interstellar Medium Conditions of Galaxies at  $z \sim 5.6 - 9$  with Ultraviolet and Optical Nebular Lines, *The Astrophysical Journal*, 971, 21, (2024)
  6. A. Calabrò,..., **S. Mascia**, et al., Evidence of extreme ionization conditions and low metallicity in GHZ2/GLASS-z12 from a combined analysis of NIR-Spec and MIRI observations, accepted, *ApJ* (2024)
  7. M. Castellano,..., **S. Mascia**, et al., JWST NIRSpec Spectroscopy of the Remarkable Bright Galaxy GHZ2/GLASS-z12 at Redshift 12.34, *The Astrophysical Journal*, 972, 143 (2024)
  8. A. Calabrò,..., **S. Mascia**, et al., The evolution of the SFR and  $\Sigma$ SFR of galaxies in cosmic morning ( $4 < z < 10$ ), *Astronomy and Astrophysics*, 690, A290 (2024)
  9. L. Napolitano,..., **S. Mascia**, et al., Peering into cosmic reionization: the Ly $\alpha$  visibility evolution from galaxies at  $z = 4.5 - 8.5$  with JWST, *Astronomy and Astrophysics*, 688, A106 (2024)
  10. J. W. Cole,..., **S. Mascia**, et al., CEERS: Increasing Scatter along the Star-Forming Main Sequence Indicates Early Galaxies Form in Bursts, accepted, *Astronomy and Astrophysics* (2023)
  11. K. Boyett,..., **S. Mascia**, et al., A massive interacting galaxy 525 million years after the Big Bang, *Nat Astron* (2024)
  12. A. Calabrò,..., **S. Mascia**, et al., Near-infrared emission line diagnostics for AGN from the local Universe to  $z \sim 3$ , *Astronomy and Astrophysics* 679, A80 (2023)
  13. L. Napolitano,..., **S. Mascia**, et al., Identifying Ly $\alpha$  emitter candidates with Random Forest: learning from galaxies in CANDELS survey, *Astronomy and Astrophysics* 677, A138 (2023)
  14. M. Castellano,..., **S. Mascia**, et al., The ionizing photon production efficiency of bright  $z \sim 2-5$  galaxies, *Astronomy and Astrophysics*, 675, A121 (2023)
  15. N. Roy,..., **S. Mascia**, et al., Early Results from GLASS-JWST XXII: Rest frame UV-optical spectral properties of Lyman-alpha emitting galaxies at  $3 < z < 6$ , *The Astrophysical Journal Letters*, 952, L14 (2023)
  16. M. Talia,..., **S. Mascia**, et al., The VANDELS ESO public spectroscopic survey: the spectroscopic measurements catalogue, *Astronomy and Astrophysics*, 678, A25 (2023)
  17. T. Jones,..., **S. Mascia**, et al., Early Results from GLASS-JWST. XXI: Rapid assembly of a galaxy at  $z = 6.23$  revealed by its C/O abundance, *The Astrophysical Journal*, 951, L17 (2023)



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18. D. Paris,..., **S. Mascia**, et al., The GLASS-JWST Early Release Science Program. II. Stage I release of NIRCам imaging and catalogs in the Abell 2744 region, *The Astrophysical Journal*, Volume 952, 20 (2023)
  19. B. Vulcani,..., **S. Mascia**, et al., Early results from GLASS-JWST. Unveiling a population of “red-excess” galaxies in Abell2744 and in the coeval field, *The Astrophysical Journal*, 948, L15 (2023)
  20. M. Castellano,..., **S. Mascia**, et al., A High Density of Bright Galaxies at  $z \approx 10$  in the A2744 region, *The Astrophysical Journal*, 948, L14 (2023)
  21. T. Morishita,..., **S. Mascia**, et al., Early results from GLASS-JWST. XVIII: A spectroscopically confirmed protocluster 650 million years after the Big Bang, *The Astrophysical Journal*, 947, L24 (2023)
  22. G. Prieto-Lyon,..., **S. Mascia**, et al., The production of ionizing photons in UV-faint  $z \sim 3-7$  galaxies, *Astronomy and Astrophysics*, 672, A186 (2023)
  23. T. Treu,..., **S. Mascia**, et al., Early results from GLASS-JWST. XII: the morphology of galaxies at the epoch of reionization, *The Astrophysical Journal*, 942, L28 (2023)
  24. P. Santini,..., **S. Mascia**, et al., Early results from GLASS-JWST. XI: Stellar masses and mass-to-light ratio of  $z > 7$  galaxies, *The Astrophysical Journal*, 942, L27 (2023)
  25. N. Leethochawalit,..., **S. Mascia**, et al., Early results from GLASS-JWST. X: Rest-frame UV-optical properties of galaxies at  $7 < z < 9$ , *The Astrophysical Journal*, 942, L26 (2023)
  26. A. Calabrò,..., **S. Mascia**, et al., The properties of the Interstellar Medium in star-forming galaxies at redshifts  $2 < z < 5$  from the VANDELS survey, *Astronomy and Astrophysics*, 667, A117 (2022)
  27. M. Castellano,..., **S. Mascia**, et al., Early results from GLASS-JWST. III: Galaxy candidates at  $z \sim 9-15$ , *The Astrophysical Journal*, 938, L15 (2022)
  28. G. Roberts-Borsani,..., **S. Mascia**, et al., Early Results from GLASS-JWST. I: Confirmation of Lensed  $z > 7$  Lyman-Break Galaxies Behind the Abell 2744 Cluster With NIRISS, *The Astrophysical Journal*, 938, L13 (2022)
  29. A. Saxena,..., **S. Mascia**, et al., Strong C IV emission from star-forming galaxies: a case for high Lyman continuum photon escape, *Monthly Notices of the Royal Astronomical Society*, 517, 1098 (2022)
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